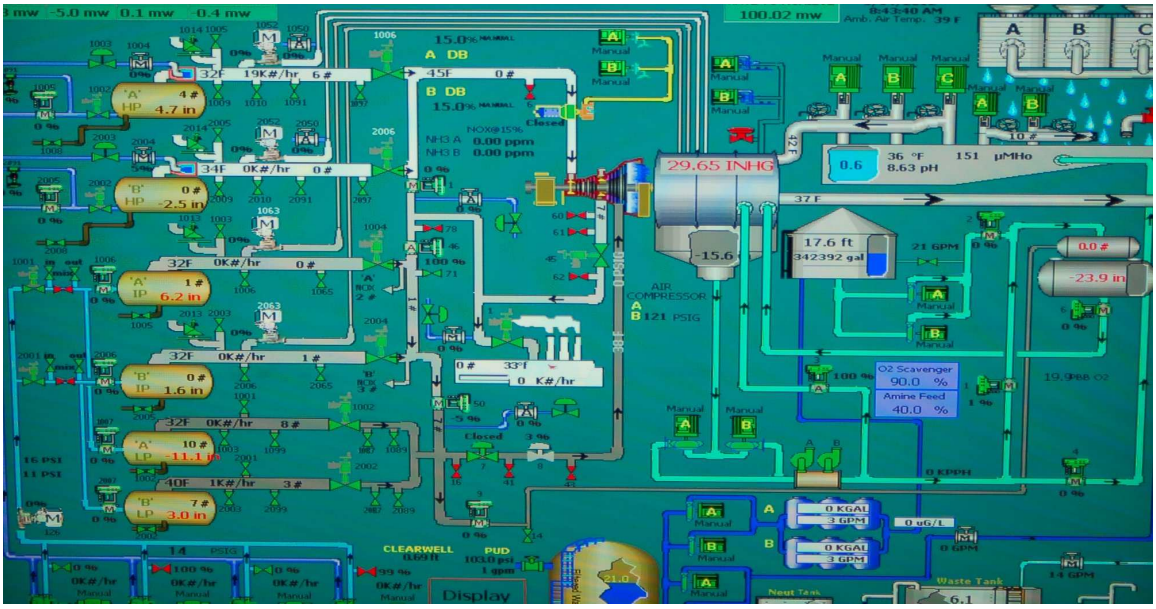
Lessons In Industrial Instrumentation



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Revision history[1]

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- Version 0.2 – released September 29, 2008 for Fall quarter student use (SafeCreative registration code `0810111072182`)
- Version 0.4 – released January 12, 2009 for Winter quarter student use (SafeCreative registration code `0901122394919`)
- Version 0.6 – released April 21, 2009 for public use (SafeCreative registration code `0904213106101`)
- Version 0.8 – released September 8, 2009 for public use (SafeCreative registration code `0909094400980`)
- Version 1.0 – released September 28, 2009 for public use (SafeCreative registration code `0909284601913`)
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- Version 1.12 – released September 20, 2010 for public use (SafeCreative registration code `1009217397803`)

[1] Version numbers ending in odd digits are developmental (e.g. 0.7, 1.23, 4.5), with only the latest revision made accessible to the public. Version numbers ending in even digits (e.g. 0.6, 1.0, 2.14) are considered "public-release" and will be archived. Version numbers beginning with zero (e.g. 0.1, 0.2, etc.) represent early editions that were substantially incomplete.

Chapter 21

Continuous temperature measurement

Temperature is the measure of average molecular kinetic energy within a substance. The concept is easiest to understand for gases under low pressure, where gas molecules randomly shuffle about. The average kinetic (motional) energy of these gas molecules defines temperature for that quantity of gas. There is even a formula expressing the relationship between average kinetic energy ($\overline{E_k}$) and temperature (T) for a monatomic (single-atom molecule) gas:

$$\overline{E_k} = \frac{3kT}{2}$$

Where,

$\overline{E_k}$ = Average kinetic energy of the gas molecules (joules)

k = Boltzmann's constant (1.38×10^{-23} joules/Kelvin)

T = Absolute temperature of gas (Kelvin)

Thermal energy is a different concept: the quantity of *total kinetic energy* for this random molecular motion. If the average kinetic energy is defined as $\frac{3kT}{2}$, then the total kinetic energy for all the molecules in a monatomic gas must be this quantity times the total number of molecules (N) in the gas sample:

$$E_{\text{thermal}} = \frac{3NkT}{2}$$

This may be equivalently expressed in terms of the number of *moles* of gas rather than the number of molecules (a staggeringly large number for any realistic sample):

$$E_{\text{thermal}} = \frac{3nRT}{2}$$

Where,

E_{thermal} = Total thermal energy for a gas sample (joules)
n = Quantity of gas in the sample (moles)
R = Ideal gas constant (8.315 joules per mole-Kelvin)
T = Absolute temperature of gas (Kelvin)

Heat is defined as the exchange of thermal energy from one sample to another, by way of conduction (direct contact), convection (transfer via a moving fluid), or radiation (emitted energy); although you will often find the terms *thermal energy* and *heat* used interchangeably.

Illustrating by way of example, a single molecule of gas moving at a constant velocity will have a definite temperature. Two or three molecules moving at the same speed will have the same *temperature*, but together represent a greater *thermal energy* than any one of them considered alone. Heat is either the reduction or increase of thermal energy by transfer of energy. If these gas molecules happen to collide with slower-moving gas molecules, the faster molecules will lose velocity while the slower molecules will gain velocity. Thus, the higher-temperature molecules cool down while the lower-temperature molecules warm up: a transfer of *heat*.

Temperature is a more easily detected quantity than either thermal energy or heat. In fact, when we need to measure either thermal energy or heat, we do so by measuring temperature and then inferring the desired variable based on the laws of thermodynamics.

There are many different ways to measure temperature, from a simple glass-bulb mercury thermometer to sophisticated infrared optical sensor systems. Like all other areas of measurement, there is no single technology that is best for all applications. Each temperature-measurement technique has its own strengths and weaknesses. One responsibility of the instrument technician is to know these pros and cons so as to choose the best technology for the application, and this knowledge is best obtained through understanding the operational principles of each technology.

21.1 Bi-metal temperature sensors

Solids tend to expand when heated. The amount that a solid sample will expand with increased temperature depends on the size of the sample, the material it is made of, and the amount of temperature rise. The following formula relates linear expansion to temperature change:

$$l = l_0(1 + \alpha\Delta T)$$

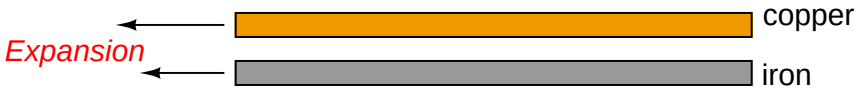
Where,
l = Length of material after heating
l_0 = Original length of material
α = Coefficient of linear expansion
ΔT = Change in temperature

Here are some typical values of α for common metals:

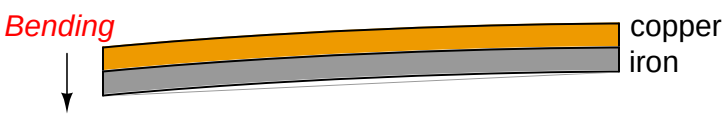
- Aluminum = 25×10^{-6} per degree C
- Copper = 16.6×10^{-6} per degree C
- Iron = 12×10^{-6} per degree C
- Tin = 20×10^{-6} per degree C
- Titanium = 8.5×10^{-6} per degree C

As you can see, the values for α are quite small. This means the amount of expansion (or contraction) for modest temperature changes are almost too small to see unless the sample size (l_0) is huge. We can readily see the effects of thermal expansion in structures such as bridges, where expansion joints must be incorporated into the design to prevent serious problems due to changes in ambient temperature. However, for a sample the size of your hand the change in length from a cold day to a warm day will be microscopic.

One way to amplify the motion resulting from thermal expansion is to bond two strips of dissimilar metals together, such as copper and iron. If we were to take two equally-sized strips of copper and iron, lay them side-by-side, and then heat both of them to a higher temperature, we would see the copper strip lengthen slightly more than the iron strip:



If we bond these two strips of metal together, this differential growth will result in a bending motion greatly exceeding the linear expansion. This device is called a *bi-metal strip*:



This bending motion is significant enough to drive a pointer mechanism, activate an electromechanical switch, or perform any number of other mechanical tasks, making this a very simple and useful *primary sensing element* for temperature. Older home thermostats often used this principle to both indicate room temperature and to actuate electrical switches for controlling room temperature. Electric hot water heater units still use this type of device (usually in the form of a convex bi-metal disk) to sense over-temperature conditions and automatically shut off power to the heater if the water temperature exceeds a pre-set limit.

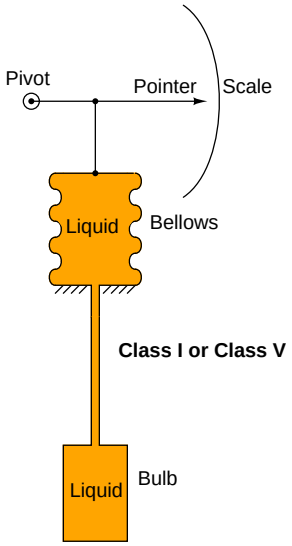
If a bi-metallic strip is twisted over a long length, it will tend to un-twist as it heats up. This twisting motion may be used to directly drive the needle of a temperature gauge. This is the operating principle of the temperature gauge shown in the following photograph:



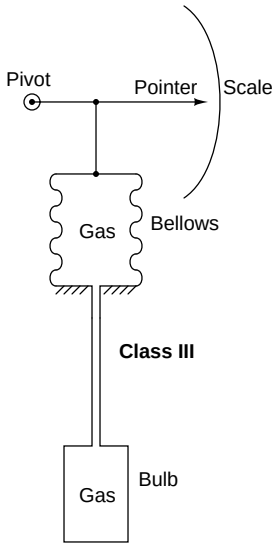
21.2 Filled-bulb temperature sensors

Filled-bulb systems exploit the principle of fluid expansion to measure temperature. If a fluid is enclosed in a sealed system and then heated, the molecules in that fluid will exert a greater pressure on the walls of the enclosing vessel. By measuring this pressure, and/or by allowing the fluid to expand under constant pressure, we may infer the temperature of the fluid.

Class I and Class V systems use a liquid fill fluid (class V is mercury). Here, the volumetric expansion of the liquid drives an indicating mechanism to show temperature:

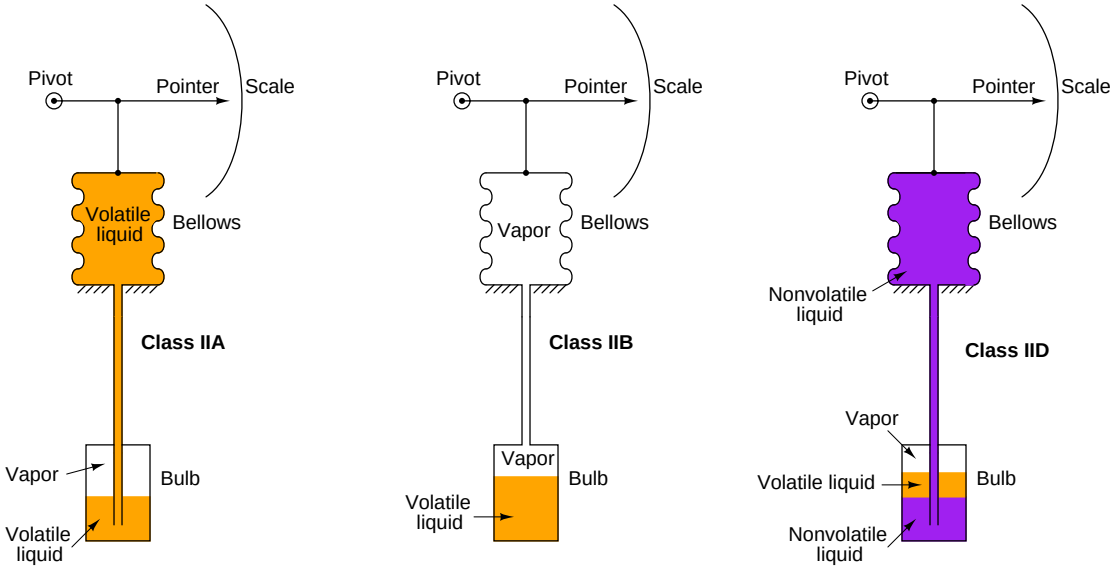


Class III systems use a gas fill fluid instead of liquid. Here, the change in pressure with temperature (as described by the Ideal Gas Law) allows us to sense the bulb's temperature:



In these systems, it is quite critical that the tube connecting the sensing bulb to the indicating element be of minimal volume, so the fluid expansion is primarily due to changes in temperature at the bulb rather than changes in temperature along the length of the tube. It is also important to realize that the fluid volume contained by the bellows (or bourdon tube or diaphragm . . .) is also subject to expansion and contraction due to temperature changes at the indicator. This means the temperature indication varies somewhat as the indicator temperature changes, which is not desirable, since we intend the device to measure temperature (exclusively) at the bulb. Various methods of compensation exist for this effect (for example, a bi-metal spring inside the indicator mechanism to automatically offset the indication as ambient temperature changes), but it may be permanently offset through a simple "zero" adjustment provided that the ambient temperature at the indicator does not change much.

A fundamentally different class of filled-bulb system is the Class II, which uses a volatile liquid/vapor combination to generate a temperature-dependent fluid expansion:

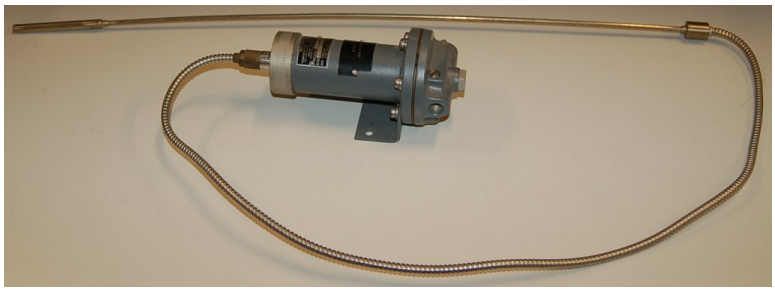


Given that the liquid and vapor are in direct contact with each other, the pressure in the system will be precisely equal to the *saturated vapor pressure* at the vapor/liquid interface. This makes the Class II system sensitive to temperature only at the bulb and nowhere else along the system's volume. Because of this phenomenon, a Class II filled-bulb system requires no compensation for temperature changes at the indicator.

Class II systems do have one notable idiosyncrasy, though: they have a tendency to switch from Class IIA to Class IIB when the temperature of the sensing bulb crosses the ambient temperature at the indicator. Simply put, the liquid tends to seek the colder portion of a Class II system while the vapor tends to seek the warmer portion. This causes problems when the indicator and sensing bulb exchange identities as warmer/colder. The rush of liquid up (or down) the capillary tubing as the system tries to reach a new equilibrium causes intermittent measurement errors. Class II filled-bulb systems designed to operate in either IIA or IIB mode are classified as *IIC*.

One calibration problem common to all systems with liquid-filled capillary tubes is an offset in temperature measurement due to hydrostatic pressure (or suction) resulting from a different in height between the measurement bulb and the indicator. This represents a "zero" shift in calibration, which may be permanently offset by a "zero" adjustment at the time of installation. Class III (gas-filled) and Class IIB (vapor-filled) systems, of course, suffer no such problem because there is no liquid in the capillary tube to generate a pressure due to height.

A photograph of a pneumatic temperature transmitter using a filled-bulb as the sensing element appears here:



This transmitter happens to be a Moore Products "Nullmatic" model. The capillary tube connecting the fluid-filled bulb to the transmitter mechanism is protected by a spiral-metal jacket. The bulb itself is located at the very end of the stainless steel "wand" which inserts into the process fluid to be measured:



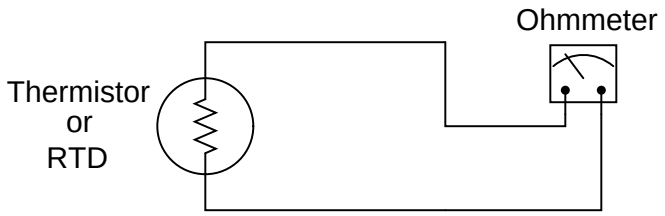
Instead of directly actuating a pointer mechanism, the fluid pressure in this instrument actuates a self-balancing pneumatic mechanism to produce a 3 to 15 PSI air pressure signal representing process temperature.

Filled-bulb temperature sensors are seldom used in industrial applications anymore, chiefly due to the superiority of electrical sensors. The only significant advantage filled-bulb sensors hold over electrical sensors is not needing electricity[1] to function, but this is usually not a serious consideration within a modern industrial facility.

[1] Even this advantage is not always true. It is possible to build self-powered thermocouple temperature indicators, where an analog meter movement is driven by the electrical energy a thermocouple sensing junction outputs. Here, no external electrical power source is required! However, the accuracy of self-powered thermocouple systems is poor, as is the ability to measure small temperature ranges.

21.3 Thermistors and Resistance Temperature Detectors (RTDs)

One of the simplest classes of temperature sensor is one where temperature effects a change in electrical resistance. With this type of primary sensing element, a simple ohmmeter is able to function as a thermometer, interpreting the resistance as a temperature measurement:



Thermistors are devices made of metal oxide which either increase in resistance with increasing temperature (a *positive temperature coefficient*) or decrease in resistance with increasing temperature (a *negative temperature coefficient*). *RTD*s are devices made of pure metal wire (usually platinum or copper) which always increase in resistance with increasing temperature. The major difference between thermistors and RTDs is linearity: thermistors are highly sensitive and nonlinear, whereas RTDs are relatively insensitive but very linear. For this reason, thermistors are typically used where high accuracy is unimportant. Many consumer-grade devices use thermistors for temperature sensors.

21.3.1 Temperature coefficient of resistance (α)

A *Resistive Temperature Detector* (RTD) is a special temperature-sensing element made of fine metal wire, the electrical resistance of which changes with temperature as approximated by the following formula:

$$R_T = R_{ref}[1 + \alpha(T - T_{ref})]$$

Where,
R_T = Resistance of RTD at given temperature T (ohms)
R_{ref} = Resistance of RTD at the reference temperature T_{ref} (ohms)
α = Temperature coefficient of resistance (ohms per ohm/degree)

The following example shows how to use this formula to calculate the resistance of a "100 ohm" platinum RTD with a temperature coefficient value of 0.00392 at a temperature of 35 degrees Celsius:

$$R_T = 100\,\Omega[1 + (0.00392)(35^o\text{ C} - 0^o\text{ C})]$$

$$R_T = 100\,\Omega[1 + 0.1372]$$

$$R_T = 100\,\Omega[1.1372]$$

$$R_T = 113.72\,\Omega$$

Due to nonlinearities in the RTD's behavior, this linear RTD formula is only an approximation. A better approximation is the *Callendar-van Dusen formula*, which introduces second, third, and fourth-degree terms for a better fit: $R_T = R_{ref}(1 + AT + BT^2 - 100CT^3 + CT^4)$ for temperatures ranging $-200\ ^o\text{C} < T < 0\ ^o\text{C}$ and $R_T = R_{ref}(1 + AT + BT^2)$ for temperatures ranging $0\ ^o\text{C} < T < 661\ ^o\text{C}$, both assuming $T_{ref} = 0\ ^o\text{C}$. The A, B, and C coefficients vary with the specific type of RTD, equivalent in role to α in the linear RTD formula.

Water's melting/freezing point is the standard reference temperature for most RTDs. Here are some typical values of α for common metals:

- Nickel = 0.00672 Ω/Ω^oC
- Tungsten = 0.0045 Ω/Ω^oC
- Silver = 0.0041 Ω/Ω^oC
- Gold = 0.0040 Ω/Ω^oC
- Platinum = 0.00392 Ω/Ω^oC
- Copper = 0.0038 Ω/Ω^oC

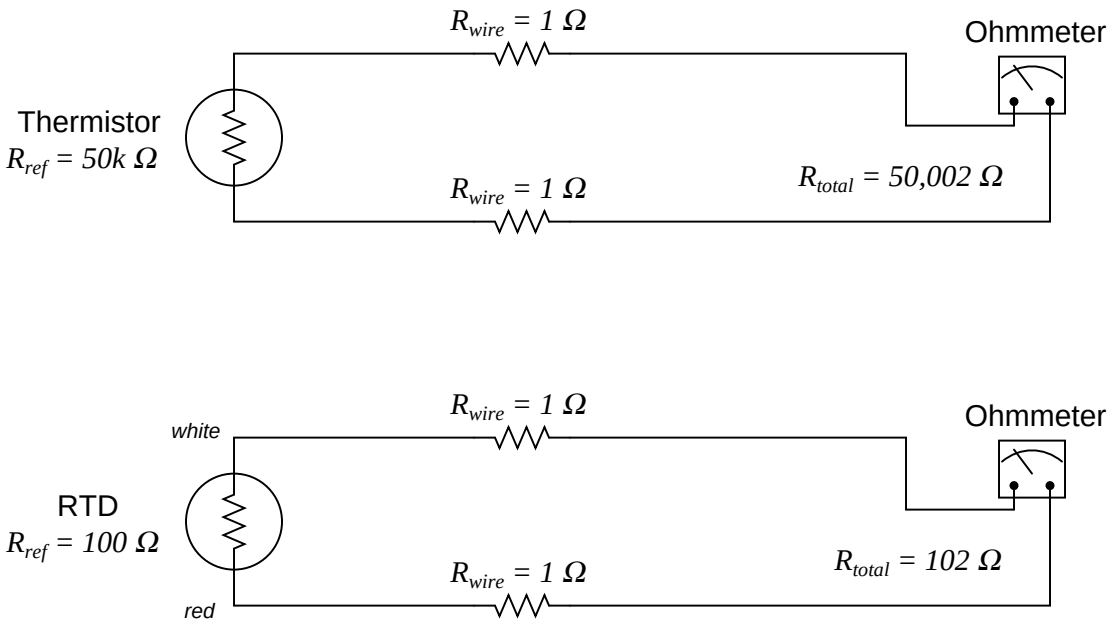
As mentioned previously, platinum is a common wire material for industrial RTD construction. The alpha (α) value for platinum varies according to the alloying of the metal. For "reference grade" platinum wire, the most common alpha value is 0.003902. Industrial-grade platinum alloy RTD wire is commonly available in two different coefficient values: 0.00385 (the "European" alpha value) and 0.00392 (the "American" alpha value), of which the "European" value of 0.00385 is most commonly used (even in the United States!).

An alternative to mathematically predicting the resistance of an RTD is to simply look up its predicted resistance versus temperature in a *table* of values published for that RTD type. Tables capture the nuances of an RTD's non-linearity without adding any mathematical complexity: simply look up the resistance corresponding to a given temperature, or vice-versa. If a value falls between two nearest entries in the table, you may *interpolate* the find the desired value, regarding the two nearest table entries as end-points defining a line segment, calculating the point you desire along that line.

100 Ω is a very common reference resistance (R_{ref} at 0 degrees Celsius) for industrial RTDs. 1000 Ω is another common reference resistance, and some industrial RTDs have reference resistances as low as 10 Ω. Compared to thermistors with their tens or even hundreds of thousands of ohms' resistance, an RTD's resistance is comparatively small. This can cause problems with measurement, since the wires connecting an RTD to its ohmmeter possess their own resistance, which will be a more substantial percentage of the total circuit resistance than for a thermistor.

21.3.2 Two-wire RTD circuits

The following schematic diagrams show the relative effects of 2 ohms total wire resistance on a thermistor circuit and on an RTD circuit:



As you can see, wire resistance *adds* to the sensing element's resistance to create a larger total circuit resistance which will be interpreted by the receiving instrument (ohmmeter) as a *falsely high* temperature reading, assuming a positive temperature coefficient of resistance for the sensing element.

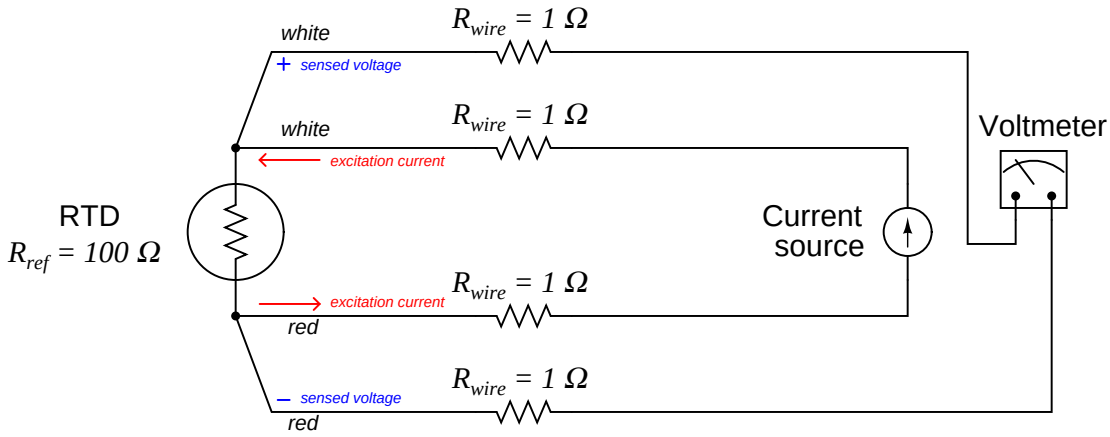
Clearly, wire resistance is more problematic for low-resistance RTDs than for high-resistance thermistors. In the RTD circuit, wire resistance constitutes 1.96% of the total circuit resistance. In the thermistor circuit, the same 2 ohms of wire resistance constitutes only 0.004% of the total circuit resistance. The thermistor's huge reference resistance value "swamps"[2] the wire resistance to the point that the latter becomes insignificant by comparison.

In HVAC (Heating, Ventilation, and Air Conditioning) systems, where the temperature measurement range is relatively narrow, the nonlinearity of thermistors is not a serious concern and their relative immunity to wire resistance error is a definite advantage over RTDs. In industrial temperature measurement applications where the temperature ranges are usually much wider, the nonlinearity of thermistors becomes a significant problem, so we must find a way to use low-resistance RTDs and deal with the (lesser) problem of wire resistance.

[2] "Swamping" is the term given to the overshadowing of one effect by another. Here, the normal resistance of the thermistor greatly overshadows ("swamps") any wire resistance in the circuit, such that wire resistance becomes negligible.

21.3.3 Four-wire RTD circuits

A very old electrical measurement technique known as the *Kelvin* or *four-wire* method is a practical solution to the problem of wire resistance. Commonly employed to make precise resistance measurements for scientific experiments in laboratory conditions, the four-wire technique uses four wires to connect the resistance under test (in this case, the RTD) to the measuring instrument, which consists of a voltmeter and a precision current source. Two wires carry "excitation" current to the RTD from the current source while the other two wires merely "sense" voltage drop across the RTD resistor element and carry that voltage signal to the voltmeter. RTD resistance is calculated using Ohm's Law: taking the measured voltage displayed by the voltmeter and dividing that figure by the regulated current value of the current source. A simple 4-wire RTD circuit is shown here for illustration:



Wire resistances are completely inconsequential in this circuit. The two "excitation" wires carrying current to the RTD will drop some voltage along their length, but this voltage drop is only "seen" by the current source and not the voltmeter. The two "sense" wires connecting the voltmeter to the RTD also possess resistance, but they drop negligible voltage because the voltmeter draws so little current through them[3]. Thus, the resistances of the current-carrying wires are of no effect because the voltmeter never senses their voltage drops, and the resistances of the voltmeter's sensing wires are of no effect because they carry practically zero current.

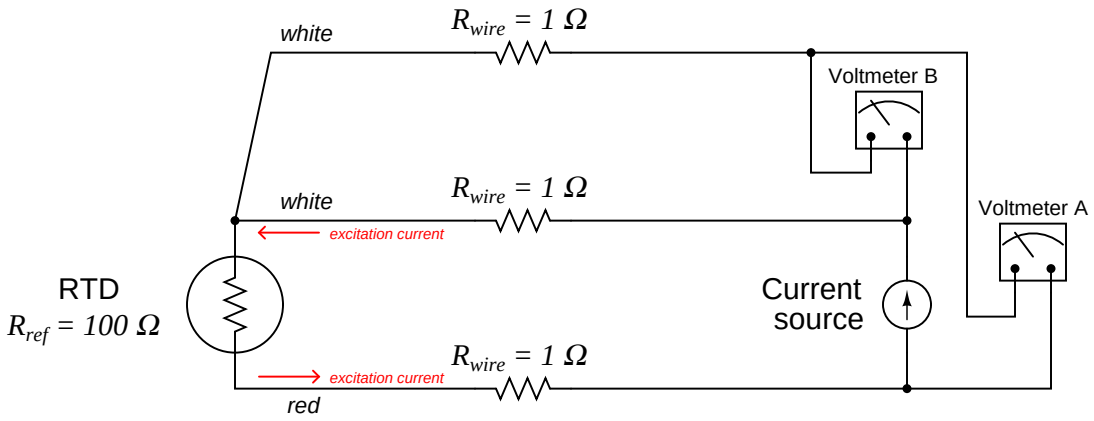
Note how wire colors (*white* and *red*) indicate which wires are common pairs at the RTD. The RTD is polarity-insensitive because it is nothing more than a resistor, which is why it doesn't matter which color is positive and which color is negative.

The only disadvantage of the four-wire method is the sheer number of wires necessary. Four wires *per RTD* can add up to a sizeable wire count when many different RTDs are installed in a process area.

[3]Remember that an ideal voltmeters has *infinite* input impedance, and modern semiconductor-amplified voltmeters have impedances of several mega-ohms or more.

21.3.4 Three-wire RTD circuits

A compromise between two-wire and four-wire RTD connections is the *three-wire* connection, which looks like this:



In a three-wire RTD circuit, voltmeter "A" measures the voltage dropped across the RTD plus the voltage dropped across the bottom current-carrying wire. Voltmeter "B" measures just the voltage dropped across the top current-carrying wire. Assuming both current-carrying wires will have (very nearly) the same resistance, subtracting the indication of voltmeter "B" from the indication given by voltmeter "A" yields the voltage dropped across the RTD:

$$V_{RTD} = V_{\text{meter(A)}} - V_{\text{meter(B)}}$$

Once again, RTD resistance is calculated from the RTD voltage and the known current source value using Ohm's Law, just as it is in a 4-wire circuit.

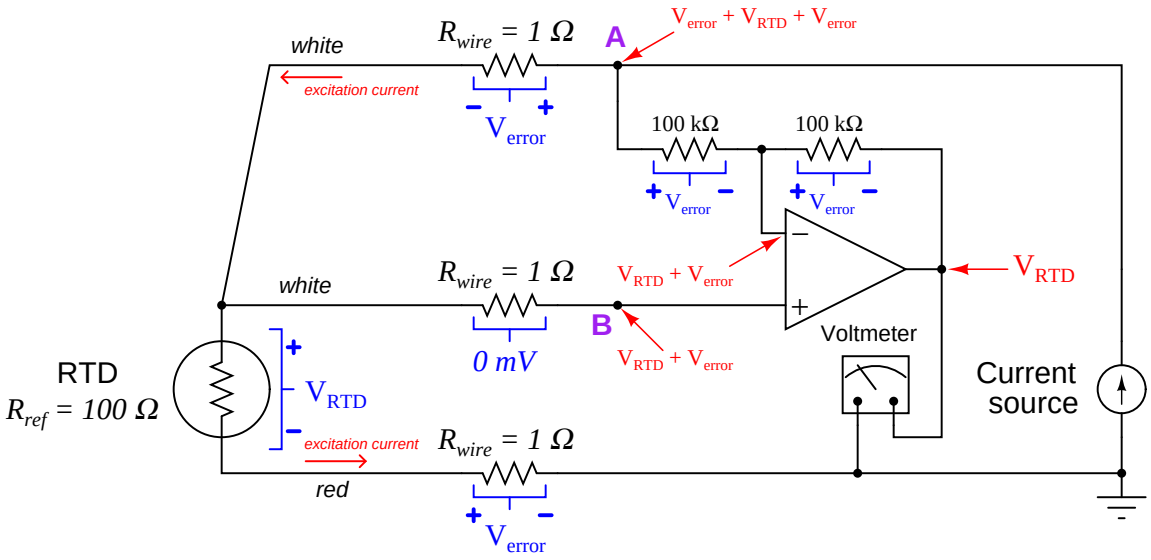
If the resistances of the two current-carrying wires are precisely identical (and this includes the electrical resistance of any connections within those current-carrying paths, such as terminal blocks), the calculated RTD voltage will be the same as the true RTD voltage, and no wire-resistance error will appear. If, however, one of those current-carrying wires happens to exhibit more resistance than the other, the calculated RTD voltage will not be the same as the actual RTD voltage, and a measurement error will result.

Thus, we see that the three-wire RTD circuit saves us wire cost over a four-wire circuit, but at the "expense" of a potential measurement error. The beauty of the four-wire design was that wire resistances were completely irrelevant: a true determination of RTD voltage (and therefore RTD resistance) could be made regardless of how much resistance each wire had, or even if the wire resistances were different from each other. The error-canceling property of the three-wire circuit, by contrast, hinges on the assumption that the two current-carrying wires have exactly the same resistance, which may or may not actually be true.

It should be understood that real three-wire RTD instruments do not employ direct-indicating voltmeters as shown in these simplified examples. Actual RTD instruments use either analog or digital "conditioning" circuits to measure the voltage drops and perform the necessary calculations

to compensate for wire resistance. The voltmeters shown in the four-wire and three-wire diagrams serve only to illustrate the basic concepts, not to showcase practical instrument designs.

A practical electronic circuit for a 3-wire RTD sensor is shown here (differential voltages shown in blue, ground-referenced voltages shown in red):



Note that the voltage appearing at point B with reference to ground is the RTD's voltage plus the voltage dropped by the lower current-carrying wire: $V_{RTD} + V_{error}$. It is this "error" voltage we must eliminate in order to achieve an accurate measurement of RTD voltage drop, essential for accurately inferring RTD temperature. The voltage appearing at point A is greater by the upper wire's voltage drop ($V_{error} + V_{RTD} + V_{error}$) because that point spans one more wire resistance in the circuit than point B[4].

Like all negative-feedback operational amplifier circuits, the amplifier does its best to maintain the two input terminals at (nearly) the same voltage. Thus, the voltage at point B is duplicated at the inverting input terminal by the amplifier's feedback action. From this we may see that the voltage drop across the left-hand 100 kΩ resistor is simply V_{error}: the potential difference between point A and point B. The feedback current driving through this resistor goes through the other 100 kΩ feedback resistor equally, causing the same voltage drop to appear there (V_{error}). We may see that the polarity of this second resistor's voltage drop ends up subtracting that quantity from the voltage appearing at the inverting input terminal. The inverting terminal voltage ($V_{RTD} + V_{error}$) minus the right-hand 100 kΩ resistor's voltage drop (V_{error}) is simply V_{RTD}, and so the voltmeter registers the true RTD voltage drop without any wire resistance error.

Like the dual-voltmeter circuit shown previously, this amplified 3-wire RTD sensing circuit "assumes" the two current-carrying wires will have the exact same resistance and therefore drop

[4]Note that the middle wire resistance is of no effect because it does not carry the RTD's current. The amount of current entering or exiting an operational amplifier is assumed to be zero for all practical purposes.

the same amount of voltage. If this is not the case, and one of these wires drops more voltage than the other, the circuit will fail to yield the exact RTD voltage (V_{RTD}) at the amplifier output. This is the fundamental limitation of any 3-wire RTD circuit: the cancellation of wire resistance is only as good as the wires' resistances are precisely equal to each other.

A photograph of a modern temperature transmitter capable of receiving input from 2-wire, 3-wire, or 4-wire RTDs (as well as thermocouples, another type of temperature sensor entirely) shows the connection points and the labeling describing how the sensor is to be connected to the appropriate terminals:

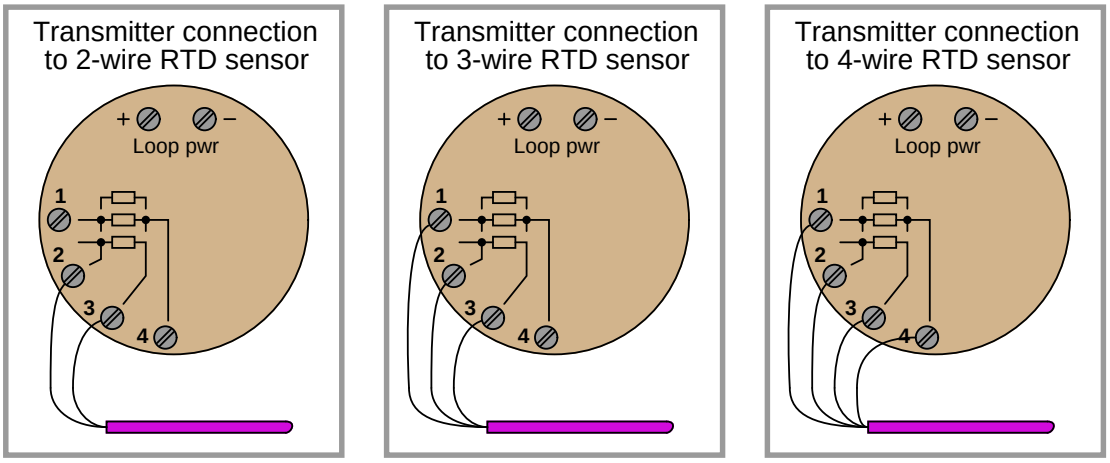


The rectangle symbol shown on the label represents the resistive element of the RTD. The symbol with the "+" and "−" marks represents a thermocouple junction, and may be ignored for the purposes of this discussion. As shown by the diagram, a two-wire RTD would connect between terminals 2 and 3. Likewise, a three-wire RTD would connect to terminals 1, 2, and 3 (with terminals 1 and 2 being the points of connection for the two common wires of the RTD). Finally, a four-wire RTD would connect to terminals 1, 2, 3, and 4 (terminals 1 and 2 being common, and terminals 3 and 4 being common, at the RTD).

Once the RTD has been connected to the appropriate terminals of the temperature transmitter, the transmitter needs to be electronically configured for that type of RTD. In the case of this particular temperature transmitter, the configuration is performed using a "smart" communicator device using the HART digital protocol to access the transmitter's microprocessor-based settings. Here, the technician would configure the transmitter for 2-wire, 3-wire, or 4-wire RTD connection.

21.3.5 Proper RTD sensor connections

Proper connections for all three types of RTD sensor (2-wire, 3-wire, and 4-wire) to a user-configurable transmitter are shown in the following illustrations:

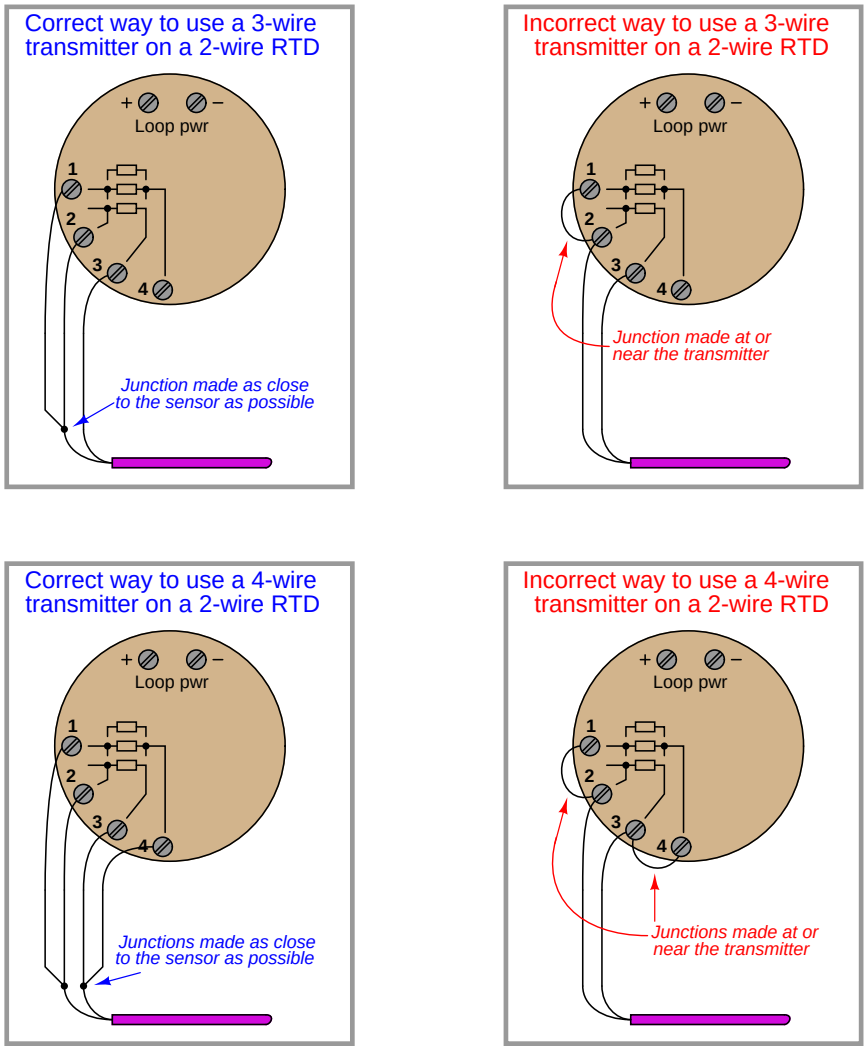


It is critically important to note that the common connections shown by the symbols for 3- and 4-wire RTD sensors represent junction points *at the sensor*; not terminals jumpered by the technician at the time of installation, and not internal jumpers inside the transmitter. The whole purpose of having 3-wire and 4-wire RTD circuits is to eliminate errors due to voltage drop along the current-carrying wires, and this can only be realized if the "sensing" wire(s) extend out to the RTD itself and connect there. If the transmitter's sensing terminal(s) are only jumpered to a current-carrying terminal, the transmitter will sense voltage dropped by the RTD *plus* voltage dropped by the current-carrying wire(s), leading to falsely high temperature indications.

Misconceptions surrounding proper RTD connections unfortunately abound both in students and in working industry professionals. With any luck, the following presentation will help you avoid such mistakes, and more importantly help you *understand why* the correct connections are best.

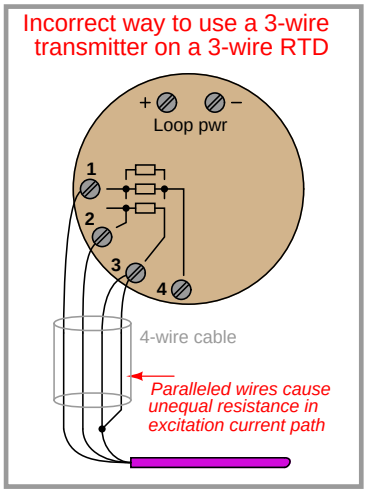
Always bear in mind the purpose of a 3-wire or a 4-wire RTD connection: *to avoid inaccuracies caused by voltage drops along the current-carrying wires.* The only way to do this is to ensure the sensing (non-current-carrying) wire(s) extend from the transmitter terminal(s) all the way to the sensor itself. This way, the transmitter is able to "look past" the voltage drops of the current-carrying wires to "see" the voltage dropped only by the RTD itself.

The following illustrations show both correct and incorrect ways to connect a 2-wire RTD to a 3- or 4-wire transmitter:



Jumpers placed at the transmitter terminals defeat the purpose of the transmitter's 3-wire or 4-wire capabilities, downgrading its performance to that of a 2-wire system.

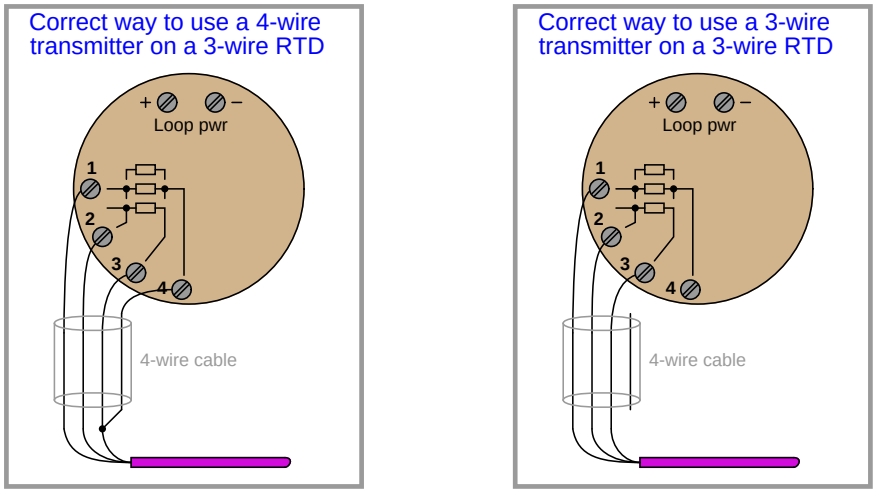
A similar problem occurs when someone tries to connect a 3-wire RTD to a 3-wire transmitter using a conveniently available 4-wire cable:



3-wire RTD measurement is based on the assumption that both current-carrying wires have exactly the same electrical resistance. By paralleling two of the four wires in the 4-wire cable, you will create unequal resistances in the current path, thus leading to measurement errors at the transmitter[5].

[5]These errors will result only if the paralleled wires carry current. If the two wires you paralleled happen to join the transmitter's sensing terminal to the RTD (the one carrying no current), no errors will result. However, many RTD transmitters do not document which of the terminals *sense* (carry no current) versus which of them *excite* (carry current to the RTD), and so there is a probability of getting it wrong if you simply guess. Given that there is no real benefit to having paralleled wires connecting the transmitter's sensing terminal to the RTD, my advice is to either use all four wires and configure the transmitter for 4-wire mode, or don't use the fourth wire at all.

Better solutions for the 3-wire RTD and 4-wire cable scenario include configuring the transmitter for 4-wire RTD input and actually using all four terminals (shown on left), or keeping the transmitter configured for 3-wire RTD input and not using the fourth wire in the cable at all (shown on right):



21.3.6 Self-heating error

One problem inherent to both thermistors and RTDs is *self-heating*. In order to measure the resistance of either device, we must pass an electric current through it. Unfortunately, this results in the generation of heat at the resistance according to Joule's Law:

$$P = I^2 R$$

This dissipated power causes the thermistor or RTD to increase in temperature beyond its surrounding environment, introducing a positive measurement error. The effect may be minimized by limiting excitation current to a bare minimum, but this results in less voltage dropped across the device. The smaller the developed voltage, the more sensitive the voltage-measuring instrument must be to accurately sense the condition of the resistive element. Furthermore, a decreased signal voltage means we will have a decreased signal-to-noise ratio, for any given amount of noise induced in the circuit from external sources.

One clever way to circumvent the self-heating problem without diminishing excitation current to the point of uselessness is to *pulse* current through the resistive sensor and digitally sample the voltage only during those brief time periods while the thermistor or RTD is powered. This technique works well when we are able to tolerate slow sample rates from our temperature instrument, which is often the case because most temperature measurement applications are slow-changing by nature. The pulsed-current technique enjoys the further advantage of reducing power consumption for the instrument, an important factor in battery-powered temperature measurement applications.

21.4 Thermocouples

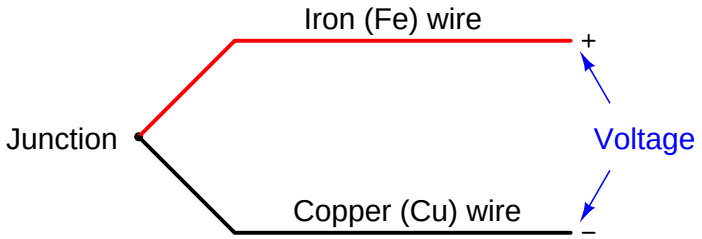
RTDs are completely passive sensing elements, requiring the application of an externally-sourced electric current in order to function as temperature sensors. Thermocouples, however, generate their own electric potential. In some ways, this makes thermocouple systems simpler because the device receiving the thermocouple's signal does not have to supply electric power to the thermocouple. It also makes thermocouple systems potentially safer than RTDs in applications where explosive compounds may exist in the atmosphere, because the power levels generated by a thermocouple tend to be less than the power levels dissipated by an RTD. The self-powering nature of thermocouples also means they do not suffer from the same "self-heating" effect as RTDs.

In other ways, however, thermocouple circuits are more complex and troublesome than RTD circuits because the generation of voltage actually occurs in *two* different locations within the circuit, not simply at the sensing point. This means the receiving circuit must "compensate" for temperature in another location in order to accurately measure temperature in the desired location.

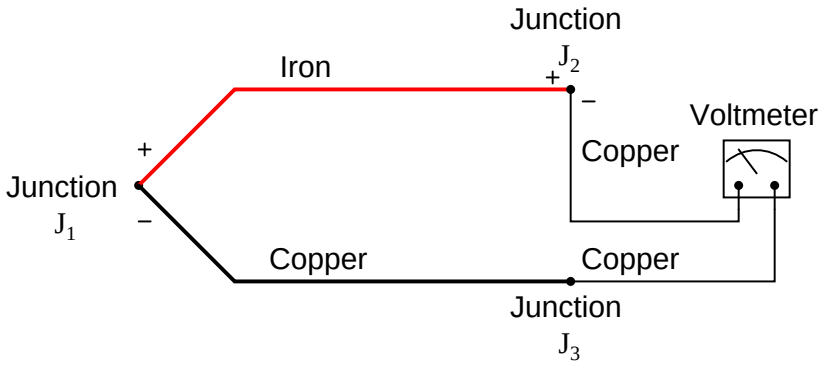
Though typically not as accurate as RTDs, thermocouples are more rugged, have greater temperature measurement spans, and are easier to manufacture in different physical forms.

21.4.1 Dissimilar metal junctions

When two dissimilar metal wires are joined together at one end, a voltage is produced at the other end that is approximately proportional to temperature. That is to say, the junction of two different metals behaves like a temperature-sensitive battery. This form of electrical temperature sensor is called a *thermocouple*:



This phenomenon provides us with a simple way to electrically infer temperature: simply measure the voltage produced by the junction, and you can tell the temperature of that junction. And it would be that simple, if it were not for an unavoidable consequence of electric circuits: when we connect any kind of electrical instrument to the thermocouple wires, we inevitably produce another junction of dissimilar metals. The following schematic shows this fact, where the iron-copper junction J_1 is necessarily complemented by a second iron-copper junction J_2 of opposing polarity:



Junction J_1 is a junction of iron and copper – two dissimilar metals – which will generate a voltage related to temperature. Note that junction J_2, which is necessary for the simple fact that we must somehow connect our copper-wired voltmeter to the iron wire, is also a dissimilar-metal junction which will also generate a voltage related to temperature. Further note how the polarity of junction J_2 stands opposed to the polarity of junction J_1 (iron = positive ; copper = negative). A third junction (J_3) also exists between wires, but it is of no consequence because it is a junction of two identical metals which does not generate a temperature-dependent voltage at all.

The presence of this second voltage-generating junction (J_2) helps explain why the voltmeter registers 0 volts when the entire system is at room temperature: any voltage generated by the iron-copper junctions will be equal in magnitude and opposite in polarity, resulting in a net (series-total) voltage of zero. Only when the two junctions J_1 and J_2 are at *different* temperatures will the voltmeter register any voltage at all.

We may express this relationship mathematically as follows:

$$V_{meter} = V_{J1} - V_{J2}$$

With the measurement (J_1) and reference (J_2) junction voltages opposed to each other, the voltmeter only "sees" the difference between these two voltages.

Thus, thermocouple systems are fundamentally *differential* temperature sensors. That is, they provide an electrical output proportional to the difference in temperature between two different points. For this reason, the wire junction we use to measure the temperature of interest is called the *measurement junction* while the other junction (which we cannot eliminate from the circuit) is called the *reference junction* (or the *cold junction*, because it is typically at a cooler temperature than the process measurement junction).

Much of the complexity of thermocouples is related to the reference junction voltage and how we must deal with that (unwanted) potential when using a thermocouple as a measuring device. For most practical applications, we just want to measure the temperature at one location, not the difference in temperature between two locations which is what a thermocouple naturally does. A number of different techniques exist to deal with this problem – forcing a differential temperature sensor to act like a single-point temperature sensor – and we will explore the most common techniques in this section.

Students and working professionals alike often find this concept of a reference junction and its effects endlessly confusing. My advice to the confused is to return to the simple iron-copper wire circuit shown previously as a "starting point," and then deduce its behavior from first principles[6]. We know that a dissimilar-metal junction creates a voltage with temperature. We also know that in order to make a complete circuit with iron and copper wire *there must be a second iron-copper junction somewhere else in that same circuit, the polarity of which is necessarily opposed to the first.* If we call the first iron-copper junction J_1 and the second junction J_2, we absolutely must conclude that the net voltage registered by the voltmeter in this circuit will be $V_{J1} - V_{J2}$.

All thermocouple circuits – no matter how simple or complex – exhibit this fundamental property. Mentally constructing a simple circuit of two dissimilar-metal wires and then performing "thought experiments" to see how that circuit will behave with those junctions at the same temperature and also at different temperatures is the best way I can suggest for any person to comprehend thermocouples. Students especially tend to cope with complexity through memorization: committing to memory catch-phrases and formulae such as $V_{meter} = V_{J1} - V_{J2}$. This is a poor coping mechanism, as it grants the illusion of understanding with none of the substance. The real secret is to know *why* a thermocouple circuit acts as it does, and that only comes through practiced reasoning. Throughout the rest of this section, as we explore reference junction compensation, how to interpret voltage measurements in thermocouple circuits, and how to simulate thermocouples at temperature, we will keep returning to this simple iron-copper wire circuit to refresh our understanding of how and why thermocouple circuits behave. If you understand this one fundamental concept, the rest will make sense to you. If you continually find yourself confused by thermocouple circuits, it means you do not yet fully understand this basic circuit, and you need to return to it and think it through until you do.

[6] By "first principles," I mean the basic laws of electric circuits. In this case, the most important law to apply is Kirchhoff's Voltage Law: the algebraic sum of voltages in any loop must be equal to zero.

21.4.2 Thermocouple types

Thermocouples exist in many different types, each with its own color codes for the dissimilar-metal wires. Here is a table showing the more common thermocouple types and their standardized colors[7], along with some distinguishing characteristics of the metal types to aid in polarity identification when the wire colors are not clearly visible:

Type	**Positive wire** *characteristic*	**Negative wire** *characteristic*	**Plug**	**Temp. range**
T	Copper (blue) *yellow colored*	Constantan (red) *silver colored*	Blue	−300 to 700 ^{o}F
J	Iron (white) *magnetic, rusty?*	Constantan (red) *non-magnetic*	Black	32 to 1400 ^{o}F
E	Chromel (violet) *shiny finish*	Constantan (red) *dull finish*	Violet	32 to 1600 ^{o}F
K	Chromel (yellow) *non-magnetic*	Alumel (red) *magnetic*	Yellow	32 to 2300 ^{o}F
N	Nicrosil (orange)	Nisil (red)	Orange	32 to 2300 ^{o}F
S	Pt90% - Rh10% (black)	Platinum (red)	Green	32 to 2700 ^{o}F
B	Pt70% - Rh30% (grey)	Pt94% - Rh6% (red)	Grey	32 to 3380 ^{o}F

Types S and B use platinum or platinum-rhodium alloy wire, with different alloying distinguishing the positive from the negative wires. Sometimes type B is colored green and red rather than grey and red.

Note how the negative (−) wire of every thermocouple type is color-coded *red*. While this may seem backward to those familiar with modern electronics (where red and black usually represent the positive and negative poles of a DC power supply, respectively), bear in mind that thermocouple color codes actually pre-date electronic power supply wire coloring!

Aside from having different usable temperature ranges, these thermocouple types also differ in terms of the atmospheres they may withstand at elevated temperatures. Type J thermocouples, for instance, by virtue of the fact that one of the wire types is *iron*, will rapidly corrode in any oxidizing[8] atmosphere. Type K thermocouples are attacked by reducing[9] atmospheres as well as sulfur and cyanide. Type T thermocouples are limited in upper temperature by the oxidation of copper (a very reactive metal when hot), but stand up to both oxidizing and reducing atmospheres quite well at lower temperatures, even when wet.

One final note on the thermocouple types shown in this table is that the temperature ranges given are approximate, and vary with the intended measurement accuracy. One may have to stay

[7] The colors in this table apply only to the United States and Canada. A stunning diversity of colors has been "standardized" for each thermocouple type per nationality. The British and Czechs use their own color code, as do the Dutch and Germans. France has its own unique color code as well. Just for fun, an "international" color code also exists which doesn't match any of the others. There are other deviations as well: the wire colors for type R and S thermocouples, for example, are standardized for *extension-grade wire* but not for *thermocouple-grade wire*.

[8] By "oxidizing," what is meant is any atmosphere containing sufficient oxygen molecules or molecules of a similar element such as chlorine or fluorine.

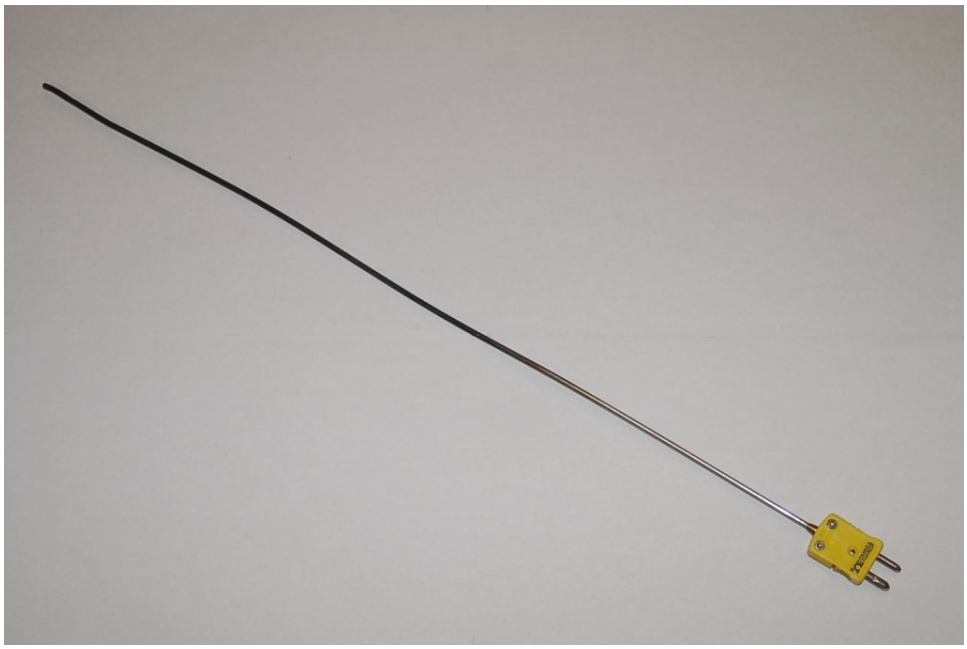
[9] "Reducing" refers to atmospheres rich in elements that readily oxidize. Practically any fuel gas (hydrogen, methane, etc.) will create a reducing atmosphere in sufficient concentration.

within a more limited range of temperature than what is shown in this table if a certain minimum level of accuracy is desired from the thermocouple. Consult manufacturers' data for details!

21.4.3 Connector and tip styles

In its simplest form, a thermocouple is nothing more than a pair of dissimilar-metal wires joined together. However, in industrial practice, we often must package thermocouples in a more rugged form than a bare metal junction. For instance, most industrial thermocouples are manufactured in such a way that the dissimilar-metal wires are protected from physical damage by a stainless steel or ceramic *sheath*, and they are often equipped with molded-plastic plugs for quick connection to and disconnection from a thermocouple-based instrument.

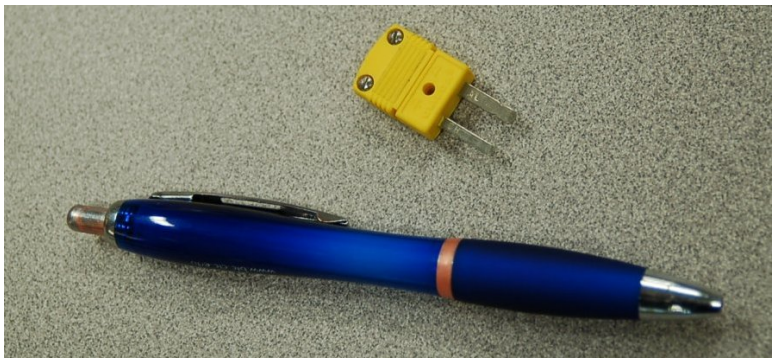
A photograph of a type K industrial thermocouple (approximately 20 inches in length) reveals this "sheathed" and "connectorized" construction:



The stainless steel sheath of this particular thermocouple shows signs of discoloration from previous service in a hot process. Note the different diameters of the plug terminals. This "polarized" design makes it difficult[10] to insert backward into a matching socket.

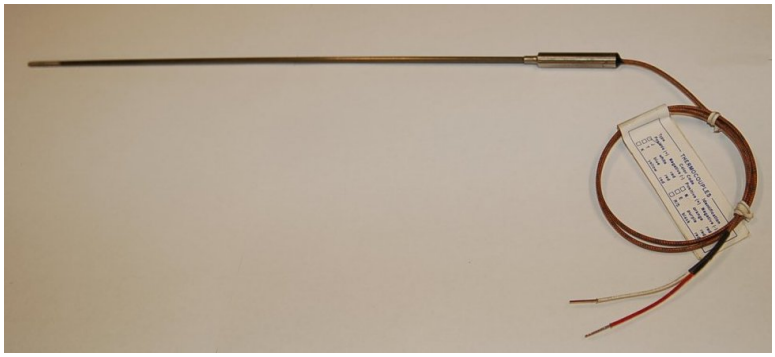
[10] It should be noted that no amount of engineering or design is able to *completely* prevent people from doing the wrong thing. I have seen this style of thermocouple plug forcibly mated the wrong way to a socket. The amount of insertion force necessary to make the plug fit backward into the socket was quite extraordinary, yet this apparently was not enough of a clue for this wayward individual to give them pause.

A miniature version of this same plug (designed to attach to thermocouple wire by screw terminals, rather than be molded onto the end of a sheathed assembly) is shown here, situated next to a ballpoint pen for size comparison:



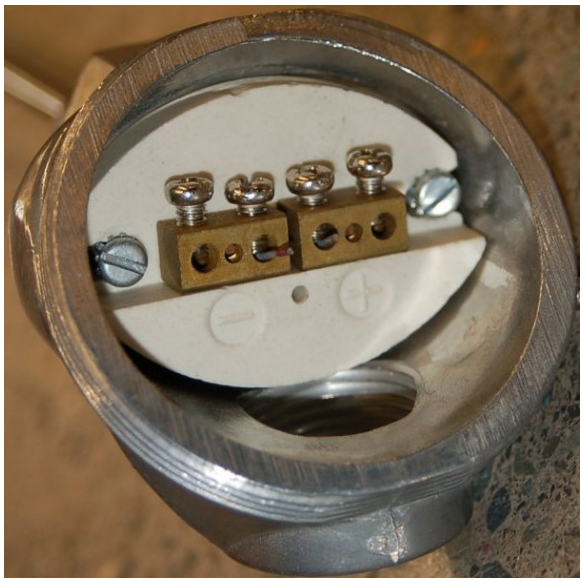
Industrial-grade thermocouples are available with this miniature style of molded plug end as an alternative to the larger (standard) plug. Miniature plug-ends are often the preferred choice for laboratory applications, while standard-sized plugs are often the preferred choice for field applications.

Some industrial thermocouples have no molded plug at all, but terminate simply in a pair of open wire ends. The following photograph shows a type J thermocouple of this construction:



If the electronic measuring instrument (e.g. temperature transmitter) is located near enough for the thermocouple's wires to reach the connection terminals, no plug or socket is needed at all in the circuit. If, however, the distance between the thermocouple and measuring instrument is too far to span with the thermocouple's own wires, a common termination technique is to attach a special terminal block and connection "head" to the top of the thermocouple allowing a pair of thermocouple extension wires to join and carry the millivoltage signal to the measuring instrument.

This next photograph shows a close-up view of such a thermocouple "head":



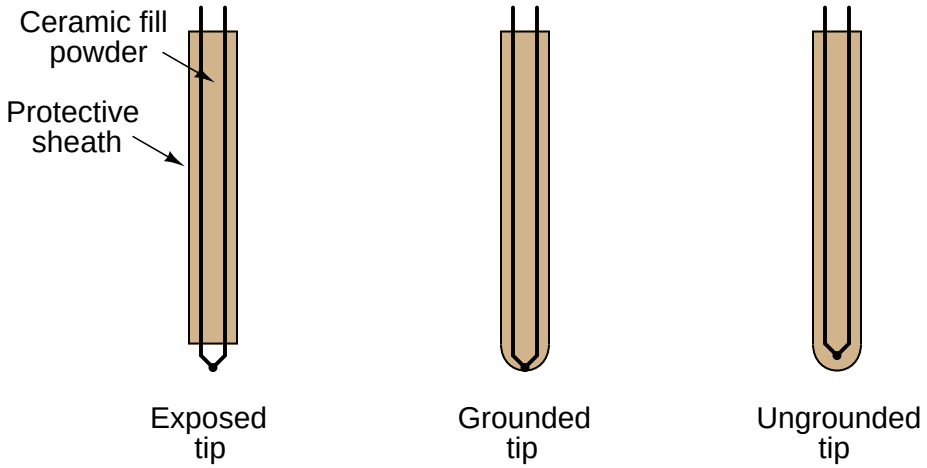
As you can see from this photograph, the screws directly press against the solid metal thermocouple wires to make a firm connection between each wire and the brass terminal block. Since the "head" attaches directly to one end of the thermocouple, the thermocouple's wires will be trimmed just long enough to engage with the terminal screws inside the head. Both brass terminal blocks are mounted on a ceramic base, the purpose of the ceramic being to help equalize the temperatures between the two brass blocks while still maintaining electrical isolation. This assembly is sometimes referred to as an *isothermal terminal block* because it acts to keep all connection points at a common temperature ("iso-thermal" = "same-temperature"). A threaded cover on the head provides easy access to these connection points for installation and maintenance, while ensuring the connections are covered and protected from ambient weather conditions the rest of the time.

Thermocouple wires are most often manufactured in solid form rather than stranded form. A common mistake made with thermocouple wires is for technicians to crimp compression-style terminals ("lugs") onto the solid wires. While this may form a usable connection at first, compression-style terminals are simply unable to maintain adequate compression when applied to solid wire of any type, thermocouple wire included. Over time, solid wires *will* loosen inside compression terminals leading to circuit problems. In the case of a thermocouple circuit, bad wire connections lead to a situation where the receiving instrument "thinks" the thermocouple has failed open. This situation is commonly called *burnout*, referring to the phenomenon where a thermocouple junction fails open from being "burned out" by excessive temperature.



You will most often find compression terminals (improperly) applied to solid thermocouple wire tips where those wires must terminate under the head of a screw. Compression terminals are correct to use in applications where stranded wire terminates at a screw head, but not solid wire. The proper termination technique for solid wire under a screw head is to wrap the solid wire in a semi-circle and directly clamp it under the screw head.

At the other end of the thermocouple, we have a choice of tip styles. For maximum sensitivity and fastest response, the dissimilar-metal junction may be unsheathed (bare). This design, however, makes the thermocouple more fragile. Sheathed tips are typical for industrial applications, available in either *grounded* or *ungrounded* forms:

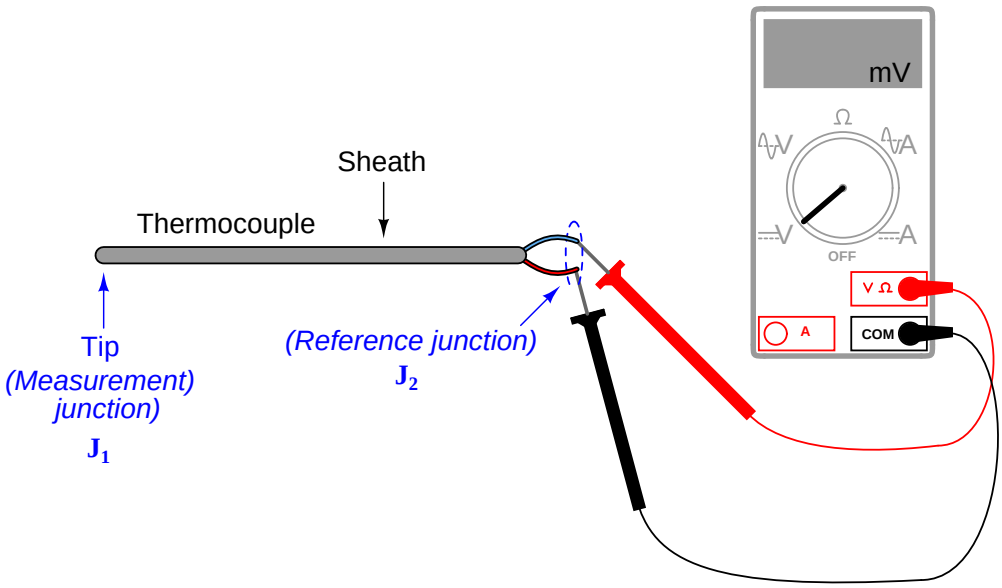


Grounded-tip thermocouples exhibit faster response times[11] and greater sensitivity than ungrounded-tip thermocouples, but they are vulnerable to *ground loops*: circuitous paths for electric current between the conductive sheath of the thermocouple and some other point in the thermocouple circuit. In order to avoid this potentially troublesome effect, most industrial thermocouples are of the ungrounded design.

[11] Grounded thermocouples often have thermal time constant values less than *half* that of comparable ungrounded thermocouples. Exposed-tip thermocouples are even faster than grounded-tip, typically by even larger ratios than grounded-tip thermocouples are to ungrounded thermocouples.

21.4.4 Manually interpreting thermocouple voltages

Recall that the amount of voltage indicated by a voltmeter connected to a thermocouple is the *difference* between the voltage produced by the measurement junction (the point where the two dissimilar metals join at the location we desire to sense temperature at) and the voltage produced by the reference junction (the point where the thermocouple wires join to the voltmeter wires):



$$V_{meter} = V_{J1} - V_{J2}$$

This makes thermocouples inherently *differential* sensing devices: they generate a measurable voltage in proportion to the *difference* in temperature between two locations. This inescapable fact of thermocouple circuits complicates the task of interpreting any voltage measurement obtained from a thermocouple.

In order to translate a voltage measurement produced by a voltmeter connected to a thermocouple, we must *add* the voltage produced by the measurement junction (V_{J2}) to the voltage indicated by the voltmeter to find the voltage being produced by the measurement junction (V_{J1}). In other words, we manipulate the previous equation into the following form:

$$V_{J1} = V_{J2} + V_{meter}$$

We may ascertain the reference junction voltage by placing a thermometer near that junction (where the thermocouple wire attaches to the voltmeter test leads) and referencing a *thermocouple table* showing temperatures and corresponding voltages for that thermocouple type. Then, we may take the voltage sum for V_{J1} and re-reference that same table, finding the temperature value corresponding to the calculated measurement junction voltage. The National Institute of Standards and Technology (NIST) in the United States publishes tables showing junction voltages

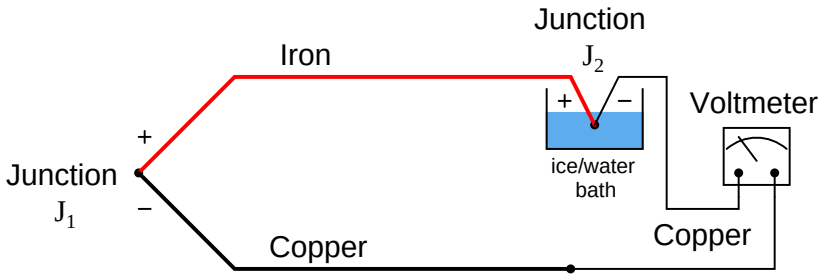
and temperatures for standardized thermocouple types. While it is possible to mathematically model a thermocouple junction's voltage in the same way we may model an RTD's resistance, the functions for thermocouples are less linear than for RTDs, and so tables are greatly preferred for practical use.

To illustrate, suppose we connected a voltmeter to a type K thermocouple and measured 14.30 millivolts. A thermometer situated near the thermocouple wire / voltmeter junction point shows an ambient temperature of 73 degrees Fahrenheit. Referencing a table of voltages for type K thermocouples (in this case, the NIST "ITS-90" reference standard), we see that a type K junction at 73 degrees Fahrenheit corresponds to 0.910 millivolts. Adding this figure to our meter measurement of 14.30 millivolts, we arrive at a sum of 15.21 millivolts for the measurement ("hot") junction. Going back to the same table of values, we see 15.21 millivolts falls between 701 and 702 degrees Fahrenheit. Linearly interpolating between the table values (15.203 mV at 701 ^{o}F and 15.226 mV at 702 ^{o}F), we may more precisely determine the measurement junction to be 701.3 degrees Fahrenheit.

The process of manually taking voltage measurements, referencing a table of millivoltage values, performing addition, then re-referencing the same table is rather tedious. Compensation for the reference junction's inevitable presence in the thermocouple circuit is something we must do, but it is not something that must always be done by a human being. The next subsection discusses ways to automatically compensate for the effect of the reference junction, which is the only practical alternative for continuous thermocouple-based temperature instruments.

21.4.5 Reference junction compensation

Multiple techniques exist to deal with the influence of the reference junction's temperature[12]. One technique is to physically fix the temperature of that junction at some constant value so it is always stable. This way, any changes in measured voltage *must* be due to changes in temperature at the measurement junction, since the reference junction has been rendered incapable of changing temperature. This may be accomplished by immersing the reference junction in a bath of ice and water, the ice/water mixture ensuring a stable temperature by means of water's *latent heat of fusion*[13]:



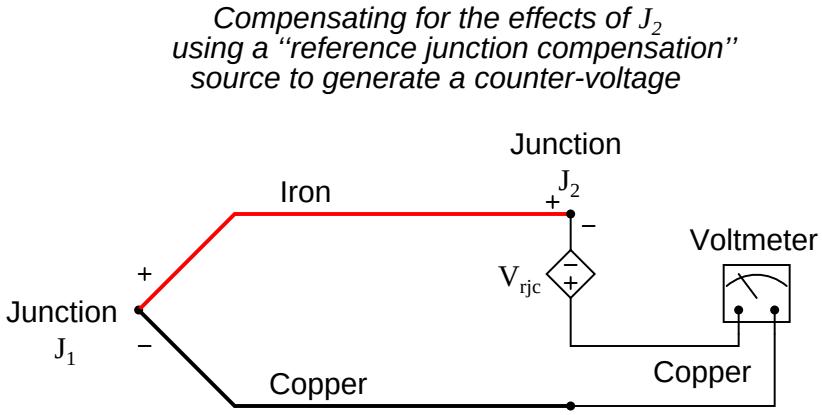
In fact, this is how thermocouple temperature/voltage tables are referenced: describing the amount of voltage produced for given temperatures at the measurement junction with the reference junction held at the freezing point of water (0 ^{o}C = 32 ^{o}F). With the reference junction maintained at the freezing point of water, and thermocouple tables referenced to that specific cold junction temperature, the voltmeter's indication will simply and directly correspond to the temperature of measurement junction J_1 at all times.

However, fixing the reference junction at the temperature of freezing water is impractical for any real thermocouple application outside of a laboratory. Instead, we need to find some other way to compensate for changes in reference junction temperature, so that we may accurately interpret the temperature of the measurement junction despite random changes in reference junction temperature.

[12] Early texts on thermocouple use describe multiple techniques for automatic compensation of the reference ("cold") junction. One design placed a mercury bulb thermometer at the reference junction, with a loop of thin platinum wire dipped into the mercury. As junction temperature rose, the mercury column would rise and short past a greater length of the platinum wire loop, causing its resistance to decrease which in turn would electrically bias the measurement circuit to offset the effects of the reference junction's voltage. Another design used a bi-metallic spring to offset the pointer of the meter movement, so that changes in temperature at the indicating instrument (where the reference junction was located) would result in the analog meter's needle becoming offset from its normal "zero" point, thus compensating for the offset in voltage created by the reference junction.

[13] For any two-phase mixture of any single substance (in this case, H_2O) the temperature of that mixture will be a strict function of pressure, the mixture possessing only one thermodynamic degree of freedom. Any addition or removal of heat from the ice/water mix results in a phase change (e.g. either more ice melts to become water, or more water freezes to become ice) rather than a temperature change. If even more precision is desired, a *triple point cell* may be used to fix the reference junction's temperature, since any substance at its triple point will possess zero degrees of thermodynamic freedom (i.e. neither its pressure nor temperature can change).

A practical way to compensate for the reference junction voltage is to include an additional voltage source within the thermocouple circuit equal in magnitude and opposite in polarity to the reference junction voltage. If this additional voltage is made continually equal to the reference junction's potential, it will precisely counter the reference junction voltage, resulting in the full (measurement junction) voltage appearing at the measuring instrument terminals. This is called a *reference junction compensation* or *cold junction*[14] *compensation* circuit:



In order for such a compensation strategy to work, the compensating voltage must continuously track the voltage produced by the reference junction. To do this, the compensating voltage source (V_{rjc} in the above schematic) uses some other temperature-sensing device such as a thermistor or RTD to sense the local temperature at the terminal block where junction J_2 is formed and produce a counter-voltage that is precisely equal and opposite to J_2's voltage ($V_{rjc} = V_{J2}$) at all times. Having canceled the effect of the reference junction, the voltmeter now only registers the voltage produced by the measurement junction J_1:

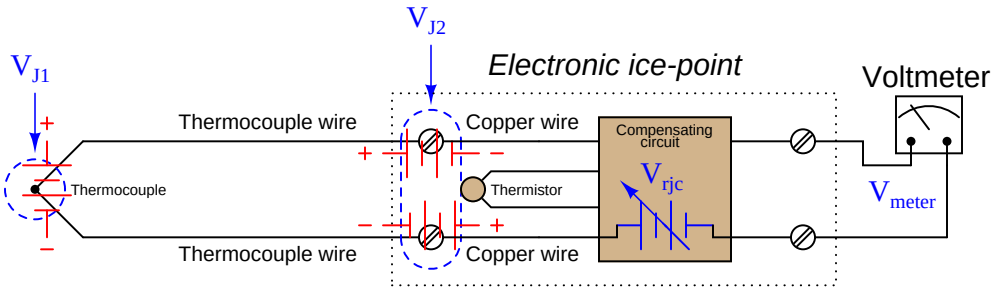
$$V_{meter} = V_{J1} - V_{J2} + V_{rjc}$$

$$V_{meter} = V_{J1} + 0 \qquad (\text{If } V_{rjc} = V_{J2})$$

$$V_{meter} = V_{J1}$$

[14] Please note that "cold junction" is just a synonymous label for "reference junction." In fact the "cold" reference junction may very well be at a warmer temperature than the so-called "hot" measurement junction! Nothing prevents anyone from using a thermocouple to measure temperatures below the freezing point of water.

Some instrument manufacturers sell *electronic ice point* modules designed to provide reference junction compensation for un-compensated instruments such as standard voltmeters. The "ice point" circuit performs the function shown by V_{rjc} in the previous diagram: it inserts a counter-acting voltage to cancel the voltage generated by the reference junction, so that the voltmeter only "sees" the measurement junction's voltage. This compensating voltage is maintained at the proper value according to the terminal temperature where the thermocouple wires connect to the ice point module, sensed by a thermistor or RTD:



Example values:

T_{J1} = 570 °F (type J) V_{J1} = 16.266 mV

T_{J2} = 69 °F (type J) V_{J2} = 1.048 mV V_{rjc} = 1.048 mV

$$V_{meter} = V_{J1} - V_{J2} + V_{rjc}$$

$$V_{meter} = 16.266 \text{ mV} - 1.048 \text{ mV} + 1.048 \text{ mV}$$

$$\mathbf{V_{meter} = 16.266 \text{ mV}} \text{ (equivalent to 570 °F)}$$

In this example, we see the measurement junction (J1) at a temperature of 570 degrees Fahrenheit, generating a voltage of 16.266 millivolts. If this thermocouple were directly connected to the meter, the meter would only register 15.218 millivolts, because the reference junction (J2, at 69 degrees Fahrenheit) opposes with its own voltage of 1.048 millivolts. With the ice point compensation circuit installed, however, the 1.048 millivolts of the reference junction is canceled by the ice point circuit's equal-and-opposite 1.048 millivolt source. This allows the full 16.266 millivolt signal from the measurement junction reach the voltmeter where it may be read and correlated to temperature by a type J thermocouple table.

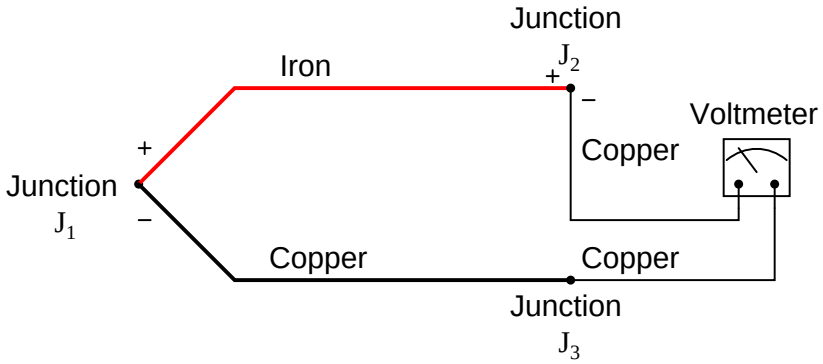
At first it may seem pointless to go through the trouble of building a reference junction compensation (ice point) circuit, when doing so requires the use of some other temperature-sensing element such as a thermistor or RTD. After all, why bother to do this just to be able to use a thermocouple to accurately measure temperature, when we could simply use this "other" device to directly measure the process temperature? In other words, isn't the usefulness of a thermocouple invalidated if we must rely on some other type of electrical temperature sensor just to compensate

for an idiosyncrasy of the thermocouple?

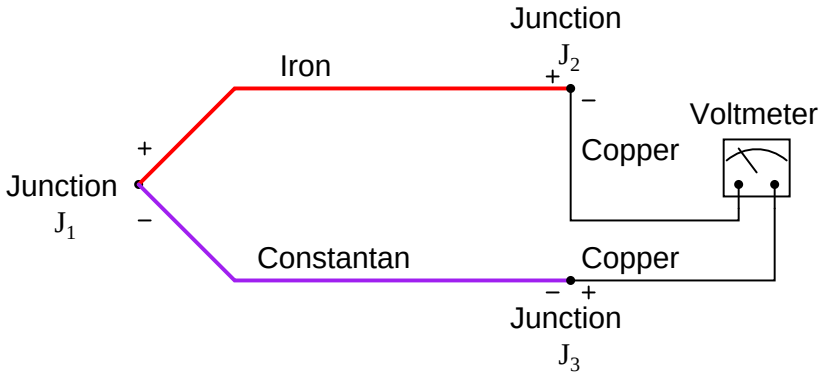
The answer to this very good question is that thermocouples enjoy certain advantages over these other sensor types. Thermocouples are extremely rugged and have far greater temperature-measurement ranges than thermistors, RTDs, and other primary sensing elements. However, if the application does not demand extreme ruggedness or large measurement ranges, a thermistor or RTD *is* most likely the better choice!

21.4.6 Law of Intermediate Metals

It is crucial to realize that the phenomenon of a "reference junction" is an inevitable effect of having to close the electric circuit loop in a circuit made of dissimilar metals. This is true regardless of the number of metals involved. In the last example, only two metals were involved: iron and copper. This formed one iron-copper junction (J_1) at the measurement end and one iron-copper junction (J_2) at the indicator end. Recall that the copper-copper junction J_3 was of no consequence because its identical metallic composition generates no thermal voltage:



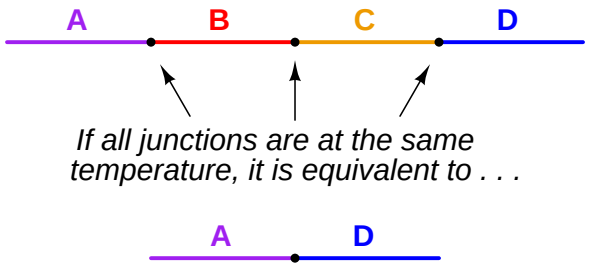
But what about more complex thermocouple circuits, involving more than two wire types? How do we define what a "reference junction" is, or how it behaves, when we have more than two dissimilar-metal junctions in the same circuit? Take for instance this example of a type J thermocouple:



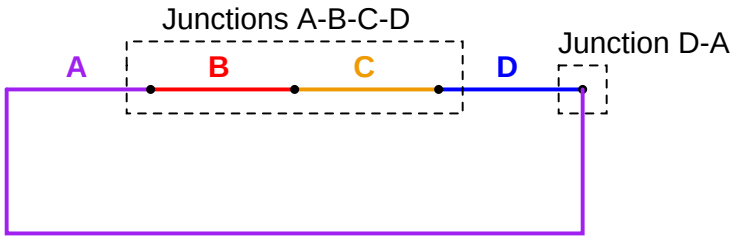
Here we have *three* voltage-generating junctions: J_1 of iron and constantan, J_2 of iron and copper, and J_3 of copper and constantan[15]. Upon first inspection it would seem we have a much more complex situation than we did with just two metals (iron and copper), but fortunately the situation is just as simple as it was before provided the temperatures of J_2 and J_3 are equal, which will be true if those two junctions are located very near each other (at the voltmeter).

[15] A junction of copper and constantan just happens to be a *type T* thermocouple junction.

A principle of thermo-electric circuits called the *Law of Intermediate Metals* helps us see this clearly. According to this law, intermediate metals in a series of junctions are of no consequence to the overall (net) voltage so long as those intermediate junctions are all at the same temperature. Representing this pictorially, the net effect of having four different metals (A, B, C, and D) joined together in series is the same as just having the first and last metal in that series (A and D) joined with one junction, if all intermediate junctions are at the same temperature:



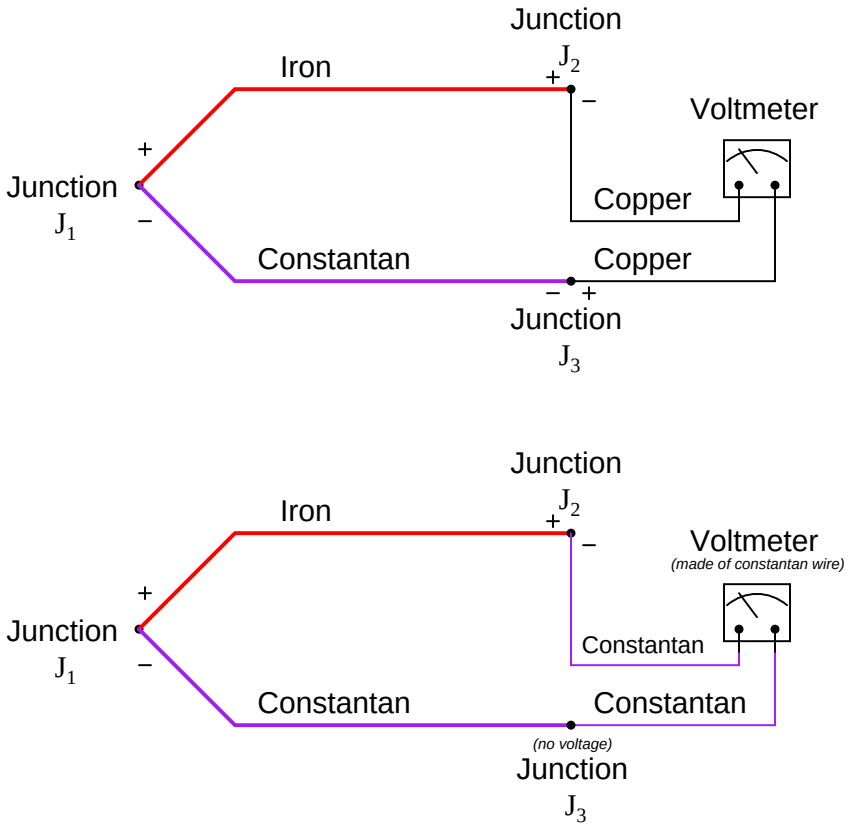
A simple proof of the Law of Intermediate Metals may be built upon the *Law of Energy Conservation*, one of the most fundamental principles in all of physics. Consider what would happen if we were to join the series of dissimilar metal wires shown above into a continuous loop:



In this diagram we see that the wire made of metal "A" connects to a string of metal junctions formed by metals "B", "C", and "D". If all these dissimilar metal junctions are at the same temperature, there will be no difference of temperature anywhere in the circuit to drive a current, and we would therefore expect the current in this circuit to be zero. This is in accordance with the Law of Energy Conservation, which forbids the passage of electric current through resistive wire without some motive power source driving it. Thus, based on the premise that energy must be conserved (i.e. that an electric current cannot flow through any resistance without a power source), we must conclude that *the net effect of all those series-connected metal junctions at the same temperature must be zero*. In other words, junctions A-B, B-C, C-D, and D-A all at the same temperature and connected in series *must* generate zero voltage, as if those junctions were all reduced to a single A-A junction which of course cannot produce any electromotive force (voltage) because it is not comprised of dissimilar metals. If the Law of Intermediate Metals were untrue, it would mean that the junctions A-B-C-D were *not* equivalent to the single junction A-D, which would mean they would produce a different voltage than the D-A junction at the right-hand end of this circuit (while at the same temperature), and therefore this circuit would produce some net voltage to drive a current continuously through resistive wire in violation of the Law of Energy Conservation. Since we know the Law of Energy Conservation to be well-founded (and we can also *build* such dissimilar metal

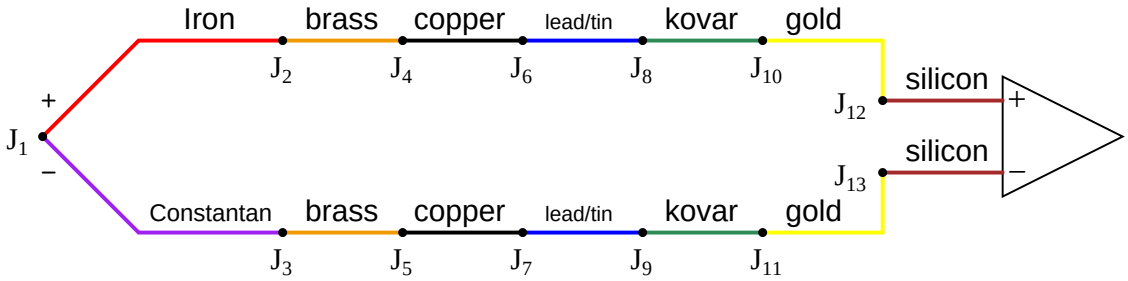
loop circuits and empirically determine their currents to be zero), we may rest assured that the Law of Intermediate Metals is true.

In our type J thermocouple circuit where iron and constantan both join to copper, we see copper as an intermediate metal between junctions J_2 and J_3. Being located next to each other on the indicating instrument, identical temperature is a reasonable assumption for J_2 and J_3, so we may invoke the Law of Intermediate Metals and simply treat junctions J_2 and J_3 as a *single* iron-constantan reference junction. In other words, the Law of Intermediate Metals tells us we can treat the following two circuits identically:



The practical importance of this Law is that we can always treat the reference junction(s) as a single junction made from the same two metal types as the measurement junction, so long as all dissimilar metal junctions at the reference location are at the same temperature.

This fact is extremely important in the age of semiconductor circuitry, where the connection of a thermocouple to an electronic amplifier involves a long series of dissimilar-material junctions. Here we see a multitude of reference junctions, formed by the necessary connections from thermocouple wire to the silicon substrate inside the amplifier chip:

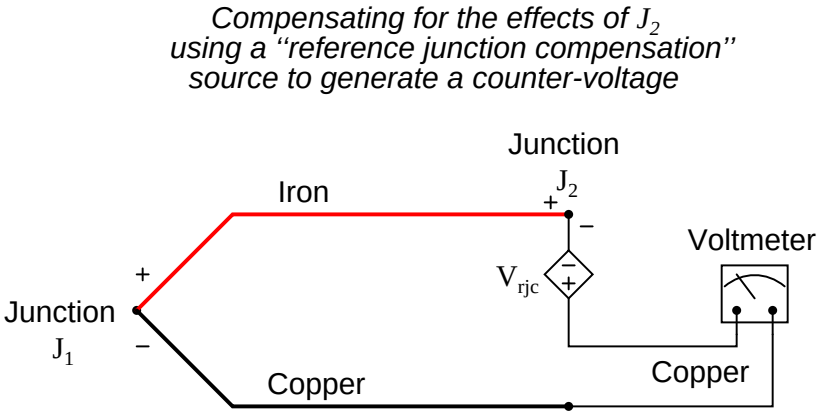


Here we see the metals of the thermocouple wire (type J – iron and constantan) joining to a pair of brass terminal screws, which in turn join to copper traces on a printed circuit board, which join to lead/tin solder, which join to thin wires made of Kovar, which terminate at gold traces on the silicon chip, which are bonded to the silicon itself.

It should be obvious that each complementary junction pair in this series loop cancel each other if each pair is at the same temperature (e.g. gold-silicon junction J_{12} cancels with silicon-gold junction J_{13} because they generate the exact same amount of voltage with opposing polarities; Kovar-gold junction J_{10} cancels with gold-Kovar junction J_{11} for the same reason; etc.). The Law of Intermediate Metals goes one step further by telling us junctions J_2 through J_{13} taken together in series are of the same effect as a single reference junction of iron and constantan. Automatic reference junction compensation is as simple as counter-acting the voltage produced by this equivalent iron-constantan junction at whatever temperature junctions J_2 through J_{13} happen to be at.

21.4.7 Software compensation

Previously, it was suggested that automatic compensation could be accomplished by intentionally inserting a temperature-dependent voltage source in series with the circuit, oriented in such a way as to oppose the reference junction's voltage:



$$V_{meter} = V_{J1} - V_{J2} + V_{rjc}$$

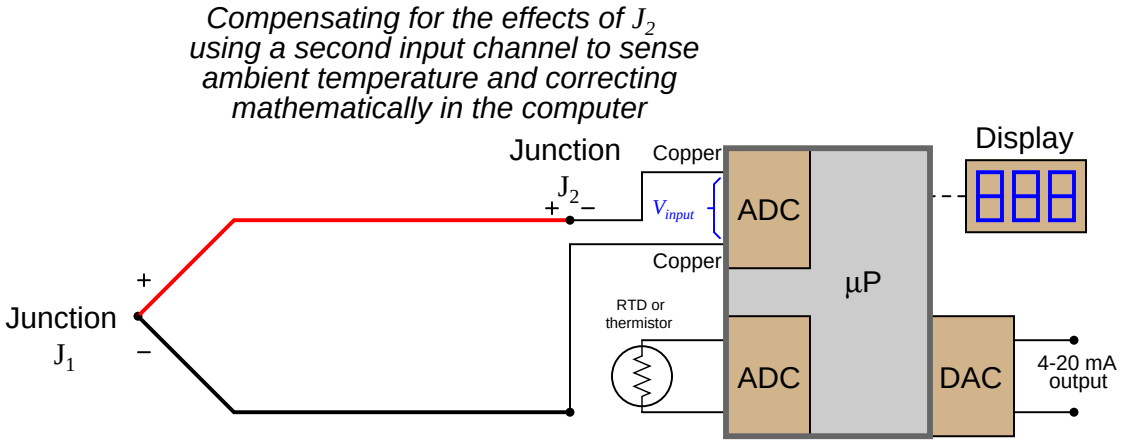
If the series voltage source V_{rjc} is exactly equal in magnitude to the reference junction's voltage (V_{J2}), those two terms cancel out of the equation and lead to the voltmeter measuring only the voltage of the measurement junction J_1:

$$V_{meter} = V_{J1} + 0$$

$$V_{meter} = V_{J1}$$

This technique is known as *hardware* compensation, and is employed in analog thermocouple temperature transmitter designs. Previously we saw an example of this called an *ice point*, the purpose of which was to electrically counter the reference junction voltage to render that junction's voltage inconsequential as though that junction were immersed in a bath of ice-water.

A modern technique for reference junction compensation more suitable to digital transmitter designs is called *software* compensation:



Instead of canceling the effect of the reference junction *electrically*, we cancel the effect *arithmetically* inside the microprocessor-based transmitter. In other words, we let the receiving analog-digital converter circuit see the difference in voltage between the measurement and reference junctions ($V_{input} = V_{J1} - V_{J2}$), but then after digitizing this voltage measurement we have the microprocessor add the equivalent voltage value corresponding to the ambient temperature sensed by the RTD or thermistor (V_{rjc}):

$$\text{Compensated total} = V_{input} + V_{rjc}$$

$$\text{Compensated total} = (V_{J1} - V_{J2}) + V_{rjc}$$

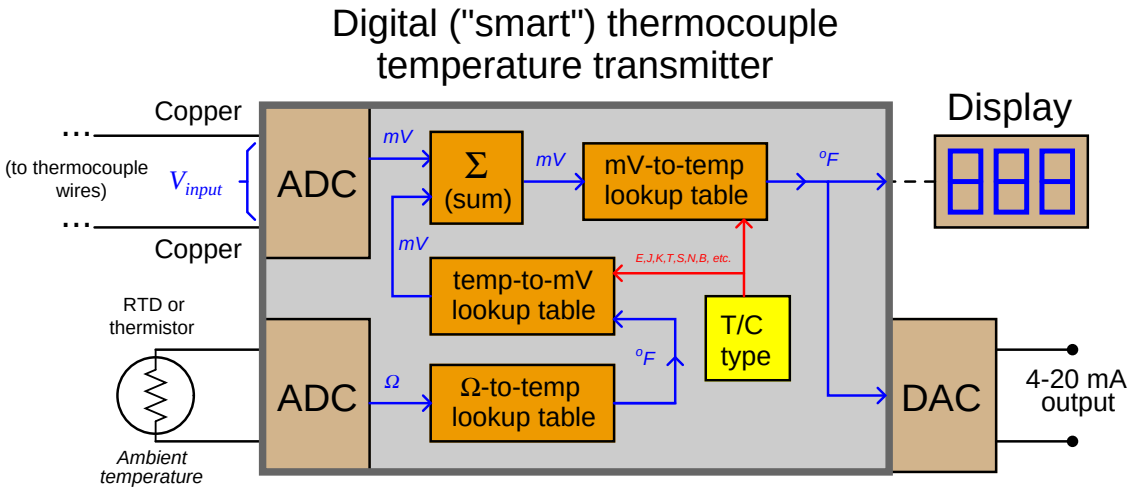
Since we know the calculated value of V_{rjc} should be equal to the real reference junction voltage (V_{J2}), the result of this digital addition should be a compensated total equal only to the measurement junction voltage V_{J1}:

$$\text{Compensated total} = V_{J1} - V_{J2} + V_{rjc}$$

$$\text{Compensated total} = V_{J1} + 0$$

$$\text{Compensated total} = V_{J1}$$

A block diagram of a thermocouple temperature transmitter with software compensation appears here:



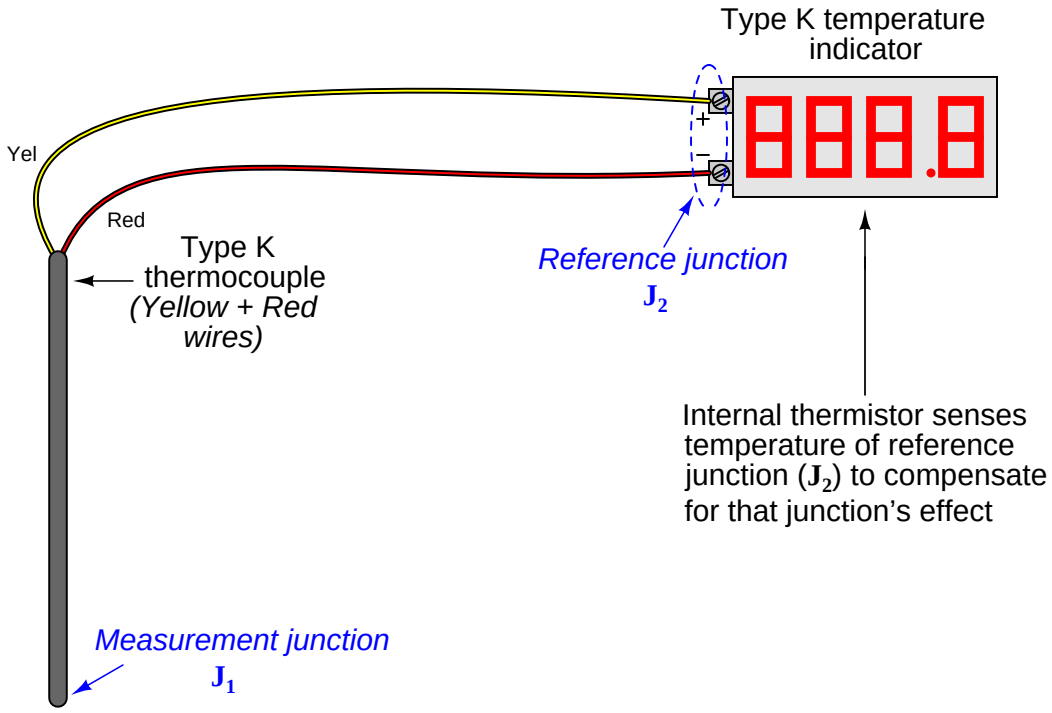
Perhaps the greatest advantage of software compensation is the flexibility to easily switch between different thermocouple types with no hardware modification. So long as the microprocessor memory is programmed with look-up tables relating voltage values to temperature values, it may accurately measure (and compensate for the reference junction of) *any* thermocouple type. Hardware-based compensation schemes (e.g. an analog "ice point" circuit) require re-wiring or replacement to accommodate different thermocouple types, since each ice-point circuit is built to generate a compensating voltage for a specific type of thermocouple.

21.4.8 Extension wire

In every thermocouple circuit there must be both a measurement junction and a reference junction: this is an inevitable consequence of forming a complete circuit (loop) using dissimilar-metal wires. As we already know, the voltage received by the measuring instrument from a thermocouple will be the *difference* between the voltages produced by the measurement and reference junctions. Since the purpose of most temperature instruments is to accurately measure temperature *at a specific location*, the effects of the reference junction's voltage must be "compensated" for by some means, either a special circuit designed to add an additional canceling voltage or by a software algorithm to digitally cancel the reference junction's effect.

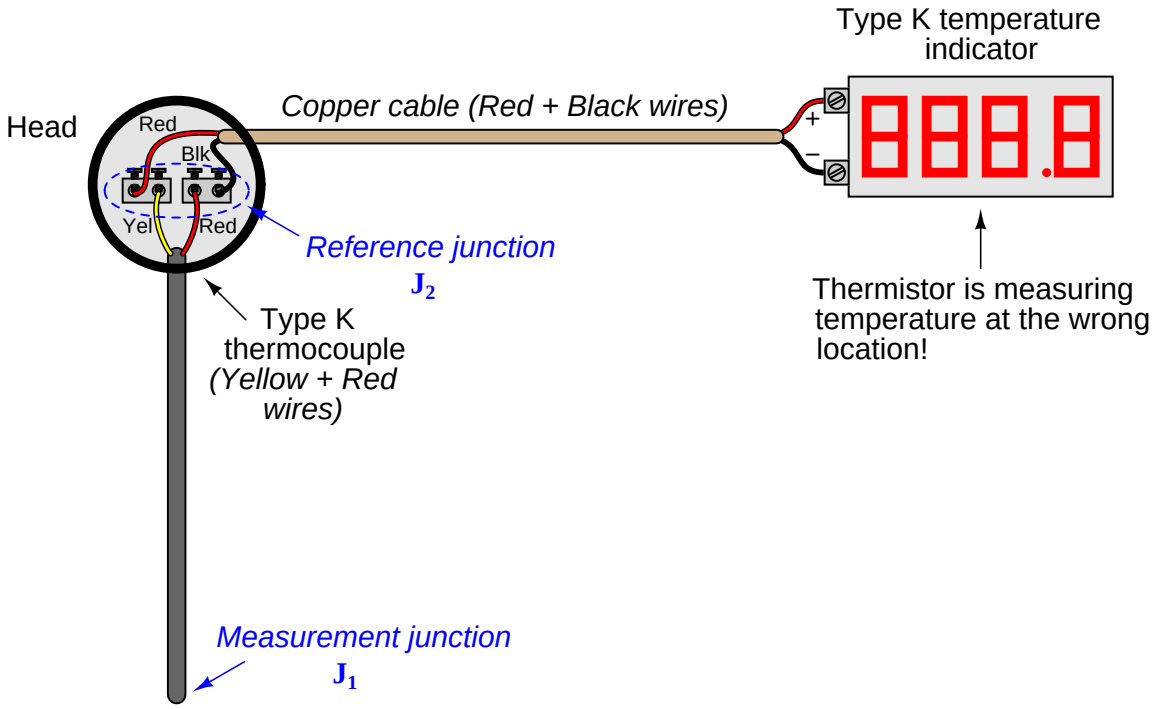
In order for reference junction compensation to be effective, the compensation mechanism must "know" the temperature of the reference junction. This fact is so obvious, it hardly requires mentioning. However, what is not so obvious is how easily this compensation may be unintentionally defeated simply by installing a different type of wire in a thermocouple circuit.

To illustrate, let us examine a simple type K thermocouple installation, where the thermocouple connects directly to a panel-mounted temperature indicator by long wires:



Like all modern thermocouple instruments, the panel-mounted indicator contains its own internal reference junction compensation, so that it is able to compensate for the temperature of the reference junction formed at its connection terminals, where the internal (copper) wires of the indicator join to the chromel and alumel wires of the thermocouple. The indicator senses this junction temperature using a small thermistor thermally bonded to the connection terminals.

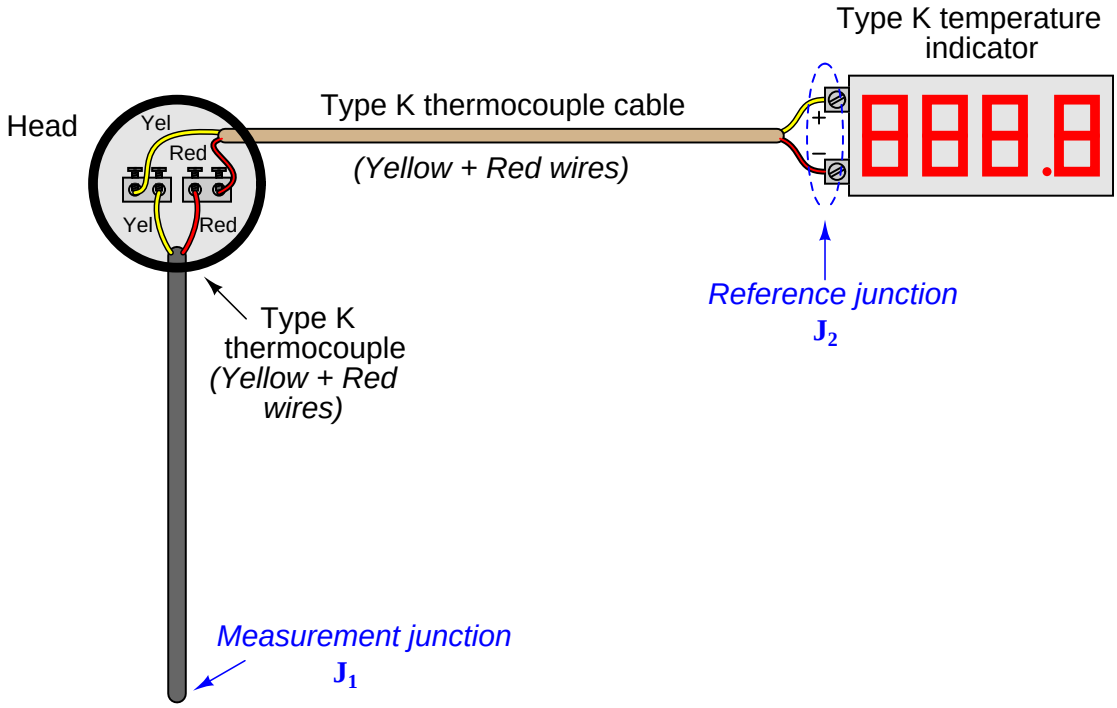
Now let us consider the same thermocouple installation with a length of copper cable (two wires) joining the field-mounted thermocouple to the panel-mounted indicator:



Even though nothing has changed in the thermocouple circuit except for the type of wires joining the thermocouple to the indicator, the reference junction has completely shifted position. What used to be a reference junction (at the indicator's terminals) is no longer, because now we have copper wires joining to copper wires. Where there is no dissimilarity of metals, there can be no thermoelectric potential. At the thermocouple's connection "head," however we now have a joining of chromel and alumel wires to copper wires, thus forming a reference junction *in a new location* at the thermocouple head. What is worse, this new location is likely to be at a different temperature than the panel-mounted indicator, which means the indicator's reference junction compensation will be compensating for the wrong temperature.

The only practical way to avoid this problem is to keep the reference junction where it belongs: at the terminals of the panel-mounted instrument where the ambient temperature is measured and the reference junction's effects accurately compensated. If we must install "extension" wire to join a thermocouple to a remotely-located instrument, that wire must be of a type that does not form another dissimilar-metal junction at the thermocouple head, but will form one at the receiving instrument.

An obvious approach is to simply use thermocouple wire of the same type as the installed thermocouple to join the thermocouple to the indicator. For our hypothetical type K thermocouple, this means a type K cable installed between the thermocouple head and the panel-mounted indicator:



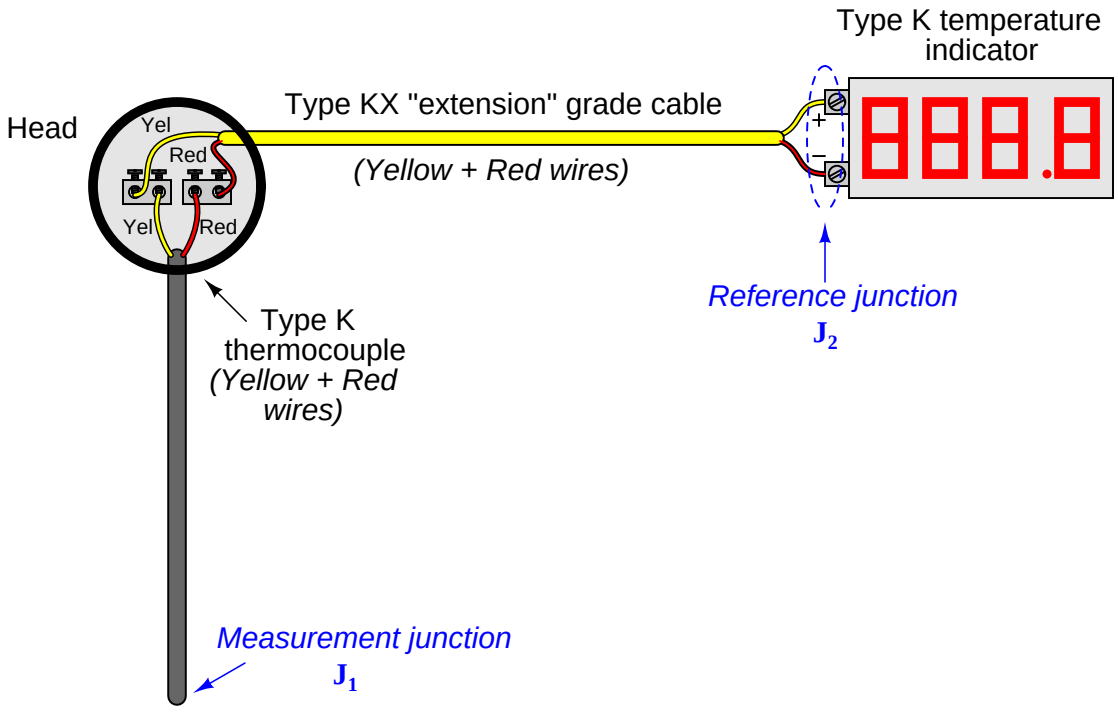
With chromel joining to chromel and alumel joining to alumel at the head, no dissimilar-metal junctions are created at the thermocouple. However, with chromel and alumel joining to copper at the indicator (again), the reference junction has been re-located to its rightful place. This means the thermocouple head's temperature will have no effect on the performance of this measurement system, and the indicator will be able to properly compensate for any ambient temperature changes at the panel as it was designed to do. The only problem with this approach is the potential expense of thermocouple-grade cable. This is especially true with some types of thermocouples, where the metals used are somewhat exotic (e.g. types R, S, and B).

A more economical alternative, however, is to use something called *extension-grade* wire to make the connection between the thermocouple and the receiving instrument. "Extension-grade" thermocouple wire is made less expensive than full "thermocouple-grade" wire by choosing metal alloys similar in thermo-electrical characteristics to the real thermocouple wires within modest temperature ranges. So long as the temperatures at the thermocouple head and receiving instrument terminals don't get too hot or too cold, the extension wire metals joining to the thermocouple wires and joining to the instrument's copper wires need not be *precisely identical* to the true thermocouple wire alloys. This allows for a wider selection of metal types, some of which are substantially less expensive than the measurement-grade thermocouple alloys. Also, extension-grade wire may use insulation with a narrower temperature rating than thermocouple-grade wire, reducing cost even

further.

An interesting historical reference to the use of extension-grade wire appears in Charles Robert Darling's 1911 text *Pyrometry – A Practical Treatise on the Measurement of High Temperatures*. On page 61, Darling describes "compensating leads" marketed under the brand-name of *Peake* designed to be used with platinum-alloy thermocouples. These "compensating" wires were made of two different copper-nickel alloys, each copper-nickel alloy matched with the respective thermocouple metal (in this case, pure platinum and a 90%-10% platinum-iridium alloy) to generate an equal and opposite millivoltage at any reasonable temperature found at the thermocouple head. Thus, the only reference junction in the thermocouple circuit is where these copper-nickel extension wires joined with the indicating instrument, rather than being located at the thermocouple head as it would be if simple copper extension wires were employed. With platinum being such an expensive metal (both then and now!), the cost savings realized by being able to use cheaper extension wire to connect the platinum thermocouple to a distant receiving instrument is significant.

Extension-grade cable is denoted by a letter "X" following the thermocouple letter. For our hypothetical type K thermocouple system, this would mean type "KX" extension cable:



Thermocouple extension cable also differs from thermocouple-grade (measurement) cable in the coloring of its outer jacket. Whereas thermocouple-grade cable is typically[16] brown in exterior color, extension-grade cable is usually colored[17] to match the thermocouple plug (yellow for type K, black for type J, blue for type T, etc.).

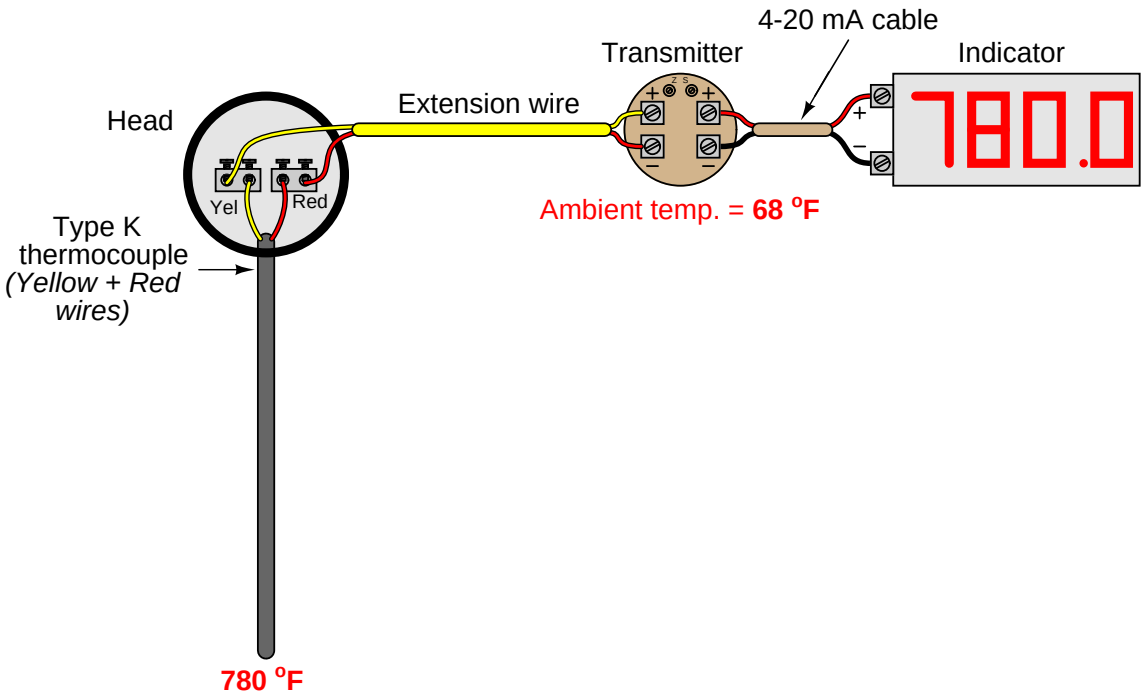
[16] No coloring standard exists in the United States for platinum thermocouple-grade wire (e.g. types R, S, etc.).

[17] The colors I list here are for thermocouples in the United States.

21.4.9 Side-effects of reference junction compensation

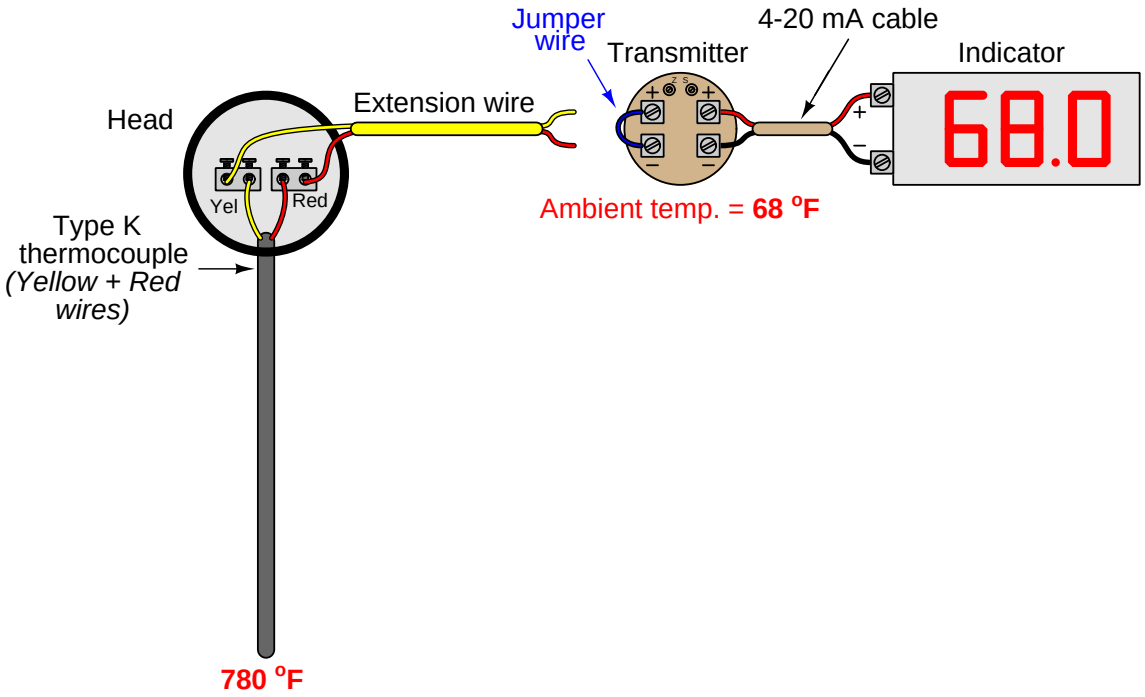
Reference junction compensation is a necessary part of any precision thermocouple circuit, due to the inescapable fact of the reference junction's existence. When you form a complete circuit of dissimilar metals, you *will* form both a measurement junction and a reference junction, with those two junctions' polarities opposed to one another. This is why reference junction compensation – whether it takes the form of a hardware circuit or an algorithm in software – must exist within every precision thermocouple instrument.

The presence of reference junction compensation in every precision thermocouple instrument results in an interesting phenomenon: *if you directly short-circuit the thermocouple input terminals of such an instrument, it will always register ambient temperature, regardless of the thermocouple type the instrument is built or configured for.* This behavior may be illustrated by example, first showing a normal operating temperature measurement system and then with that same system short-circuited. Here we see a temperature indicator receiving a 4-20 mA current signal from a temperature transmitter, which is receiving a millivoltage signal from a type "K" thermocouple sensing a process temperature of 780 degrees Fahrenheit:



The transmitter's internal reference junction compensation feature compensates for the ambient temperature of 68 degrees Fahrenheit. If the ambient temperature rises or falls, the compensation will automatically adjust for the change in reference junction potential, such that the output will still register the process (measurement junction) temperature of 780 degrees F. This is what the reference junction compensation is designed to do.

Now, we disconnect the thermocouple from the temperature transmitter and short-circuit the transmitter's input:



With the input short-circuited, the transmitter "sees" no voltage at all from the thermocouple circuit. There is no measurement junction nor a reference junction to compensate for, just a piece of wire making both input terminals electrically common. This means the reference junction compensation inside the transmitter no longer performs a useful function. However, the transmitter does not "know" it is no longer connected to the thermocouple, so the compensation keeps on working even though it has nothing to compensate for. Recall the voltage equation relating measurement, reference, and compensation voltages in a hardware-compensated thermocouple instrument:

$$V_{meter} = V_{J1} - V_{J2} + V_{rjc}$$

Disconnecting the thermocouple wire and connecting a shorting jumper to the instrument eliminates the V_{J1} and V_{J2} terms, leaving only the compensation voltage to be read by the meter[18]:

$$V_{meter} = 0 + V_{rjc}$$

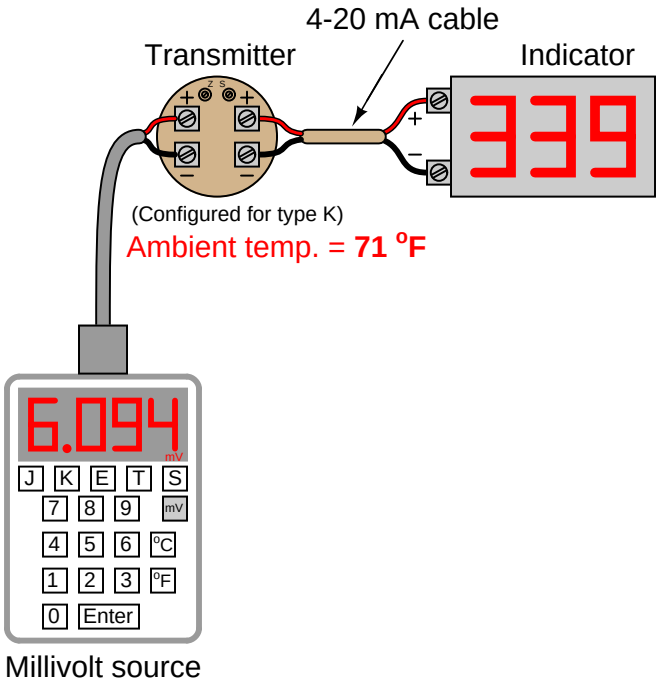
$$V_{meter} = V_{rjc}$$

[18]The effect will be exactly the same for an instrument with software compensation rather than hardware compensation. With software compensation, there is no literal V_{rjc} voltage source, but the equivalent millivolt value is digitally added to the zero input measured at the thermocouple connection terminals, resulting in the same effect of measuring ambient temperature.

This is why the instrument registers the equivalent temperature created by the reference junction compensation feature: this is the only signal it "sees" with its input short-circuited. This phenomenon is true regardless of which thermocouple type the instrument is configured for, which makes it a convenient "quick test" of instrument function in the field. If a technician short-circuits the input terminals of any thermocouple instrument, it should respond as though it is sensing ambient temperature.

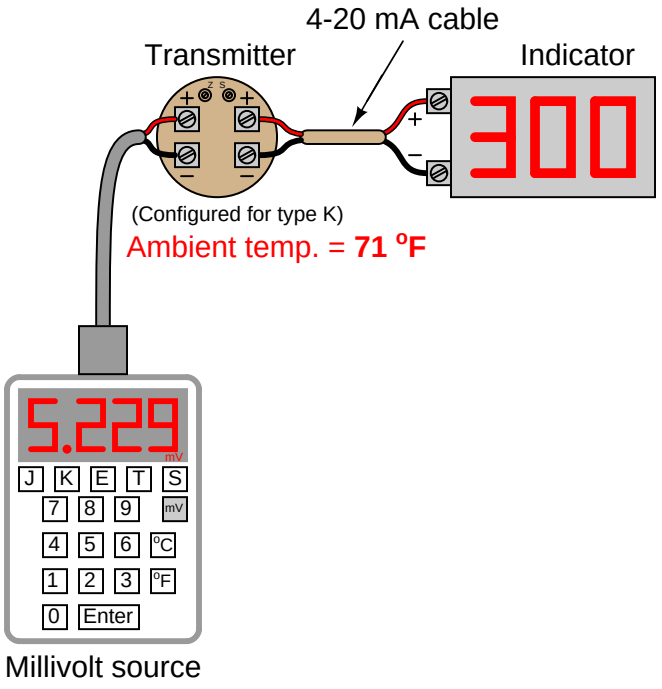
While this interesting trait is a somewhat useful side-effect of reference junction compensation in thermocouple instruments, there are other effects that are not quite so useful. The presence of reference junction compensation becomes quite troublesome, for example, if one tries to simulate a thermocouple using a precision millivoltage source. Simply setting the millivoltage source to the value corresponding to the desired (simulation) temperature given in a thermocouple table will yield an incorrect result for any ambient temperature other than the freezing point of water!

Suppose, for example, a technician wished to simulate a type K thermocouple at 300 degrees Fahrenheit by setting a millivolt source to 6.094 millivolts (the voltage corresponding to 300 ^{o}F for type K thermocouples according to the ITS-90 standard). Connecting the millivolt source to the instrument will *not* result in an instrument response appropriate for 300 degrees F:



Instead, the instrument registers 339 degrees because its internal reference junction compensation feature is still active, compensating for a reference junction voltage that no longer exists. The millivolt source's output of 6.094 mV gets *added* to the compensation voltage (inside the transmitter) of 0.865 mV – the necessary millivolt value to compensate for a type K reference junction at 71 ^{o}F – with the result being a larger millivoltage (6.959 mV) interpreted by the transmitter as a temperature of 339 ^{o}F.

One way to use a millivoltage source to simulate a desired temperature is for the instrument technician to "out-think" the transmitter's compensation feature by specifying a millivolt signal that is offset by the amount of equivalent voltage generated by the transmitter's compensation. In other words, instead of setting the millivolt source to a value of 6.094 mV, the technician should set the source to only 5.229 mV so the transmitter's compensation will add 0.865 mV to this value to arrive at 6.094 mV and properly register as 300 degrees Fahrenheit:

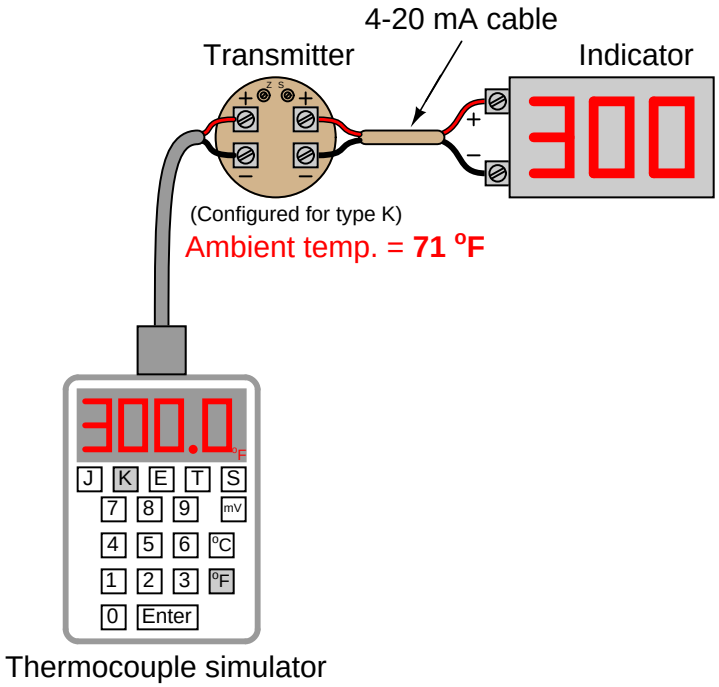


Years ago, the only suitable piece of test equipment available for generating the precise millivoltage signals necessary to calibrate thermocouple instruments was a device called a *precision potentiometer*. These "potentiometers" used a stable *mercury cell* battery (sometimes called a *standard cell*) as a voltage reference and a potentiometer with a calibrated knob to output low-voltage signals. Photographs of two vintage precision potentiometers are shown here:



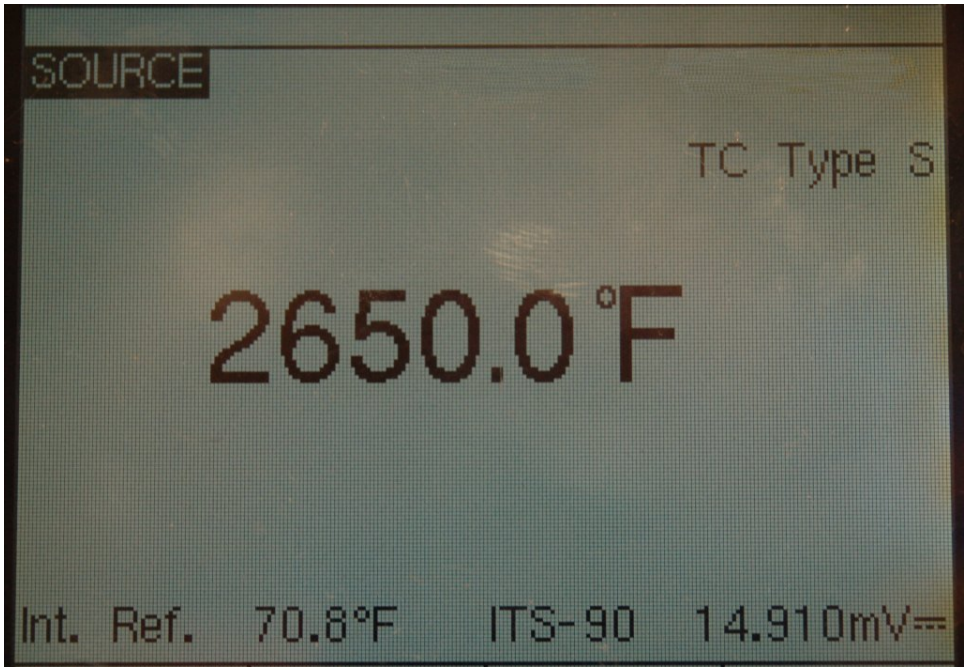


Of course, modern thermocouple calibrators also provide direct entry of temperature and automatic compensation to "un-compensate" the transmitter such that any desired temperature may be easily simulated:



In this example, when the technician sets the calibrator for 300 ^{o}F (type K), it measures the ambient temperature and automatically subtracts 0.865 mV from the output signal, so only 5.229 mV is sent to the transmitter terminals instead of the full 6.094 mV. The transmitter's internal reference junction compensation adds the 0.865 mV offset value (thinking it must compensate for a reference junction that in reality is not there) and "sees" a total signal voltage of 6.094 mV, interpreting this properly as 300 degrees Fahrenheit.

The following photograph shows the display of a modern thermocouple calibration device (a Fluke model 744 documenting process calibrator) being used to generate a thermocouple signal. In this particular example, the thermocouple type is set to type "S" (Platinum-Rhodium/Platinum) at a temperature of 2650 degrees Fahrenheit:



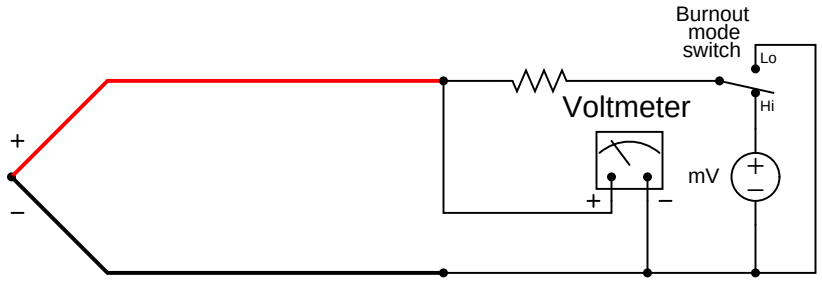
The ITS-90 thermocouple standard declares a millivoltage signal value of 15.032 mV for a type S thermocouple junction at 2650 degrees F (with a reference junction temperature of 32 degrees F). Note how the calibrator does *not* output 15.032 mV even though the simulated temperature has been set to 2650 degrees F. Instead, it outputs 14.910 mV, which is 0.122 mV less than 15.032 mV. This offset of 0.122 mV corresponds to the calibrator's local temperature of 70.8 degrees F (according to the ITS-90 standard for type S thermocouples).

When the calibrator's 14.910 mV signal reaches the thermocouple instrument being calibrated (be it an indicator, transmitter, or even a controller equipped with a type S thermocouple input), the instrument's own internal reference junction compensation will add 0.122 mV to the received signal of 14.910 mV, "thinking" it needs to compensate for a real reference junction. The result will be a perceived measurement junction signal of 15.032 mV, which is exactly what we want the instrument to "think" it sees if our goal is to simulate connection to a real type S thermocouple at a temperature of 2650 degrees F.

21.4.10 Burnout detection

Another consideration for thermocouples is *burnout detection*. The most common failure mode for thermocouples is to fail open, otherwise known as "burning out." An open thermocouple is problematic for any voltage-measuring instrument with high input impedance because the lack of a complete circuit on the input makes it possible for electrical noise from surrounding sources (power lines, electric motors, variable-frequency motor drives) to be detected by the instrument and falsely interpreted as a wildly varying temperature.

For this reason it is prudent to design into the thermocouple instrument some provision for generating a consistent state in the absence of a complete circuit. This is called the *burnout mode* of a thermocouple instrument. A simple thermocouple circuit equipped with burnout detection is shown in this diagram:



The resistor in this circuit provides a connection to a stable voltage[19] in the event of an open thermocouple. It is sized in the mega-ohm range to minimize its effect during normal operation when the thermocouple circuit is complete. Only when the thermocouple fails open will the miniscule current through the resistor have any substantial effect on the voltmeter's indication. The SPDT switch provides a selectable burnout mode: in the event of a burnt-out thermocouple, we can configure the meter to either read high temperature (sourced by the instrument's internal millivoltage source) or low temperature (grounded), depending on what failure mode we deem safest[20] for the application.

[19] For those readers familiar with digital logic gate circuits, this resistor fulfills the same function as a *pullup* or *pulldown* resistor on the input of a digital gate: providing a stable logic state in the event of a floating input condition.

[20] This is a good application of *fail-safe design*, where we choose the transmitter's failure mode based on the safest outcome. For example, if our temperature transmitter were being used to sense the temperature of a furnace where excessive temperature was more dangerous than insufficient temperature, we would want to configure it for "high" burnout. This way if the thermocouple fails open, the transmitter will report a dangerous (but false) measurement of furnace temperature to the controller, which in turn will automatically act to *decrease* the furnace's actual temperature (i.e. the safer condition.)

21.5 Non-contact temperature sensors

Virtually any mass above absolute zero temperature emits electromagnetic radiation (photons, or light) as a function of that temperature. This basic fact makes possible the measurement of temperature by analyzing the light emitted by an object. The *Stefan-Boltzmann Law* of radiated energy quantifies this fact, declaring that the rate of heat lost by radiant emission from a hot object is proportional to the fourth power of the absolute temperature:

$$\frac{dQ}{dt} = e\sigma AT^4$$

Where,
$\frac{dQ}{dt}$ = Radiant heat loss rate (watts)
e = Emissivity factor (unitless)
σ = Stefan-Boltzmann constant (5.67×10^{-8} W / m^2 $\cdot$ K^4)
A = Surface area (square meters)
T = Absolute temperature (Kelvin)

The primary advantage of non-contact thermometry (or *pyrometry* as high-temperature measurement is often referred) is rather obvious: with no need to place a sensor in direct contact with the process, a wide variety of temperature measurements may be made that are either impractical or impossible to make using any other technology.

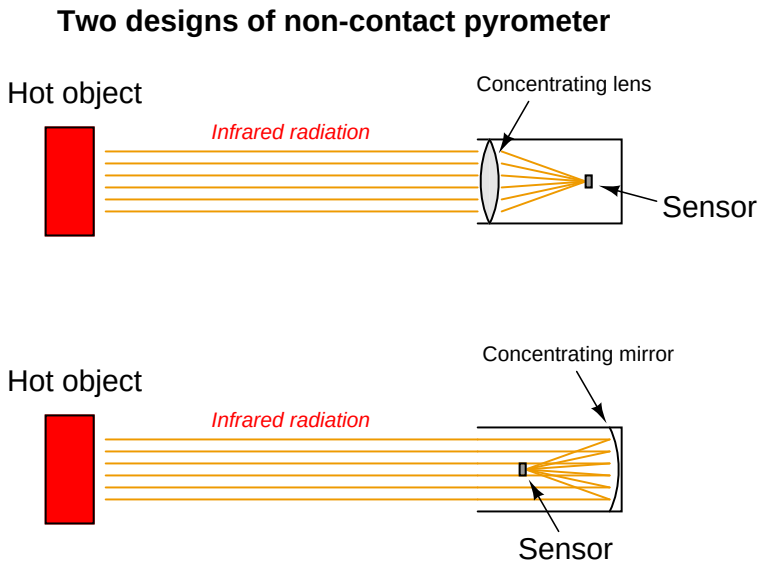
A major disadvantage of non-contact thermometry is that it only reveals the *surface temperature* of an object. Sensing the thermal radiation emanated from a pipe, for instance, only tells you the surface temperature of that pipe and not the true temperature of the fluid within the pipe. Another example is when doctors use non-contact thermometry to assess irregularities in body temperature: what they detect is just skin temperature. While it may be true that "hot spots" beneath the surface of an object may be detectable this way, it is only because the surface temperature of that object differs as a consequence of the hot spot(s) beneath. If a hotter-than-normal region inside of an object fails to transfer enough thermal energy to the surface to manifest as a hotter surface temperature, that region will be invisible to non-contact thermometry.

It may surprise some readers to discover that non-contact pyrometry is nearly as old as thermocouple technology[21], the first non-contact pyrometer being constructed in 1892.

[21] Although Seebeck discovered thermo-electricity in 1822, the technique of measuring temperature by sensing the voltage produced at a dissimilar-metal junction was delayed in practical development until 1886 when rugged and accurate electrical meters became available for industrial use.

21.5.1 Concentrating pyrometers

A time-honored design for non-contact pyrometers is to concentrate[22] incident light from the surface of a heated object onto a small temperature-sensing element. A rise in temperature at the sensor reveals the intensity of the infrared optical energy falling upon it, which as discussed previously is a function of the target object's surface temperature (absolute temperature to the fourth power):

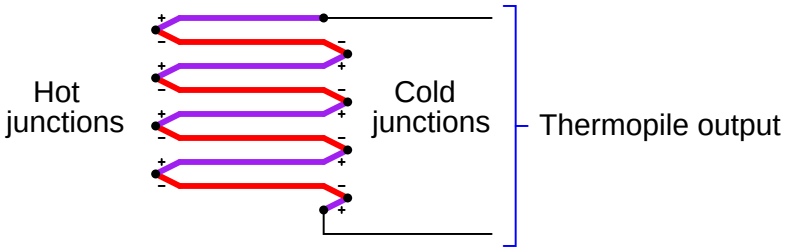


The fourth-power characteristic of Stefan-Boltzmann's law means that a doubling of absolute temperature at the hot object results in sixteen times as much radiant energy falling on the sensor, and therefore a sixteen-fold increase in the sensor's temperature rise over ambient. A tripling of target temperature (absolute) yields *eighty one times as much radiant energy*, and therefore an 81-fold increase in sensor temperature rise. This extreme nonlinearity limits the practical application of non-contact pyrometry to relatively narrow ranges of target temperature wherever good accuracy is required.

Thermocouples were the first type of sensor used in non-contact pyrometers, and they still find application in modern versions of the same technology. Since the sensor does not become nearly as hot as the target object, the output of any single thermocouple junction at the sensor area will be quite small. For this reason, instrument manufacturers often employ a series-connected array of thermocouples called a *thermopile* to generate a stronger electrical signal.

[22] Anyone who has ever used a magnifying glass (a concentrating lens) to concentrate sunlight knows how this works. If you were to use a magnifying glass to concentrate sunlight onto a thermocouple-type sensor, you could (at least in principle) infer the temperature of the sun in this manner.

The basic concept of a thermopile is to connect multiple thermocouple junctions in series so their voltages will add:



Examining the polarity marks of each junction (type E thermocouple wires are assumed in this example: chromel and constantan), we see that all the "hot" junctions' voltages aid each other, as do all the "cold" junctions' voltages. Like all thermocouple circuits, though, the each "cold" junction voltage opposes each the "hot" junction voltage. The example thermopile shown in this diagram, with four hot junctions and four cold junctions, will generate four times the potential difference that a single type E thermocouple hot/cold junction pair would generate, assuming all the hot junctions are at the same temperature and all the cold junctions are at the same temperature.

When used as the detector for a non-contact pyrometer, the thermopile is oriented so all the concentrated light falls on the hot junctions (the "focal point" where the light focuses to a small spot), while the cold junctions face away from the focal point to a region of ambient temperature. Thus, the thermopile acts like a multiplied thermocouple, generating more voltage than a single thermocouple junction could under the same temperature conditions.

A popular design of non-contact pyrometer manufactured for years by Honeywell was the *Radiamatic*[23], using ten thermocouple junction pairs arrayed in a circle. All the "hot" junctions were placed at the center of this circle where the focal point of the concentrated light fell, while all the "cold" junctions were situated around the circumference of the circle away from the heat of the focal point. A table of values showing the approximate relationship between target temperature and millivolt output for one model of Radiamatic sensing unit reveals the fourth-power function:

Target temperature (K)	**Millivolt output**
4144 K	34.8 mV
3866 K	26.6 mV
3589 K	19.7 mV
3311 K	14.0 mV
3033 K	9.9 mV
2755 K	6.6 mV
2478 K	4.2 mV
2200 K	2.5 mV
1922 K	1.4 mV
1644 K	0.7 mV

We may test the basic[24] validity of the Stefan-Boltzmann law by finding the ratio of temperatures for any two temperature values in this table, raising that ratio to the fourth power, and seeing if the millivolt output signals for those same two temperatures match the new ratio. The operating theory here is that increases in target temperature will produce fourth-power increases in sensor temperature rise, since the sensor's temperature rise should be a direct function of radiation *power* impinging on it.

For example, if we were to take 4144 K and 3033 K as our two test temperatures, we find that the ratio of these two temperature values is 1.3663. Raising this ratio to the fourth power gives us 3.485 for a predicted ratio of millivolt values. Multiplying the 3033 K millivoltage value of 9.9 mV by 3.485 gives us 34.5 mV, which is quite close to the value of 34.8 mV advertised by Honeywell:

$$\frac{4144 \text{ K}}{3033 \text{ K}} = 1.3663$$

$$\left(\frac{4144 \text{ K}}{3033 \text{ K}}\right)^4 = 1.3663^4 = 3.485$$

$$(3.485)(9.9 \text{ mV}) \approx 34.8 \text{ mV}$$

If accuracy is not terribly important, and if the range of measured temperatures for the process is modest, we may take the millivoltage output of such a sensor and interpret it *linearly*. When

[23] Later versions of the Radiamatic (dubbed the *Radiamatic II*) were more than just a bare thermopile and optical concentrator, containing electronic circuitry to output a linearized 4-20 mA signal representing target temperature.

[24] Comparing temperature ratios versus thermopile millivoltage ratios assumes linear thermocouple behavior, which we know is not exactly true. Even if the thermopile focal point temperatures precisely followed the ratios predicted by the Stefan-Boltzmann law, we would still expect some inconsistencies due to the non-linearities of thermocouple voltages. There will also be variations from predicted values due to shifts in radiated light frequencies, changes in emissivity factor, thermal losses within the sensing head, and other factors that refuse to remain constant over wide ranges of received radiation intensity. The lesson here is to not expect perfect agreement with theory!

used in this fashion, a non-contact pyrometer is often referred to as an *infrared thermocouple*, with the output voltage intended to connect directly to a thermocouple-input instrument such as an indicator, transmitter, recorder, or controller. An example of this usage is the OS-36 line of infrared thermocouples manufactured by Omega.

Infrared thermocouples are manufactured for a narrow range of temperature (most OS-36 models limited to a calibration span of 100 ^{o}F or less), their thermopiles designed to produce millivolt signals corresponding to a standard thermocouple type (T, J, K, etc.) over that narrow range.

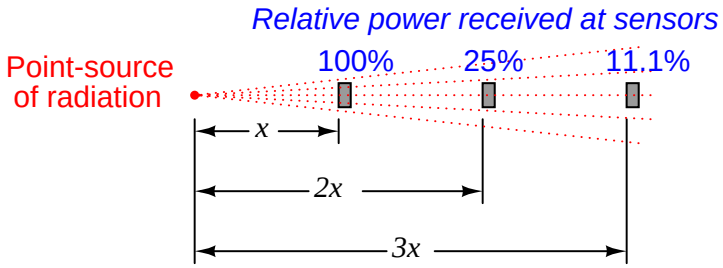
21.5.2 Distance considerations

An interesting and useful characteristic of non-contact pyrometers is that their calibration does *not* depend on the distance separating the sensor from the target object's surface[25]. This is counter-intuitive to anyone who has ever stood near an intense radiative heat source: standing in close proximity to a bonfire, for example, results in much hotter skin temperature than standing far away from it. Why wouldn't a non-contact pyrometer register cooler target temperatures when it was far away, given the fact that infrared radiation from the object spreads out with increased separation distance? The fact that an infrared pyrometer does not suffer from this limitation is good for our purposes in measuring temperature, but it doesn't seem to make sense at first.

One key to understanding this paradox is to quantify the bonfire experience, where perceived temperature falls off with increased distance. In physics, this is known as the *inverse square law*: the intensity of radiation falling on an object from a point-source decreases with the *square* of the distance separating the radiation source from the object. Backing away to twice the distance from a bonfire results in a *four-fold* decrease in received infrared radiation; backing way to three times the distance results in a *nine-fold* decrease in received radiation.

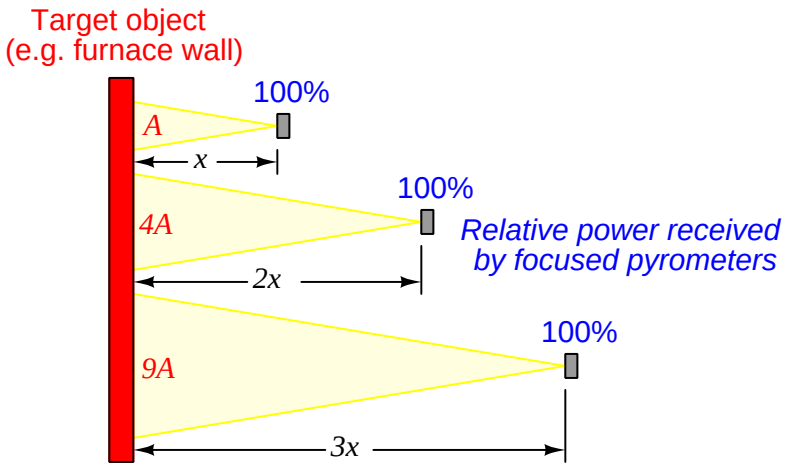
[25] An important caveat to this rule is *so long as the target object completely fills the sensor's field of view (FOV)*. The reason for this caveat will become clear at the conclusion of the explanation.

Placing a sensor at three integer distances (x, $2x$, and $3x$) from a radiation point-source results in relative power levels of 100%, 25% (one-quarter), and 11.1% (one-ninth) falling upon the sensor at those locations, respectively:



This is a basic physical principle for all kinds of radiation, grounded in simple geometry. If we examine the radiation flux emanating from a point-source, we find that it must spread out as it travels in straight lines, and that the spreading-out happens at a rate defined by the square of the distance. An analogy for this phenomenon is to imagine a spherical latex balloon expanding as air is blown into it. The surface area of the balloon is proportional to the square of its radius. Likewise, the radiation flux emanating from a point-source spreads out in straight lines, in all directions, reaching a total area proportional to the square of the distance from the point (center). The total flux measured as a sphere will be the same no matter what the distance from the point-source, but the area it is divided over increases with the square of the distance, and so any object of fixed area backing away from a point-source of radiation encounters a smaller and smaller fraction of that flux.

If non-contact pyrometers really were "looking" at a point-source of infrared radiation, their signals *would* indeed decrease with distance. The saving grace here is that non-contact pyrometers are focused-optic devices, with a definite *field of view*, and that field of view should always be completely filled by the target object (assumed to be at a uniform temperature). As distance between the pyrometer and the target object changes, the cone-shaped field of view covers a surface area on that object proportional to the square of the distance[26] Backing the pyrometer away to twice the distance increases the viewing area on the target object by a factor of four; backing away to three times the distance increases the viewing area nine times:

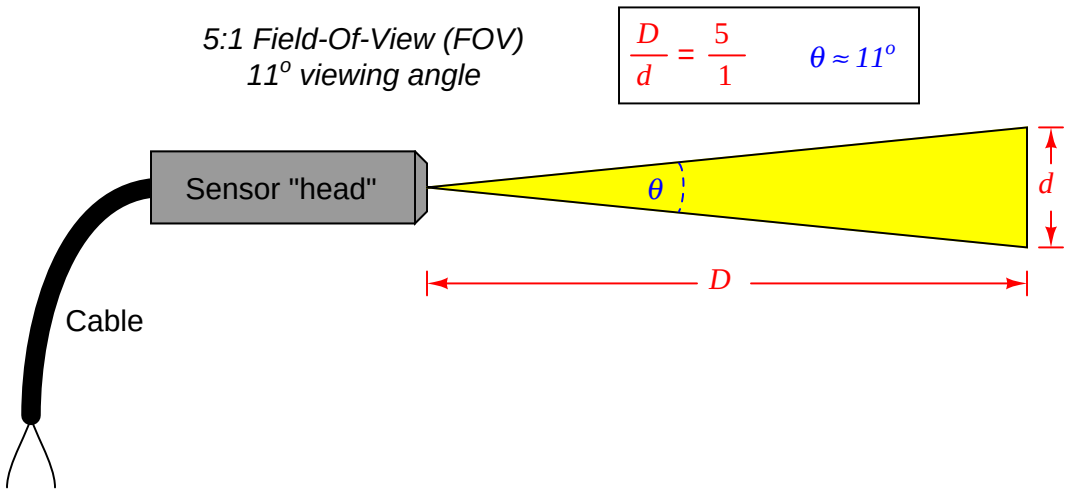


So, even though the inverse square law correctly declares that radiation emanating from the hot wall (which may be thought of as a collection of point-sources) decreases in intensity with the square of the distance, this attenuation is perfectly balanced by an increased viewing area of the pyrometer. Doubling the separation distance *does* result in the flux from any given point on the wall spreading out by a factor of four, but the pyrometer now sees *four times as many similar points on the wall* as it did previously. So long as all the points within the field of view are uniform in temperature, the result is a perfect cancellation with the pyrometer providing the exact same temperature measurement at *any* distance from the target.

If the sensor's field of view expands far enough to capture objects *other* than the one whose temperature we intend to measure, measurement errors will result. The sensor will now yield a weighted average of all objects within its field of view, and so it is important to ensure that field is limited to cover just the object we intend to measure.

[26] The field of view (a circle where the viewing "cone" intercepts the flat surface of the object) increases linearly in diameter with increases in distance between the sensor and the object. However, since the area of a circle is proportional to the square of its diameter ($A = \frac{\pi D^2}{4}$ or $A = \pi r^2$), we may say that the viewing area increases with the square of the distance between the sensor and object.

Non-contact sensor fields-of-view are typically specified either as an angle, as a distance ratio, or both. For example, the following illustration shows a non-contact temperature sensor with a 5:1 (approximately 11°) field of view:



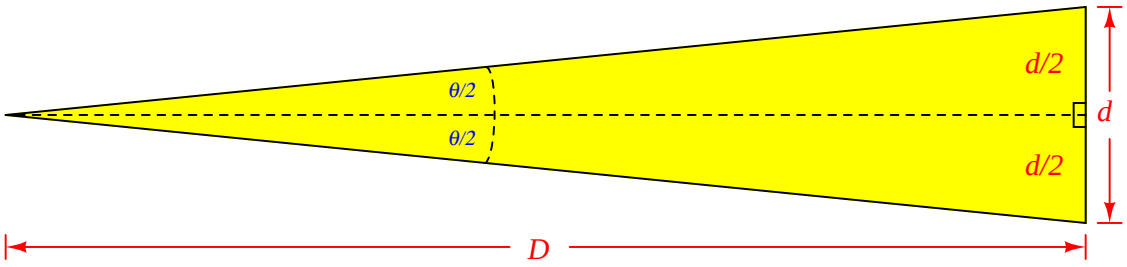
The mathematical relationship between viewing angle (θ) and distance ratio (D/d) follows the tangent function:

$$\frac{D}{d} = \frac{1}{2\tan\left(\frac{\theta}{2}\right)} \qquad \theta = 2\tan^{-1}\left(\frac{d}{2D}\right)$$

A sampling of common field-of-view distance ratios and approximate viewing angles appears in this table:

Distance ratio	**Angle** (approximate)
1:1	53°
2:1	30°
3:1	19°
5:1	11°
7:1	8°
10:1	6°

A trigonometric explanation for these equations is shown in the following diagram, where the isosceles field-of-vision triangle is split into two "right" triangles, each one having an adjacent side length of D and an opposite side length of $d/2$ for angle $\theta/2$:



Since we know the tangent function is the ratio of opposite to adjacent side lengths for a right triangle, this means the tangent of the half-angle ($\theta/2$) will be equal to the ratio of the opposite side length ($d/2$) to the adjacent side length (D):

$$\tan\left(\frac{\theta}{2}\right) = \frac{d/2}{D} = \frac{d}{2D}$$

Solving for the length ratio D/d is now just a matter of algebraically manipulating this equation:

$$\tan\left(\frac{\theta}{2}\right) = \frac{d}{2D}$$

$$\frac{2D}{d} = \frac{1}{\tan\left(\frac{\theta}{2}\right)}$$

$$\frac{D}{d} = \frac{1}{2\tan\left(\frac{\theta}{2}\right)}$$

Solving for the viewing angle (θ) requires another form of manipulation on the basic tangent equation, where we "un-do" the tangent function by using the *inverse* tangent (or "arctangent") function:

$$\tan\left(\frac{\theta}{2}\right) = \frac{d}{2D}$$

$$\tan^{-1}\left[\tan\left(\frac{\theta}{2}\right)\right] = \tan^{-1}\left(\frac{d}{2D}\right)$$

$$\frac{\theta}{2} = \tan^{-1}\left(\frac{d}{2D}\right)$$

$$\theta = 2\tan^{-1}\left(\frac{d}{2D}\right)$$

21.5.3 Emissivity

Aside from their inherent nonlinearity, perhaps the main disadvantage of non-contact temperature sensors is their inaccuracy. The emissivity factor (e) in the Stefan-Boltzmann equation varies with the composition of a substance, but beyond that there are several other factors (surface finish, shape, etc.) that affect the amount of radiation a sensor will receive from an object. For this reason, emissivity is not a very practical way to gauge the effectiveness of a non-contact pyrometer. Instead, a more comprehensive measure of an object's "thermal-optical measureability" is *emittance*.

A perfect emitter of thermal radiation is known as a *blackbody*. Emittance for a blackbody is unity (1), while emittance figures for any real object is a value between 1 and 0. The only certain way to know the emittance of an object is to test that object's thermal radiation at a known temperature. This assumes we have the ability to measure that object's temperature by direct contact, which of course renders void one of the major purposes of non-contact thermometry: to be able to measure an object's temperature without having to touch it. Not all hope is lost, though: all we have to do is obtain an emittance value for that object *one time*, and then we may calibrate any non-contact pyrometer for that object's particular emittance so as to measure its temperature in the future without contact.

Beyond the issue of emittance, other idiosyncrasies plague non-contact pyrometers. Objects also have the ability to *reflect* and *transmit* radiation from other bodies, which taints the accuracy of any non-contact device sensing the radiation from that body. An example of the former is trying to measure the temperature of a silver mirror using an optical pyrometer: the radiation received by the pyrometer is mostly from other objects, merely *reflected* by the mirror. An example of the latter is trying to measure the temperature of a gas or a clear liquid, and instead primarily measuring the temperature of a solid object in the background (*through* the gas or liquid).

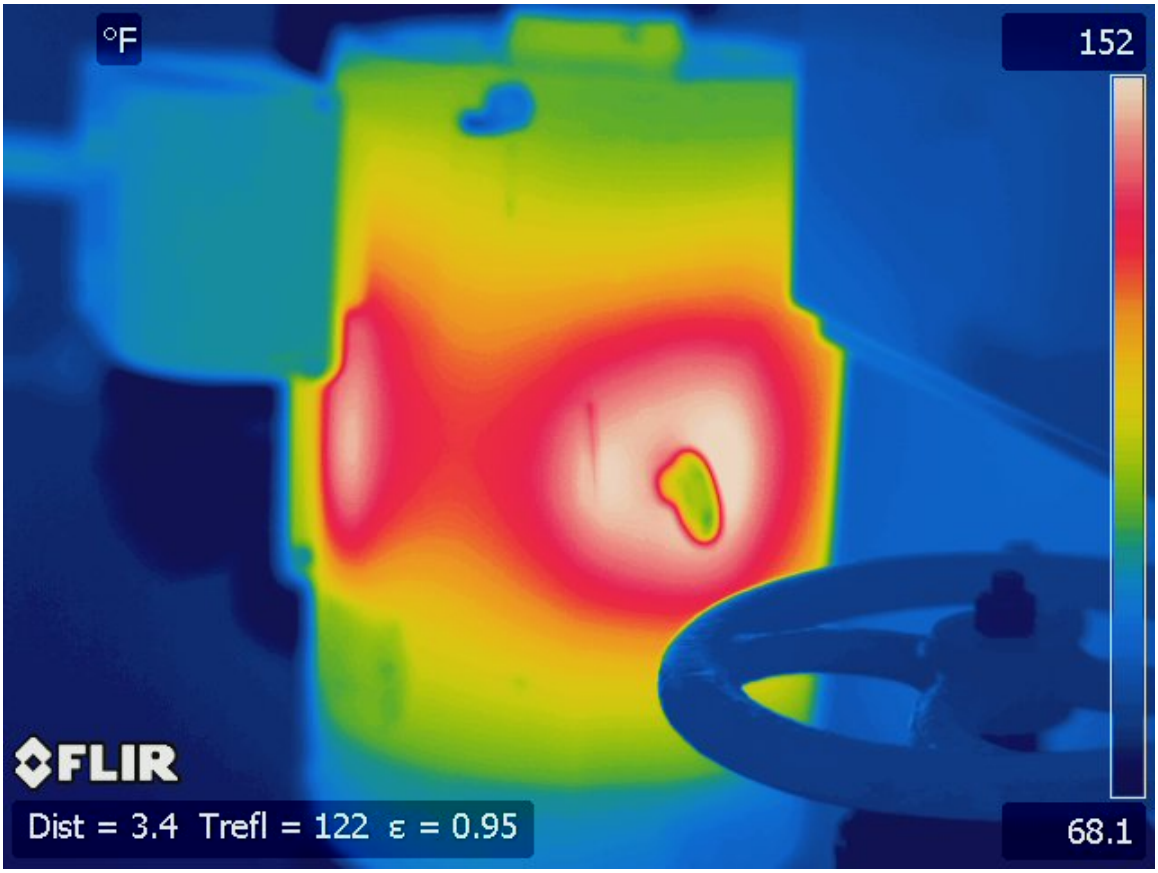
Nevertheless, non-contact pyrometers have been and will continue to be useful in specific applications where other, contact-based temperature measurement techniques are impractical.

21.5.4 Thermal imaging

A very useful application of non-contact sensor technology is *thermal imaging*, where a dense array of infrared radiation sensors provides a graphic display of objects in its view according to their temperatures. Each object shown on the digital display of a thermal imager is artificially colored in the display on a chromatic scale that varies with temperature, hot objects typically registering as red tones and cold objects typically registering as blue tones. Thermal imaging is very useful in the electric power distribution industry, where technicians may inspect power line insulators and other objects at elevated potential for "hot spots" without having to make physical contact with those objects. Thermal imaging is also useful in performing "energy audits" of buildings and other heated structures, providing a means of revealing points of heat escape through walls, windows, and roofs. In such applications, relative differences in temperature are often more important to detect than specific temperature values. "Hot spots" readily appear on a thermal imager display, and may give useful data on the test subject even in the absence of accurate temperature measurement at any one spot.

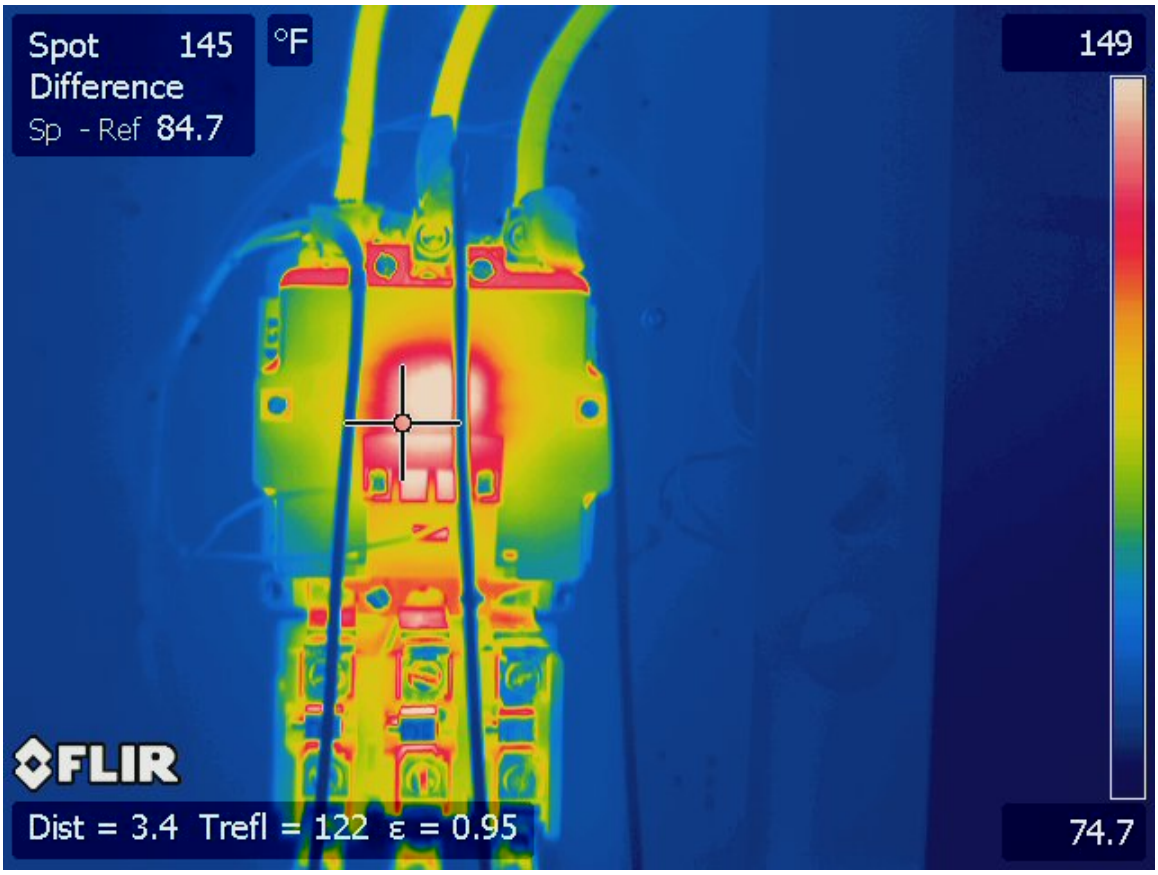
Again, it is important to stress that thermal imaging only provides an assessment of the object's *surface* temperature, and not the temperature *within* that object. Variations in surface temperature detectable by thermal imaging are a secondary effect of temperature variations within the object, and as such may not accurately depict the true thermal gradient(s) within the object.

A digital image taken with a thermal imaging instrument by maintenance personnel at a municipal water treatment facility shows "hot spots" on an electric motor. A color scale on the right-hand side of the image serves as a "legend" for interpreting color as temperature. In this particular shot, dark blue is 68.1 ^{o}F and white is 152 ^{o}F:



This particular electric motor is in a vertical orientation, with the electrical connection box in the upper-left corner and two prominent hot spots on both the near and the left-facing sides of the case. A manual valve handle appears in the foreground, silhouetted in dark blue against a lighter blue (warmer) background. A lifting "eye" on the motor case appears as a green (cooler) shape in the middle of a white (warmer) area. The two "hot spots" correspond directly to stator windings and magnetic pole faces inside the motor, which are close enough to the motor's casing to cause variations in surface temperature.

Thermal imaging is particularly useful for detecting hot spots on equipment unsafe to directly touch, as is the case with many "live" electrical components. This next thermal image shows an operating three-phase motor starter (contactor and overload block):



The bright spot in the center of the contactor is the higher temperature of the electromagnetic coil, providing magnetic force to actuate the contactor mechanism. Perhaps the most telling detail of this thermal image, however, is the difference in temperature between the overload heater connections (the six screws located near the bottom of the starter assembly). Note how the middle heater's screws register slightly higher temperatures than the screws on either of the other two heater elements. Large temperature differences may indicate poor electrical connections (i.e. greater resistance) at the hot spots, or imbalances in phase current. It is also possible that the elevated temperature of this particular overload heater is simply due to it having less open surface area for it to radiate heat, since the two overload heaters flanking it enjoy the advantage of having more air cooling. If three people pack themselves into a narrow bench seat, the middle person is going to be warmer than either of the outer two!

Another noteworthy detail in this image is the "Spot Difference" measurement provided by the

thermal imager. A cross-hair cursor on the display serves to locate a particular spot in the image, which in this case is contrasted against a reference spot chosen in an earlier step. The instrument automatically subtracts the current and reference spot temperatures to give a ΔT indication, in this particular case 84.7 ^{o}F.

A thermal image of a three-phase circuit breaker shows a much more even distribution of temperature:



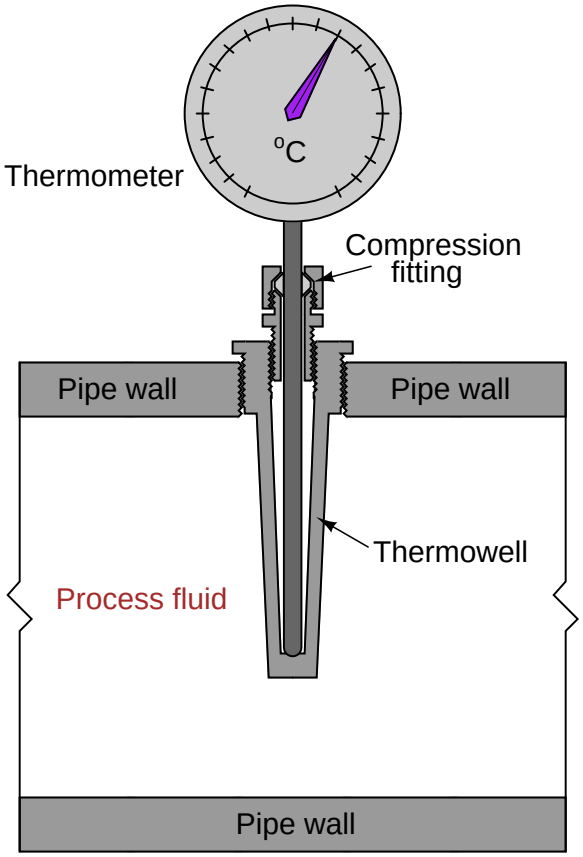
The hottest objects in this image are the three load screw terminals, appearing as white/red against a blue/green background. Note the range of the temperature scale on the right of the image: this particular image only spans a temperature range of 61.3 ^{o}F to 68.6 ^{o}F. This narrow temperature range tells us the differences in temperature shown by colors in this image are nothing to worry about.

Here, the instrument provides a single-point temperature measurement of 68.4 ^{o}F at the cursor ("Spot") location rather than a differential temperature measurement between two points.

21.6 Temperature sensor accessories

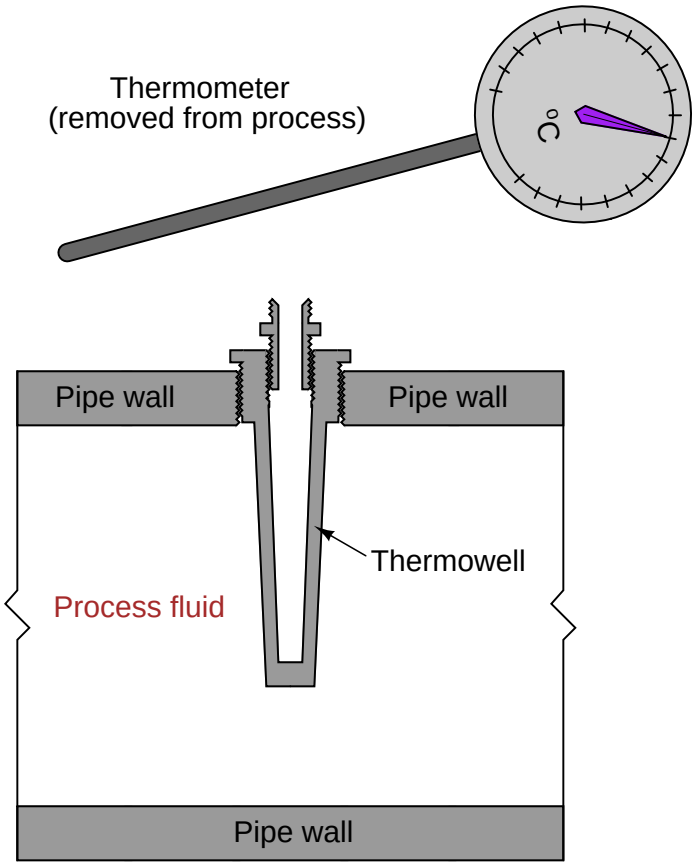
One of the most important accessories for any temperature-sensing element is a pressure-tight sheath known as a *thermowell*. This may be thought of as a thermally conductive protrusion into a process vessel or pipe[27] allowing a temperature-sensitive instrument to detect process temperature without opening a hole in the vessel or pipe. Thermowells are critically important for installations where the temperature element (RTD, thermocouple, thermometer, etc.) must be replaceable without de-pressurizing the process.

Thermowells may be made out of any material that is thermally conductive, pressure-tight, and not chemically reactive with the process. Most thermowells are formed out of either metal (stainless steel or other alloy) or ceramic materials. A simple diagram showing a thermowell in use with a temperature gauge is shown here:

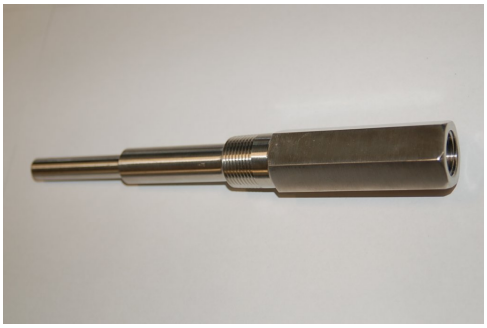


[27] In general, it is better to install a thermowell in a pipe rather than in a vessel because the greater fluid turbulence of flow in a pipe expedites heat transfer by convection as well as helps to clean solid fouling off of the thermowell's surface.

If the temperature gauge is removed for maintenance or replacement, the thermowell maintains pressure integrity of the pipe (no process fluid leaking out, and no air leaking in):



Photographs of a real (stainless steel) thermowell are shown here, the left-hand photo showing the entire length of the thermowell, and the right-hand photo showing the end where the temperature-sensing device is inserted:





A photo of a complete RTD assembly (connection head, RTD, and thermowell) appears in the next photograph:



Another photo shows an RTD installed in a thermowell on the side of a commercial freezer, using a Rosemount model 3044C temperature transmitter to output a 4-20 mA signal to an operator display:



As useful as thermowells are, they are not without their caveats. All thermowells, no matter how well they may be installed, increase the first-order time lag of the temperature sensor by virtue of their mass and specific heat value. It should be intuitively obvious that a few pounds of metal will not heat up and cool down as fast as a few ounces' worth of RTD or thermocouple, and therefore the addition of a thermowell to the sensing element will decrease the responsiveness of any temperature-sensing element. What is not so obvious is that such time lags, if severe enough, may compromise the stability of feedback control. A control system receiving a "delayed" temperature measurement will not see the live temperature of the process in real time due to this lag.

A potential problem with thermowells is incorrect installation of the temperature-sensing element. The element *must* be inserted with full contact at the bottom of the thermowell's blind hole. If any air gap is allowed to exist between the end of the temperature element and the bottom of the thermowell's hole, this will add a *second* time lag to the measurement system[28]. Some thermowells include a spring clip in the bottom of the blind hole to help maintain constant metal-to-metal contact between the sensing element and the thermowell wall.

[28] The air gap acts as a thermal *resistance* while the mass of the element itself acts as a thermal *capacitance*. Thus, the inclusion of an air gap forms a thermal "RC time constant" delay network secondary to the thermal delay incurred by the thermowell. This adds another "order" of lag to the system, not just an increase in its thermal time constant. Generally speaking, multiple orders of lag are detrimental to process control because they increase phase shift in a feedback loop and may lead to oscillation.

21.7 Process/instrument suitability

The primary consideration for selecting a proper temperature sensing element for any application is the expected temperature range. Mechanical (bi-metal) and filled-system temperature sensors are limited to relatively low process temperatures, and cannot relay signals very far from the point of measurement.

Thermocouples are by far the most rugged and wide-ranging of the contact-type temperature sensors. Accuracies vary with thermocouple type and installation quality.

RTDs are more fragile than thermocouples, but they require no reference compensation and are inherently more linear.

Optical sensors lack the ability to measure temperature of fluids inside vessels unless a transparent window is provided in the vessel for light emissions to reach the sensor. Otherwise, the best an optical sensor can do is report the skin temperature of a vessel. For monitoring surface temperatures of solid objects, especially objects that would be impractical or even dangerous to contact (e.g. electrical insulators on high-voltage power lines), optical sensors are the only appropriate solution.

Chemical reactivity is a concern for contact-type sensors. If the sensing element is held inside a thermowell, that thermowell must be selected for minimum reaction with the process fluid(s). Bare thermocouples are particularly vulnerable to chemical reactions given the nature of most thermocouple metals (iron, nickel, copper, etc.), and must be carefully chosen for the particular process chemistry to avoid reliability problems later.

21.8 Review of fundamental principles

Shown here is a partial listing of principles applied in the subject matter of this chapter, given for the purpose of expanding the reader's view of this chapter's concepts and of their general interrelationships with concepts elsewhere in the book. Your abilities as a problem-solver and as a life-long learner will be greatly enhanced by mastering the applications of these principles to a wide variety of topics, the more varied the better.

- **Kirchhoff's Voltage Law**: the algebraic sum of all voltages in a loop is equal to zero. Relevant to thermocouple circuit calculations, where the reference junction voltage always opposes the measurement junction voltage. Also relevant to RTD circuits when determining the amount of voltage sensed by the instrument compared to the amount of voltage dropped by the RTD resistance element.
- **Ideal Gas Law**: $PV = nRT$, describing the relationship between gas pressure, chamber volume, gas quantity (in moles), and gas temperature. Relevant to Class III filled-bulb temperature sensors, where the increased pressure of an enclosed gas corresponds to the temperature of that gas.
- **Self-balancing opamp circuits**: all self-balancing operational amplifier circuits work on the principle of negative feedback maintaining a nearly zero differential input voltage to the opamp. Making the "simplifying assumption" that the opamp's differential input voltage is exactly zero assists in circuit analysis, as does the assumption that the input terminals draw negligible current.
- **Stefan-Boltzmann Law**: $\frac{dQ}{dt} = e\sigma AT^4$, that all objects warmer than absolute zero radiate thermal energy (photons). Relevant to non-contact pyrometry, where the intensity of the received radiation is proportional to the fourth power of the object's absolute temperature.
- **Time constant**: (τ), defined as the amount of time it takes a system to change 63.2% of the way from where it began to where it will eventually stabilize. The system will be within 1% of its final value after 5 time constants' worth of time has passed (5τ). Relevant to temperature lags caused by sensor mass and thermowells.
- **Inverse square law**: the strength of a field radiating away from a point-source diminishes proportionately to the square of the distance from the source. Relevant to determining the amount of radiant energy intercepted by a sensor when monitoring a point-source of heat.

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Chapter 22

Continuous fluid flow measurement

The measurement of fluid flow is arguably the single most complex type of process variable measurement in all of industrial instrumentation[1]. Not only is there a bewildering array of technologies one might use to measure fluid flow – each one with its own limitations and idiosyncrasies – but the very nature of the variable itself lacks a singular definition. "Flow" may refer to volumetric flow (the number of fluid *volumes* passing by per unit time), mass flow (the number of fluid mass units passing by per unit time), or even *standardized* volumetric flow (the number of gas volumes flowing, supposing different pressure and temperature values than what the actual process line operates at). Flowmeters configured to work with gas or vapor flows often are unusable on liquid flows. The dynamic properties of the fluids themselves change with flow rates. Most flow measurement technologies cannot achieve respectable measurement linearity from the maximum rated flow all the way to zero flow, no matter how well matched they might be to the process application.

Furthermore, the performance of most flowmeter technologies critically depends on proper installation. One cannot simply hang a flowmeter at any location in a piping system and expect it to function as designed. This is a constant source of friction between piping (mechanical) engineers and instrumentation (controls) engineers on large industrial projects. What might be considered excellent piping layout from the perspective of process equipment function and economy is often poor (at best) for good flow measurement, and vice-versa. In many cases the flowmeter equipment gets installed improperly and the instrument technicians have to deal with the resulting measurement problems during process unit start-up.

Even after a flowmeter has been properly selected for the process application and properly installed in the piping, problems may arise due to changes in process fluid properties (density, viscosity, conductivity), or the presence of impurities in the process fluid. Flowmeters are also subject to far more "wear and tear" than most other primary sensing elements, given the fact that a flowmeter's sensing element(s) must lie directly in the path of potentially abrasive fluid streams.

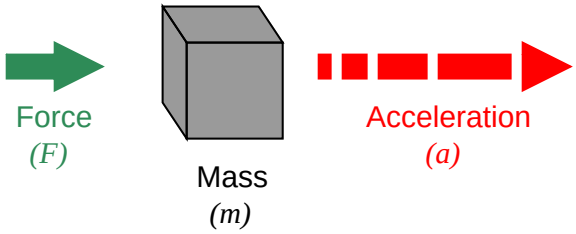
Given all these complications, it is imperative for instrumentation professionals to understand the complexities of flow measurement. What matters most is that you thoroughly understand the *physical principles* upon which each flowmeter depends. If the "first principles" of each technology are understood, the appropriate applications and potential problems become much easier

[1] Analytical (chemical composition) measurement is undeniably more complex and diverse than flow measurement, but analytical measurement covers a great deal of specific measurement types. As a *single* process variable, flow measurement is probably the most complex.

to recognize.

22.1 Pressure-based flowmeters

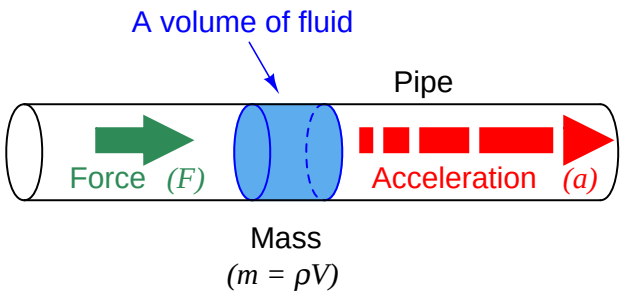
All masses require force to accelerate (we can also think of this in terms of the mass generating a reaction force as a result of being accelerated). This is quantitatively expressed by Newton's Second Law of Motion:



Newton's Second Law formula

$F = ma$

All fluids possess mass, and therefore require force to accelerate just like solid masses. If we consider a quantity of fluid confined inside a pipe[2], with that fluid quantity having a mass equal to its volume multiplied by its mass density ($m = \rho V$, where ρ is the fluid's mass per unit volume), the force required to accelerate that fluid "plug" would be calculated just the same as for a solid mass:



Newton's Second Law formula

$F = ma$ $F = \rho Va$

[2]Sometimes referred to as a *plug* of fluid.

Since this accelerating force is applied on the cross-sectional area of the fluid plug, we may express it as a *pressure*, the definition of pressure being force per unit area:

$$F = \rho V a$$

$$\frac{F}{A} = \rho \frac{V}{A} a$$

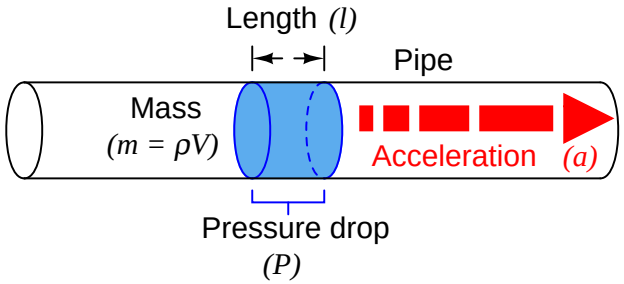
$$P = \rho \frac{V}{A} a$$

Since the rules of algebra required we divide *both* sides of the force equation by area, it left us with a fraction of volume over area ($\frac{V}{A}$) on the right-hand side of the equation. This fraction has a physical meaning, since we know the volume of a cylinder divided by the area of its circular face is simply the length of that cylinder:

$$P = \rho \frac{V}{A} a$$

$$P = \rho l a$$

When we apply this to the illustration of the fluid mass, it makes sense: the pressure described by the equation is actually a *differential*[3] pressure drop from one side of the fluid mass to the other, with the length variable (l) describing the spacing between the differential pressure ports:

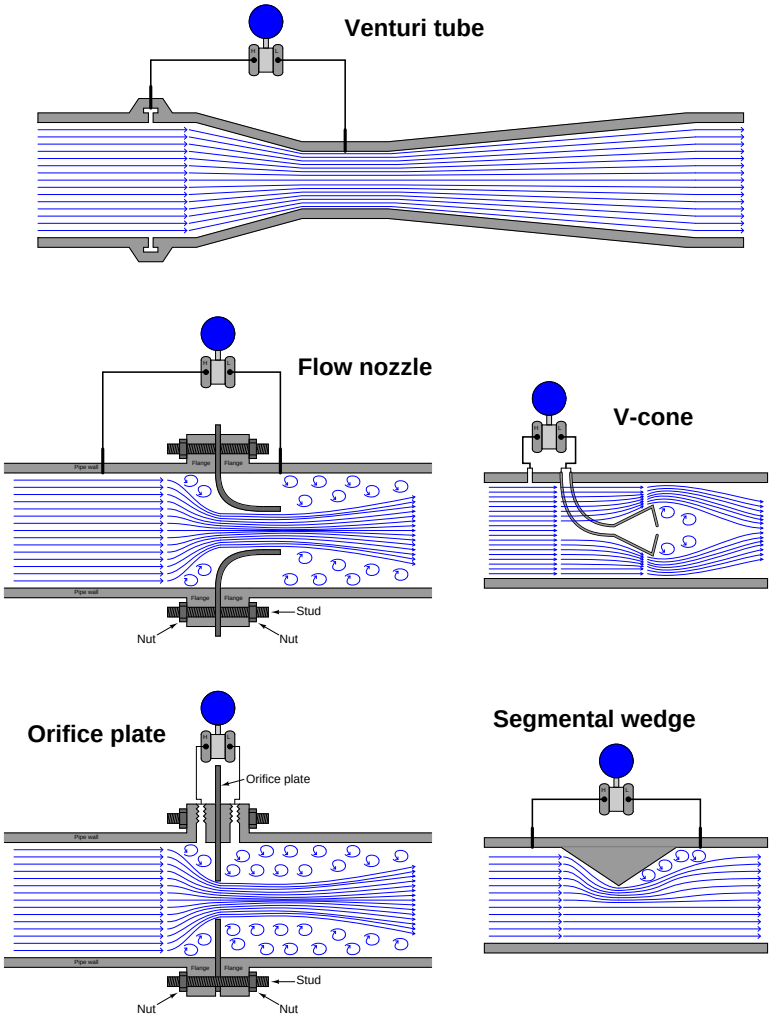


This tells us we can accelerate a "plug" of fluid by applying a difference of pressure across its length. The amount of pressure we apply will be in direct proportion to the density of the fluid and its rate of acceleration. Conversely, we may measure a fluid's rate of acceleration by measuring the pressure developed across a distance over which it accelerates.

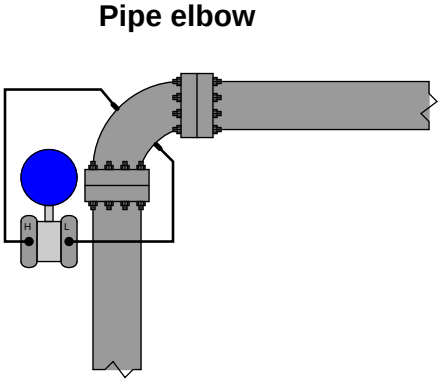
We may easily force a fluid to accelerate by altering its natural flow path. The difference of pressure generated by this acceleration will indirectly indicate the rate of acceleration. Since the acceleration we see from a change in flow path is a direct function of how fast the fluid was originally moving, the acceleration (and therefore the pressure drop) indirectly indicates fluid flow rate.

[3] What really matters in Newton's Second Law equation is the *resultant* force causing the acceleration. This is the vector sum of all forces acting on the mass. Likewise, what really matters in this scenario is the *resultant* pressure acting on the fluid plug, and this resultant pressure is the difference of pressure between one face of the plug and the other, since those two pressures impart two forces on the fluid mass in direct opposition to each other.

A very common way to cause linear acceleration in a moving fluid is to pass the fluid through a constriction in the pipe, thereby increasing its velocity (remember that the definition of acceleration is a change in velocity). The following illustrations show several devices used to linearly accelerate moving fluids when placed in pipes, with differential pressure transmitters connected to measure the pressure drop resulting from this acceleration:

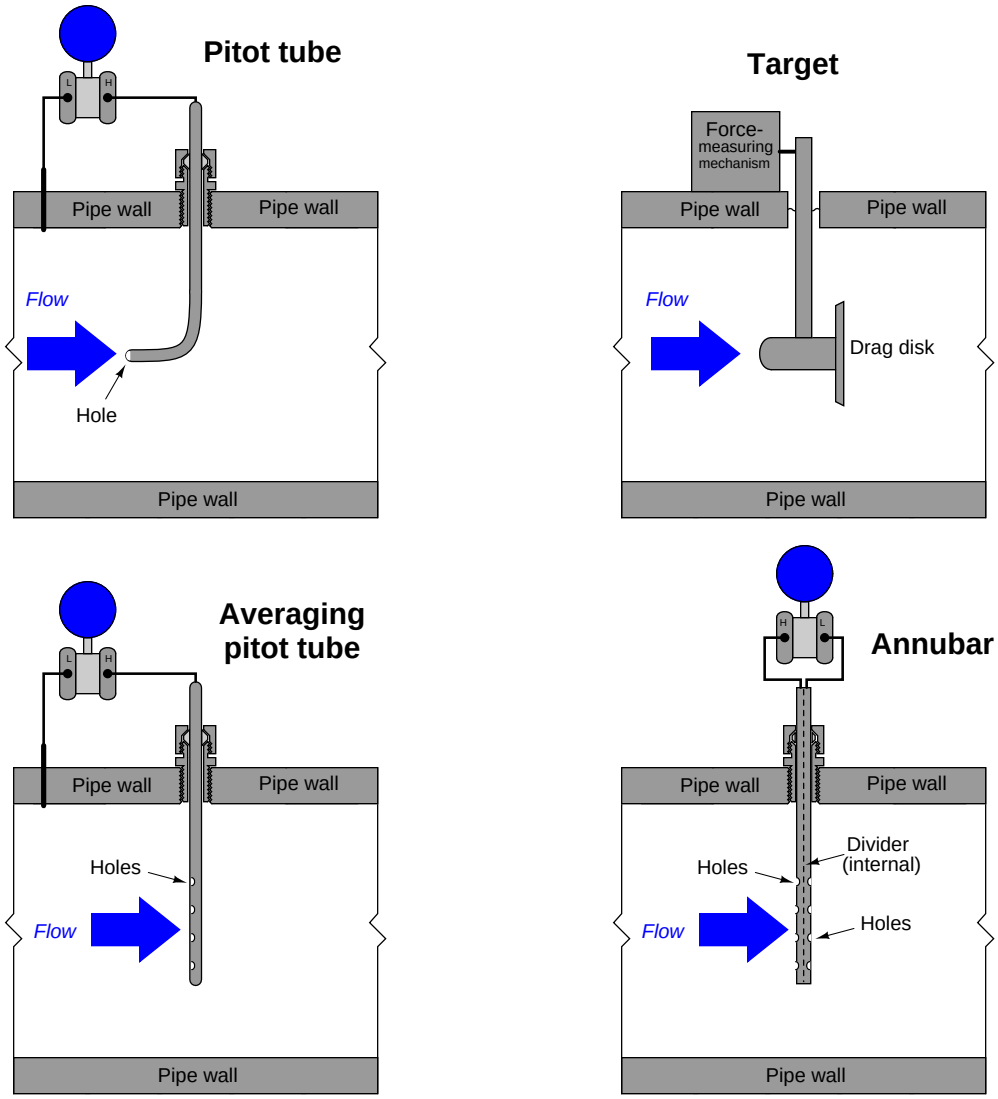


Another way we may accelerate a fluid is to force it to turn a corner through a pipe fitting called an *elbow*. This will generate radial acceleration, causing a pressure difference between the outside and inside of the elbow which may be measured by a differential pressure transmitter:



The pressure tap located on the outside of the elbow's turn registers a greater pressure than the tap located on the inside of the elbow's turn, due to the inertial force of the fluid's mass being "flung" to the outside of the turn as it rounds the corner.

Yet another way to cause a change in fluid velocity is to force it to *decelerate* by bringing a portion of it to a full stop. The pressure generated by this deceleration (called the *stagnation pressure*) tells us how fast it was originally flowing. A few devices working on this principle are shown here:



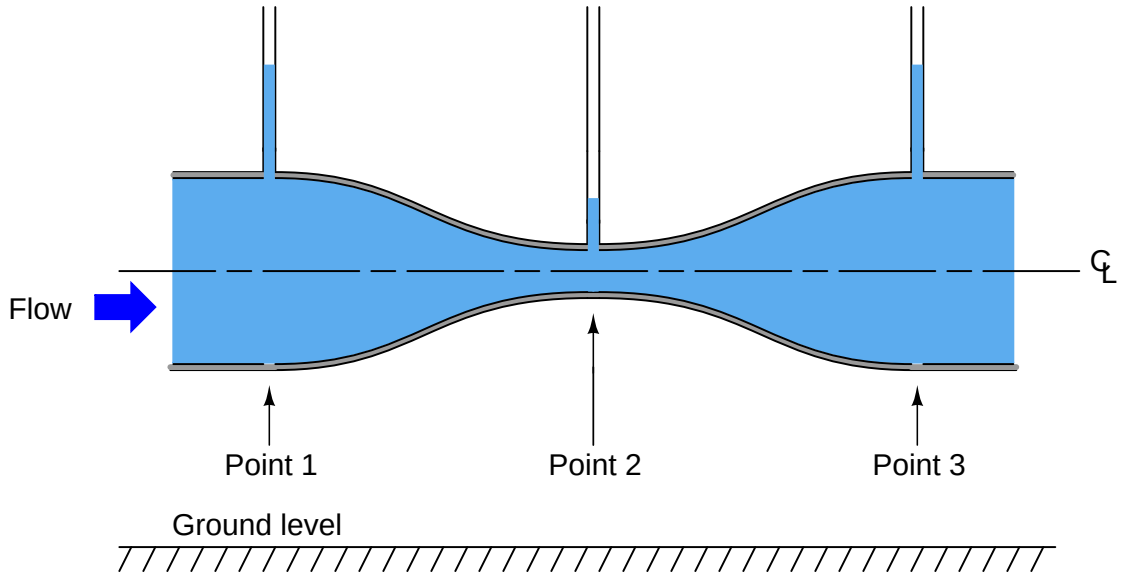
The following subsections in this flow measurement chapter explore different primary sensing elements (PSE's) used to generate differential pressure in a moving fluid stream. Despite their very different designs, they all operate on the same fundamental principle: causing a fluid to accelerate or decelerate by forcing a change in its flow path, and thus generating a measurable pressure difference. The following subsection will introduce a device called a *venturi tube* used to measure fluid flow

rates, and derive mathematical relationships between fluid pressure and flow rate starting from basic physical conservation laws.

22.1.1 Venturi tubes and basic principles

The standard "textbook example" flow element used to create a pressure change by accelerating a fluid stream is the *venturi tube*: a pipe purposefully narrowed to create a region of low pressure. As shown previously, venturi tubes are not the only structure capable of producing a flow-dependent pressure drop. You should keep this in mind as we proceed to derive equations relating flow rate with pressure change: although the venturi tube is the canonical form, the exact same mathematical relationship applies to all flow elements generating a pressure drop by accelerating fluid, including orifice plates, flow nozzles, V-cones, segmental wedges, pipe elbows, pitot tubes, etc.

If the fluid going through the venturi tube is a liquid under relatively low pressure, we may vividly show the pressure at different points in the tube by means of *piezometers*[4], which are transparent tubes allowing us to view liquid column heights. The greater the height of liquid column in the piezometer, the greater the pressure at that point in the flowstream:



As indicated by the piezometer liquid heights, pressure at the constriction (point 2) is the least, while pressures at the wide portions of the venturi tube (points 1 and 3) are the greatest. This is a counter-intuitive result, but it has a firm grounding in the physics of mass and energy conservation. If we assume no energy is added (by a pump) or lost (due to friction) as fluid travels through this pipe, then the Law of Energy Conservation describes a situation where the fluid's energy must remain constant at all points in the pipe as it travels through. If we assume no fluid joins this flowstream from another pipe, or is lost from this pipe through any leaks, then the Law of Mass Conservation describes a situation where the fluid's mass flow rate must remain constant at all points in the pipe as it travels through.

[4] Think of a piezometer tube as nothing more than a manometer tube: the greater the fluid pressure at the bottom of the tube, the higher the liquid will rise inside the tube.

So long as fluid density remains fairly constant[5], fluid velocity must increase as the cross-sectional area of the pipe decreases, as described by the Law of Continuity (see section 2.11.11 beginning on page 205 for more details on this concept):

$$A_1 \overline{v_1} = A_2 \overline{v_2}$$

Rearranging variables in this equation to place velocities in terms of areas, we get the following result:

$$\frac{\overline{v_2}}{\overline{v_1}} = \frac{A_1}{A_2}$$

This equation tells us that the ratio of fluid velocity between the narrow throat (point 2) and the wide mouth (point 1) of the pipe is the same ratio as the mouth's area to the throat's area. So, if the mouth of the pipe had an area 5 times as great as the area of the throat, then we would expect the fluid velocity in the throat to be 5 times as great as the velocity at the mouth. Simply put, the narrow throat causes the fluid to accelerate from a lower velocity to a higher velocity.

[5]This is a very sound assumption for liquids, and a fair assumption for gases when pressure changes through the venturi tube are modest.

We know from our study of energy in physics that kinetic energy is proportional to the square of a mass's velocity ($E_k = \frac{1}{2}mv^2$). If we know the fluid molecules increase velocity as they travel through the venturi tube's throat, we may safely conclude that those molecules' kinetic energies must increase as well. However, we also know that the total energy at any point in the fluid stream must remain constant, because no energy is added to or taken away from the stream in this simple fluid system[6]. Therefore, if kinetic energy increases at the throat, potential energy must correspondingly decrease to keep the total amount of energy constant at any point in the fluid.

Potential energy may be manifest as height above ground, and/or as pressure in a fluid system. Since this venturi tube is level with the ground, there cannot be a height change to account for a change in potential energy. Therefore, there *must* be a change of pressure (P) as the fluid travels through the venturi throat. The Laws of Mass and Energy Conservation invariably lead us to this conclusion: fluid pressure must decrease as it travels through the narrow throat of the venturi tube[7].

Conservation of energy at different points in a fluid stream is neatly expressed in *Bernoulli's Equation* as a constant sum of elevation, pressure, and velocity "heads" (see section 2.11.13 beginning on page 208 for more details on this concept):

$$z_1\rho g + \frac{v_1^2\rho}{2} + P_1 = z_2\rho g + \frac{v_2^2\rho}{2} + P_2$$

Where,

z = Height of fluid (from a common reference point, usually ground level)
ρ = Mass density of fluid
g = Acceleration of gravity
v = Velocity of fluid
P = Pressure of fluid

[6] One of the simplifying assumptions we make in this derivation is that friction plays no significant role in the fluid's behavior as it moves through the venturi tube. In truth, no industrial fluid flow is totally frictionless (especially through more primitive flow elements such as orifice plates), and so our "theoretical" equations must be adjusted a bit to match real life.

[7] To see a graphical relationship between fluid acceleration and fluid pressures in a venturi tube, examine the illustration found in section B beginning on page 3179.

We will use Bernoulli's equation to develop a precise mathematical relationship between pressure and flow rate in a venturi tube. To simplify our task, we will hold to the following assumptions for our venturi tube system:

- No energy lost or gained in the venturi tube (all energy is conserved)
- No mass lost or gained in the venturi tube (all mass is conserved)
- Fluid is incompressible
- Venturi tube centerline is level (no height changes to consider)

Applying the last two assumptions to Bernoulli's equation, we see that the "elevation head" term drops out of both sides, since z, ρ, and g are equal at all points in the system:

$$\frac{v_1^2\rho}{2} + P_1 = \frac{v_2^2\rho}{2} + P_2$$

Now we will algebraically re-arrange this equation to show pressures at points 1 and 2 in terms of velocities at points 1 and 2:

$$\frac{v_2^2\rho}{2} - \frac{v_1^2\rho}{2} = P_1 - P_2$$

Factoring $\frac{\rho}{2}$ out of the velocity head terms:

$$\frac{\rho}{2}(v_2^2 - v_1^2) = P_1 - P_2$$

The Continuity equation shows us the relationship between velocities v_1 and v_2 and the areas at those points in the venturi tube, assuming constant density (ρ):

$$A_1 v_1 = A_2 v_2$$

Specifically, we need to re-arrange this equation to define v_1 in terms of v_2 so we may substitute into Bernoulli's equation:

$$v_1 = \left(\frac{A_2}{A_1}\right) v_2$$

Performing the algebraic substitution:

$$\frac{\rho}{2}(v_2^2 - \left[\left(\frac{A_2}{A_1}\right) v_2\right]^2) = P_1 - P_2$$

Distributing the "square" power:

$$\frac{\rho}{2}(v_2^2 - \left(\frac{A_2}{A_1}\right)^2 v_2^2) = P_1 - P_2$$

Factoring v_2^2 out of the outer parentheses set:

$$\frac{\rho v_2^2}{2}(1 - \left(\frac{A_2}{A_1}\right)^2) = P_1 - P_2$$

Solving for v_2, step by step:

$$\frac{\rho v_2^2}{2} = \left(\frac{1}{1 - \left(\frac{A_2}{A_1}\right)^2}\right)(P_1 - P_2)$$

$$\rho v_2^2 = 2\left(\frac{1}{1 - \left(\frac{A_2}{A_1}\right)^2}\right)(P_1 - P_2)$$

$$v_2^2 = 2\left(\frac{1}{1 - \left(\frac{A_2}{A_1}\right)^2}\right)\left(\frac{P_1 - P_2}{\rho}\right)$$

$$v_2 = \sqrt{2}\frac{1}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}}\sqrt{\frac{P_1 - P_2}{\rho}}$$

The result shows us how to solve for fluid velocity at the venturi throat (v_2) based on a difference of pressure measured between the mouth and the throat ($P_1 - P_2$). We are only one step away from a volumetric flow equation here, and that is to convert velocity (v) into flow rate (Q). Velocity is expressed in units of length per time (feet or meters per second or minute), while volumetric flow is expressed in units of volume per time (cubic feet or cubic meters per second or minute). Simply multiplying throat velocity (v_2) by throat area (A_2) will give us the result we seek:

General flow/area/velocity relationship:

$$Q = Av$$

Equation for throat velocity:

$$v_2 = \sqrt{2}\frac{1}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}}\sqrt{\frac{P_1 - P_2}{\rho}}$$

Multiplying both sides of the equation by throat area:

$$A_2 v_2 = \sqrt{2}A_2\frac{1}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}}\sqrt{\frac{P_1 - P_2}{\rho}}$$

Now we have an equation solving for volumetric flow (Q) in terms of pressures and areas:

$$Q = \sqrt{2}A_2 \frac{1}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \sqrt{\frac{P_1 - P_2}{\rho}}$$

Please note how many constants we have in this equation. For any given venturi tube, the mouth and throat areas (A_1 and A_2) will be fixed. This means nearly half the variables found within this rather long equation are actually constant for any given venturi tube, and therefore do not change with pressure, density, or flow rate. Knowing this, we may re-write the equation as a simple proportionality:

$$Q \propto \sqrt{\frac{P_1 - P_2}{\rho}}$$

To make this a more precise mathematical statement, we may insert a *constant of proportionality* (k) and once more have a true equation to work with:

$$Q = k\sqrt{\frac{P_1 - P_2}{\rho}}$$

22.1.2 Volumetric flow calculations

As we saw in the previous subsection, we may derive a relatively simple equation for predicting flow through a fluid-accelerating element given the pressure drop generated by that element and the density of the fluid flowing through it:

$$Q = k\sqrt{\frac{P_1 - P_2}{\rho}}$$

This equation is a simplified version of the one derived from the physical construction of a venturi tube:

$$Q = \sqrt{2}A_2 \frac{1}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \sqrt{\frac{P_1 - P_2}{\rho}}$$

As you can see, the constant of proportionality (k) shown in the simpler equation is nothing more than a condensation of the first half of the longer equation: k represents the geometry of the venturi tube. If we define k by the mouth and throat areas (A_1, A_2) of any particular venturi tube, we must be very careful to express the pressures and densities in compatible units of measurement. For example, with k strictly defined by flow element geometry (tube areas measured in square *feet*), the calculated flow rate (Q) must be in units of cubic *feet* per second, the pressure values P_1 and P_2 must be in units of pounds per square *foot*, and mass density must be in units of *slugs* per cubic *foot*. We cannot arbitrarily choose different units of measurement for these variables, because the units must "agree" with one another. If we wish to use more convenient units of measurement such as inches of water column for pressure and specific gravity (unitless) for density, the original (longer) equation simply will not work.

However, if we happen to know the differential pressure produced by any particular flow element tube with any particular fluid density at a specified flow rate (real-life conditions), we may *calculate* a value for k in the short equation that makes all those measurements "agree" with one another. In other words, we may use the constant of proportionality (k) as a *unit-of-measurement correction factor* as well as a definition of element geometry. This is a useful property of all proportionalities: simply insert values (expressed in any unit of measurement) determined by physical experiment and solve for the proportionality constant's value to satisfy the expression as an equation. If we do this, the value we arrive at for k will automatically compensate for whatever units of measurement we arbitrarily choose for pressure and density.

For example, if we know a particular orifice plate develops 45 inches of water column differential pressure at a flow rate of 180 gallons per minute of water (specific gravity = 1), we may insert these values into the equation and solve for k:

$$Q = k\sqrt{\frac{P_1 - P_2}{\rho}}$$

$$180 = k\sqrt{\frac{45}{1}}$$

$$k = \frac{180}{\sqrt{\frac{45}{1}}} = 26.83$$

Now we possess a value for k (26.83) that yields a flow rate in units of "gallons per minute" given differential pressure in units of "inches of water column" and density expressed as a specific gravity for this particular orifice plate. From the known fact of all accelerating flow elements' behavior (flow rate proportional to the square root of pressure divided by density) and from a set of values experimentally determined for this particular orifice plate, we now have an equation useful for calculating flow rate given any set of pressure and density values we may happen to encounter with this particular orifice plate:

$$\left[\frac{\text{gal}}{\text{min}}\right] = 26.83\sqrt{\frac{[\text{"W.C.}]}{\text{Specific gravity}}}$$

This k value lets us predict flow for any given pressure difference – and vice-versa – for this particular orifice plate. For example, if we wished to know the water flow rate corresponding to a pressure difference of 60 inches water column, we could use this equation to calculate a flow rate of 207.8 gallons per minute:

$$Q = 26.83\sqrt{\frac{60}{1}}$$

$$Q = 207.8 \text{ GPM}$$

As another example, a measured differential pressure of 110 inches water column across this orifice plate generated by a flow of gasoline (specific gravity = 0.657) would correspond to a gasoline flow rate of 347 gallons per minute:

$$Q = 26.83\sqrt{\frac{110}{0.657}}$$

$$Q = 347 \text{ GPM}$$

Suppose, though, we wished to have an equation for calculating the flow rate through this same orifice plate given pressure and density data in different units (say, kPa instead of inches water column, and kilograms per cubic meter instead of specific gravity). In order to do this, we would need to re-calculate the constant of proportionality (k) to accommodate those new units of measurement. To do this, all we would need is a single set of experimental data for the orifice plate relating flow in GPM, pressure in kPa, and density in kg/m^3.

Applying this to our original data where a water flow rate of 180 GPM resulted in a pressure drop of 45 inches water column, we could convert the pressure drop of 45 "W.C. into 11.21 kPa and express the density as 1000 kg/m^3 to solve for a new value of k:

$$Q = k\sqrt{\frac{P_1 - P_2}{\rho}}$$

$$180 = k\sqrt{\frac{11.21}{1000}}$$

$$k = \frac{180}{\sqrt{\frac{11.21}{1000}}} = 1700$$

Nothing about the orifice plate's geometry has changed from before, only the units of measurement we have chosen to work with. Now we possess a value for k (1700) for the same orifice plate yielding a flow rate in units of "gallons per minute" given differential pressure in units of "kilopascals" and density in units of "kilograms per cubic meter."

$$\left[\frac{\text{gal}}{\text{min}}\right] = 1700\sqrt{\frac{[\text{kPa}]}{\text{kg/m}^3}}$$

If we were to be given a pressure drop in kPa and a fluid density in kg/m^3 for this orifice plate, we could calculate the corresponding flow rate (in GPM) with our new value of k (1700) just as easily as we could with the old value of k (26.83) given pressure in "W.C. and specific gravity.

22.1.3 Mass flow calculations

Measurements of *mass* flow are preferred over measurements of *volumetric* flow in process applications where mass balance (monitoring the rates of mass entry and exit for a process) is important. Whereas volumetric flow measurements express the fluid flow rate in such terms as *gallons per minute* or *cubic meters per second*, mass flow measurements always express fluid flow rate in terms of actual mass units over time, such *pounds (mass) per second* or *kilograms per minute*. Applications for mass flow measurement include custody transfer (where a fluid product is bought or sold by its mass), chemical reaction processes (where the mass flow rates of reactants must be maintained in precise proportion in order for the desired chemical reactions to occur), and steam boiler control systems (where the out-flow of vaporous steam must be balanced by an equivalent in-flow of liquid water to the boiler – here, volumetric comparisons of steam and water flow would be useless because one cubic foot of steam is certainly not the same number of H_2O molecules as one cubic foot of water).

If we wish to calculate *mass* flow instead of volumetric flow, the equation does not change much. The relationship between volume (V) and mass (m) for a sample of fluid is its mass density (ρ):

$$\rho = \frac{m}{V}$$

Similarly, the relationship between a volumetric *flow rate* (Q) and a mass *flow rate* (W) is also the fluid's mass density (ρ):

$$\rho = \frac{W}{Q}$$

Solving for W in this equation leads us to a product of volumetric flow rate and mass density:

$$W = \rho Q$$

A quick dimensional analysis check using common metric units confirms this fact. A mass flow rate in kilograms per second will be obtained by multiplying a mass density in kilograms per cubic meter by a volumetric flow rate in cubic meters per second:

$$\left[\frac{\text{kg}}{\text{s}}\right] = \left[\frac{\text{kg}}{\text{m}^3}\right]\left[\frac{\text{m}^3}{\text{s}}\right]$$

Therefore, all we have to do to turn our general volumetric flow equation into a mass flow equation is multiply both sides by fluid density (ρ):

$$Q = k\sqrt{\frac{P_1 - P_2}{\rho}}$$

$$\rho Q = k\rho\sqrt{\frac{P_1 - P_2}{\rho}}$$

$$W = k\rho\sqrt{\frac{P_1 - P_2}{\rho}}$$

It is generally considered "inelegant" to show the same variable more than once in an equation if it is not necessary, so let's try to consolidate the two densities (ρ) using algebra. First, we may write ρ as the product of two square-roots:

$$W = k\sqrt{\rho}\sqrt{\rho}\sqrt{\frac{P_1 - P_2}{\rho}}$$

Next, we will break up the last radical into a quotient of two separate square roots:

$$W = k\sqrt{\rho}\sqrt{\rho}\frac{\sqrt{P_1 - P_2}}{\sqrt{\rho}}$$

Now we see how one of the square-rooted ρ terms cancels out the one in the denominator of the fraction:

$$W = k\sqrt{\rho}\sqrt{P_1 - P_2}$$

It is also considered "inelegant" to have multiple radicands in an equation where one will suffice, so we will re-write our equation for esthetic improvement[8]:

$$W = k\sqrt{\rho(P_1 - P_2)}$$

As with the volumetric flow equation, all we need in order to arrive at a suitable k value for any particular flow element is a set of values taken from that real element in service, expressed in whatever units of measurement we desire.

[8] This re-write is solidly grounded in the rules of algebra. We know that $\sqrt{a}\sqrt{b} = \sqrt{ab}$, which is what allows us to do the re-write.

For example, if we had a venturi tube generating a differential pressure of 2.30 kilopascals (kPa) at a mass flow rate of 500 kilograms per minute of naphtha (a petroleum product having a density of 0.665 kilograms per liter), we could solve for the k value of this venturi tube as such:

$$W = k\sqrt{\rho(P_1 - P_2)}$$

$$500 = k\sqrt{(0.665)(2.3)}$$

$$k = \frac{500}{\sqrt{(0.665)(2.3)}}$$

$$k = 404.3$$

Now that we know a value of 404.3 for k will yield kilograms per minute of liquid flow through this venturi tube given pressure in kPa and density in kilograms per liter, we may readily predict the mass flow rate through this tube for any other pressure drop and fluid density we might happen to encounter. The value of 404.3 for k relates the disparate units of measurement for us:

$$\left[\frac{\text{kg}}{\text{min}}\right] = 404.3\sqrt{\left[\frac{\text{kg}}{\text{l}}\right][\text{kPa}]}$$

As with volumetric flow calculations, the calculated value for k neatly accounts for any set of measurement units we may arbitrarily choose. The key is first knowing the proportional relationship between flow rate, pressure drop, and density. Once we combine that proportionality with a specific set of data experimentally gathered from a particular flow element, we have a true equation properly relating all the variables together in our chosen units of measurement.

If we happened to measure 6.1 kPa of differential pressure across this same venturi tube as it flowed sea water (density = 1.03 kilograms per liter), we could calculate the mass flow rate quite easily using the same equation (with the k factor of 404.3):

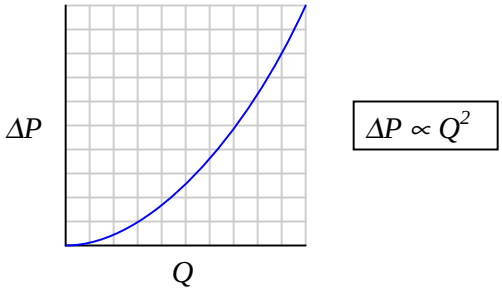
$$W = 404.3\sqrt{(1.03)(6.1)}$$

$$W = 1013.4\ \frac{\text{kg}}{\text{min}}$$

22.1.4 Square-root characterization

It should be apparent by now that the relationship between flow rate (whether it be volumetric or mass) and differential pressure for any fluid-accelerating flow element is non-linear: a doubling of flow rate will *not* result in a doubling of differential pressure. Rather, a doubling of flow rate will result in a *quadrupling* of differential pressure. Likewise, a tripling of flow rate results in *nine times* as much differential pressure developed by the fluid-accelerating flow element.

When plotted on a graph, the relationship between flow rate (Q) and differential pressure (ΔP) is quadratic, like one-half of a parabola. Differential pressure developed by a venturi, orifice plate, pitot tube, or any other acceleration-based flow element is proportional to the *square* of the flow rate:

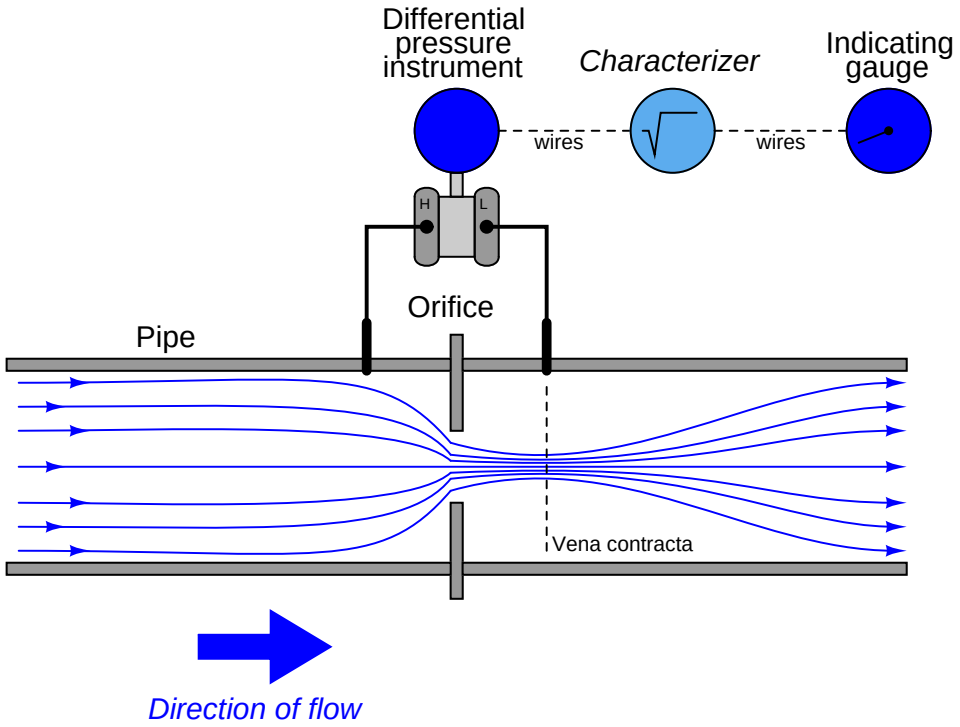


An unfortunate consequence of this quadratic relationship is that a pressure-sensing instrument connected to such a flow element will *not* directly sense flow rate. Instead, the pressure instrument will be sensing what is essentially the square of the flow rate. The instrument may register correctly at the 0% and 100% range points if correctly calibrated for the flow element it connects to, but it will fail to register linearly in between. Any indicator, recorder, or controller connected to the pressure-sensing instrument will likewise register incorrectly at any point between 0% and 100% of range, because the pressure signal is not a direct representation of flow rate.

In order that we may have indicators, recorders, and controllers that actually do register linearly with flow rate, we must mathematically "condition" or "characterize" the pressure signal sensed by the differential pressure instrument. Since the mathematical function inherent to the flow element is quadratic (square), the proper conditioning for the signal must be the inverse of that: *square root*. Just as taking the square-root of the square of a number yields the original number[9], taking the square-root of the differential pressure signal – which is itself a function of flow squared – yields a signal directly representing flow.

[9] For positive numbers only!

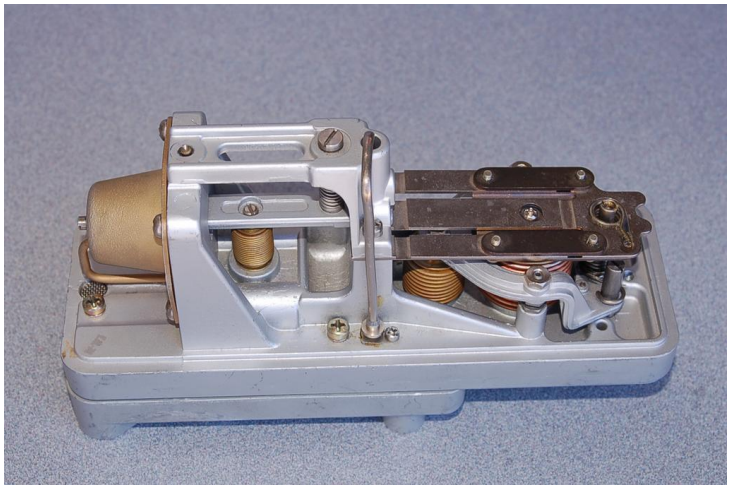
The traditional means of implementing the necessary signal characterization was to install a "square root" function relay between the transmitter and the flow indicator, as shown in the following diagram:



The modern solution to this problem is to incorporate square-root signal characterization either inside the transmitter or inside the receiving instrument (e.g. indicator, recorder, or controller). Either way, the square-root function must be implemented *somewhere*[10] in the loop in order that flow may be accurately measured throughout the operating range.

[10] With so many modern instruments being capable of digitally implementing this square-root function, one must be careful to ensure it is only done *once* in the loop. I have personally witnessed flow-measurement installations where both the pressure transmitter and the indicating device were configured for square-root characterization. This essentially performed a *fourth* root characterization on the signal, which is just as bad as no characterization at all! Like anything else technical, the key to successful implementation is a correct understanding of how the system is supposed to work. Simply memorizing that "the instrument must be set up with square-root to measure flow" and blindly applying that mantra is a recipe for failure.

In the days of pneumatic instrumentation, this square-root function was performed in a separate device called a *square root extractor*. The Foxboro model 557 (left) and Moore Products model 65 (right) pneumatic square root extractors are classic examples of this technology[11]:





Pneumatic square root extraction relays approximated the square-root function by means of triangulated force or motion. In essence, they were *trigonometric* function relays, not square-root relays per se. However, for small angular motions, certain trigonometric functions were close enough to a square-root function that the relays were able to serve their purpose in characterizing the output signal of a pressure sensor to yield a signal representing flow rate.

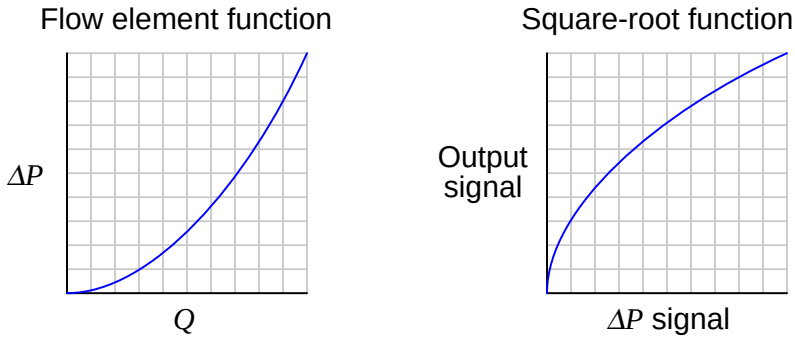
[11] Despite the impressive craftsmanship and engineering that went into the design of pneumatic square root extractors, their obsolescence is mourned by no one. These devices were notoriously difficult to set up and calibrate accurately, especially as they aged.

The following table shows the ideal response of a pneumatic square root relay:

Input signal	Input %	Output %	Output signal
3 PSI	0%	0%	3 PSI
4 PSI	8.33%	28.87%	6.464 PSI
5 PSI	16.67%	40.82%	7.899 PSI
6 PSI	25%	50%	9 PSI
7 PSI	33.33%	57.74%	9.928 PSI
8 PSI	41.67%	64.55%	10.75 PSI
9 PSI	50%	70.71%	11.49 PSI
10 PSI	58.33%	76.38%	12.17 PSI
11 PSI	66.67%	81.65%	12.80 PSI
12 PSI	75%	86.60%	13.39 PSI
13 PSI	83.33%	91.29%	13.95 PSI
14 PSI	91.67%	95.74%	14.49 PSI
15 PSI	100%	100%	15 PSI

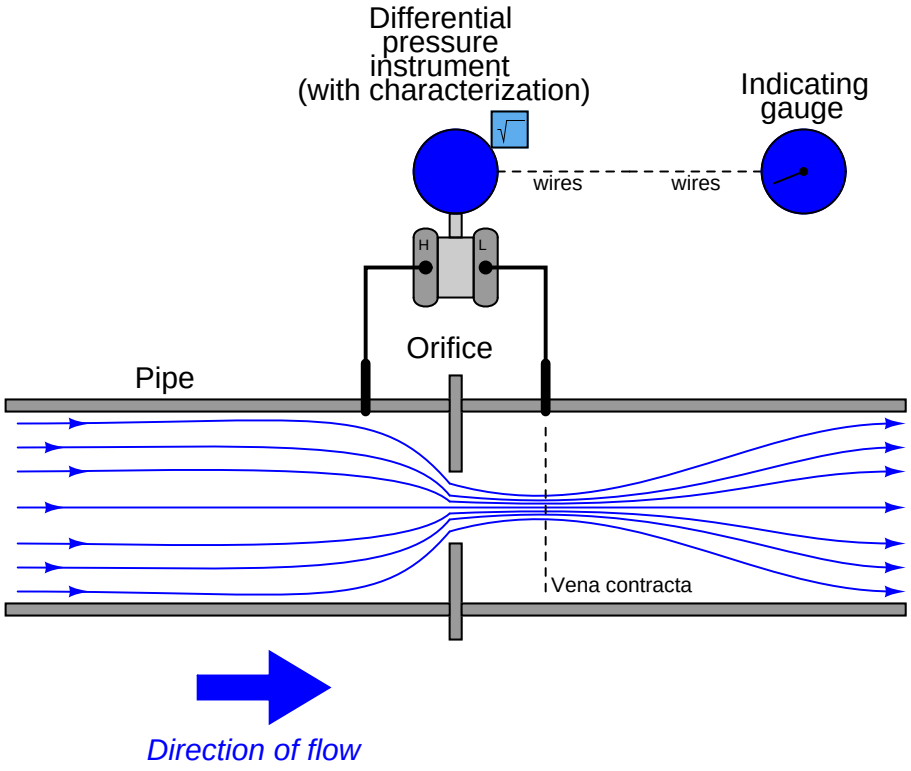
As you can see from the table, the square-root relationship is most evident in comparing the input and output *percentage* values. For example, at an input signal pressure of 6 PSI (25%), the output signal percentage will be the square root of 25%, which is 50% ($0.5 = \sqrt{0.25}$) or 9 PSI as a pneumatic signal. At an input signal pressure of 10 PSI (58.33%), the output signal percentage will be 76.38%, because $0.7638 = \sqrt{0.5833}$, yielding an output signal pressure of 12.17 PSI.

When graphed, the function of a square-root extractor is precisely opposite (inverted) of the quadratic function of a flow-sensing element such as an orifice plate, venturi, or pitot tube:



When cascaded – the square-root function placed immediately after the flow element's "square" function – the result is an output signal that tracks linearly with flow rate (Q). An instrument connected to the square root relay's signal will therefore register flow rate as it should.

Although analog electronic square-root relays have been built and used in industry for characterizing the output of 4-20 mA electronic transmitters, a far more common implementation of electronic square-root characterization occurs in DP transmitters designed with the square-root function built in. This way, no external relay device is necessary to characterize the DP transmitter's signal into a flow rate signal:



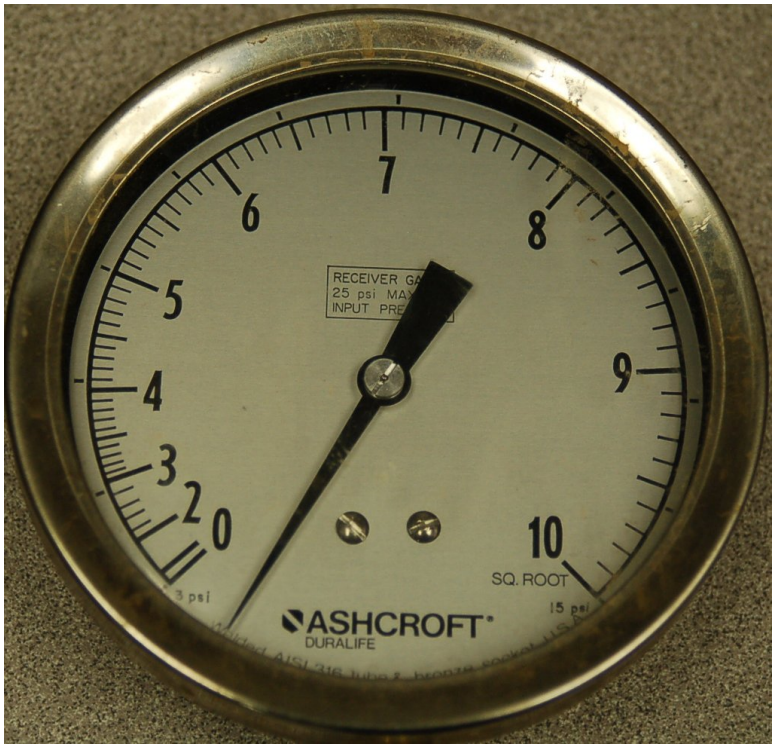
Using a characterized DP transmitter, any 4-20 mA sensing instrument connected to the transmitter's output wires will directly interpret the signal as flow rate rather than as pressure. A calibration table for such a DP transmitter (with an input range of 0 to 150 inches water column) is shown here:

ΔP	**Input %**	**Output %** $= \sqrt{\textbf{Input \%}}$	**Output signal**
0 "W.C.	0%	0%	4 mA
37.5 "W.C.	25%	50%	12 mA
75 "W.C.	50%	70.71%	15.31 mA
112.5 "W.C.	75%	86.60%	17.86 mA
150 "W.C.	100%	100%	20 mA

Once again, we see how the square-root relationship is most evident in comparing the input and output *percentages*. Note how the percentages in this table precisely match the percentages in the

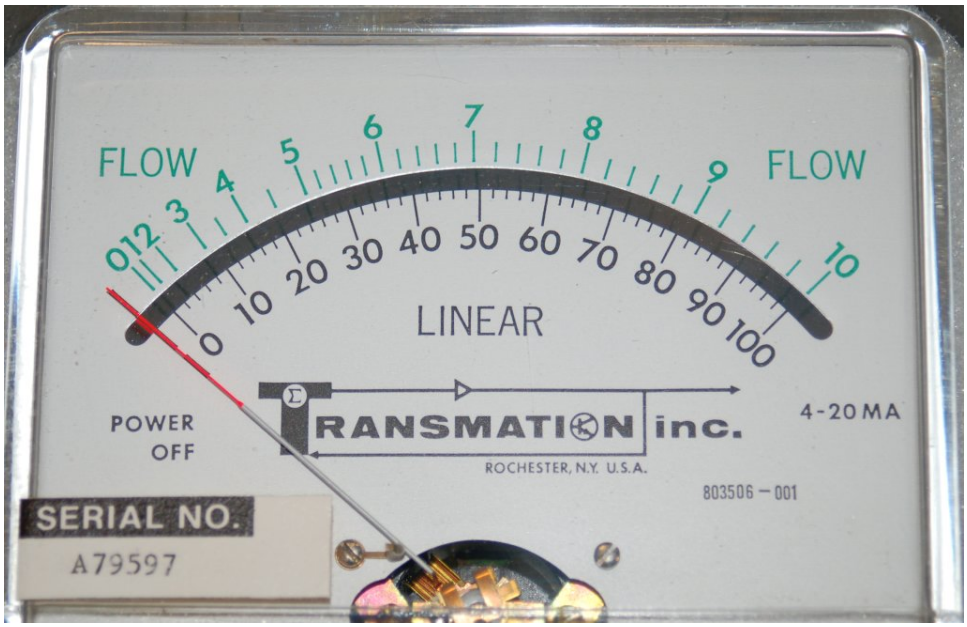
pneumatic relay table: 0% input gives 0% output; 25% input gives 50% output, 50% input gives 70.71% output, etc.

An ingenious solution to the problem of square-root characterization, commonly seen in pneumatic flow-measurement systems where the DP transmitter lacks square-root characterization, is to use an indicating device with a square-root indicating scale. For example, the following photograph shows a 3-15 PSI "receiver gauge" designed to directly sense the output of a pneumatic DP transmitter:



Note how the gauge mechanism responds directly and linearly to a 3-15 PSI input signal range (note the "3 PSI" and "15 PSI" labels in small print at the extremes of the scale, and the linearly-spaced marks around the outside of the scale arc representing 1 PSI each), but how the flow markings (0 through 10 on the inside of the scale arc) are spaced in a non-linear fashion.

An electronic variation on this theme is to draw a square-root scale on the face of a meter movement driven by the 4-20 mA output signal of an electronic DP transmitter:



As with the square-root receiver gauge, the meter movement's response to the transmitter signal is linear. Note the linear scale (drawn in black text labeled "LINEAR") on the bottom and the corresponding square-root scale (in green text labeled "FLOW") on the top. This makes it possible for a human operator to read the scale in terms of (characterized) flow units. Instead of using complicated mechanisms or circuitry to characterize the transmitter's signal, a non-linear scale "performs the math" necessary to interpret flow.

A major disadvantage to the use of these non-linear indicator scales is that the transmitter signal itself remains un-characterized. Any other instrument receiving this un-characterized signal will either require its own square-root characterization or simply not interpret the signal in terms of flow at all. An un-characterized flow signal input to a process controller can cause loop instability at high flow rates, where small changes in actual flow rate result in huge changes in differential pressure sensed by the transmitter. A fair number of flow control loops operating without characterization have been installed in industrial applications (usually with square-root scales drawn on the face of the indicators, and square-root paper installed in chart recorders), but these loops are notorious for achieving good flow control at only one setpoint value. If the operator raises or lowers the setpoint value, the "gain" of the control loop changes thanks to the nonlinearities of the flow element, resulting in either under-responsive or over-responsive action from the controller.

Despite the limited practicality of non-linear indicating scales, they hold significant value as teaching tools. Closely examine the scales of both the receiver gauge and the 4-20 mA indicating meter, comparing the linear and square-root values at common points on each scale. A couple of examples are highlighted on the electric meter's scale:



A few correlations between the linear and square-root scales of either the pneumatic receiver gauge or the electric indicating meter verify the fact that the square-root function is *encoded* in the spacing of the numbers on each instrument's non-linear scale.

Another valuable lesson we may take from the faces of these indicating instruments is how uncertain the flow measurement becomes at the low end of the scale. Note how for each indicating instrument (both the receiver gauge and the meter movement), the square-root scale is "compressed" at the low end, to the point where it becomes impossible to interpret fine increments of flow at that end of the scale. At the high end of each scale, it's a different situation entirely: the numbers are spaced so far apart that it's easy to read fine distinctions in flow values (e.g. 94% flow versus 95% flow). However, the scale is so crowded at the low end that it's really impossible to clearly distinguish two different flow values such as 4% from 5%.

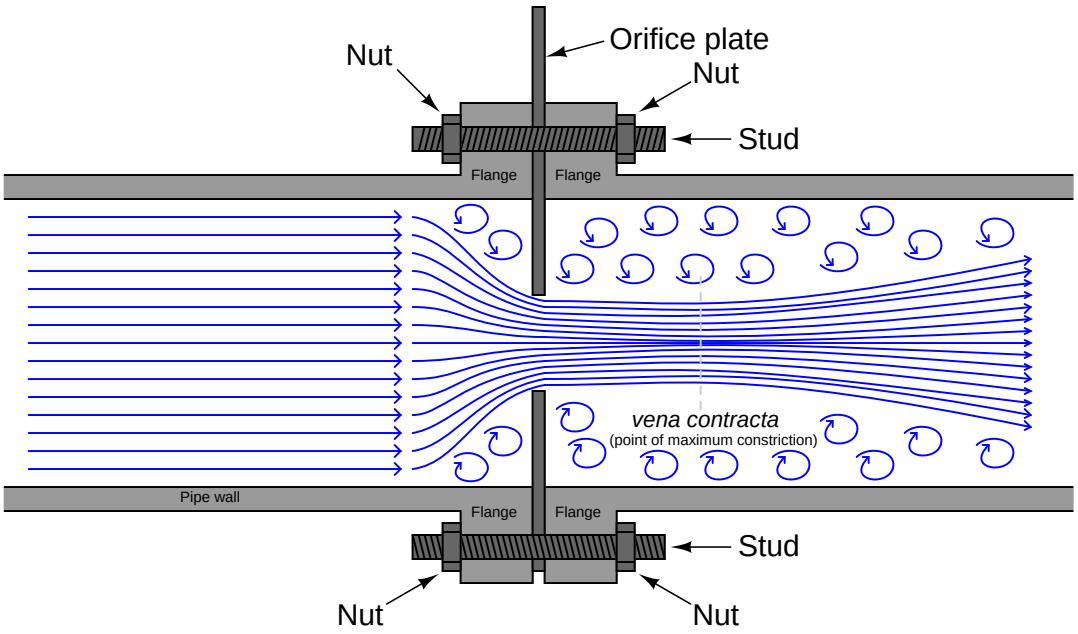
This "crowding" is not just an artifact of a visual scale; it is a reflection of a fundamental limitation in measurement certainty with this type of flow measurement. The amount of differential pressure separating different low-range values of flow for a flow element is so little, even small amounts of pressure-measurement error equate to large amounts of flow-measurement error. In other words, it becomes more and more difficult to precisely interpret flow rate as the flow rate decreases toward the low end of the scale. The "crowding" we see on these indicator's square-root scales is a visual reflection of this fundamental problem: even a small error in interpreting the pointer's position at the low end of the scale can yield major errors in flow interpretation.

A technical term used to quantify this problem is *turndown*. "Turndown" refers to the ratio of high-range measurement to low-range measurement possible for an instrument while maintaining reasonable accuracy. For pressure-based flowmeters, which must deal with the non-linearities of Bernoulli's Equation, the practical turndown is often no more than 3 to 1 (3:1). This means a flowmeter ranged for 0 to 300 GPM might only read with reasonable accurately down to a flow

of 100 GPM. Below that, the accuracy becomes so poor that the measurement is almost useless. Advances in DP transmitter technology have pushed this ratio further, perhaps as far as 10:1 for certain installations. However, the fundamental problem is not transmitter resolution, but rather the nonlinearity of the flow element itself. This means *any* source of pressure-measurement error – whether originating in the transmitter's pressure sensor or not – compromises our ability to accurately measure flow at low rates. Even with a *perfectly* calibrated transmitter, errors resulting from wear of the flow element (e.g. a dulled edge on an orifice plate) or from uneven liquid columns in the impulse tubes connecting the transmitter to the element, will cause large flow-measurement errors at the low end of the instrument's range where the flow element produces only small differential pressures. Everyone involved with the technical details of flow measurement needs to understand this fact: the fundamental problem of limited turndown is grounded in the physics of turbulent flow and potential/kinetic energy exchange for these flow elements. Technological improvements will help, but they cannot overcome the limitations imposed by physics. If better turndown is required for a particular flow-measurement application, an entirely different flowmeter technology should be considered.

22.1.5 Orifice plates

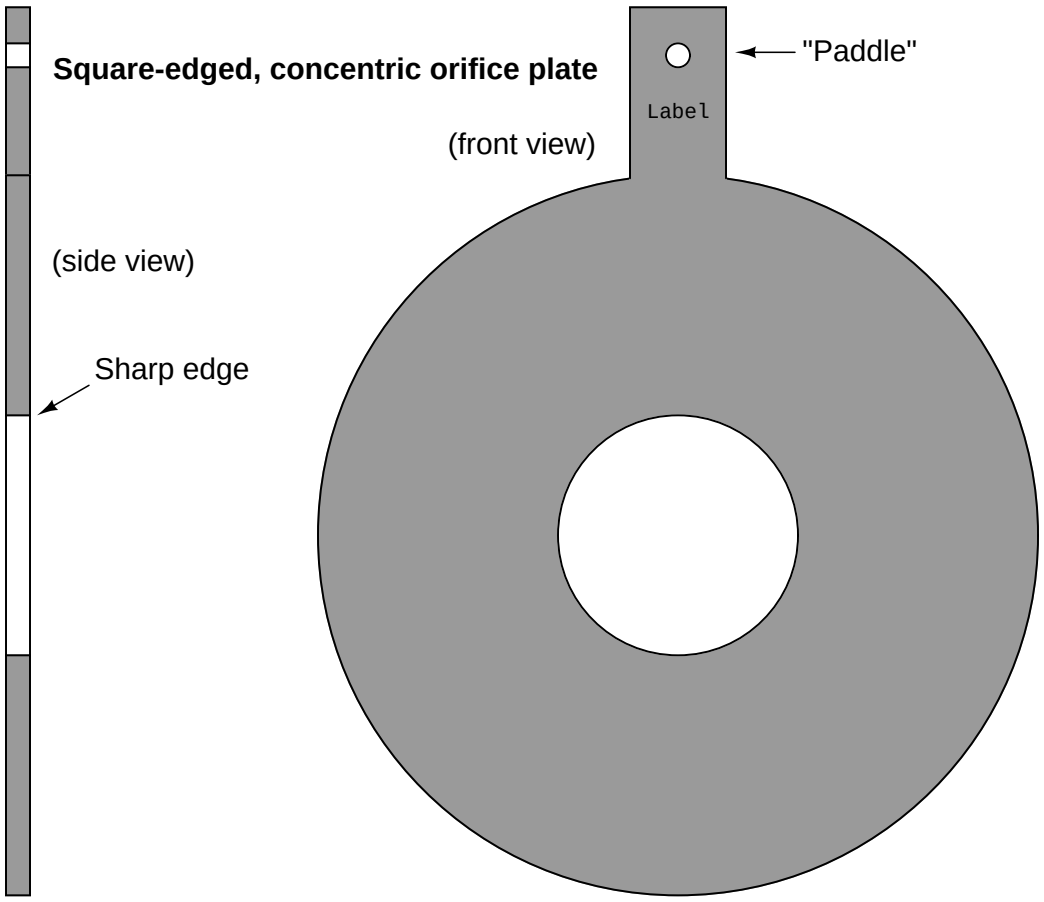
Of all the pressure-based flow elements in existence, the most common is the *orifice plate*. This is simply a metal plate with a hole in the middle for fluid to flow through. Orifice plates are typically sandwiched between two flanges of a pipe joint, allowing for easy installation and removal:



The point where the fluid flow profile constricts to a minimum cross-sectional area after flowing through the orifice is called the *vena contracta*, and it is the area of minimum fluid pressure. The vena contracta corresponds to the narrow throat of a venturi tube. The precise location of the vena contracta for an orifice plate installation will vary with flow rate, and also with the *beta ratio* (β) of the orifice plate, defined as the ratio of bore diameter (d) to inside pipe diameter (D):

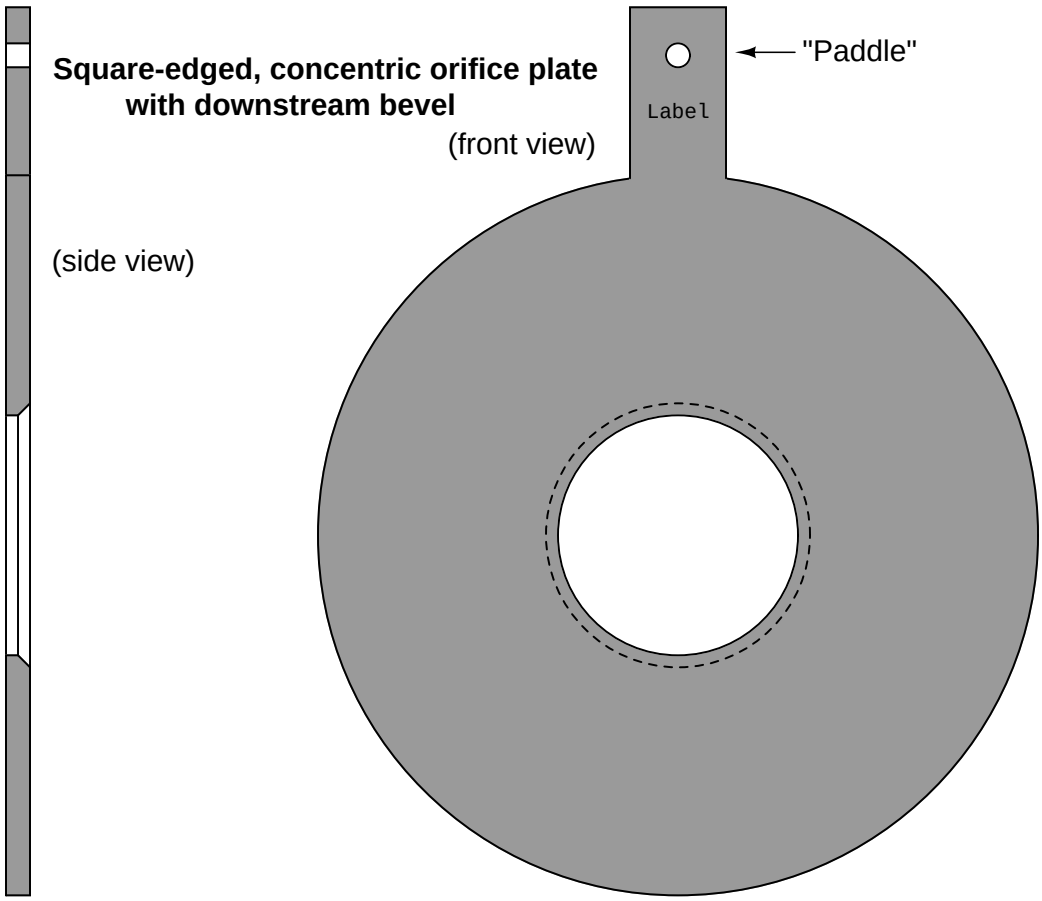
$$\beta = \frac{d}{D}$$

The simplest design of orifice plate is the *square-edged, concentric* orifice. This type of orifice plate is manufactured by machining a precise, straight hole in the middle of a thin metal plate. Looking at a side view of a square-edged concentric orifice plate reveals sharp edges (90° corners) at the hole:



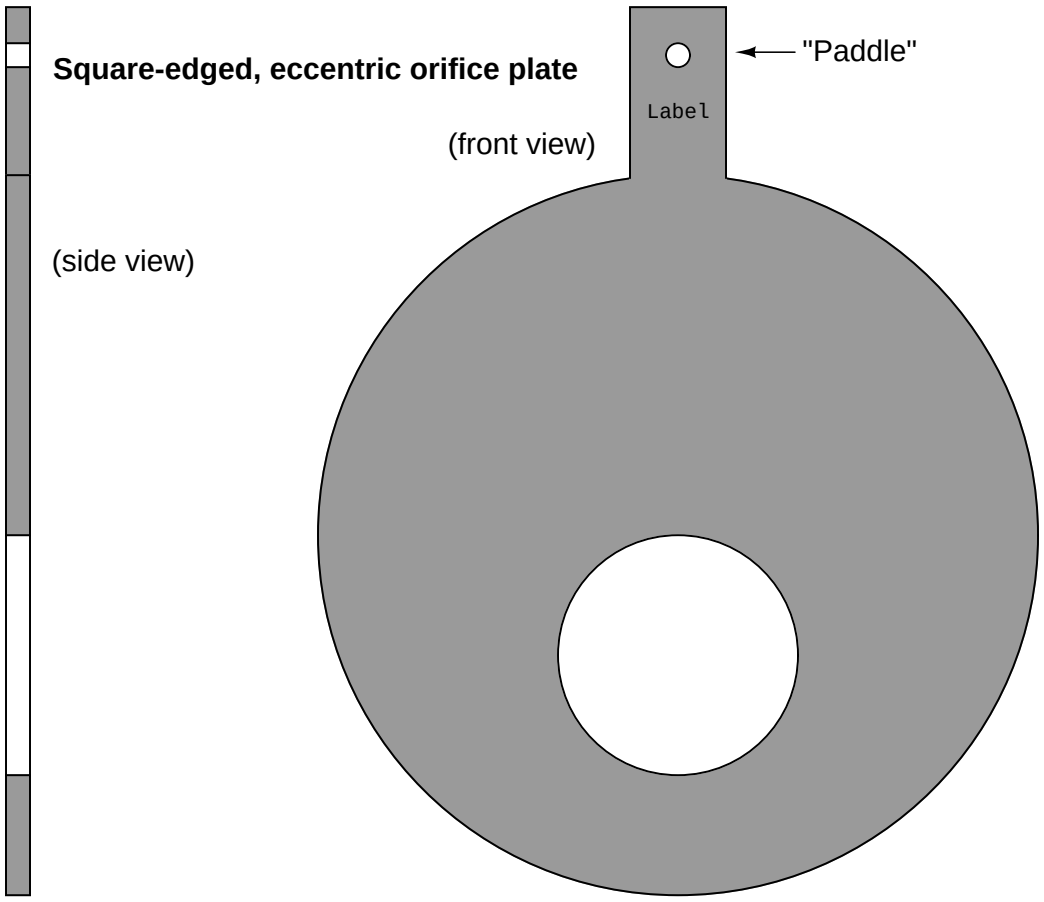
Square-edged orifice plates may be installed in either direction, since the orifice plate "appears" exactly the same from either direction of fluid approach. In fact, this allows square-edged orifice plates to be used for measuring bidirectional flow rates (where the fluid flow direction reverses itself from time to time). A text label printed on the "paddle" of any orifice plate customarily identifies the upstream side of that plate, but in the case of the square-edged orifice plate it does not matter.

The purpose of having a square edge on the hole in an orifice plate is to minimize contact with the fast-moving moving fluid stream going through the hole. Ideally, this edge will be knife-sharp. If the orifice plate is relatively thick (1/8 or an inch or more), it may be necessary to bevel the downstream side of the hole to further minimize contact with the fluid stream:



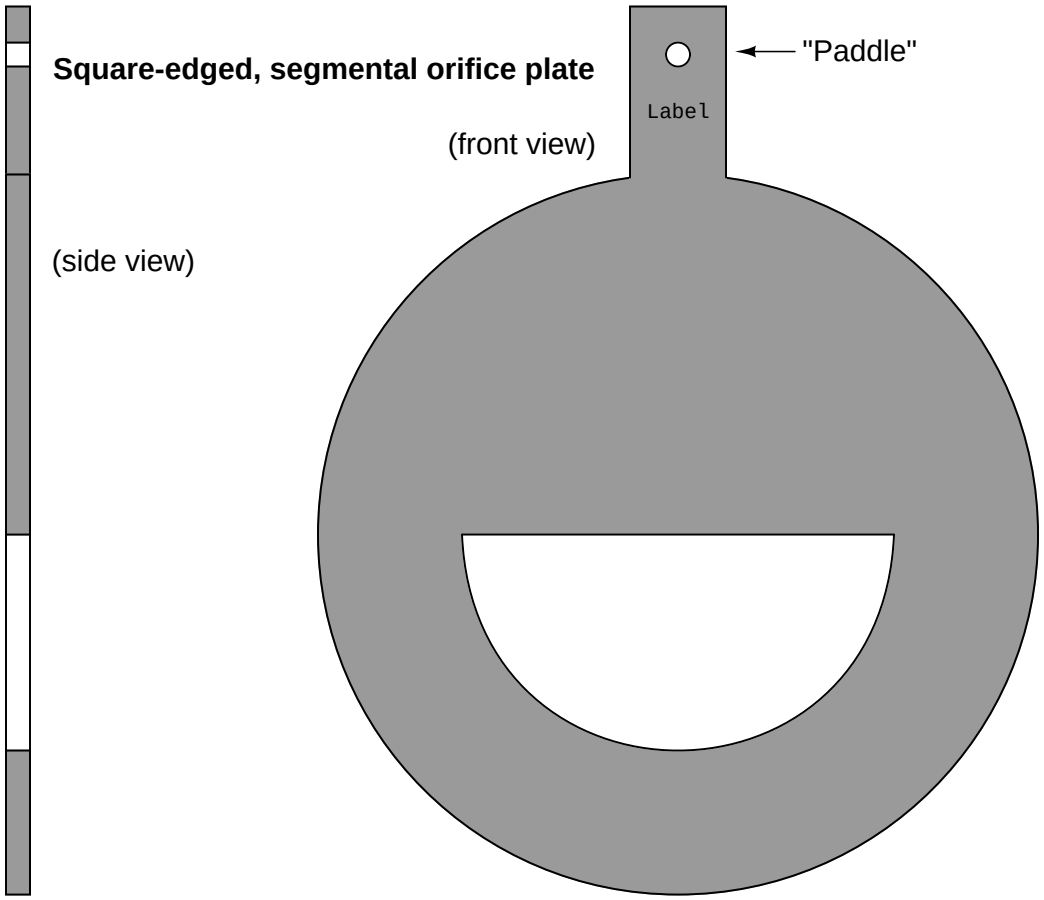
Looking at the side-view of this orifice plate, the intended direction of flow is left-to-right, with the sharp edge facing the incoming fluid stream and the bevel providing a non-contact outlet for the fluid. Beveled orifice plates are obviously uni-directional, and *must* be installed with the paddle text facing upstream.

Other square-edged orifice plates exist to address conditions where gas bubbles or solid particles may be present in liquid flows, or where liquid droplets or solid particles may be present in gas flows. The first of this type is called the *eccentric* orifice plate, where the hole is located off-center to allow the undesired portions of the fluid to pass through the orifice rather than build up on the upstream face:



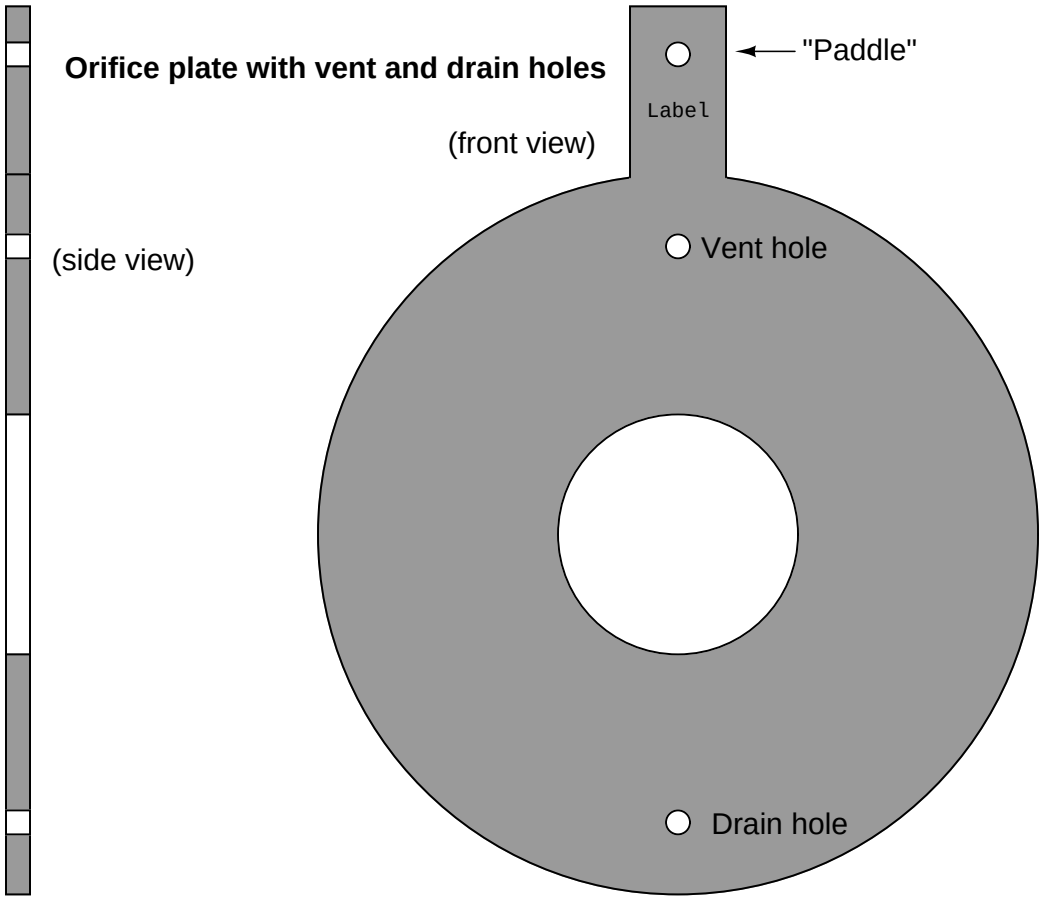
For gas flows, the hole should be offset downward, so any liquid droplets or solid particles may easily pass through. For liquid flows, the hole should be offset upward to allow gas bubbles to pass through and offset downward to allow heavy solids to pass through.

The second off-center orifice plate type is called the *segmental orifice plate*, where the hole is not circular but rather just a segment of a concentric circle:



As with the eccentric orifice plate design, the segmental hole should be offset downward in gas flow applications and either upward or downward in liquid flow applications depending on the type of undesired material(s) in the flowstream.

An alternative to offsetting or re-shaping the bore hole of an orifice plate is to simply drill a small hole near the edge of the plate, flush with the inside diameter of the pipe, allowing undesired substances to pass through the plate rather than collect on the upstream side. If such a hole is oriented upward to pass vapor bubbles, it is called a *vent hole*. If the hole is oriented downward to pass liquid droplets or solids, it is called a *drain hole*. Vent and drain holes are useful when the concentration of these undesirable substances is not significant enough to warrant either an eccentric or segmental orifice:



The addition of a vent or drain hole should have negligible impact on the performance of an orifice plate due to its small size relative to the main bore. If the quantity of undesirable material in the flowstream (bubbles, droplets, or solids) is excessive, an eccentric or segmental orifice plate might be a better choice[12].

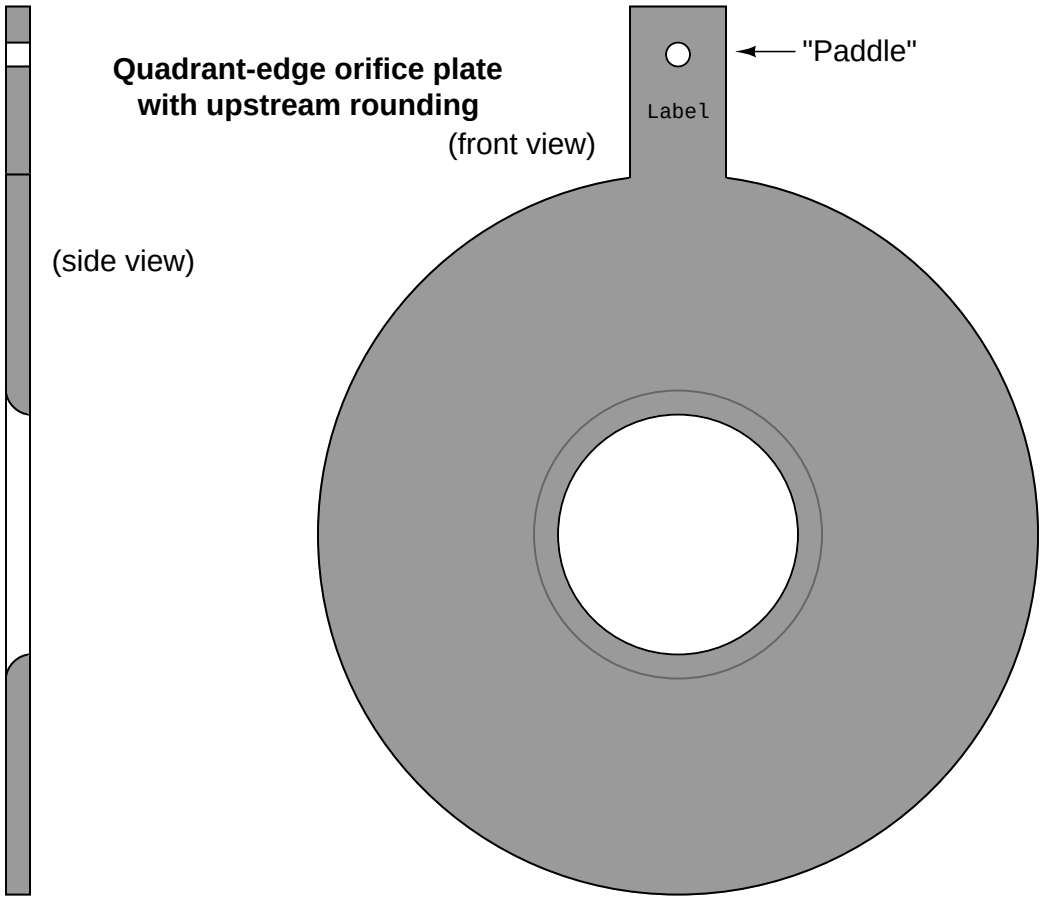
[12] L.K. Spink, in his book *Principles and Practice of Flow Meter Engineering*, notes that drain holes intended to pass solid objects may be useless in small pipe sizes, where the hole is so small it will probably become plugged with solid debris and cease to provide benefit. In such installations he recommends re-orienting the pipe vertically instead

Some orifice plates employ non-square-edged holes for the purpose of improving performance at low Reynolds number[13] values, where the effects of fluid viscosity are more apparent. These orifice plate types employ rounded- or conical-entrance holes in an effort to minimize the effects of fluid viscosity. Experiments have shown that decreased Reynolds number causes the flowstream to not contract as much when traveling through an orifice, thus limiting fluid acceleration and decreasing the amount of differential pressure produced by the orifice plate. However, experiments have also shown that decreased Reynolds number in a venturi-type flow element causes an *increase* in differential pressure due to the effects of friction against the entrance cone walls. By manufacturing an orifice plate in such a way that the hole exhibits "venturi-like" properties (i.e. a dull edge where the fast-moving fluid stream has more contact with the plate), these two effects tend to cancel each other, resulting in an orifice plate that maintains consistent accuracy at lower flow rates and/or higher viscosities than the simple square-edged orifice.

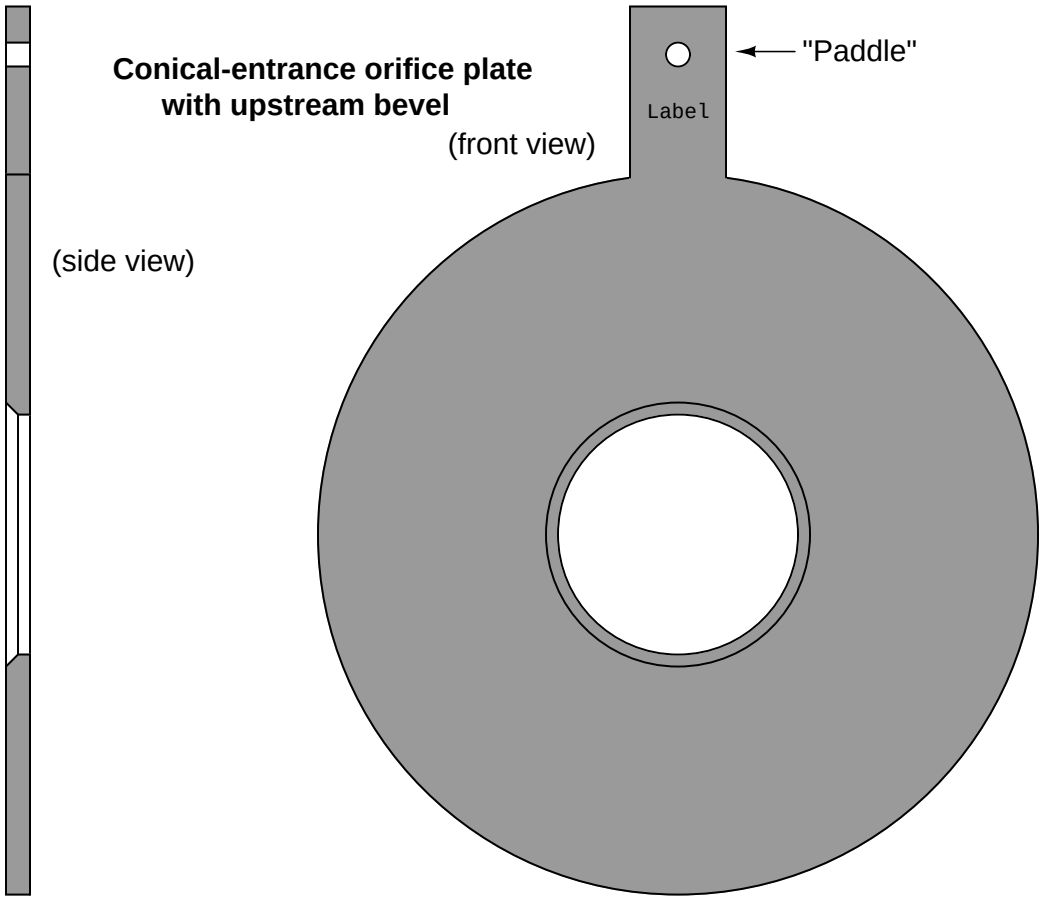
of horizontally. This allows solids to pass through the main bore of the orifice without "damming" on the upstream side of the orifice plate. I would add the suggestion to consider a different primary element entirely, such as a venturi tube. The small size of the line will limit the cost of such an element, and the performance is likely to be far better than an orifice plate anyway.

[13]To read more about the concept of Reynolds number, refer to section 2.11.10 beginning on page 201.

Two common non-square-edge orifice plate designs are the *quadrant-edge* and *conical-entrance* orifices. The quadrant-edge is shown first:

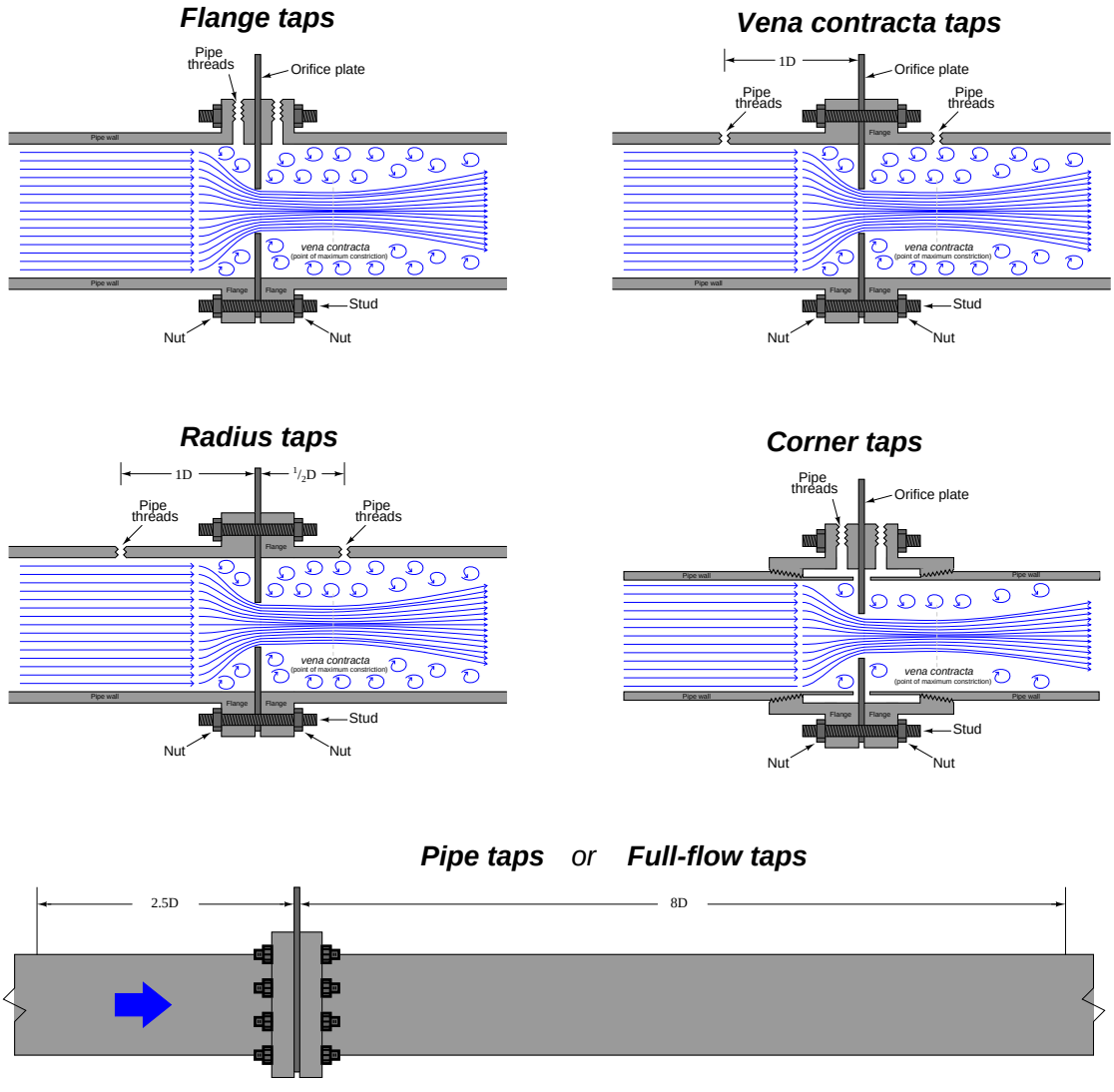


The conical-entrance orifice plate looks like a beveled square-edge orifice plate installed backwards, with flow entering the conical side and exiting the square-edged side:



Here, is it vitally important to pay attention to the paddle's text label. This is the only sure indication of which direction an orifice plate needs to be installed. One can easily imagine an instrument technician mistaking a conical-entrance orifice plate for a square-edged, beveled orifice plate and installing it backward!

Several standards exist for pressure tap locations. Ideally, the upstream pressure tap will detect fluid pressure at a point of minimum velocity, and the downstream tap will detect pressure at the vena contracta (maximum velocity). In reality, this ideal is never perfectly achieved. An overview of the most popular tap locations for orifice plates is shown in the following illustration:



Flange taps are the most popular tap location for orifice meter runs on large pipes in the United States. Flanges may be manufactured with tap holes pre-drilled and finished before the flange is even welded to the pipe, making this a very convenient pressure tap configuration. Most of the other tap configurations require drilling into the pipe after installation, which is not only labor-intensive, but may possibly weaken the pipe at the locations of the tap holes.

Vena contracta taps offer the greatest differential pressure for any given flow rate, but require precise calculations to properly locate the downstream tap position. *Radius taps* are an approximation of vena contracta taps for large pipe sizes (one-half pipe diameter downstream for the low-pressure tap location). An unfortunate characteristic of both these taps is the requirement of drilling through the pipe wall. Not only does this weaken the pipe, but the practical necessity of drilling the tap holes in the installed location rather than in a controlled manufacturing environment means there is considerable room for installation error[14].

Corner taps must be used on small pipe diameters where the vena contracta is so close to the downstream face of the orifice plate that a downstream flange tap would sense pressure in the highly turbulent region (too far downstream). Corner taps obviously require special (i.e. expensive) flange fittings, which is why they tend to be used only when necessary.

Care should be taken to avoid measuring downstream pressure in the highly turbulent region following the vena contracta. This is why the *pipe tap* (also known as *full-flow tap*) standard calls for a downstream tap location eight pipe diameters away from the orifice: to give the flow stream room to stabilize for more consistent pressure readings[15].

Wherever the taps are located, it is vitally important that the tap holes be completely flush with the inside wall of the pipe or flange. Even the smallest recess or burr left from drilling will cause measurement errors, which is why tap holes are best drilled in a controlled manufacturing environment rather that at the installation site where the task will likely be performed by non-experts.

[14] One significant source of error for customer-drilled tap holes is the interior finish of the holes. Even a small "burr" of metal left where the hole penetrates the inner surface of the pipe wall will cause substantial flow measurement errors!

[15] What this means is that a "pipe tap" installation is actually measuring permanent pressure loss, which also happens to scale with the square of flow rate because the primary mechanism for energy loss in turbulent flow conditions is the translation of linear velocity to angular (swirling) velocity in the form of eddies. This kinetic energy is eventually dissipated in the form of heat as the eddies eventually succumb to viscosity.

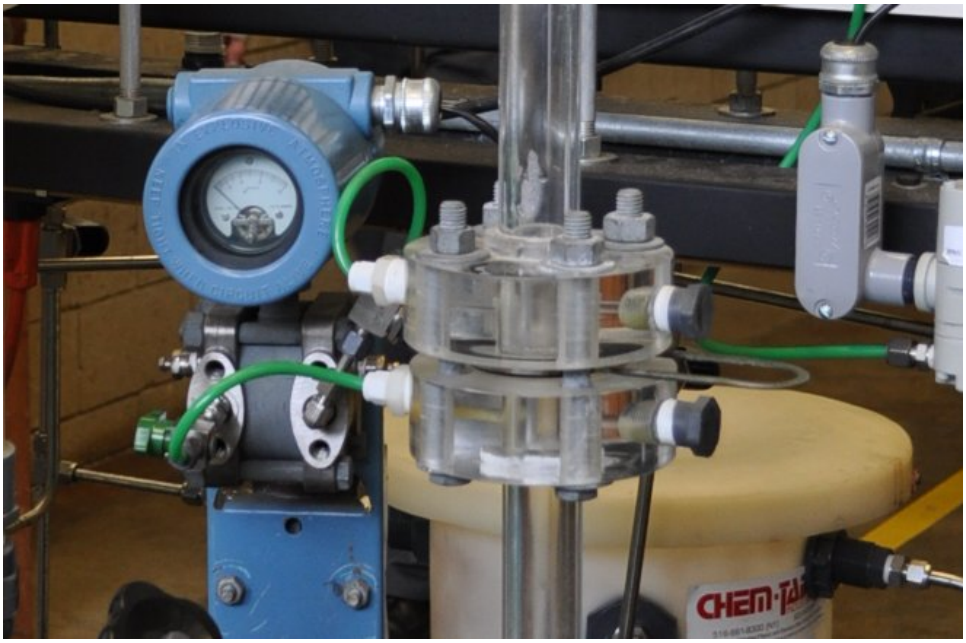
A photograph of an orifice plate used to measure the flow of natural gas to a large turbine engine is shown here, with a Rosemount model 3051 differential pressure transmitter sensing the pressure drop generated by the orifice:



Flange taps are used in this orifice installation, with the taps and pressure transmitter located above the pipe centerline to avoid picking up any liquid droplets that may pass through the pipe. The direction of gas flow in this particular installation is from left to right, making the left-hand pressure tap the "high pressure" side and the right-hand pressure tap the "low pressure" side.

As you can see by the pressure gauge's indication in the photograph, the static line pressure of the natural gas inside the pipe is over 300 PSI. The amount of pressure drop generated by the orifice plate at full flow, however, is likely only a few PSI (100 inches water column is typical for many orifice plate installations). This is why we must use a *differential* pressure transmitter to measure the orifice plate's pressure drop: only a DP transmitter will sense the difference in pressure across the orifice while rejecting the static (common-mode) line pressure inside the pipe.

A photograph of another orifice plate with flange taps appears here, shown on a vertical pipe. In this example, the pipe and flanges are formed of acrylic (transparent plastic):

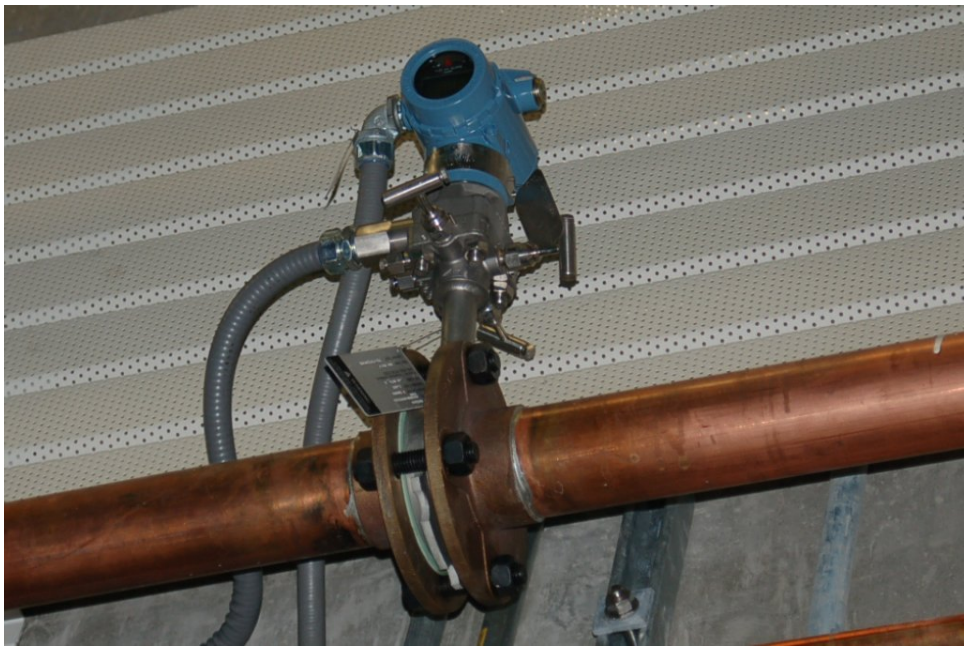


As is customary with orifice plates mounted in vertical pipes, the direction of flow is upward (from bottom to top), making the bottom tap the "high pressure" side and the top tap the "low pressure" side. Since this is a liquid application, the transmitter is located below[16] the taps in order to avoid collecting bubbles of air or other gases in the impulse lines.

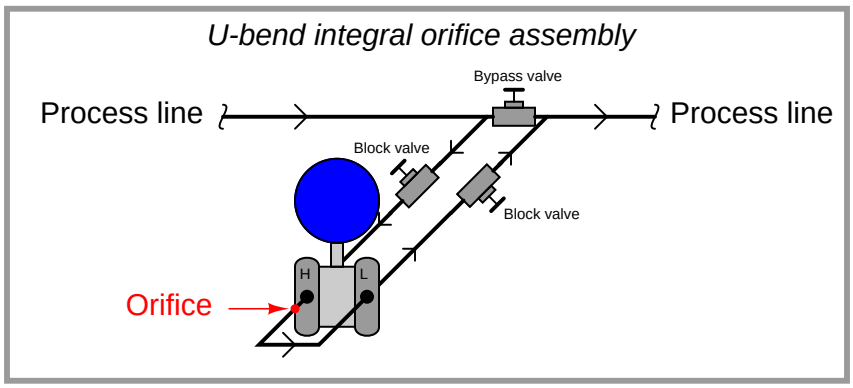
This particular orifice and flow transmitter (Rosemount model 1151) is used on a process "trainer" unit at Brazosport College in Lake Jackson, Texas. The transparent flanges, pipes, and process vessels make it easier for students to visualize the fluid motion.

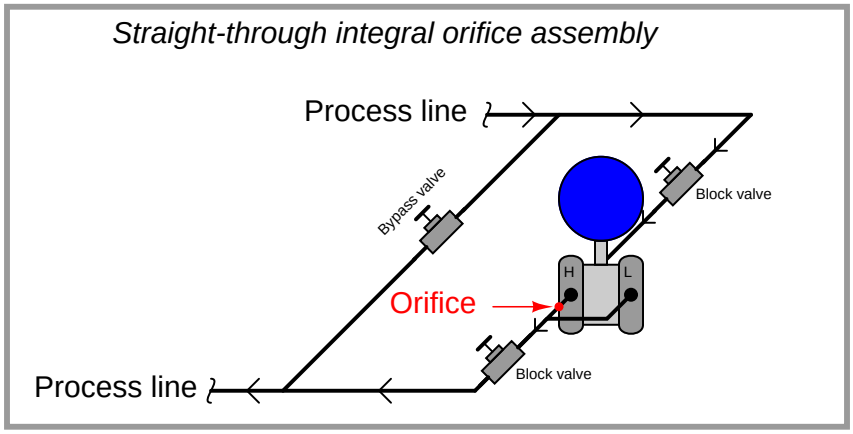
[16] One installation error seen in this photograph is a green plastic impulse tube with a bend extending above the upper flange tap. Any elevated portion of the impulse tube system will tend to collect gas bubbles over time, possibly causing measurement errors. A better installation would ensure the impulse tubes never extend above the flange tap they connect to on the liquid-bearing pipe.

For relatively low flow rates, an alternative arrangement is the *integral orifice plate*. This is where a small orifice plate directly attaches to the differential pressure-sensing element, eliminating the need for impulse lines. A photograph of an integral orifice plate and transmitter is shown here, in an application measuring the flow of purified oxygen gas through a copper pipe:



Even smaller integral orifices exist for the measurement of very low flow rates, another type of integral orifice is available. This style uses the pass-through nature of a typical differential pressure transmitter's capsule flanges to advantage. Note the paths of fluid flow, as well as the unusual orientation of the three-valve manifold, in these two example illustrations:

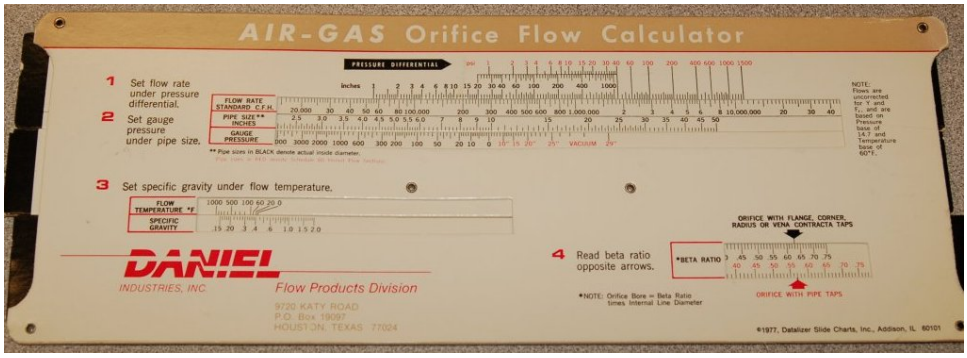


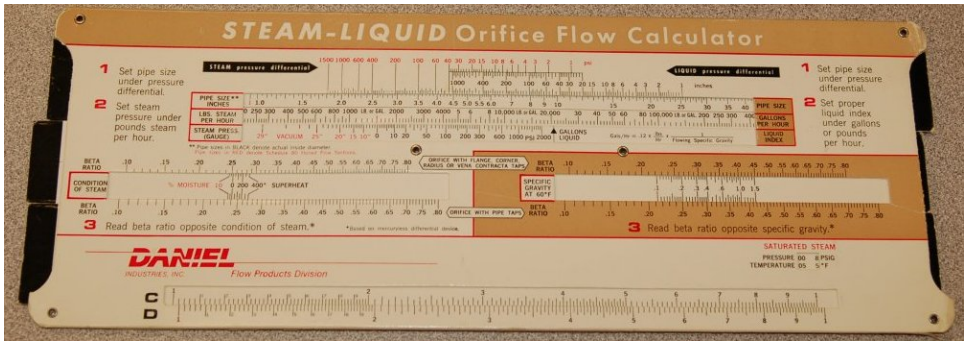


In both examples, the third hand valve serves the purpose of bypassing process fluid flow rather than equalizing both ports of the differential pressure transmitter. Process fluid actually flows *through* at least one chamber of the DP transmitter's body. Clearly, this kind of integral orifice plate is practical only for very low rates of flow, typically where the process flow line is no larger than the size of the pressure ports on the transmitter's body.

The task of properly sizing an orifice plate for any given application is complex enough to recommend the use of special orifice sizing computer software provided by orifice plate manufacturers. There are a number of factors to consider in orifice plate sizing, and these software packages account for all of them. Best of all, the software provided by manufacturers is often linked to data for that manufacturer's product line, helping to assure installed results in close agreement with predictions.

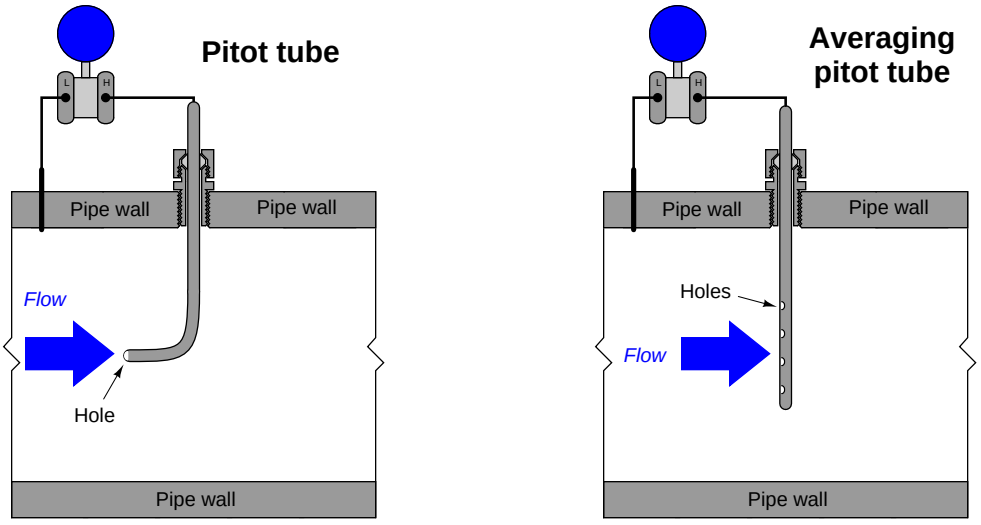
In the days before ubiquitous personal computers and the Internet, some orifice plate manufacturers provided customers with paper "slide rule" calculators to help them select appropriate orifice plate sizes from known process parameters. The following photographs show the front and back sides of one such slide rule:



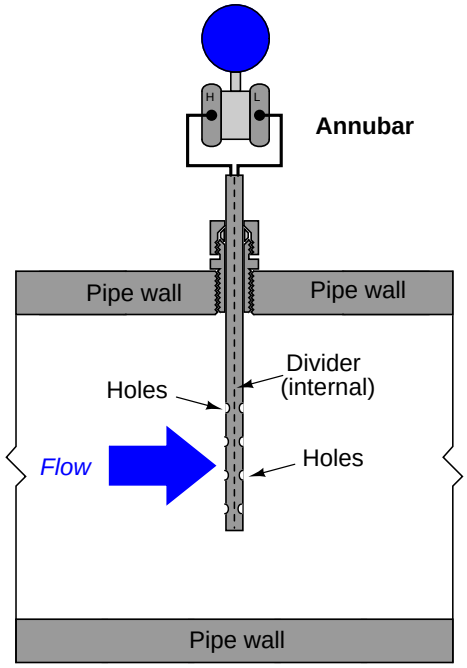


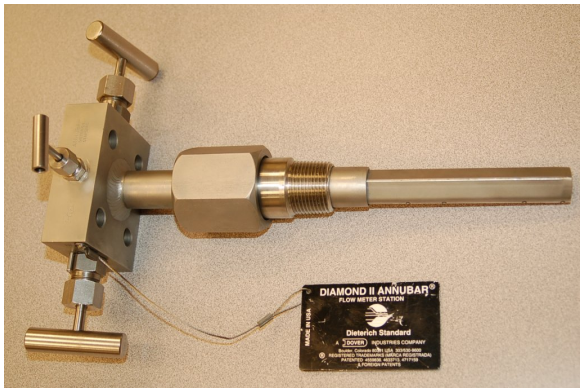
22.1.6 Other differential producers

Other pressure-based flow elements exist as alternatives to the orifice plate. The *Pitot tube*, for example, senses pressure as the fluid stagnates (comes to a complete stop) against the open end of a forward-facing tube. A shortcoming of the classic single-tube Pitot assembly is sensitivity to fluid velocity at just one point in the pipe, so a more common form of Pitot tube seen in industry is the *averaging* Pitot tube consisting of several stagnation holes sensing velocity at multiple points across the width of the flow:



A variation on the latter theme is the *Annubar* flow element, a trade name of the Dieterich Standard corporation. An "Annubar" is an averaging pitot tube consolidating high and low pressure-sensing ports in a single probe assembly:

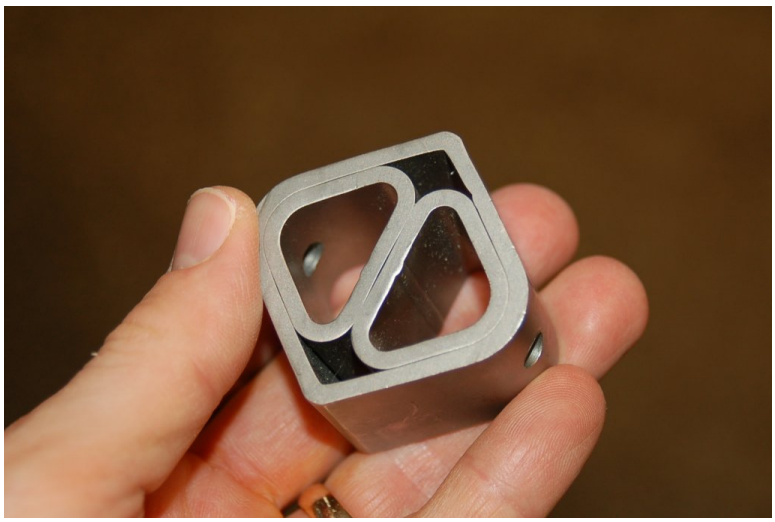




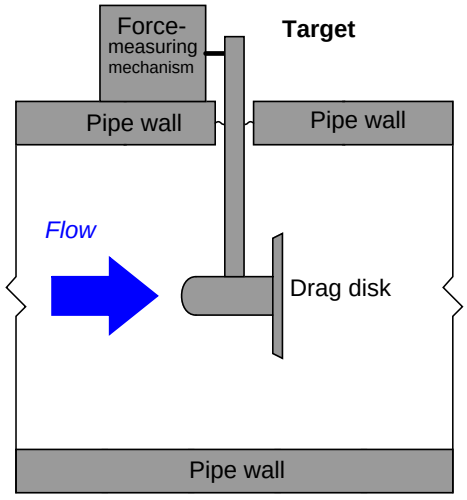
What appears at first glance to be a single, square-shaped tube inserted into the pipe is actually a double-ported tube with holes on both the upstream and downstream edges:



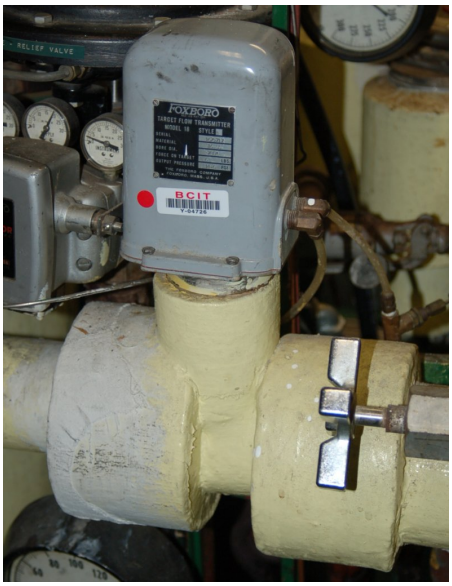
A section of Annubar tube clearly shows the porting and dual chambers, designed to bring upstream (stagnation) and downstream pressures out of the pipe to a differential pressure-sensing instrument:



A less sophisticated realization of the stagnation principle is the *target* flow sensor, consisting of a blunt "paddle" (or "drag disk") inserted into the flowstream. The force exerted on this paddle by the moving fluid is sensed by a special transmitter mechanism, which then outputs a signal corresponding to flow rate (proportional to the square of fluid velocity, just like an orifice plate):

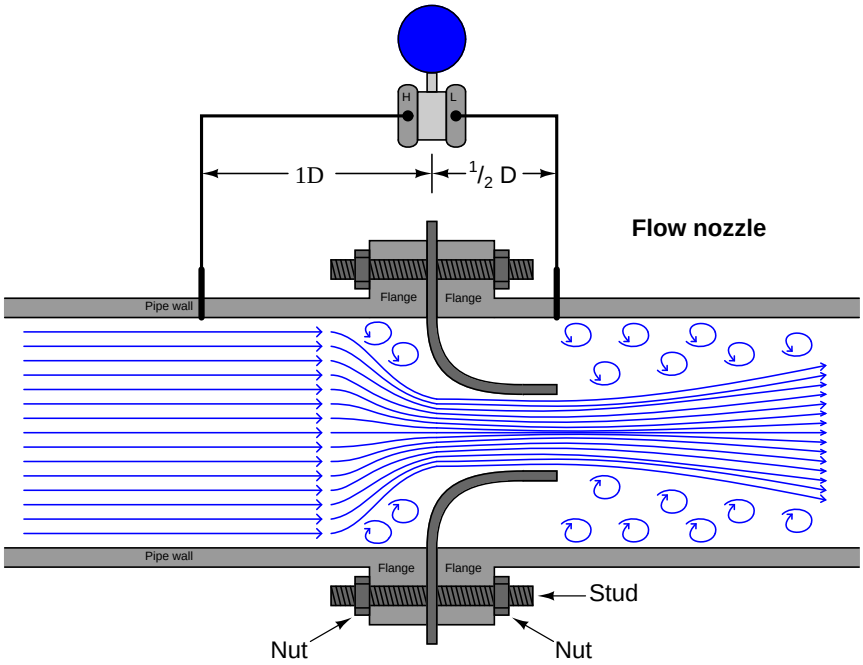


The following photograph shows a Foxboro model 18 target flow transmitter installed on a line conveying liquid wood pulp:

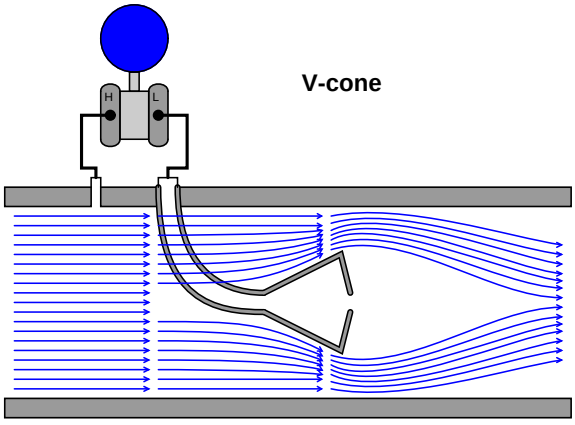


The classic venturi tube pioneered by Clemens Herschel in 1887 has been adapted in a variety of forms broadly classified as *flow tubes*. All flow tubes work on the same principle: developing a differential pressure by channeling fluid flow from a wide tube to a narrow tube. The differ from the classic venturi only in construction details, the most significant detail being a significantly shorter length than the classic venturi tube. Examples of flow tube designs include the *Dall* tube, *Lo-Loss* flow tube, *Gentile* or *Bethlehem* flow tube, and the *B.I.F. Universal Venturi*.

Another variation on the venturi theme is called a *flow nozzle*, designed to be clamped between the faces of two pipe flanges in a manner similar to an orifice plate. The goal here is to achieve simplicity of installation approximating that of an orifice plate while improving performance (less permanent pressure loss) over orifice plates:



Two more variations on the venturi theme are the *V-cone* and *Segmental wedge* flow elements. The V-cone (or "venturi cone," a trade name of the McCrometer division of the Danaher corporation) may be thought of as a venturi tube or orifice plate in reverse[17]: instead of narrowing the tube's diameter to cause fluid acceleration, fluid must flow around a cone-shaped obstruction placed in the middle of the tube. The tube's effective area will be reduced by the presence of this cone, causing fluid to accelerate through the restriction just as it would through the throat of a classic venturi tube:

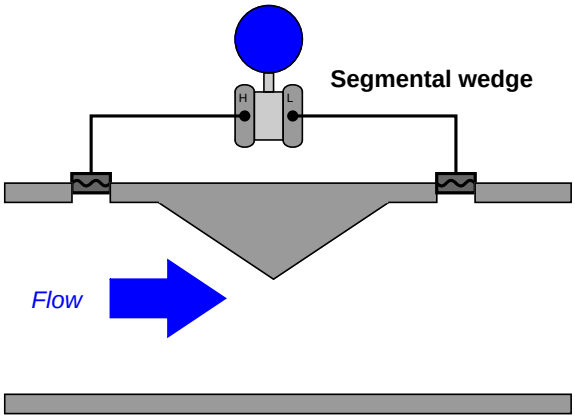


This cone is hollow, with a pressure-sensing port on the downstream side allowing for easy detection of fluid pressure near the vena contracta. Upstream pressure is sensed by another port in the pipe wall upstream of the cone. The following photograph shows a V-cone flow tube, cut away for demonstration purposes:

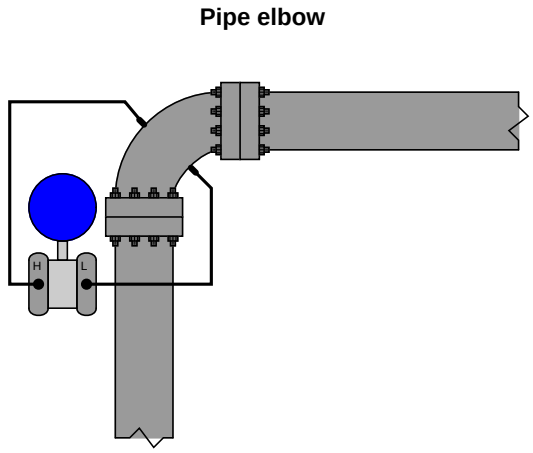


[17] If an orifice plate is a "donut," the V-cone is a "donut hole."

Segmental wedge elements are special pipe sections with wedge-shaped restrictions built in. These devices, albeit crude, are useful for measuring the flow rates of *slurries*[18], especially when pressure is sensed by the transmitter through remote-seal diaphragms (to eliminate the possibility of impulse tube plugging):



Finally, the lowly pipe elbow may be pressed into service as a flow-measuring element, since fluid turning a corner[19] in the elbow experiences radial acceleration and therefore generates a differential pressure along the axis of acceleration:



Pipe elbows should be considered for flow measurement only as a last resort. Their inaccuracies tend to be extreme, owing to the non-precise construction of most pipe elbows and the relatively

[18] A "slurry" is a suspension of solid particles within a liquid. *Mud* is a common example of a slurry.

[19] This phenomenon may be observed when watching the flow of water through a turn in a river, especially if the river is fast-moving. Water level at the far (outside) bank of the turn will be higher than the water level at the near (inside) bank of the turn, due to radial acceleration of the water and the pressure difference that acceleration generates. In fact, that difference in water height may even be used to estimate the river's flow rate!

weak differential pressures generated[20].

A final point should be mentioned on the subject of differential-producing elements, and that is their energy dissipation. Orifice plates are simple and relatively inexpensive to install, but their permanent pressure loss is high compared with other primary elements such as venturi tubes. Permanent pressure loss is permanent energy loss from the flowstream, which usually represents a loss in energy invested into the process by pumps, compressors, and/or blowers. Fluid energy dissipated by an orifice plate thus (usually) translates into a requirement of greater energy input to that process[21].

With the financial and ecological costs of energy being non-trivial in our modern world, it is important to consider energy loss as a significant factor in choosing the appropriate primary element for a pressure-based flowmeter. It might very well be that an "expensive" venturi tube saves more money in the long term than a "cheap" orifice plate, while delivering greater measurement accuracy as an added benefit.

[20] The fact that a pipe elbow generates small differential pressure is an accuracy concern because other sources of pressure become larger by comparison. Noise generated by fluid turbulence in the elbow, for example, becomes a significant portion of the pressure sensed by the transmitter when the differential pressure is so low (i.e. the signal-to-noise ratio becomes smaller). Errors caused by differences in elbow tap elevation and different impulse line fill fluids, for example, become more significant as well.

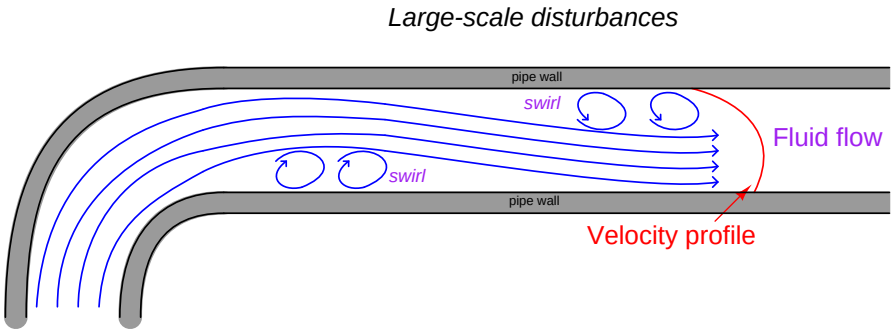
[21] This is not always the case, as primary elements are often found on throttled process lines. In such cases where a control valve normally throttles the flow rate, any energy dissipated by the orifice plate is simply less energy that the valve would otherwise be required to dissipate. Therefore, the presence or absence of an orifice plate has no net impact on energy dissipation when used on a process flow throttled by a control valve.

22.1.7 Proper installation

Perhaps the most common way in which the flow measurement accuracy of any flowmeter becomes compromised is incorrect installation, and pressure-based flowmeters are no exception to this rule. The following list shows some of the details one must consider in installing a pressure-based flowmeter element:

- Necessary upstream and downstream straight-pipe lengths
- Beta ratio (ratio of orifice bore diameter to pipe diameter: $\beta = \frac{d}{D}$)
- Impulse tube tap locations
- Tap finish
- Transmitter location in relation to the pipe

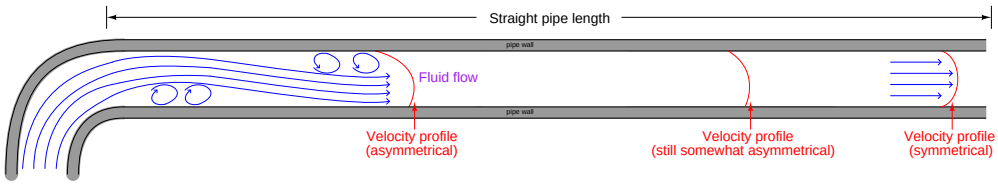
Sharp turns in piping networks introduce large-scale turbulence[22] into the flowstream. Elbows, tees, valves, fans, and pumps are some of the most common causes of large-scale turbulence in piping systems. Successive pipe elbows in different planes are some of the worst offenders in this regard. When the natural flow path of a fluid is disturbed by such piping arrangements, the velocity profile of that fluid becomes distorted; e.g. the velocity gradient from one wall boundary of the pipe to the other will not be orderly. Large eddies in the flowstream (called *swirl*) will appear. This may cause problems for pressure-based flow elements which rely on linear acceleration (change in velocity in one dimension) to measure fluid flow rate. If the flow profile is distorted enough, the acceleration detected at the element may be too great or too little, and therefore not properly represent the full fluid flowstream[23]. For this reason, pressure-based flowmeters should always be located *upstream* of major disturbances such as control valves and pipe elbows where possible.



[22] This is not to be confused with micro-turbulence in the fluid, which cannot be eliminated at high Reynolds number values. In fact, "fully-developed turbulent flow" is desirable for head-based meter elements such as orifice plates because it means the flow profile will be relatively flat (even velocities across the pipe's diameter) and frictional forces (viscosity) will be negligible. The thing we are trying to avoid is *large-scale* turbulent effects such as eddies, swirl, and asymmetrical flow profiles, which compromise the ability of most flowmeters to accurate measure flow rate.

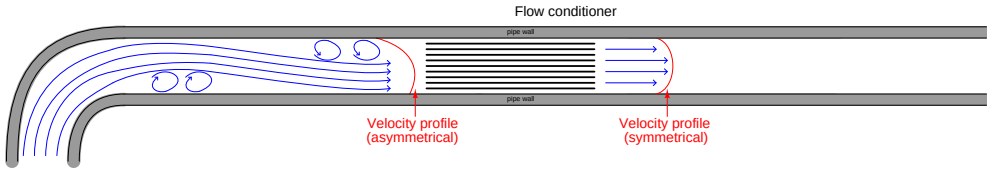
[23] L.K. Spink mentions in his book *Principles and Practice of Flow Meter Engineering* that certain tests have shown flow measurement errors induced from severe disturbances as far as 60 to 100 pipe diameters upstream of the primary flow element. The April 2000 update of API standard 14.3 (for custody-transfer measurement of natural gas using orifice plates) calls for upward of *145 pipe diameters* of straight-length pipe upstream of the orifice plate!

Even disturbances located *downstream* of the flow element may affect measurement accuracy if the disturbances are severe enough and/or close enough to the flow element. Unfortunately, both upstream and downstream flow disturbances are unavoidable on all but the simplest fluid systems. This means we must devise ways to stabilize a flowstream's velocity profile near the flow element in order to achieve accurate measurements of flow rate. A very simple and effective way to stabilize a flow profile is to provide adequate lengths of straight pipe ahead of (and behind) the flow element. Given enough time, even the most chaotic flowstream will "settle down" to a symmetrical profile all on its own. The following illustration shows the effect of a pipe elbow on a flowstream, and how the velocity profile returns to a normal (symmetrical) form after traveling through a sufficient length of straight pipe:



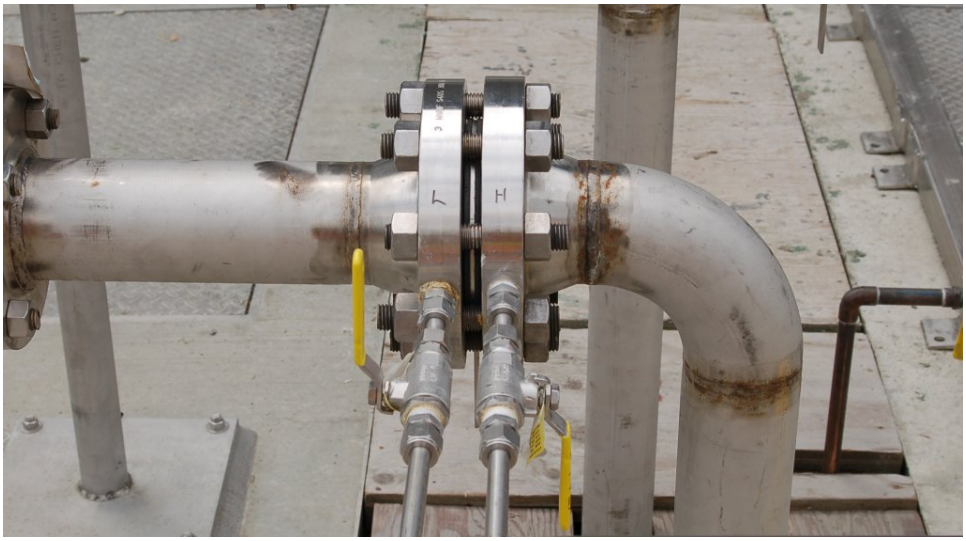
Recommendations for minimum upstream and downstream straight-pipe lengths vary significantly with the nature of the turbulent disturbance, piping geometry, and flow element. As a general rule, elements having a smaller beta ratio (ratio of throat diameter d to pipe diameter D) are more tolerant of disturbances, with profiled flow devices (e.g. venturi tubes, flow tubes, V-cones) having the greatest tolerance[24]. Ultimately, you should consult the flow element manufacturer's documentation for a more detailed recommendation appropriate to any specific application.

In applications where sufficient straight-run pipe lengths are impractical, another option exists for "taming" turbulence generated by piping disturbances. Devices called *flow conditioners* may be installed upstream of the flow element to help the flow profile achieve symmetry in a far shorter distance than simple straight pipe could do alone. Flow conditioners take the form of a series of tubes or vanes installed inside the pipe, parallel to the direction of flow. These tubes or vanes force the fluid molecules to travel in straighter paths, thus stabilizing the flowstream prior to entering a flow element:



[24] Flow elements with low beta ratio values tolerate greater disturbance in the flow pattern because they accelerate the flowstream to a greater degree. This may be best visualized by a thought experiment where we imagine an orifice plate with a very large beta ratio (i.e. one where the bore size is nearly as large as the pipe diameter): such an orifice plate would hardly accelerate the fluid at all, which would mean a mis-shapen flow profile entering the bore would probably remain mis-shapen exiting it. The acceleration imparted to a flowstream by a low-beta element tends to overshadow any asymmetries in the flow profile. However, there are disadvantages to using low-beta elements, one of them being increased permanent pressure loss which may translate to increased operating costs due to energy loss.

This next photograph shows a *very poor* orifice plate installation, where the straight-run pipe requirements were completely ignored:



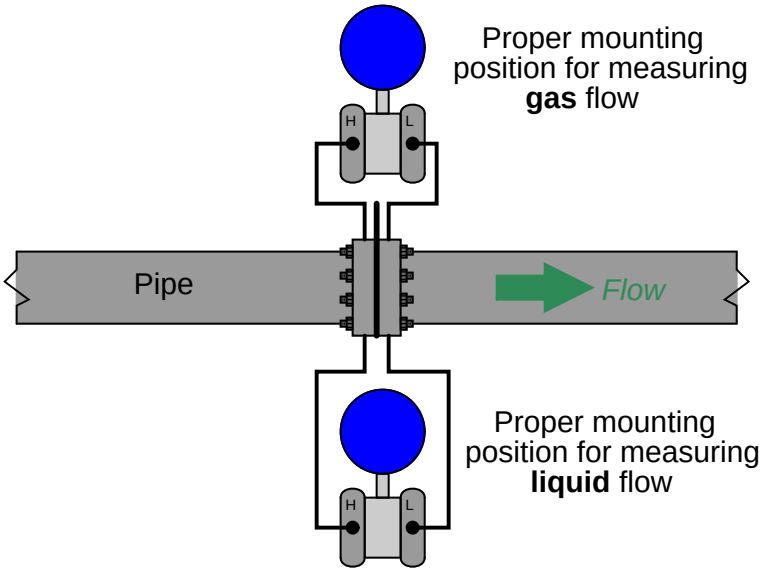
Not only is the orifice plate placed much too close to an elbow, the elbow happens to be on the upstream side of the orifice plate, where disturbances have the greatest effect! The saving grace of this installation is that it is not used for critical monitoring or control: it is simply a manual indication of air flow rate where accuracy is not terribly important. Nevertheless, it is sad to see how an orifice meter installation could have been so easily improved with just a simple re-location of the orifice plate along the piping length.

Poor installations such as this are surprisingly common, owing to the ignorance many piping designers have of flowmeter design and operating principles. Of all the criteria which must be balanced when designing a piping layout, optimum flowmeter location is often placed low in the order of importance (if it appears at all!). In applications where accuracy is important, flowmeter location needs to be a very high priority even if it means a more expensive, cumbersome, and/or unattractive[25] piping design.

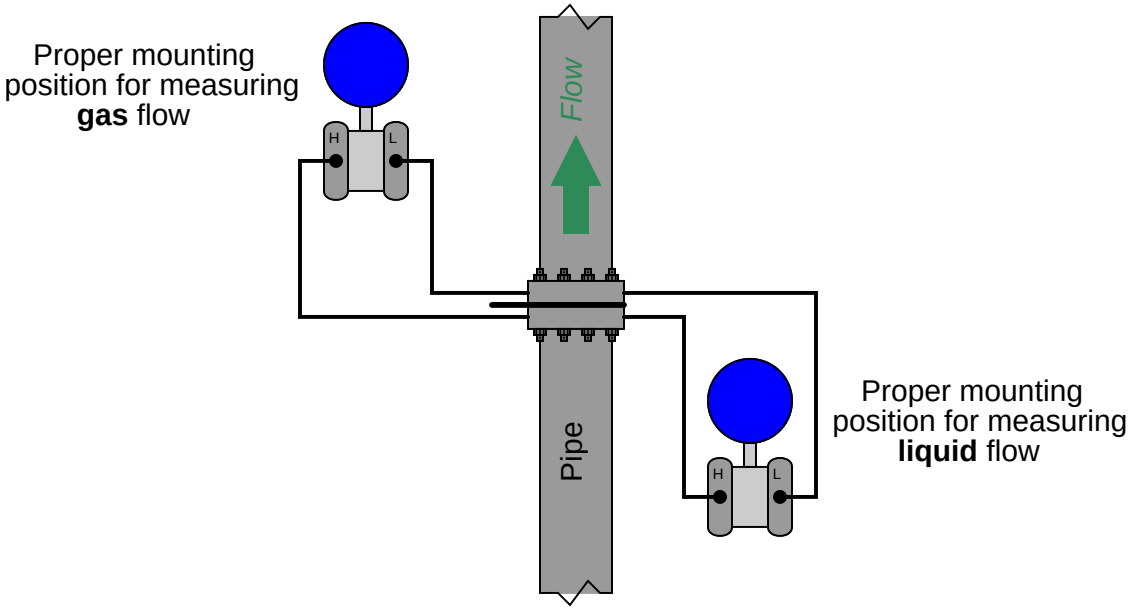
[25] Beauty is truly in the eye of the beholder. While a piping designer might see straight-run lengths of pipe in awkward locations – necessitating more pipe and/or more bends elsewhere in the system to accommodate – as wasteful and ugly, the instrument engineer sees it as a thing of beauty.

Another common source of trouble for pressure-based flowmeters is improper transmitter location. Here, the type of process fluid flow being measured dictates how the pressure-sensing instrument should be located in relation to the pipe. For gas and vapor flows, it is important that no stray liquid droplets collect in the impulse lines leading to the transmitter, lest a vertical liquid column begin to collect in those lines and generate an error-producing pressure. For liquid flows, it is important that no gas bubbles collect in the impulse lines, or else those bubbles may displace liquid from the lines and thereby cause unequal vertical liquid columns, which would (again) generate an error-producing differential pressure.

In order to let gravity do the work of preventing these problems, we must locate the transmitter *above* the pipe for gas flow applications and *below* the pipe for liquid flow applications. This illustration shows both installations for a horizontal pipe:



This next illustration shows both installations on a vertical pipe:



Condensible vapor applications (such as steam flow measurement) have traditionally been treated similarly to liquid measurement applications. Here, condensed liquid will collect in the transmitter's impulse lines so long as the impulse lines are cooler than the vapor flowing through the pipe (which is typically the case). Placing the transmitter below the pipe allows vapors to condense and fill the impulse lines with liquid (condensate), which then acts as a natural seal protecting the transmitter from exposure to hot process vapors.

In such applications it is important for the technician to pre-fill both impulse lines with condensed liquid prior to placing the flowmeter into service. "Tee" fittings with removable plugs or fill valves are provided to do this. Failure to pre-fill the impulse lines will likely result in measurement errors during initial operation, as condensed vapors will inevitably fill the impulse lines at slightly different rates and cause a difference in vertical liquid column heights within those lines.

It should be noted that some steam flow element installations, however, will work well if the impulse lines are above the pipe. If such an installation is possible, the advantage of not having to deal with pre-filling impulse lines (or waiting for steam to condense to equal levels in both lines) are significant. For more information, I recommend consulting the Rosemount whitepaper entitled "Top Mount Installation for DP Flowmeters in Steam Service" (document 00870-0200-4809 first published August 2009).

If tap holes must be drilled into the pipe (or flanges) at the process site, great care must be taken to properly drill and de-burr the holes. A pressure-sensing tap hole should be flush with the inner pipe wall, with no rough edges or burrs to create turbulence. Also, there should be no reliefs or countersinking near the hole on the inside of the pipe. Even small irregularities at the tap holes may generate surprisingly large flow-measurement errors.

22.1.8 High-accuracy flow measurement

When we derived a formula for predicting flow rate from pressure dropped by a venturi tube (or orifice), we had to make many assumptions, chief among them being a total lack of friction (i.e. no energy dissipated due to friction) within the moving fluid and perfect stream-line flow (i.e. complete lack of turbulence). Suffice it to say, the flow formulae you have seen so far in this chapter are only approximations of reality. Orifice plates are some of the worst offenders in this regard, since the fluid encounters such abrupt changes in geometry passing through the orifice. Venturi tubes are nearly ideal, since the machined contours of the tube ensure gradual changes in fluid pressure and minimize turbulence.

However, in the real world we must often do the best we can with imperfect technologies. Orifice plates, despite being less than perfect as flow-sensing elements, are convenient and economical to install in flanged pipes. Orifice plates are also the easiest type of flow element to replace in the event of damage or routine servicing. In applications such as custody transfer (also called "fiscal" measurement), where the flow of fluid represents product being bought and sold, flow measurement accuracy is paramount. It is therefore important to figure out how to coax the most accuracy from the common orifice plate in order that we may measure fluid flows both accurately and economically.

If we compare the true flow rate through a pressure-generating primary sensing element against the theoretical flow rate predicted by an idealized equation, we may notice a substantial discrepancy[26]. Causes of this discrepancy include, but are not limited to:

- Energy losses due to turbulence and viscosity
- Energy losses due to friction against the pipe and element surfaces
- Unstable location of *vena contracta* with changes in flow
- Uneven velocity profiles caused by irregularities in the pipe
- Fluid compressibility
- Thermal expansion (or contraction) of the element and piping
- Non-ideal pressure tap location(s)
- Excessive turbulence caused by rough internal pipe surfaces

The ratio between true flow rate and theoretical flow rate for any measured amount of differential pressure is known as the *discharge coefficient* of the flow-sensing element, symbolized by the variable C. Since a value of 1 represents a theoretical ideal, the actual value of C for any real pressure-generating flow element will be less than 1:

$$C = \frac{\text{True flow}}{\text{Theoretical flow}}$$

[26] Richard W. Miller, in his outstanding book *Flow Measurement Engineering Handbook*, states that venturi tubes may come within 1 to 3 percent of ideal, while a square-edged orifice plate may perform as poorly as only 60 percent of theoretical!

For gas and vapor flows, true flow rate deviates even more from the theoretical (ideal) flow value than liquids do, for reasons that have to do with the compressible nature of gases and vapors. A *gas expansion factor* (Y) may be calculated for any flow element by comparing its discharge coefficient for gases against its discharge coefficient for liquids. As with the discharge coefficient, values of Y for any real pressure-generating element will be less than 1:

$$Y = \frac{C_{gas}}{C_{liquid}}$$

$$Y = \frac{\left(\frac{\text{True gas flow}}{\text{Theoretical gas flow}}\right)}{\left(\frac{\text{True liquid flow}}{\text{Theoretical liquid flow}}\right)}$$

Incorporating these factors into the ideal volumetric flow equation developed on in section 22.1.1, we arrive at the following formulation:

$$Q = \sqrt{2}\frac{CYA_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}}\sqrt{\frac{P_1 - P_2}{\rho}}$$

If we wished, we could even add another factor to account for any necessary unit conversions (N), getting rid of the constant $\sqrt{2}$ in the process:

$$Q = N\frac{CYA_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}}\sqrt{\frac{P_1 - P_2}{\rho}}$$

Sadly, neither the discharge coefficient (C) nor the gas expansion factor (Y) will remain constant across the entire measurement range of any given flow element. These variables are subject to some change with flow rate, which further complicates the task of accurately inferring flow rate from differential pressure measurement. However, if we know the values of C and Y for typical flow conditions, we may achieve good accuracy most of the time.

Likewise, the fact that C and Y change with flow places limits on the accuracy obtainable with the "proportionality constant" formulae seen earlier. Whether we are measuring volumetric or mass flow rate, the k factor calculated at one particular flow condition will not hold constant for *all* flow conditions:

$$Q = k\sqrt{\frac{P_1 - P_2}{\rho}}$$

$$W = k\sqrt{\rho(P_1 - P_2)}$$

This means after we have calculated a value for k based on a particular flow condition, we can only trust the results of the equation for flow conditions not too different from the one we used to calculate k.

As you can see in both flow equations, the density of the fluid (ρ) is an important factor. If fluid density is relatively stable, we may treat ρ as a constant, incorporating its value into the proportionality factor (k) to make the two formulae even simpler:

$$Q = k_Q \sqrt{P_1 - P_2}$$

$$W = k_W \sqrt{P_1 - P_2}$$

However, if fluid density is subject to change over time, we will need some means to continually calculate ρ so our inferred flow measurement will remain accurate. Variable fluid density is a typical state of affairs in gas flow measurement, since all gases are compressible by definition. A simple change in static gas pressure within the pipe is all that is needed to make ρ change, which in turn affects the relationship between flow rate and differential pressure drop.

The American Gas Association (AGA) provides a formula for calculating volumetric flow of any gas using orifice plates in their #3 Report, compensating for changes in gas pressure and temperature. A variation of that formula is shown here (consistent with previous formulae in this section):

$$Q = N \frac{CYA_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \sqrt{\frac{Z_s P_1 (P_1 - P_2)}{G_f Z_{f1} T}}$$

Where,

Q = Volumetric flow rate (SCFM = standard cubic feet per minute)

N = Unit conversion factor

C = Discharge coefficient (accounts for energy losses, Reynolds number corrections, pressure tap locations, etc.)

Y = Gas expansion factor

A_1 = Cross-sectional area of mouth

A_2 = Cross-sectional area of throat

Z_s = Compressibility factor of gas under standard conditions

Z_{f1} = Compressibility factor of gas under flowing conditions, upstream

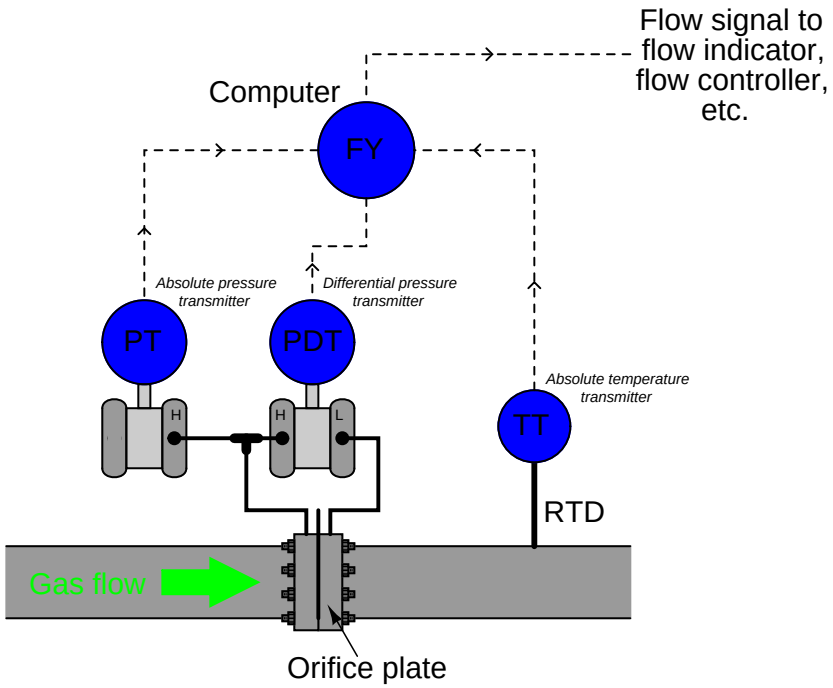
G_f = Specific gravity of gas (density compared to ambient air)

T = Absolute temperature of gas

P_1 = Upstream pressure (absolute)

P_2 = Downstream pressure (absolute)

This equation implies the continuous measurement of absolute gas pressure (P_1) and absolute gas temperature (T) inside the pipe, in addition to the differential pressure produced by the orifice plate ($P_1 - P_2$). These measurements may be taken by three separate devices, their signals routed to a gas flow computer:



Note the location of the RTD (thermowell), positioned downstream of the orifice plate so the turbulence it generates will not create additional turbulence at the orifice plate. The American Gas Association (AGA) allows for upstream placement of the thermowell, but only if located at least three feet upstream of a flow conditioner[27].

In order to best control all the physical parameters necessary for good orifice metering accuracy, it is standard practice for custody transfer flowmeter installations to use *honed meter runs* rather than standard pipe and pipe fittings. A "honed run" is a complete piping assembly consisting of a manufactured fitting to hold the orifice plate and sufficient straight lengths of pipe upstream and downstream, the interior surfaces of that pipe machined ("honed") to have a glass-smooth surface with precise and symmetrical dimensions. Honed runs ensure minimum disruption to the flowing gas or liquid, thus improving measurement accuracy by avoiding unnecessary turbulence and/or distorted flow profiles. Such piping "runs" are quite expensive, but necessary to achieve flow measurement accuracy worthy of custody transfer.

[27]Specified in Part 2 of the AGA Report #3, section 2.6.5, page 22. A major reason for this is von Kármán vortex shedding caused by the gas having to flow around the width of the thermowell. The "street" of vortices shed by the thermowell will cause serious pressure fluctuations at the orifice plate unless mitigated by a flow conditioner, or by locating the thermowell downstream so that the vortices do not reach the orifice.

This photograph shows a set of AGA3-compliant orifice meter runs measuring the flow of natural gas:

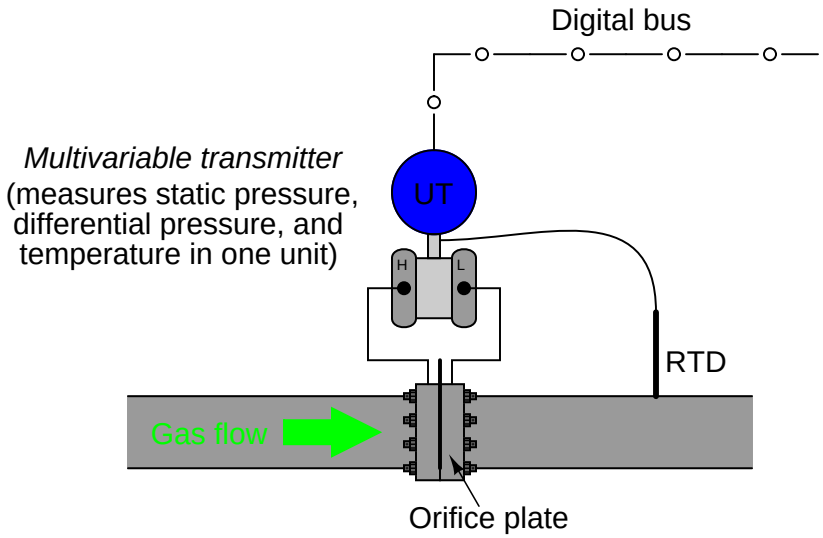


Note the special transmitter manifolds, built to accept both the differential pressure and absolute pressure (Rosemount model 3051) transmitters. Also note the quick-change fittings (the ribbed metal housings) holding the orifice plates, to facilitate convenient change-out of the orifice plates which is periodically necessary due to wear. It is not unheard of to replace orifice plates on a daily basis in some industries to ensure the sharp orifice edges necessary for accurate measurement[28].

Although not visible in this photograph, these meter runs are connected together by a network of shut-off valves directing the flow of natural gas through as few meter runs as desired. When the total gas flow rate is great, all meter runs are placed into service and their respective flow rates summed to yield a total flow measurement. When the total flow rate decreases, individual meter runs are shut off, resulting in increased flow rates through the remaining meter runs. This "staging" of meter runs expands the effective *turndown* or *rangeability* of the orifice plate as a flow-sensing element, resulting in much more accurate flow measurement over a wide range of flow rates than if a single (large) orifice meter run were used.

[28] This is especially true in the gas exploration industry, where natural gas coming out of the well is laden with mineral debris.

An alternative to multiple instruments (differential pressure, absolute pressure, and temperature) installed on each meter run is to use a single *multi-variable* transmitter capable of measuring gas temperature as well as both static and differential pressures. This approach enjoys the advantage of simpler installation over the multi-instrument approach:



The Rosemount model 3095MV and Yokogawa model EJX910 are examples of multi-variable transmitters designed to perform compensated gas flow measurement, equipped with multiple pressure sensors, a connection port for an RTD temperature sensor, and sufficient digital computing power to continuously calculate flow rate based on the AGA equation. Such multi-variable transmitters may provide an analog output for computed flow rate, or a digital output where all three primary variables *and* the computed flow rate may be transmitted to a host system (as shown in the previous illustration). The Yokogawa EJX910A provides an interesting signal output option: a digital *pulse* signal, where each pulse represents a specific quantity (either volume or mass) of fluid. The frequency of this pulse train represents flow rate, while the total number of pulses counted over a period of time represents the total amount of fluid that has passed through the orifice plate over that amount of time.

This photograph shows a Rosemount 3095MV transmitter used to measure mass flow on a pure oxygen (gas) line. The orifice plate is an "integral" unit immediately below the transmitter body, sandwiched between two flange plates on the copper line. A three-valve manifold interfaces the model 3095MV transmitter to the integral orifice plate structure:



The temperature-compensation RTD may be clearly seen on the left-hand side of the photograph, installed at the elbow fitting in the copper pipe.

Liquid flow measurement applications may also benefit from compensation, because liquid density changes with temperature. Static pressure is not a concern here, because liquids are considered incompressible for all practical purposes[29]. Thus, the formula for compensated liquid flow measurement does not include any terms for static pressure, just differential pressure and temperature:

$$Q = N \frac{CYA_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \sqrt{(P_1 - P_2)[1 + k_T(T - T_{ref})]}$$

The constant k_T shown in the above equation is the proportionality factor for liquid expansion with increasing temperature. The difference in temperature between the measured condition (T) and the reference condition (T_{ref}) multiplied by this factor determines how much less dense the liquid is compared to its density at the reference temperature. It should be noted that some liquids – notably hydrocarbons – have thermal expansion factors significantly greater than water. This makes temperature compensation for hydrocarbon liquid flow measurement very important if the measurement principle is volumetric rather than mass-based.

[29] Liquids can and do compress, the measurement of their "compressibility" being what is called the *bulk modulus*. However, this compressibility is too slight to be of any consequence in most flow measurement applications. A notable exception is the metering of diesel fuel through a high-pressure injection pump, where liquid pressures range in the *tens of thousands* of PSI, and the compressibility of the liquid diesel fuel may affect the precise timing of individual injections into the engine cylinders.

22.1.9 Equation summary

Volumetric flow rate (Q) full equation:

$$Q = N \frac{CYA_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \sqrt{\frac{P_1 - P_2}{\rho_f}}$$

Volumetric flow rate (Q) simplified equation:

$$Q = k\sqrt{\frac{P_1 - P_2}{\rho_f}}$$

Mass flow rate (W):

$$W = N \frac{CYA_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \sqrt{\rho_f (P_1 - P_2)}$$

Mass flow rate (W) simplified equation:

$$W = k\sqrt{\rho_f (P_1 - P_2)}$$

Where,

Q = Volumetric flow rate (e.g. gallons per minute, flowing cubic feet per second)

W = Mass flow rate (e.g. kilograms per second, slugs per minute)

N = Unit conversion factor

C = Discharge coefficient (accounts for energy losses, Reynolds number corrections, pressure tap locations, etc.)

Y = Gas expansion factor ($Y = 1$ for liquids)

A_1 = Cross-sectional area of mouth

A_2 = Cross-sectional area of throat

ρ_f = Fluid density at flowing conditions (actual temperature and pressure at the element)

k = Constant of proportionality (determined by experimental measurements of flow rate, pressure, and density)

The beta ratio (β) of a differential-producing element is the ratio of throat diameter to mouth diameter ($\beta = \frac{d}{D}$). This is the primary factor determining acceleration as the fluid increases velocity entering the constricted throat of a flow element (venturi tube, orifice plate, wedge, etc.). The following expression is often called the *velocity of approach factor* (commonly symbolized as E_v), because it relates the velocity of the fluid through the constriction to the velocity of the fluid as it approaches the flow element:

$$E_v = \frac{1}{\sqrt{1 - \beta^4}} = \text{ Velocity of approach factor}$$

This same velocity approach factor may be expressed in terms of mouth and throat areas (A_1 and A_2, respectively):

$$E_v = \frac{1}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} = \text{ Velocity of approach factor}$$

Beta ratio has a significant impact on the number of straight-run pipe lengths needed to condition the flow profile upstream and downstream of the flow element. Large beta ratios (where the bore diameter approaches the flowtube's inside diameter) are more sensitive to piping disturbances, since there is less acceleration of the flowstream through the element, and therefore flow profile asymmetries caused by piping disturbances are significant in comparison to the fluid's through-bore velocity. Small beta ratio values correspond to larger acceleration factors, where disturbances in the flow profile become "swamped[30]" by the high throat velocities created by the element's constriction. A disadvantage of small beta ratio values is that the flow element exhibits a greater permanent pressure loss, which is an operational cost if the flow is provided by a machine such as an engine- or motor-driven pump (more energy required to turn the pump, equating to a greater operating cost to run the process).

[30] "Swamping" is a term commonly used in electrical engineering, where a bad effect is overshadowed by some other effect much larger in magnitude, to the point where the undesirable effect is negligible in comparison.

When computing the volumetric flow of a gas in *standard* volume units (e.g. SCFM), the equation becomes much more complex than the simple (flowing) volumetric rate equation. Any equation computing flow in standard units must predict the effective expansion of the gas if it were to transition from flowing conditions (the actual pressure and temperature it experiences flowing through the pipe) to standard conditions (one atmosphere pressure at 60 degrees Fahrenheit). The compensated gas flow measurement equation published by the American Gas Association (AGA Report #3) in 1992 for orifice plates with flange taps calculates this expansion to standard conditions with a series of factors accounting for flowing and standard ("base") conditions, in addition to the more common factors such as velocity of approach and gas expansion. Most of these factors are represented in the AGA3 equation by different variables beginning with the letter F:

$$Q = F_n(F_c + F_{sl})Y F_{pb} F_{tb} F_{tf} F_{gr} F_{pv} \sqrt{h_W P_{f1}}$$

Where,

Q = Volumetric flow rate (standard cubic feet per hour – SCFH)

F_n = Numeric conversion factor (accounts for certain numeric constants, unit-conversion coefficients, and the velocity of approach factor E_v)

F_c = Orifice calculation factor (a polynomial function of the orifice plate's β ratio and Reynolds number), appropriate for flange taps

F_{sl} = Slope factor (another polynomial function of the orifice plate's β ratio and Reynolds number), appropriate for flange taps

$F_c + F_{sl} = C_d$ = Discharge coefficient, appropriate for flange taps

Y = Gas expansion factor (a function of β, differential pressure, static pressure, and specific heats)

F_{pb} = Base pressure factor = $\frac{14.73 \text{ PSI}}{P_b}$, with pressure in PSIA (absolute)

F_{tb} = Base temperature factor = $\frac{T_b}{519.67}$, with temperature in degrees Rankine

F_{tf} = Flowing temperature factor = $\sqrt{\frac{519.67}{T_f}}$, with temperature in degrees Rankine

F_{gr} = Real gas relative density factor = $\sqrt{\frac{1}{G_r}}$

F_{pv} = Supercompressibility factor = $\sqrt{\frac{Z_b}{Z_{f1}}}$

h_W = Differential pressure produced by orifice plate (inches water column)

P_{f1} = Flowing pressure of gas at the upstream tap (PSI absolute)

22.2 Laminar flowmeters

A unique form of differential pressure-based flow measurement deserves its own section in this flow measurement chapter, and that is the *laminar* flowmeter.

Laminar flow is a condition of fluid motion where viscous (internal fluid friction) forces greatly overshadow inertial (kinetic) forces. A flowstream in a state of laminar flow exhibits no turbulence, with each fluid molecule traveling in its own path, with limited mixing and collisions with adjacent molecules. The dominant mechanism for resistance to fluid motion in a laminar flow regime is friction with the pipe or tube walls. Laminar flow is qualitatively predicted by low values of Reynolds number.

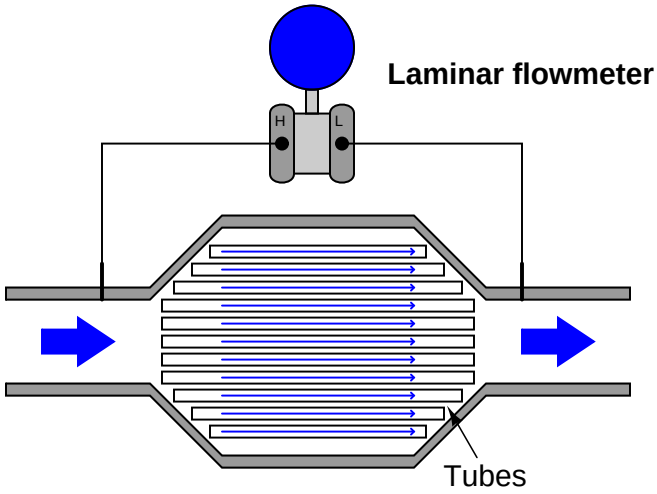
This pressure drop created by fluid friction in a laminar flowstream is quantifiable, and is expressed in the Hagen-Poiseuille equation:

$$Q = k\left(\frac{\Delta P D^4}{\mu L}\right)$$

Where,

Q = Flow rate
ΔP = Pressure dropped across a length of pipe
D = Pipe diameter
μ = Fluid viscosity
L = Pipe length
k = Coefficient accounting for units of measurement

Laminar flowmeter elements generally consist of one or more tubes whose length greatly exceeds the inside diameter, arranged in such a way as to produce a slow-moving flow velocity. An example is shown here:



The expanded diameter of the flow element ensures a lower fluid velocity than in the pipes entering and exiting the element. This decreases the Reynolds number to the point where the flow

regime exhibits laminar behavior. The large number of small-diameter tubes packed in the wide area of the element provide adequate wall surface area for the fluid's viscosity to act upon, creating an overall pressure drop from inlet to outlet which is measured by the differential pressure transmitter. This pressure drop is permanent (no recovery of pressure downstream) because the mechanism of pressure drop is friction: total dissipation (loss) of energy in the form of heat.

Another common form of laminar flow element is simply a coiled *capillary tube*: a long tube with a very small inside diameter. The small inside diameter of such a tube makes wall-boundary effects dominant, such that the flow regime will remain laminar over a wide range of flow rates. The extremely restrictive nature of a capillary tube, of course, limits the use of such flow elements to very low flow rates such as those encountered in the sampling networks of certain analytical instruments.

A unique advantage of the laminar flowmeter is its linear relationship between flow rate and developed pressure drop. It is the only pressure-based flow measurement device for filled pipes that exhibits a linear pressure/flow relationship. This means no "square-root" characterization is necessary to obtain linear flow measurements with a laminar flowmeter. The big disadvantage of this meter type is its dependence on fluid viscosity, which in turn is strongly influenced by fluid temperature. Thus, all laminar flowmeters require temperature compensation in order to derive accurate measurements, and some even use temperature *control* systems to force the fluid's temperature to be constant as it moves through the element[31].

Laminar flow elements find their widest application inside pneumatic instruments, where a linear pressure/flow relationship is highly advantageous (behaving like a "resistor" for instrument air flow) and the viscosity of the fluid (instrument air) is relatively constant. Pneumatic controllers, for instance, use laminar restrictors as part of the derivative and integral calculation modules, the combination of "resistance" from the restrictor and "capacitance" from volume chambers forming a sort of pneumatic time-constant (τ) network.

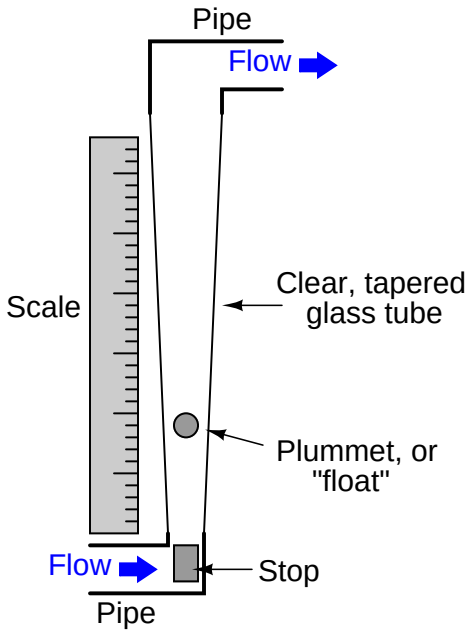
22.3 Variable-area flowmeters

An *variable-area* flowmeter is one where the fluid must pass through a restriction whose area increases with flow rate. This stands in contrast to flowmeters such as orifice plates and venturi tubes where the cross-sectional area of the flow element remains fixed.

[31] This includes elaborate oil-bath systems where the laminar flow element is submerged in a temperature-controlled oil bath, the purpose of which is to hold temperature inside the laminar element constant despite sudden changes in the measured fluid's temperature.

22.3.1 Rotameters

The simplest example of a variable-area flowmeter is the *rotameter*, which uses a solid object (called a *plummet* or *float*) as a flow indicator, suspended in the midst of a tapered tube:



As fluid flows upward through the tube, a pressure differential develops across the plummet. This pressure differential, acting on the effective area of the plummet body, develops an upward force ($F = PA$). If this force exceeds the weight of the plummet, the plummet moves up. As the plummet moves farther up in the tapered tube, the area between the plummet and the tube walls (through which the fluid must travel) grows larger. This increased flowing area allows the fluid to make it past the plummet without having to accelerate as much, thereby developing less pressure drop across the plummet's body. At some point, the flowing area reaches a point where the pressure-induced force on the plummet body exactly matches the weight of the plummet. This is the point in the tube where the plummet stops moving, indicating flow rate by it position relative to a scale mounted (or etched) on the outside of the tube.

The following rotameter uses a spherical plummet, suspended in a flow tube machined from a solid block of clear plastic. An adjustable valve at the bottom of the rotameter provides a means for adjusting gas flow:



The same basic flow equation used for pressure-based flow elements holds true for rotameters as well:

$$Q = k\sqrt{\frac{P_1 - P_2}{\rho}}$$

However, the difference in this application is that the value inside the radicand is constant, since the pressure difference will remain constant[32] and the fluid density will likely remain constant as well. Thus, k will change in proportion to Q. The only variable within k relevant to plummet position is the flowing area between the plummet and the tube walls.

[32] If we know that the plummet's weight will remain constant, its drag area will remain constant, and that the force generated by the pressure drop will always be in equilibrium with the plummet's weight for any steady flow rate, then the relationship $F = PA$ dictates a constant pressure. Thus, we may classify the rotameter as a *constant-pressure*, *variable-area* flowmeter. This stands in contrast to devices such as orifice plates, which are *variable-pressure*, *constant-area*.

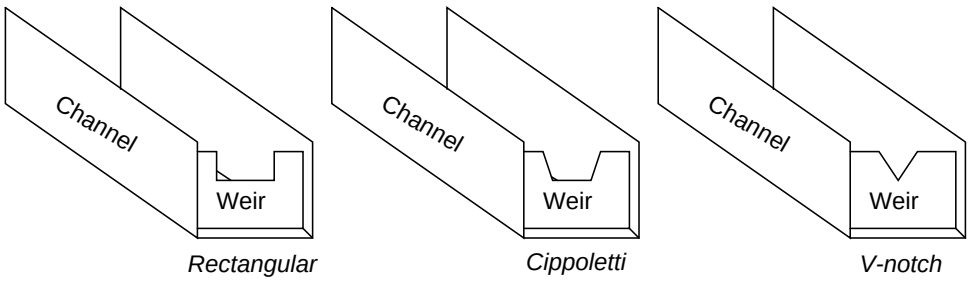
Most rotameters are indicating devices only. They may be equipped to transmit flow information electronically by adding sensors to detect the plummet's position in the tube, but this is not common practice.

Rotameters are very commonly used as purge flow indicators for pressure and level measurement systems requiring a constant flow of purge fluid (see sections 19.6.6 and 20.3.2 for practical examples). Such rotameters are usually equipped with hand-adjustable needle valves for manual regulation of purge fluid flow rate.

22.3.2 Weirs and flumes

A very different style of variable-area flowmeter is used extensively to measure flow rate through open channels, such as irrigation ditches. If an obstruction is placed within a channel, any liquid flowing through the channel must rise on the upstream side of the obstruction. By measuring this liquid level rise, it is possible to infer the rate of liquid flow past the obstruction.

The first form of open-channel flowmeter is the *weir*, which is nothing more than a dam obstructing passage of liquid through the channel. Three styles of weir are shown in the following illustration; the *rectangular*, *Cippoletti*, and *V-notch*:



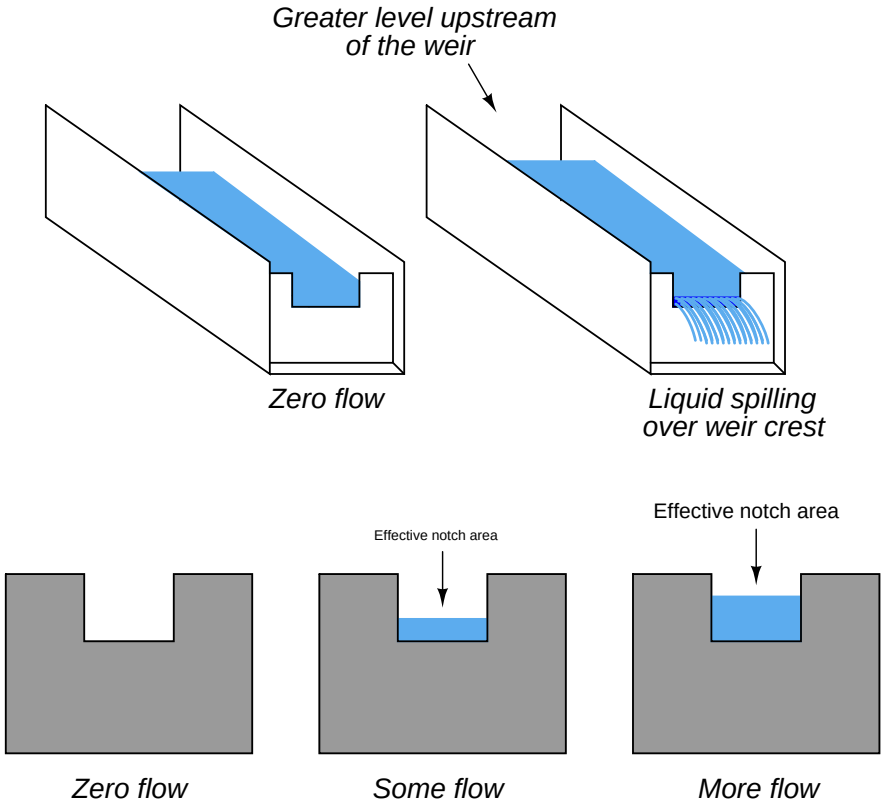
A rectangular weir has a notch of simple rectangular shape, as the name implies. A Cippoletti weir is much like a rectangular weir, except that the vertical sides of the notch have a 4:1 slope (rise of 4, run of 1; approximately a 14 degree angle from vertical). A V-notch weir has a triangular notch, customarily measuring either 60 or 90 degrees.

The following photograph shows water flowing through a Cippoletti weir made of 1/4 inch steel plate:



At a condition of zero flow through the channel, the liquid level will be at or below the crest (lowest point on the opening) of the weir. As liquid begins to flow through the channel, it must spill over the crest of the weir in order to get past the weir and continue downstream in the channel. In order for this to happen, the level of the liquid upstream of the weir must rise above the weir's crest height. This height of liquid upstream of the weir represents a hydrostatic pressure, much the same as liquid heights in piezometer tubes represent pressures in a liquid flowstream through an enclosed pipe (see section 2.11.15 for examples of this). The height of liquid above the crest of a weir is

analogous to the pressure differential generated by an orifice plate. As liquid flow is increased even more, a greater pressure (head) will be generated upstream of the weir, forcing the liquid level to rise. This effectively increases the cross-sectional area of the weir's "throat" as a taller stream of liquid exits the notch of the weir[33].



[33]Orifice plates are *variable-pressure*, *constant-area* flowmeters. Rotameters are *constant-pressure*, *variable-area* flowmeters. Weirs are *variable-pressure*, *variable-area* flowmeters. As one might expect, the mathematical functions describing each of these flowmeter types is unique!

This dependence of notch area on flow rate creates a very different relationship between flow rate and liquid height (measured above the crest) than the relationship between flow rate and differential pressure in an orifice plate:

$$Q = 3.33(L - 0.2H)H^{1.5} \qquad \text{Rectangular weir}$$

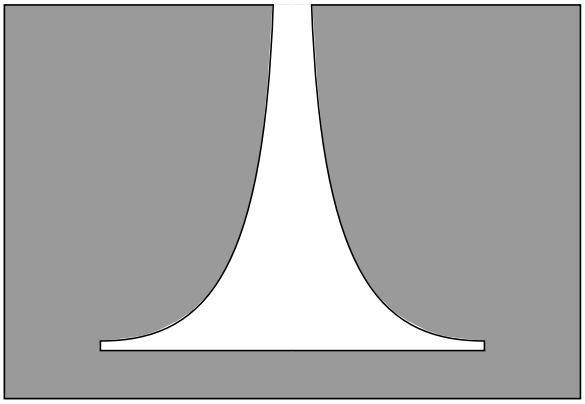
$$Q = 3.367LH^{1.5} \qquad \text{Cippoletti weir}$$

$$Q = 2.48\left(\tan\frac{\theta}{2}\right)H^{2.5} \qquad \text{V-notch weir}$$

Where,

Q = Volumetric flow rate (cubic feet per second – CFS)
L = Width of crest (feet)
θ = V-notch angle (degrees)
H = Head (feet)

As you can see from a comparison of characteristic flow equations between these three types of weirs, the shape of the weir's notch has a dramatic effect on the mathematical relationship between flow rate and head (liquid level upstream of the weir, measured above the crest height). This implies that it is possible to create almost any characteristic equation we might like just by carefully shaping the weir's notch in some custom form. A good example of this is the so-called *proportional* or *Sutro* weir, which is designed to have a linear relationship between head and flow rate:



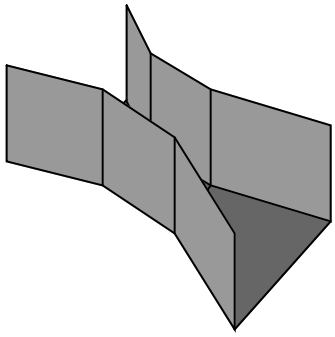
Sutro weirs are not used very often, due to their inherently weak structure and tendency to clog with debris.

A rare example of a Sutro weir appears in the following photograph, discharging flow from a lake into a stream:



The metal plates forming the weir's shape are quite thick (about 1/2 inch) to give the weir sufficient strength. A good construction practice seen on this Sutro weir, but recommended on *all* weir designs, is to bevel the downstream edge of the weir plate much like a standard orifice plate profile. The beveled edge provides a minimum-friction passageway for the liquid as it spills through the weir's opening.

A variation on the theme of a weir is another open-channel device called a *flume*. If weirs may be thought of as open-channel orifice plates, then flumes may be thought of as open-channel venturi tubes:



Like weirs, flumes generate upstream liquid level height changes indicative of flow rate. One of the most common flume design is the *Parshall flume*, named after its inventor R.L. Parshall when it was developed in the year 1920.

The following formulae relate head (upstream liquid height) to flow rate for free-flowing Parshall flumes[34]:

$$Q = 0.992H^{1.547} \qquad \text{3-inch wide throat Parshall flume}$$

$$Q = 2.06H^{1.58} \qquad \text{6-inch wide throat Parshall flume}$$

$$Q = 3.07H^{1.53} \qquad \text{9-inch wide throat Parshall flume}$$

$$Q = 4LH^{1.53} \qquad \text{1-foot to 8-foot wide throat Parshall flume}$$

$$Q = (3.6875L + 2.5)H^{1.53} \qquad \text{10-foot to 50-foot wide throat Parshall flume}$$

Where,

Q = Volumetric flow rate (cubic feet per second – CFS)

L = Width of flume throat (feet)

H = Head (feet)

Flumes are generally less accurate than weirs, but they do enjoy the advantage of being inherently self-cleaning. If the liquid stream being measured is drainage- or waste-water, a substantial amount of solid debris may be present in the flow that could cause repeated clogging problems for weirs. In such applications, flumes are often the more practical flow element for the task (and more accurate

[34] It is also possible to operate a Parshall flume in fully *submerged* mode, where liquid level must be measured at both the upstream and throat sections of the flume. Correction factors must be applied to these equations if the flume is submerged.

over the long term as well, since even the finest weir will not register accurately once fouled by debris).

Once a weir or flume has been installed in an open channel to measure the flow of liquid, some method must be employed to sense upstream liquid level and translate this level measurement into a flow measurement. Perhaps the most common technology for weir/flume level sensing is *ultrasonic* (see section 20.5.1 beginning on page 1472 for more information on how this technology works). Ultrasonic level sensors are completely non-contact, which means they cannot become fouled by the process liquid (or debris in the process liquid). However, they may be "fooled" by foam or debris floating on top of the liquid, as well as waves on the liquid surface.

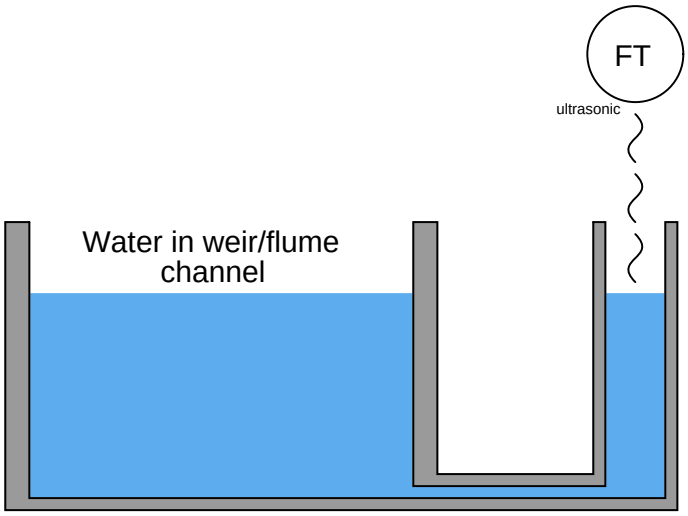
The following photograph shows a Parshall flume measuring effluent flow from a municipal sewage treatment plant, with an ultrasonic transducer mounted above the middle of the flume to detect water level flowing through:



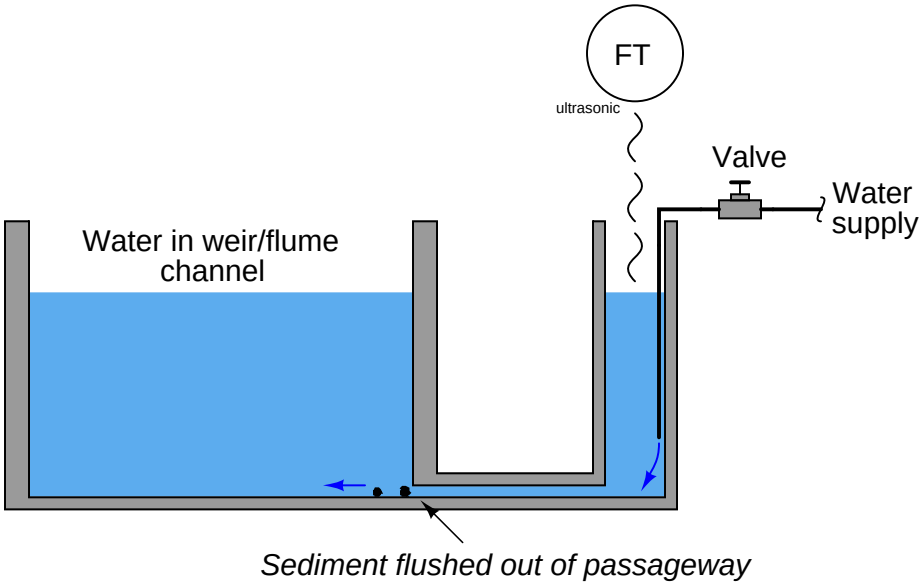
Once the liquid level is successfully measured, a computing device is used to translate that level measurement into a suitable flow measurement (and in some cases even integrate that flow measurement with respect to time to arrive at a value for total liquid volume passed through the element, in accordance with the calculus relationship $V = \int Q\,dt + C$).

A technique for providing a clean and "quiet" (still) liquid surface to measure the level of is called a *stilling well*. This is an open-top chamber connected to the weir/flume channel by a pipe, so the liquid level in the stilling well matches the liquid level in the channel. The following illustration shows a stilling well connected to a weir/flume channel, with the direction of liquid flow in the

channel being perpendicular to the page (i.e. either coming toward your eyes or going away from your eyes):

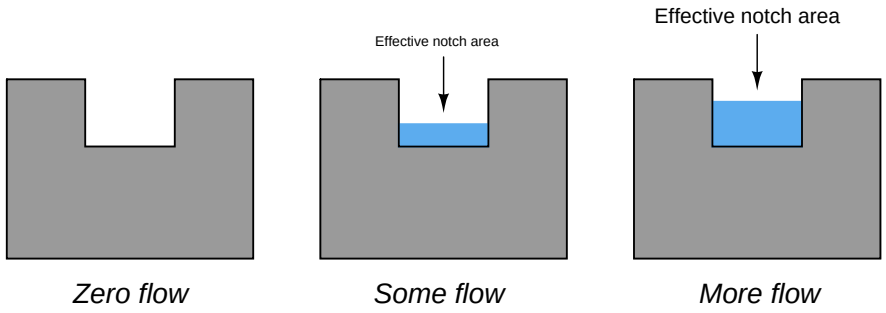


To discourage plugging of the passageway connecting the stilling well to the channel, a small flow rate of clean water may be introduced into the well. This forms a constant *purge flow* into the channel, flushing out debris that might otherwise find its way into the connecting passageway to plug it up. Note how the purge water enters the stilling well through a submerged tube, so it does not cause splashing on the water's surface inside the well which could cause measurement problems for the ultrasonic sensor:



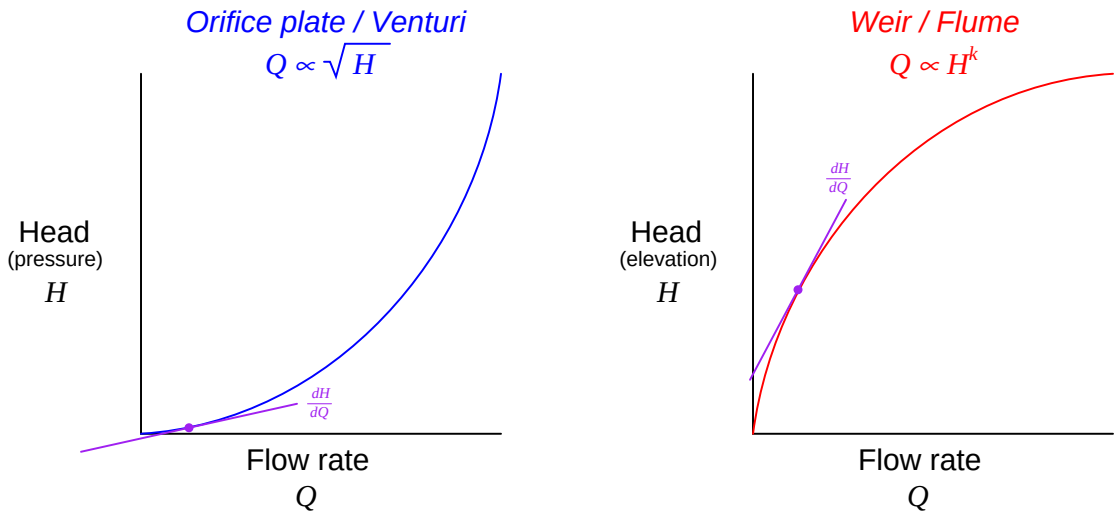
A significant advantage that weirs and flumes have over other forms of flow measurement is exceptionally high *rangeability*: the ability to measure very wide ranges of flow with a modest pressure (height) span. Another way to state this is to say that the accuracy of a weir or flume is quite high even at low flow rates.

Earlier in this section you saw a three-image representation of liquid flow through a rectangular weir. As fluid flow rate increased, so did the height (head) of the liquid upstream of the weir:



The height of liquid upstream of the weir depends on the flow rate (volumetric Q or mass W) as well as the effective area of the notch through which the fluid must pass. Unlike an orifice plate, this area changes with flow rate in both weirs and flumes. One way to envision this by comparison is to imagine a weir as acting like an elastic orifice plate, whose bore area increases with flow rate. This flow-dependent notch area exhibited by both weirs and flumes means that these devices become *more sensitive* to changes in flow as the flow rate becomes smaller.

A comparison of transfer function graphs for closed-pipe head elements such as orifice plates and venturi tubes versus weirs and flumes shows this striking difference in characteristics:



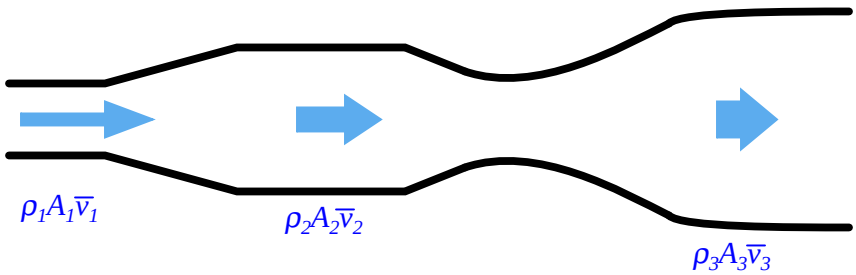
Looking at the orifice plate / venturi tube graph near the lower-left corner, you can see how small changes in flow result in extremely small changes in head (differential pressure), because the

function has a very low slope (small $\frac{dH}{dQ}$) at that end. By comparison, a weir or flume produces relatively large changes in head (liquid elevation) for small changes in flow near the bottom end of the range, because the function has a very steep slope (large $\frac{dH}{dQ}$) at that end.

The practical advantage this gives weirs and flumes is the ability to maintain high accuracy of flow measurement at very low flow rates – something a fixed-orifice element simply cannot do. It is commonly understood in industry that traditional orifice plate flowmeters do not maintain good measurement accuracy much below a third of their full-range flow (a rangeability or *turndown* of 3:1), whereas weirs (especially the V-notch design) can achieve far greater turndown (up to 500:1 according to some sources[35]).

22.4 Velocity-based flowmeters

The Law of Continuity for fluids states that the product of mass density (ρ), cross-sectional pipe area (A) and average velocity ($\overline{v}$) must remain constant through any continuous length of pipe:



If the density of the fluid is not subject to change as it travels through the pipe (a very good assumption for liquids), we may simplify the Law of Continuity by eliminating the density terms from the equation:

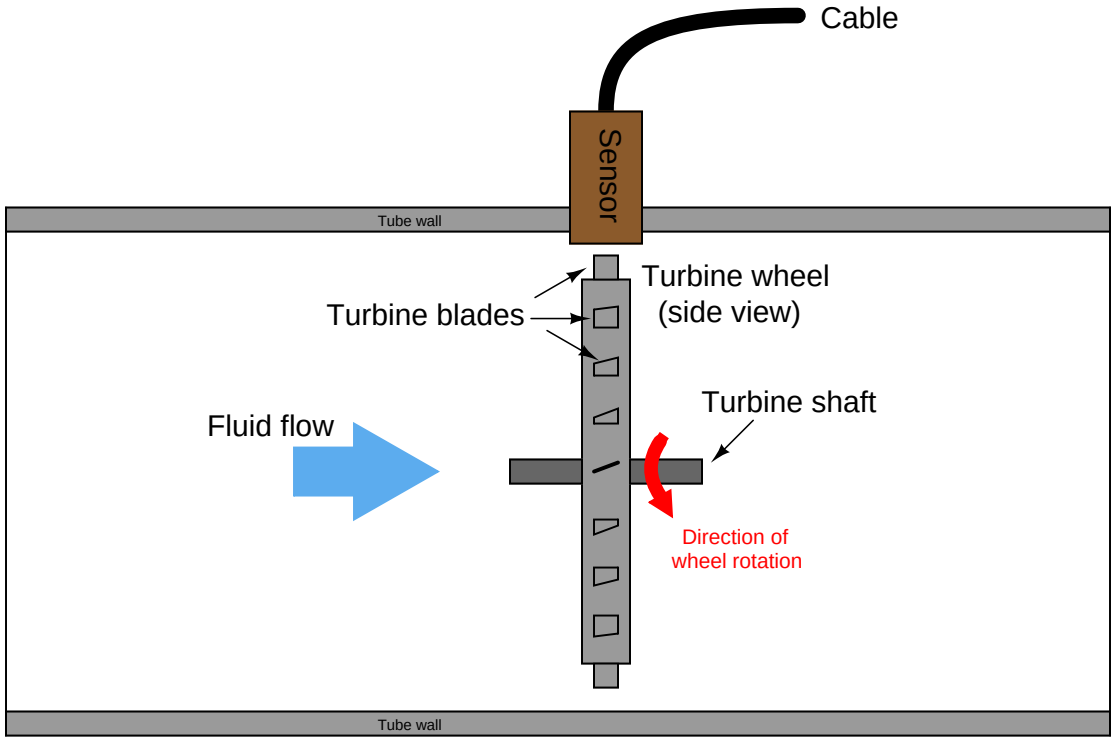
$$A_1 \overline{v_1} = A_2 \overline{v_2}$$

The product of cross-sectional pipe area and average fluid velocity is the volumetric flow rate of the fluid through the pipe ($Q = A\overline{v}$). This tells us that fluid velocity will be directly proportional to volumetric flow rate given a known cross-sectional area and a constant density for the fluid flowstream. Any device able to directly measure fluid velocity is therefore capable of inferring volumetric flow rate of fluid in a pipe. This is the basis for *velocity-based* flowmeter designs.

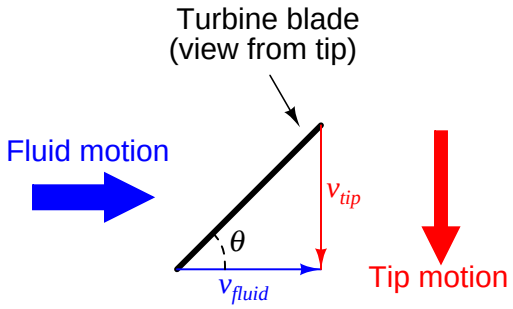
[35] These figures are reported in Béla Lipták's excellent reference book *Instrument Engineers' Handbook – Process Measurement and Analysis Volume I* (Fourth Edition). To be fair to closed-pipe elements such as orifice plates and venturi tubes, much improvement in the classic 3:1 rangeability limitation has been achieved through the use of microprocessor-based differential pressure sensors. Lipták reports rangeabilities for orifice plates as great as 10:1 through the use of such modern differential pressure instruments. However, even this pales in comparison to the rangeability of a typical weir or flume, which Lipták reports to be 75:1 for "most devices" in this category.

22.4.1 Turbine flowmeters

Turbine flowmeters use a free-spinning turbine wheel to measure fluid velocity, much like a miniature windmill installed in the flow stream. The fundamental design goal of a turbine flowmeter is to make the turbine element as free-spinning as possible, so no torque will be required to sustain the turbine's rotation. If this goal is achieved, the turbine blades will achieve a rotating (tip) velocity directly proportional to the linear velocity of the fluid, whether that fluid is a gas or a liquid:



The mathematical relationship between fluid velocity and turbine tip velocity – assuming frictionless conditions – is a ratio defined by the *tangent* of the turbine blade angle:

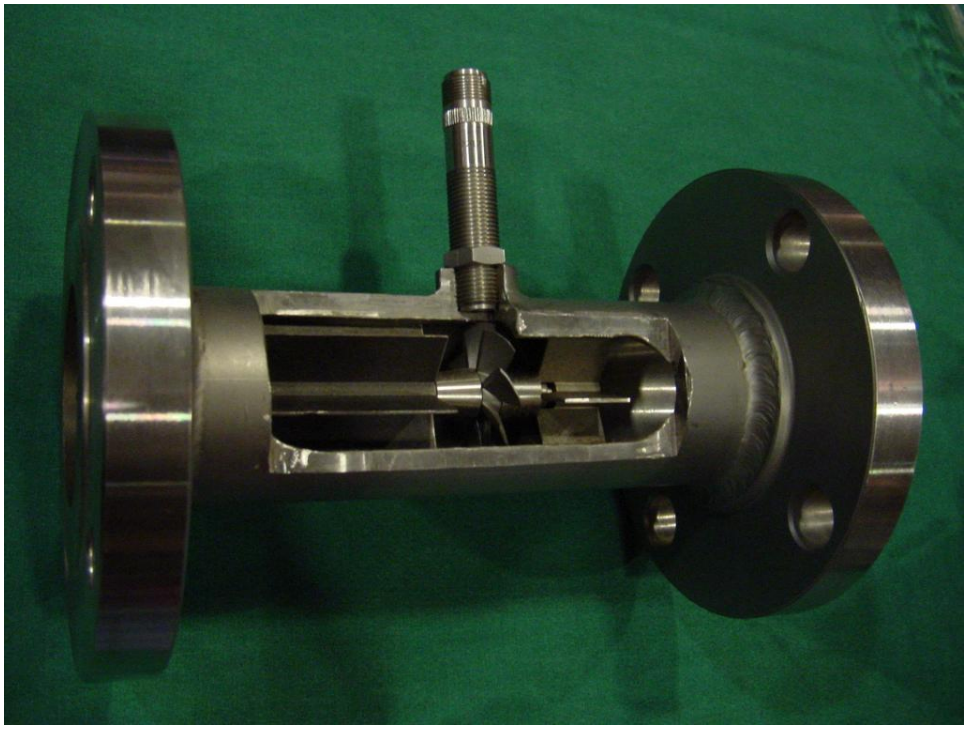


$$\tan \theta = \frac{v_{tip}}{v_{fluid}}$$

For a 45° blade angle, the relationship is 1:1, with tip velocity equaling fluid velocity. Smaller blade angles (each blade closer to parallel with the fluid velocity vector) result in the tip velocity being a fractional proportion of fluid velocity.

Turbine tip velocity is quite easy to sense using a magnetic sensor, generating a voltage pulse each time one of the ferromagnetic turbine blades passes by. Traditionally, this sensor is nothing more than a coil of wire in proximity to a stationary magnet, called a *pickup coil* or *pickoff coil* because it "picks" (senses) the passing of the turbine blades. Magnetic flux through the coil's center increases and decreases as the passing of the steel turbine blades presents a varying reluctance ("resistance" to magnetic flux), causing voltage pulses equal in frequency to the number of blades passing by each second. It is the *frequency* of this signal that represents fluid velocity, and therefore volumetric flow rate.

A cut-away demonstration model of a turbine flowmeter is shown in the following photograph. The blade sensor may be seen protruding from the top of the flowtube, just above the turbine wheel:



Note the sets of "flow conditioner" vanes immediately before and after the turbine wheel in the photograph. As one might expect, turbine flowmeters are very sensitive to *swirl* in the process fluid flowstream. In order to achieve high accuracy, the flow profile must not be swirling in the vicinity of the turbine, lest the turbine wheel spin faster or slower than it should to represent the velocity of a straight-flowing fluid. A minimum straight-pipe length of 20 pipe diameters upstream and 5 pipe diameters downstream is typical for turbine flowmeters in order to dissipate swirl from piping disturbances.

Mechanical gears and rotating cables have also been historically used to link a turbine flowmeter's turbine wheel to indicators. These designs suffer from greater friction than electronic ("pickup coil") designs, potentially resulting in more measurement error (less flow indicated than there actually is, because the turbine wheel is slowed by friction). One advantage of mechanical turbine flowmeters, though, is the ability to maintain a running total of gas usage by turning a simple odometer-style

totalizer. This design is often used when the purpose of the flowmeter is to track total fuel gas consumption (e.g. natural gas used by a commercial or industrial facility) for billing.

In an electronic turbine flowmeter, volumetric flow is directly and linearly proportional to pickup coil output frequency. We may express this relationship in the form of an equation:

$$f = kQ$$

Where,

f = Frequency of output signal (Hz, equivalent to pulses per second)

Q = Volumetric flow rate (e.g. gallons per second)

k = "K" factor of the turbine element (e.g. pulses per gallon)

Dimensional analysis confirms the validity of this equation. Using units of GPS (gallons per second) and pulses per gallon, we see that the product of these two quantities is indeed pulses per second (equivalent to cycles per second, or Hz):

$$\left[\frac{\text{Pulses}}{\text{s}}\right] = \left[\frac{\text{Pulses}}{\text{gal}}\right]\left[\frac{\text{gal}}{\text{s}}\right]$$

Using algebra to solve for flow (Q), we see that it is the quotient of frequency and k factor that yields a volumetric flow rate for a turbine flowmeter:

$$Q = \frac{f}{k}$$

The inherent linearity of a turbine flowmeter is a tremendous advantage over nonlinear flow elements such as venturi tubes and orifice plates because this linearity results in a much greater turndown ratio for accurate flow measurement. Contrasted against common orifice-type meters which are usually limited to turndown ratios of 4:1 at best, turbine meters commonly exceed turndown ratios of 10:1.

If pickup signal frequency directly represents volumetric flow rate, then the total number of pulses accumulated in any given time span will represent the amount of fluid volume (V) passed through the turbine meter over that same time span. We may express this algebraically as the product of average flow rate ($\overline{Q}$), average frequency ($\overline{f}$), k factor, and time:

$$V = \overline{Q}t = \frac{\overline{f}t}{k}$$

A more sophisticated way of calculating total volume passed through a turbine meter requires calculus, representing change in volume as the time-integral of instantaneous signal frequency and k factor over a period of time from $t = 0$ to $t = T$:

$$\Delta V = \int_0^T Q\, dt \qquad \text{or} \qquad \Delta V = \int_0^T \frac{f}{k}\, dt$$

We may achieve approximately the same result simply by using a digital counter circuit to totalize pulses output by the pickup coil and a microprocessor to calculate volume in whatever unit of measurement we deem appropriate.

As with the orifice plate flow element, standards have been drafted for the use of turbine flowmeters as precision measuring instruments in gas flow applications, particularly the custody transfer[36] of natural gas. The American Gas Association has published a standard called the Report #7 specifying the installation of turbine flowmeters for high-accuracy gas flow measurement, along with the associated mathematics for precisely calculating flow rate based on turbine speed, gas pressure, and gas temperature.

Pressure and temperature compensation is relevant to turbine flowmeters in gas flow applications because the density of the gas is a strong function of both pressure and temperature. The turbine wheel itself only senses gas *velocity*, and so these other factors must be taken into consideration to accurately calculate mass flow (or *standard* volumetric flow; e.g. SCFM).

In high-accuracy applications, it is important to individually determine the k factor for a turbine flowmeter's calibration. Manufacturing variations from flowmeter to flowmeter make precise duplication of k factor challenging, and so a flowmeter destined for high-accuracy measurement should be tested against a "flow prover" in a calibration laboratory to empirically determine its k factor. If possible, the best way to test the flowmeter's k factor is to connect the prover to the meter on site where it will be used. This way, the any effects due to the piping before and after the flowmeter will be incorporated in the measured k factor.

[36] "Custody transfer" refers to measurement applications where a product is exchanging ownership. In other words, someone is selling, and someone else is buying, quantities of fluid as part of a business transaction. It is not difficult to understand why accuracy is important in such applications, as both parties have a vested interest in a fair exchange. Government institutions also have a stake in accurate metering, as taxes are typically levied on the sale of commodity fluids such as natural gas.

The following photograph shows three AGA7-compliant installations of turbine flowmeters for measuring the flow rate of natural gas:



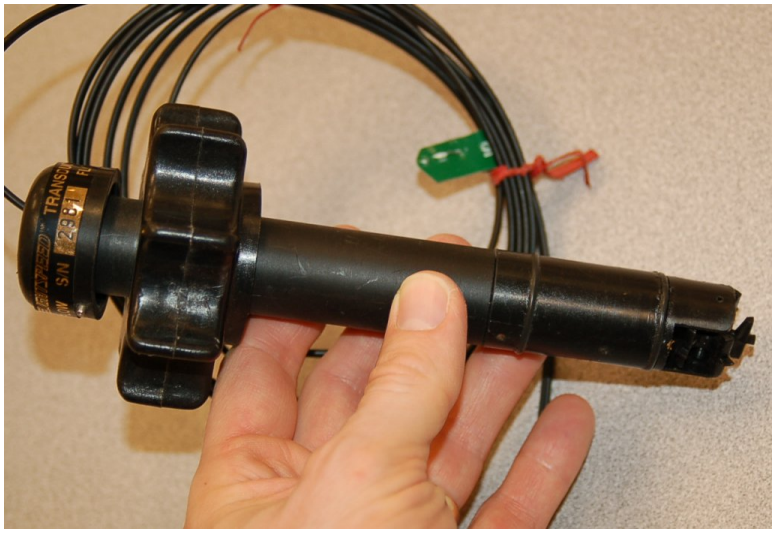
Note the pressure-sensing and temperature-sensing instrumentation installed in the pipe, reporting gas pressure and gas temperature to a flow-calculating computer (along with turbine pulse frequency) for the calculation of natural gas flow rate.

Less-critical gas flow measurement applications may use a "compensated" turbine flowmeter that mechanically performs the same pressure- and temperature-compensation functions on turbine speed to achieve true gas flow measurement, as shown in the following photograph:



The particular flowmeter shown in the above photograph uses a filled-bulb temperature sensor (note the coiled, armored capillary tube connecting the flowmeter to the bulb) and shows total gas flow by a series of pointers, rather than gas flow *rate*.

A variation on the theme of turbine flow measurement is the *paddlewheel* flowmeter, a very inexpensive technology usually implemented in the form of an insertion-type sensor. In this instrument, a small wheel equipped with "paddles" parallel to the shaft is inserted in the flowstream, with half the wheel shrouded from the flow. A photograph of a plastic paddlewheel flowmeter appears here:



A surprisingly sophisticated method of "pickup" for the plastic paddlewheel shown in the photograph uses *fiber-optic cables* to send and receive light. One cable sends a beam of light to the edge of the paddlewheel, and the other cable receives light on the other side of the paddlewheel. As the paddlewheel turns, the paddles alternately block and pass the light beam, resulting in a pulsed light beam at the receiving cable. The frequency of this pulsing is, of course, directly proportional to volumetric flow rate.

The external ends of the two fiber optic cables appear in this next photograph, ready to connect to a light source and light pulse sensor to convert the paddlewheel's motion into an electronic signal:



A problem common to all turbine flowmeters is that of the turbine "coasting" when the fluid flow suddenly stops. This is more often a problem in batch processes than continuous processes, where the fluid flow is regularly turned on and shut off. This problem may be minimized by configuring the measurement system to ignore turbine flowmeter signals any time the automatic shutoff valve reaches the "shut" position. This way, when the shutoff valve closes and fluid flow immediately halts, any coasting of the turbine wheel will be irrelevant. In processes where the fluid flow happens to pulse for reasons other than the control system opening and shutting automatic valves, this problem is more severe.

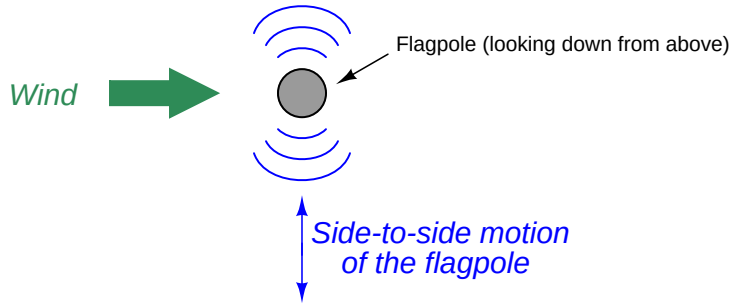
Another problem common to all turbine flowmeters is lubrication of the turbine bearings. Frictionless motion of the turbine wheel is essential for accurate flow measurement, which is a daunting design goal for the flowmeter manufacturing engineers. The problem is not as severe in applications where the process fluid is naturally lubricating (e.g. diesel fuel), but in applications such as natural gas flow where the fluid provides no lubrication to the turbine bearings, external lubrication must be supplied. This is often a regular maintenance task for instrument technicians: using a hand pump to inject light-weight "turbine oil" into the bearing assemblies of turbine flowmeters used in gas service.

Process fluid viscosity is another source of friction for the turbine wheel. Fluids with high viscosity (e.g. heavy oils) will tend to slow down the turbine's rotation even if the turbine rotates on frictionless bearings. This effect is especially pronounced at low flow rates, which leads to a *minimum linear flow* rating for the flowmeter: a flowrate below which it refuses to register proportionately to fluid flow rate.

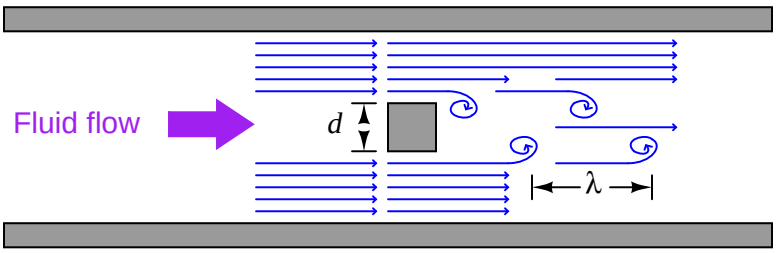
22.4.2 Vortex flowmeters

When a fluid moves with high Reynolds number past a stationary object (a "bluff body"), there is a tendency for the fluid to form *vortices* on either side of the object. Each vortex will form, then detach from the object and continue to move with the flowing gas or liquid, one side at a time in alternating fashion. This phenomenon is known as *vortex shedding*, and the pattern of moving vortices carried downstream of the stationary object is known as a *vortex street*.

It is commonplace to see the effects of vortex shedding on a windy day by observing the motion of flagpoles, light poles, and tall smokestacks. Each of these objects has a tendency to oscillate perpendicular to the direction of the wind, owing to the pressure variations caused by the vortices as they alternately form and break away from the object:



This alternating series of vortices was studied by Vincenc Strouhal in the late nineteenth century and later by Theodore von Kármán in the early twentieth century. It was determined that the distance between successive vortices downstream of the stationary object is relatively constant, and directly proportional to the width of the object, for a wide range of Reynolds number values[37]. If we view these vortices as crests of a continuous wave, the distance between vortices may be represented by the symbol customarily reserved for wavelength: the Greek letter "lambda" (λ).

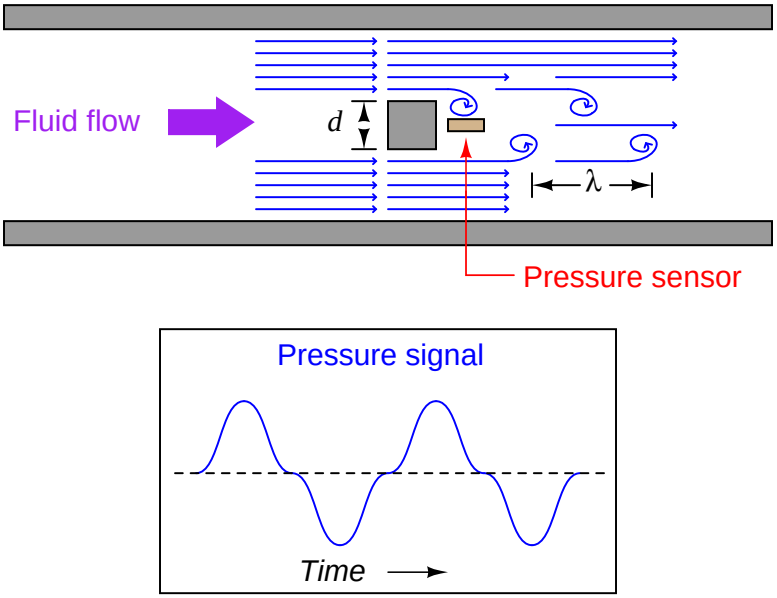


The proportionality between object width (d) and vortex street wavelength (λ) is called the *Strouhal number* (S), approximately equal to 0.17:

$$\lambda S = d \qquad\qquad \lambda \approx \frac{d}{0.17}$$

[37] It is important to note that the vortex-shedding phenomenon ceases altogether if the Reynolds number is too low. Laminar flow produces no vortices, but rather stream-line flow around any object placed in its way.

If a differential pressure sensor is installed immediately downstream of the stationary object in such an orientation that it detects the passing vortices as pressure variations, an alternating signal will be detected:



The *frequency* of this alternating pressure signal is directly proportional to fluid velocity past the object, since the wavelength is constant. This follows the classic frequency-velocity-wavelength formula common to all traveling waves ($\lambda f = v$). Since we know the wavelength will be equal to the bluff body's width divided by the Strouhal number (approximately 0.17), we may substitute this into the frequency-velocity-wavelength formula to solve for fluid velocity (v) in terms of signal frequency (f) and bluff body width (d).

$$v = \lambda f$$

$$v = \frac{d}{0.17} f$$

$$v = \frac{df}{0.17}$$

Thus, a stationary object and pressure sensor installed in the middle of a pipe section constitute a form of flowmeter called a *vortex flowmeter*. Like a turbine flowmeter with an electronic "pickup" sensor to detect the passage of rotating turbine blades, the output frequency of a vortex flowmeter is linearly proportional to volumetric flow rate.

The pressure sensors used in vortex flowmeters are not standard differential pressure transmitters, since the vortex frequency is too high to be successfully detected by such bulky instruments. Instead, the sensors are typically piezoelectric crystals. These pressure sensors need not be calibrated, since the amplitude of the pressure waves detected is irrelevant. Only the frequency of the waves matter

for measuring flow rate, and so nearly any pressure sensor with a fast enough response time will suffice.

Like turbine meters, the relationship between sensor frequency (f) and volumetric flow rate (Q) may be expressed as a proportionality, with the letter k used to represent the constant of proportionality for any particular flowmeter:

$$f = kQ$$

Where,

f = Frequency of output signal (Hz)
Q = Volumetric flow rate (e.g. gallons per second)[38]
k = "K" factor of the vortex shedding flowtube (e.g. pulses per gallon)

This means vortex flowmeters, like electronic turbine meters, each have a particular "k factor" relating the number of pulses generated per unit volume passed through the meter[39]. Counting the total number of pulses over a certain time span yields total fluid volume passed through the meter over that same time span, making the vortex flowmeter readily adaptable for "totalizing" fluid volume just like turbine meters. The direct proportion between vortex frequency and volumetric flow rate also means vortex flowmeters are *linear-responding* instruments just like turbine flowmeters. Unlike orifice plates which exhibit a quadratic response, turbine and vortex flowmeters alike enjoy a wider range (turndown) of flow measurement and do not require special signal characterization to function properly.

Since vortex flowmeters have no moving parts, they do not suffer the problems of wear and lubrication facing turbine meters. There is no moving element to "coast" as in a turbine flowmeter if fluid flow suddenly stops, which means vortex flowmeters are better suited to measuring erratic flows.

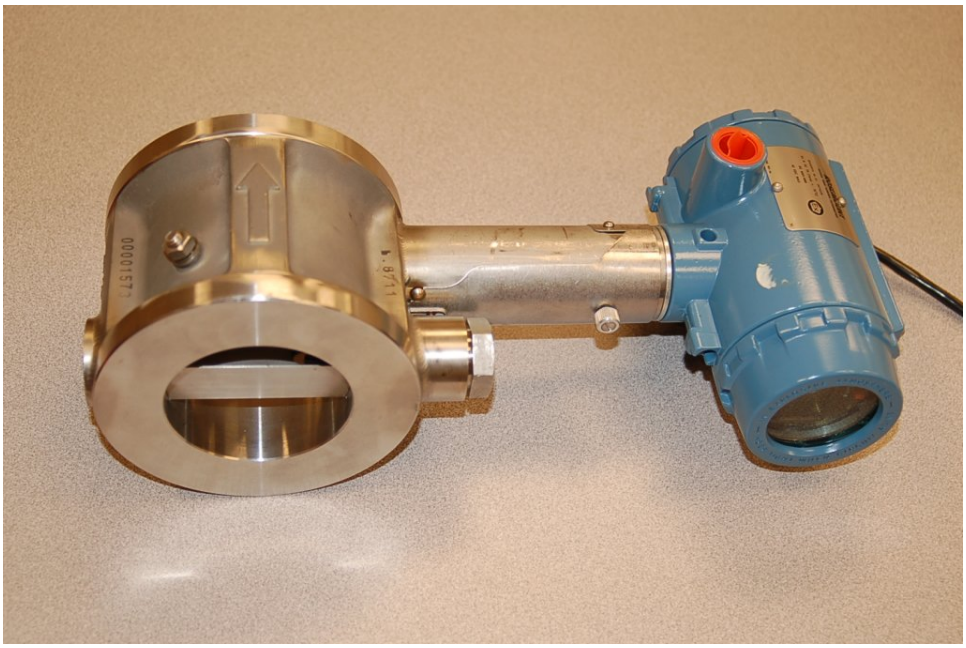
A significant disadvantage of vortex meters is a behavior known as *low flow cutoff*, where the flowmeter simply stops working below a certain flow rate. The reason for this is *laminar* flow: at low flow rates (i.e. low Reynolds number values) the effects of fluid viscosity overwhelm fluid momentum, preventing vortices from forming. This cessation of vortices causes the vortex flowmeter to register absolutely no flow at all even when there is still some (laminar) flow through the pipe. At high flow rates (i.e. high Reynolds number values), fluid momentum is enough to overcome viscosity and produce vortices, and the vortex flowmeter works just fine.

The phenomenon of low-flow cutoff for a vortex flowmeter at first seems analogous to the *minimum linear flow* limitation of a turbine flowmeter. However, vortex flowmeter low-flow cutoff is actually a far more severe problem. If the volumetric flow rate through a turbine flowmeter falls below the minimum linear value, the turbine continues to spin, albeit slower than it should. If the volumetric flow rate through a vortex flowmeter falls below the low-flow cutoff value, however, the flowmeter's signal *goes completely to zero*, indicating no flow at all. This idiosyncrasy makes vortex flowmeters entirely unsuitable in applications where the desired flow measurement range extends all the way down to zero.

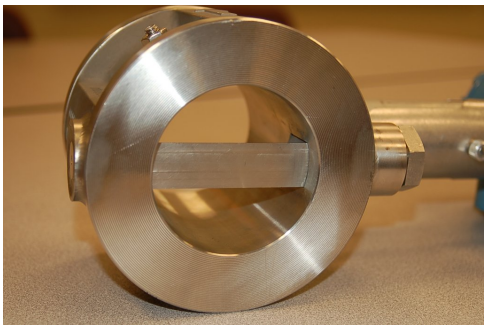
[38] Note that if flow rate is to be expressed in units of gallons per *minute* as is customary, the equation must contain a factor for minutes-to-seconds conversion: $f = \frac{kQ}{60}$

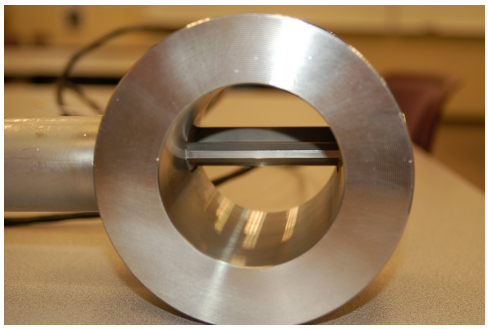
[39] This k factor is empirically determined for each flowmeter by the manufacturer using water as the test fluid (a factory "wet-calibration"), to ensure optimum accuracy.

The following photograph shows a Rosemount model 8800C vortex flow transmitter:



The next two photographs show close-up views of the flowtube assembly, front (left) and rear (right):





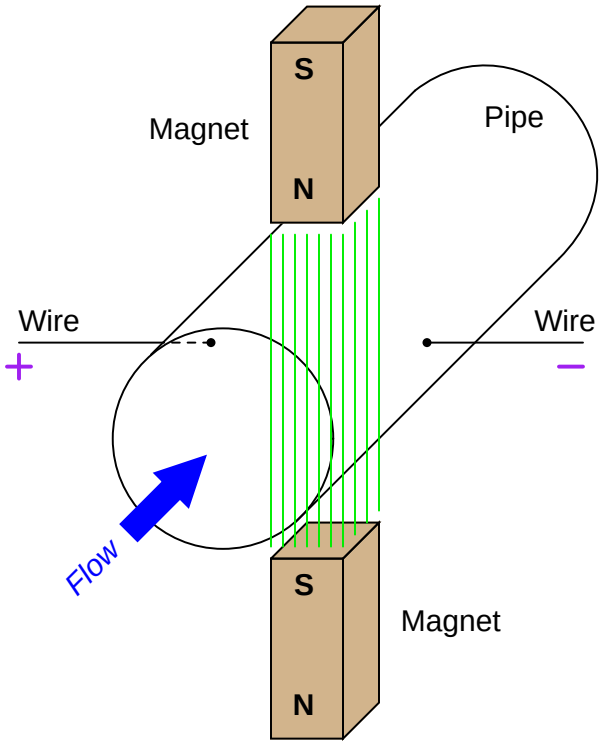
Vortex flowmeters, like other velocity-based meters, are affected by large-scale turbulence in the fluid stream and therefore require some length of straight pipe both upstream and downstream of the flowmeter to properly characterize the flow. It is typical to install vortex flowmeters with 10 pipe diameters of straight-length pipe upstream and 5 pipe diameters downstream.

22.4.3 Magnetic flowmeters

When an electrical conductor moves perpendicular to a magnetic field, a voltage is induced in that conductor perpendicular to both the magnetic flux lines and the direction of motion. This phenomenon is known as *electromagnetic induction*, and it is the basic principle upon which all electro-mechanical generators operate.

In a generator mechanism, the conductor in question is typically a coil (or set of coils) made of copper wire. However, there is no reason the conductor must be made of copper wire. *Any* electrically conductive substance in motion is sufficient to electromagnetically induce a voltage, even if that substance is a liquid[40]. Therefore, electromagnetic induction is a technique applicable to the measurement of liquid flow rates.

Consider water flowing through a pipe, with a magnetic field passing perpendicularly through the pipe:



The direction of liquid flow cuts perpendicularly through the lines of magnetic flux, generating a voltage along an axis perpendicular to both. Metal electrodes opposite each other in the pipe wall intercept this voltage, making it readable to an electronic circuit.

[40] In a practical sense, only liquid flows are measurable using this technique. Gases must be super-heated into a *plasma* state before they are able to conduct electricity, and so electromagnetic flowmeters cannot be used with most industrial gas flowstreams.

A voltage induced by the linear motion of a conductor through a magnetic field is called *motional EMF*, the magnitude of which is predicted by the following formula (assuming perfect perpendicularity between the direction of velocity, the orientation of the magnetic flux lines, and the axis of voltage measurement):

$$\mathcal{E} = Blv$$

Where,
$\mathcal{E}$ = Motional EMF (volts)
B = Magnetic flux density (Tesla)
l = Length of conductor passing through the magnetic field (meters)
v = Velocity of conductor (meters per second)

Assuming a fixed magnetic field strength (constant B) and an electrode spacing equal to the fixed diameter of the pipe (constant $l = d$), the only variable capable of influencing the magnitude of induced voltage is velocity (v). In our example, v is not the velocity of a wire segment, but rather the average velocity of the liquid flowstream ($\overline{v}$). Since we see that this voltage will be proportional to average fluid velocity, it must also be proportional to volumetric flow rate, since volumetric flow rate is also proportional to average fluid velocity[41]. Thus, what we have here is a type of flowmeter based on electromagnetic induction. These flowmeters are commonly known as *magnetic flowmeters* or simply *magflow meters*.

We may state the relationship between volumetric flow rate (Q) and motional EMF ($\mathcal{E}$) more precisely by algebraic substitution. First, we will write the formula relating volumetric flow to average velocity, and then manipulate it to solve for average velocity:

$$Q = A\overline{v}$$

$$\frac{Q}{A} = \overline{v}$$

Next, we re-state the motional EMF equation, and then substitute $\frac{Q}{A}$ for $\overline{v}$ to arrive at an equation relating motional EMF to volumetric flow rate (Q), magnetic flux density (B), pipe diameter (d), and pipe area (A):

$$\mathcal{E} = Bd\overline{v}$$

$$\mathcal{E} = Bd\frac{Q}{A}$$

$$\mathcal{E} = \frac{BdQ}{A}$$

[41] This is an application of the transitive property in mathematics: if two quantities are both equal to a common third quantity, they must also be equal to each other. This property applies to proportionalities as well as equalities: if two quantities are proportional to a common third quantity, they must also be proportional to each other.

Since we know this is a circular pipe, we know that area and diameter are directly related to each other by the formula $A = \frac{\pi d^2}{4}$. Thus, we may substitute this definition for area into the last equation, to arrive at a formula with one less variable (only d, instead of both d and A):

$$\mathcal{E} = \frac{BdQ}{\frac{\pi d^2}{4}}$$

$$\mathcal{E} = \frac{BdQ}{1}\frac{4}{\pi d^2}$$

$$\mathcal{E} = \frac{4BQ}{\pi d}$$

If we wish to have a formula defining flow rate Q in terms of motional EMF ($\mathcal{E}$), we may simply manipulate the last equation to solve for Q:

$$Q = \frac{\pi d\mathcal{E}}{4B}$$

This formula will successfully predict flow rate only for absolutely perfect circumstances. In order to compensate for inevitable imperfections, a "proportionality constant" (k) is usually included in the formula[42]:

$$Q = k\frac{\pi d\mathcal{E}}{4B}$$

Where,
Q = Volumetric flow rate (cubic meters per second)
$\mathcal{E}$ = Motional EMF (volts)
B = Magnetic flux density (Tesla)
d = Diameter of flowtube (meters)
k = Constant of proportionality

Note the linearity of this equation. Nowhere do we encounter a power, root, or other non-linear mathematical function in the equation for a magnetic flowmeter. This means no special characterization is required to calculate volumetric flow rate.

A few conditions must be met for this formula to successfully infer volumetric flow rate from induced voltage:

- The liquid must be a reasonably good conductor of electricity (*note: it is okay if the conducting fluid contains some non-conducting solids; the conductive fluid surrounding the non-conducting solid matter still provides electrical continuity between the electrodes necessary for induction*)
- The pipe must be completely filled with liquid to ensure contact with both probes as well as to ensure flow across the entire cross-section of the pipe
- The flowtube must be properly grounded to avoid errors caused by stray electric currents in the liquid

[42] The colloquial term in the United States for this sort of thing is *fudge factor*.

The first condition is met by careful consideration of the process liquid prior to installation. Magnetic flowmeter manufacturers will specify the minimum conductivity value of the liquid to be measured. The second and third conditions are met by correct installation of the magnetic flowtube in the pipe. The installation must be done in such a way as to guarantee full flooding of the flowtube (no gas pockets). The flowtube is usually installed with electrodes across from each other horizontally (never vertically!) so even a momentary gas bubble will not break electrical contact between an electrode tip and the liquid flowstream. The following photograph shows how *not* to install magnetic flowmeters:



Note in this example how the electrodes are vertically oriented instead of horizontal, because the pipes for these two magnetic flowmeters were placed too close[43] to allow proper clearance for the protruding electrodes to lie horizontally. Sadly, poor flowmeter installation is all too common in new projects, as many piping designers and pipefitters are ignorant of flowmeter operating principles. This is one way instrument engineers and technicians may deter operational problems: by involving themselves in the design phase of a piping system, and helping to educate piping designers.

Magnetic flowmeters exhibit several advantages over other types of flowmeter. They are fairly tolerant of swirl and other large-scale turbulent fluid behavior, because the induced voltage is proportional only to the *perpendicular* velocity of the conductor, in this case the velocity of the fluid along the centerline of the flowtube. As such, magnetic flowmeters do not require the long straight-runs of pipe upstream and downstream that orifice plates do, which is a great advantage in

[43] The obvious solution to this problem – relocating the pipes to give more clearance between flowmeters – would be quite expensive given the large pipe sizes involved. A "compromise" solution is to tilt the magnetic flowtubes as far as possible without the electrodes touching the adjacent flowtube. Horizontal electrode installation is ideal for horizontal pipes, but an angled installation will be better than a vertical installation.

many piping systems. Upstream straight-pipe requirements of 5 diameters and downstream straight-pipe requirements of 3 diameters is typical[44].

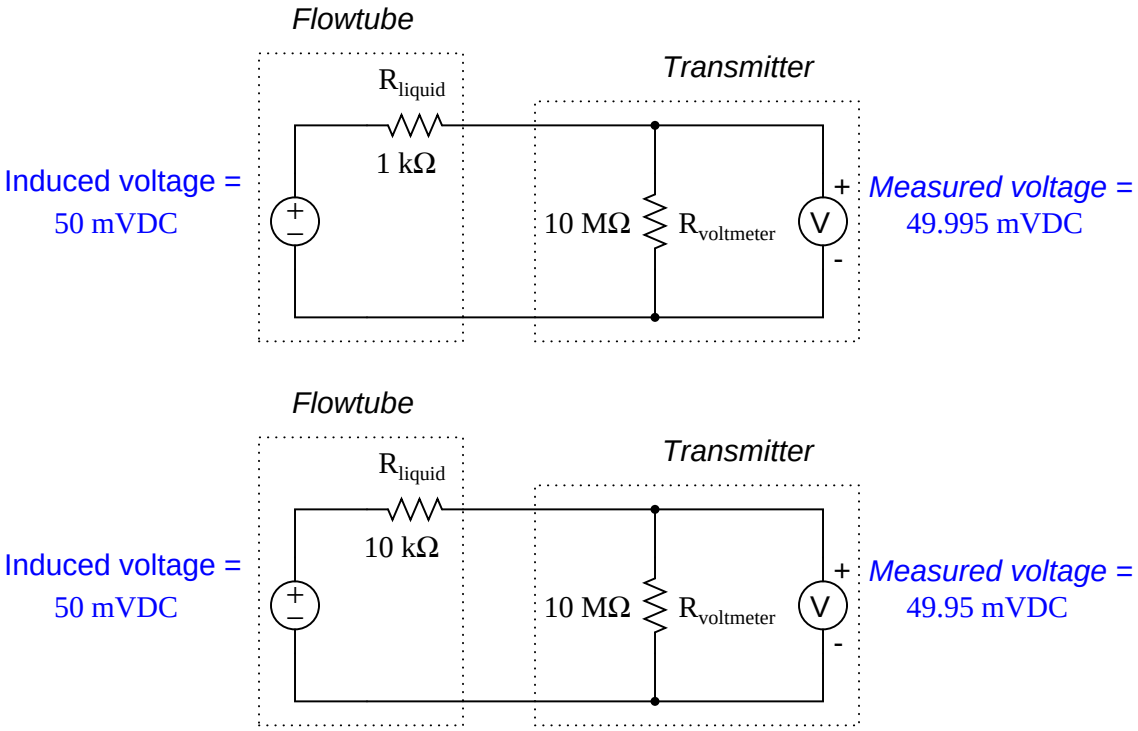
Additionally, the wide-open bore of a magnetic flowmeter's tube means there is absolutely nothing to restrict the flow, resulting in extremely low permanent pressure loss. The lack of any obstruction within the path of fluid flow means magnetic flowmeters are quite tolerant of solids[45] within the liquid flowstream, making them well-suited for measuring such process liquids as wastewater, slurries, wood pulp, and food products which might clog other types of flowmeters. In fact, magnetic flowmeters are the dominant flowmeter technology used in wastewater, wood pulping, and food processing industries for this very reason.

[44] As always, check the manufacturer's literature for specific requirements, as variations do exist for different models and sizes of magtube.

[45] Even electrically *non-conducting* solid matter is tolerated well by magnetic flowmeters, since the conducting liquid surrounding the solids still provides continuity from one electrode to the other.

Electrical conductivity of the process liquid must meet a certain minimum value, but that is all. It is surprising to some technicians that changes in liquid conductivity have little to no effect on flow measurement accuracy. It is not as though a doubling of liquid conductivity will result in a doubling of induced voltage! Motional EMF is strictly a function of physical dimensions, magnetic field strength, and fluid velocity.

Liquids with poor conductivity present a greater electrical resistance in the voltage-measuring circuit than liquids with good conductivity, but this is of little consequence because the input impedance of the detection circuitry is phenomenally high. The effect of liquid conductivity on flowmeter operation may be modeled by the following DC circuits:



Here, a ten-fold (one order of magnitude) change in liquid resistance barely affects the measured voltage (49.995 mV versus 49.95 mV) because the flow transmitter's voltage-sensing electronic circuit has such a high input impedance. The liquid's equivalent resistance value must increase dramatically beyond the values shown in this example before it will have any significant effect on flow measurement accuracy.

In fact, the only time fluid conductivity is a problem with magnetic flowmeters is when the fluid in question has negligible conductivity. Such fluids include deionized water (e.g. steam boiler feedwater, ultrapure water for pharmaceutical and semiconductor manufacturing) and oils. Most aqueous (water-based) fluids work fine with magnetic flowmeters.

Proper grounding of the flowtube is very important for magnetic flowmeters. The motional EMF generated by most liquid flowstreams is very weak (1 millivolt or less!), and therefore may be easily overshadowed by noise voltage present as a result of stray electric currents in the piping and/or liquid. To combat this problem, magnetic flowmeters are usually equipped with grounding conductors placed to shunt (bypass) stray electric currents around the flowtube so the only voltage intercepted by the electrodes will be the motional EMF produced by liquid flow, and not voltage drops created by stray currents through the resistance of the liquid. The following photograph shows a Rosemount model 8700 magnetic flowtube, with two braided[46]-wire grounding straps clearly visible:

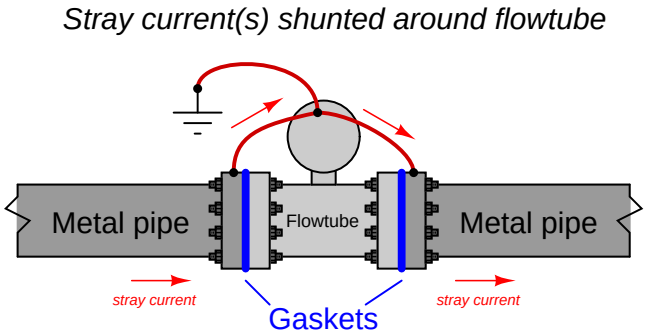


Note how both grounding straps attach to a common junction point on the flowtube housing. This common junction point should also be bonded to a functional earth ground when the flowtube is installed in the process line. On this particular flowtube you can see a stainless steel *grounding ring* on the face of the near flange, connected to one of the braided grounding straps. An identical grounding ring lays on the other flange, but it is not clearly visible in this photograph. These rings

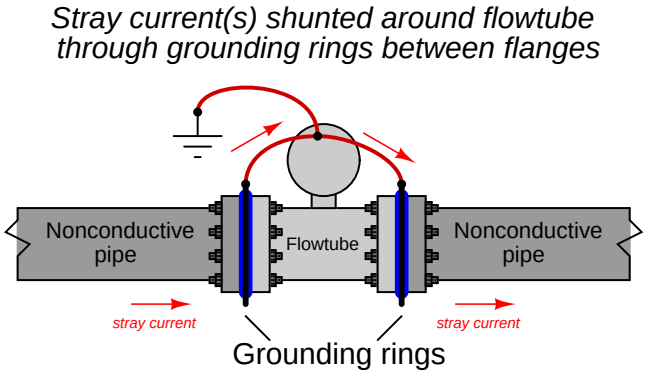
[46] Braided conductors do a better job of shunting radio-frequency currents, because at very high frequencies the *skin effect* makes the surface area of a conductor a greater factor in its conductivity than its cross-sectional area.

provide points of electrical contact with the liquid in installations where the pipe is made of plastic, or where the pipe is metal but lined with a plastic material for corrosion resistance.

If the pipe connecting to a magnetic flowmeter's flowtube is conductive (e.g. metal), grounding may be accomplished by joining the metal pipes' flanges together with grounding straps to a common grounding point on the flowtube body as such:



If the pipe connecting to a magnetic flowmeter's flowtube is non-conductive (e.g. plastic) or conductive with an insulating lining (e.g. metal pipe with plastic lining), grounding to the pipe flanges will be pointless. In order for flowtube grounding to be effective, the grounding conductors must have electrical continuity to the fluid itself. Special *grounding rings* may be sandwiched between the flanges of non-conducting pipes to provide points of electrical contact with the fluid. These grounding rings are then joined together with grounding straps to a common grounding point on the flowtube body as such:

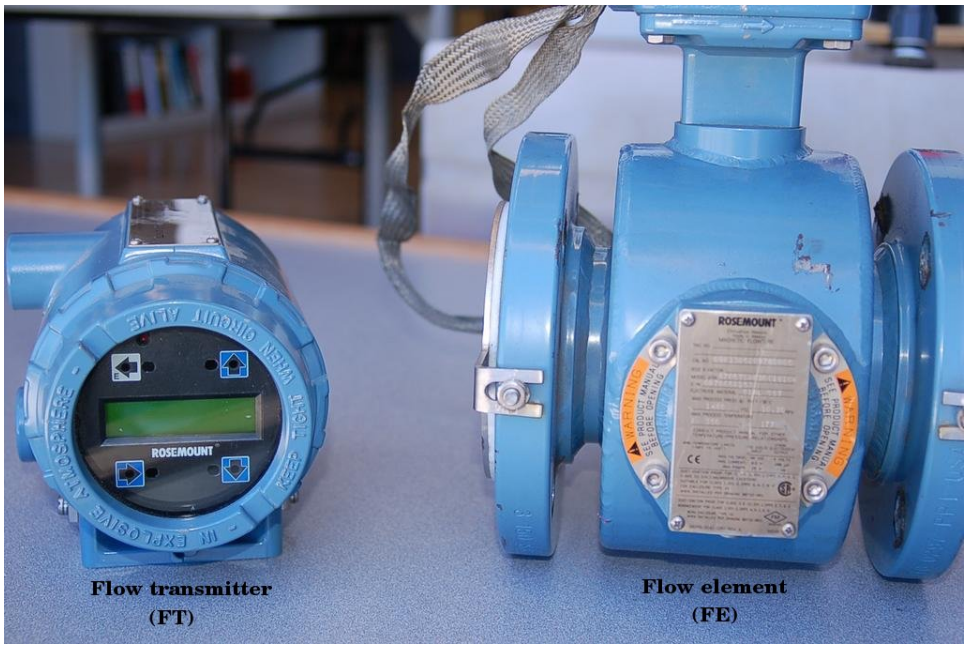


Some magnetic flowmeters have their signal conditioning electronics located integral to the flowtube assembly. A couple of examples are shown here (a pair of small Endress+Hauser flowmeters on the left and a large Toshiba flowmeter on the right):





Other magnetic flowmeters have separate electronics and flowtube assemblies, connected together by shielded cable. In these installations, the electronics assembly is referred to as the flow transmitter (FT) and the flowtube as the flow element (FE):



This next photograph shows an enormous (36 inch diameter!) magnetic flow element (black) and flow transmitter (blue, behind the person's hand shown for scale) used to measure wastewater flow at a municipal sewage treatment plant:

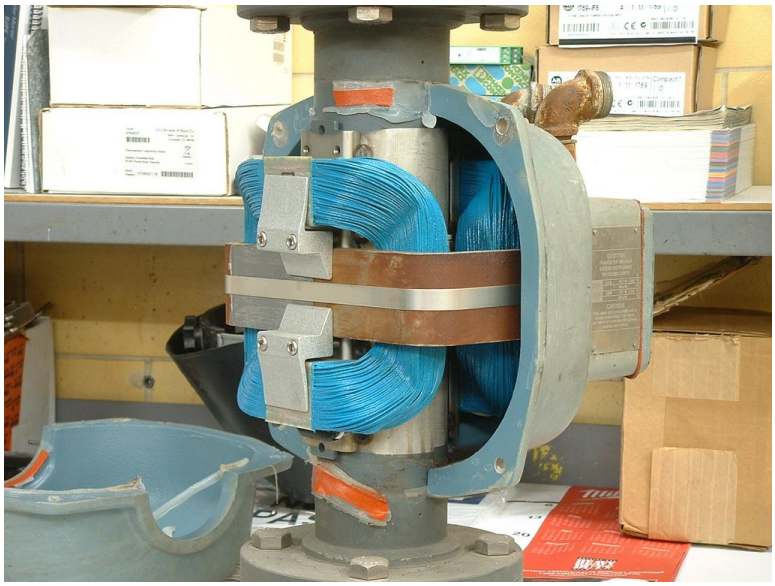


Note the vertical pipe orientation, ensuring constant contact between the electrodes and the water during flowing conditions.

While in theory a permanent magnet should be able to provide the necessary magnetic flux for a magnetic flowmeter to function, this is never done in industrial practice. The reason for this has to do with a phenomenon called *polarization* which occurs when a DC voltage is impressed across a liquid containing ions (electrically charged molecules). Electrically-charged molecules (ions) tend to collect near poles of opposite charge, which in this case would be the flowmeter electrodes. This "polarization" would soon interfere with detection of the motional EMF if a magnetic flowmeter were to use a constant magnetic flux such as that produced by a permanent magnet. A simple solution to this problem is to alternate the polarity of the magnetic field, so the motional EMF polarity also alternates and never gives the fluid ions enough time to polarize.

This is why magnetic flowmeter tubes always employ electromagnet *coils* to generate the magnetic flux instead of permanent magnets. The electronics package of the flowmeter energizes these coils with currents of alternating polarity, so as to alternate the polarity of the induced voltage across the moving fluid. Permanent magnets, with their unchanging magnetic polarities, would only be able to create an induced voltage with constant polarity, leading to ionic polarization and subsequent flow measurement errors.

A photograph of a Foxboro magnetic flowtube with one of the protective covers removed shows these wire coils clearly (in blue):



Perhaps the simplest form of coil excitation is when the coil is energized by 60 Hz AC power taken from the line power source, such as the case with this Foxboro flowtube. Since motional EMF is proportional to fluid velocity and to the flux density of the magnetic field, the induced voltage for such a coil will also be a 60 Hz sine wave whose amplitude varies[47] with volumetric flow rate.

Unfortunately, if there is any stray electric current traveling through the liquid to produce erroneous voltage drops between the electrodes, chances are it will be 60 Hz AC as well[48]. With the coil energized by 60 Hz AC, any such noise voltage may be falsely interpreted as fluid flow because the sensor electronics has no way to distinguish between 60 Hz noise in the fluid and a 60 Hz motional EMF caused by fluid flow.

A more sophisticated solution to this problem uses a *pulsed* excitation power source for the flowtube coils. This is called *DC* excitation by magnetic flowmeter manufacturers, which is a bit misleading because these "DC" excitation signals often reverse polarity, appearing more like an AC square wave on an oscilloscope display. The motional EMF for one of these flowmeters will exhibit the same waveshape, with amplitude once again being the indicator of volumetric flow rate. The sensor electronics can more easily reject any AC noise voltage because the frequency and waveshape of the noise (60 Hz, sinusoidal) will not match that of the flow-induced motional EMF signal.

The most significant disadvantage of pulsed-DC magnetic flowmeters is slower response time to changing flow rates. In an effort to achieve a "best-of-both-worlds" result, some magnetic flowmeter manufacturers produce *dual-frequency* flowmeters which energize their flowtube coils with two mixed frequencies: one below 60 Hz and one above 60 Hz. The resulting voltage signal intercepted by the electrodes is demodulated and interpreted as a flow rate.

[47] For example, in a condition of no liquid flow through the tube, the electrodes will intercept no voltage at all when the magnetic excitation is 60 Hz AC. When liquid moves slowly in the forward direction through the tube, a low-amplitude 60 Hz millivoltage signal will be detected at the electrodes. When liquid moves rapidly in the forward direction through the tube, the induced 60 Hz AC millivoltage will be greater in amplitude. Any liquid motion in the reverse direction induces a proportional 60 Hz AC voltage signal whose phase is 180° shifted from the excitation signal driving the magnetic coils of the flowtube.

[48] We know this because the largest electrical noise sources in industry are electric motors, transformers, and other power devices operating on the exact same frequency (60 Hz in the United States, 50 Hz in Europe) as the flowtube coils.

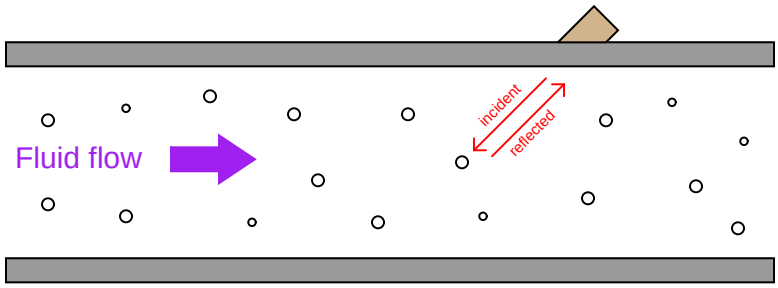
22.4.4 Ultrasonic flowmeters

Ultrasonic flowmeters measure fluid velocity by passing high-frequency sound waves along the fluid flow path. Fluid motion influences the propagation of these sound waves, which may then be measured to infer fluid velocity. Two major sub-types of ultrasonic flowmeters exist: *Doppler* and *transit-time*. Both types of ultrasonic flowmeter work by transmitting a high-frequency sound wave into the fluid stream (the *incident* pulse) and analyzing the received pulse.

Doppler flowmeters exploit the *Doppler effect*, which is the shifting of frequency resulting from waves emitted by or reflected by a moving object. A common realization of the Doppler effect is the perceived shift in frequency of a horn's report from a moving vehicle: as the vehicle approaches the listener, the pitch of the horn seems higher than normal; when the vehicle passes the listener and begins to move away, the horn's pitch appears to suddenly "shift down" to a lower frequency. In reality, the horn's frequency never changes, but the velocity of the approaching vehicle relative to the stationary listener acts to "compress" the sonic vibrations in the air. When the vehicle moves away, the sound waves are "stretched" from the perspective of the listener.

The same effect takes place if a sound wave is aimed at a moving object, and the echo's frequency is compared to the transmitted (incident) frequency. If the reflected wave returns from a bubble advancing toward the ultrasonic transducer[49], the reflected frequency will be greater than the incident frequency. If the flow reverses direction and the reflected wave returns from a bubble traveling away from the transducer, the reflected frequency will be less than the incident frequency. This matches the phenomenon of a vehicle's horn pitch seemingly increasing as the vehicle approaches a listener and seemingly decreasing as the vehicle moves away from a listener.

A Doppler flowmeter bounces sound waves off of bubbles or particulate material in the flow stream, measuring the frequency shift and inferring fluid velocity from the magnitude of that shift.



The requirement for there to be objects in the flow stream large enough to reflect sound waves limits Doppler ultrasonic flowmeters to liquid applications. "Dirty" liquids such as slurries and wastewater, or liquids carrying a substantial number of gas bubbles (e.g. carbonated beverages) are good candidate fluids for this technology. It is unrealistic to expect that any gas stream will be carrying liquid droplets or solid matter large enough to reflect strong echoes, and so Doppler flowmeters cannot be used to measure gas flow.

[49] In the industrial instrumentation world, the word "transducer" usually has a very specific meaning: a device used to process or convert standardized instrumentation signals, such as 4-20 mA converted into 3-15 PSI, etc. In the general scientific world, however, the word "transducer" describes any device converting one form of energy into another. It is this latter definition of the word that I am using when I describe an ultrasonic "transducer" – a device used to convert electrical energy into ultrasonic sound waves, and vice-versa.

The mathematical relationship between fluid velocity (v) and the Doppler frequency shift (Δf) is as follows, for fluid velocities much less than the speed of sound through that fluid ($v << c$):

$$\Delta f = \frac{2vf \cos\theta}{c}$$

Where,

Δf = Doppler frequency shift
v = Velocity of fluid (actually, of the particle reflecting the sound wave)
f = Frequency of incident sound wave
θ = Angle between transducer and pipe centerlines
c = Speed of sound in the process fluid

Note how the Doppler effect yields a direct measurement of fluid *velocity* from each echo received by the transducer. This stands in marked contrast to measurements of *distance* based on time-of-flight (time domain reflectometry – where the amount of *time* between the incident pulse and the returned echo is proportional to distance between the transducer and the reflecting surface), such as in the application of ultrasonic liquid level measurement. In a Doppler flowmeter, the time delay between the incident and reflected pulses is irrelevant. Only the *frequency shift* between the incident and reflected signals matters. This frequency shift is also directly proportional to the velocity of flow, making the Doppler ultrasonic flowmeter a linear measurement device.

Re-arranging the Doppler frequency shift equation to solve for velocity (again, assuming $v << c$),

$$v = \frac{c\Delta f}{2f \cos\theta}$$

Knowing that volumetric flow rate is equal to the product of pipe area and the average velocity of the fluid ($Q = A\overline{v}$), we may re-write the equation to directly solve for calculated flow rate (Q):

$$Q = \frac{Ac\Delta f}{2f \cos\theta}$$

A very important consideration for Doppler ultrasonic flow measurement is that the calibration of the flowmeter varies with the speed of sound through the fluid (c). This is readily apparent by the presence of c in the above equation: as c increases, Δf must proportionately decrease for any fixed volumetric flow rate Q. Since the flowmeter is designed to directly interpret flow rate in terms of Δf, an increase in c causing a decrease in Δf will thus register as a decrease in Q. This means the speed of sound for a fluid must be precisely known in order for a Doppler ultrasonic flowmeter to accurately measure flow.

The speed of sound through any fluid is a function of that medium's density and bulk modulus (how easily it compresses):

$$c = \sqrt{\frac{B}{\rho}}$$

Where,

c = speed of sound in a material (meters per second)

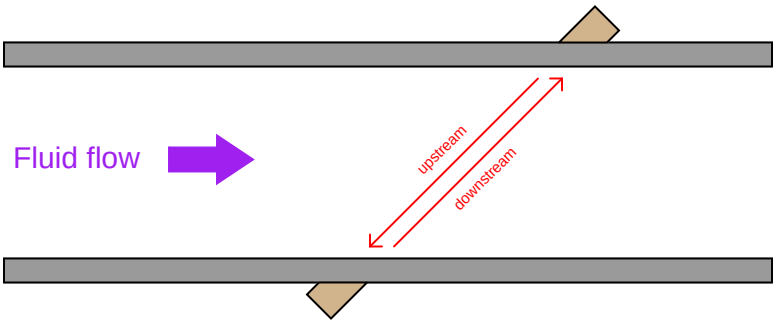
B = Bulk modulus (pascals, or newtons per square meter)

ρ = Mass density of fluid (kilograms per cubic meter)

Temperature affects liquid density, and composition (the chemical constituency of the liquid) affects bulk modulus. Thus, temperature and composition both are influencing factors for Doppler ultrasonic flowmeter calibration. Pressure is not a concern here, since pressure only affects the density of gases, and we already know Doppler flowmeters only function with liquids.

Following on the theme of requiring bubbles or particles of sufficient size, another limitation of Doppler ultrasonic flowmeters is their inability to measure flow rates of liquids that are too clean and too homogeneous. In such applications, the sound-wave reflections will be too weak to reliably measure. Such is also the case when the solid particles have a speed of sound too close to the that of the liquid, since reflection happens only when a sound wave encounters a material with a markedly different speed of sound. Doppler-type ultrasonic flowmeters are useless in applications where we cannot obtain strong sound-wave reflections.

Transit-time flowmeters, sometimes called *counterpropagation* flowmeters, are an alternative to Doppler ultrasonic flowmeters. A transit-time ultrasonic flowmeter uses a pair of opposed sensors to measure the time difference between a sound pulse traveling with the fluid flow versus a sound pulse traveling against the fluid flow. Since the motion of fluid tends to carry a sound wave along, the sound pulse transmitted downstream will make the journey faster than a sound pulse transmitted upstream[50]:



The rate of volumetric flow through a transit-time flowmeter is a simple function of the upstream and downstream propagation times:

$$Q = k\frac{t_{up} - t_{down}}{(t_{up})(t_{down})}$$

Where,

Q = Calculated volumetric flow rate

k = Constant of proportionality

t_{up} = Time for sound pulse to travel from downstream location to upstream location (upstream, against the flow)

t_{down} = Time for sound pulse to travel from upstream location to downstream location (downstream, with the flow)

An interesting characteristic of transit-time velocity measurement is that the ratio of transit time difference over transit time product remains constant with changes in the speed of sound through the fluid[51]. When this equation is cast into terms of path length (L), fluid velocity (v), and sound velocity (c), the equation simplifies to $Q = \frac{2kv}{L}$, proving that the transit-time flowmeter is linear just like the Doppler flowmeter, with the advantage of being immune to changes in the fluid's speed of sound. Changes in bulk modulus resulting from changes in fluid composition, or changes in density resulting from compositional, temperature, or pressure variations therefore have little effect on a transit-time flowmeter's accuracy.

[50] This phenomenon is analogous to paddling a canoe across the width of a river, with the canoe bow angled upstream versus angled downstream. Angled upstream, the canoeist must overcome the velocity of the river and therefore takes longer to reach the other side. Angled downstream, the river's velocity aids the canoeist's efforts and therefore the trip takes less time.

[51] If you would like to prove this to yourself, you may do so by substituting path length (L), fluid velocity (v), and sound velocity (c) for the times in the flow formula. Use $t_{up} = \frac{L}{c-v}$ and $t_{down} = \frac{L}{c+v}$ as your substitutions, then algebraically reduce the flow formula until you find that all the c terms cancel. Your final result should be $Q = \frac{2kv}{L}$.

Not only are transit-time ultrasonic flowmeters immune to changes in the speed of sound, but they are also able to measure that sonic velocity independent of the flow rate. The equation for calculating speed of sound based on upstream and downstream propagation times is as follows:

$$c = \frac{L}{2}\left(\frac{t_{up} + t_{down}}{(t_{up})(t_{down})}\right)$$

Where,

c = Calculated speed of sound in fluid

L = Path length

t_{up} = Time for sound pulse to travel from downstream location to upstream location (upstream, against the flow)

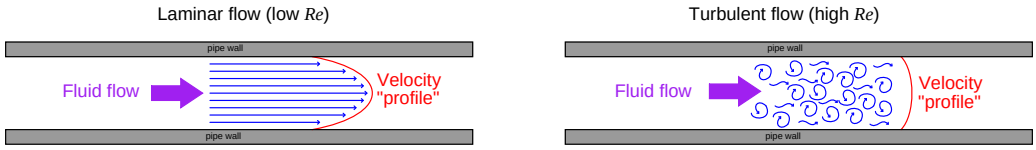
t_{down} = Time for sound pulse to travel from upstream location to downstream location (downstream, with the flow)

While not necessary or even particularly relevant for the direct purpose of flow measurement, this inference of the fluid's speed of sound is nevertheless useful as a diagnostic tool. If the true speed of sound for the fluid is known either by direct laboratory measurement of a sample or by chemical analysis[52] of a sample, this speed may be compared against the flowmeter's reported speed of sound to check the flowmeter's absolute transit time measurement accuracy. Certain problems within the sensors or within the sensor electronics may be detected in this way.

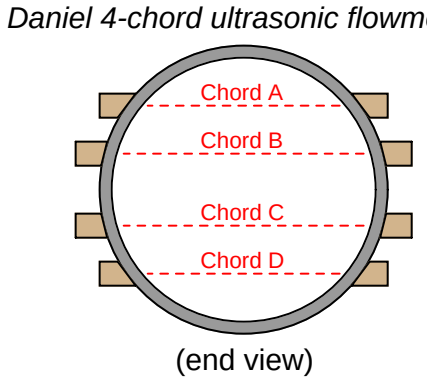
A requirement for reliable operation of a transit-time ultrasonic flowmeter is that the process fluid be free from gas bubbles or solid particles which might scatter or obstruct the sound waves. Note that this is precisely the opposite requirement of Doppler ultrasonic flowmeters, which *require* bubbles or particles to reflect sound waves. These opposing requirements neatly distinguish applications suitable for transit-time flowmeters from applications suitable for Doppler flowmeters, and also raise the possibility of using transit-time ultrasonic flowmeters on gas flowstreams as well as on liquid flowstreams.

[52] An instrument called a *gas chromatograph* is able to provide live measurement of gas composition, with a computer calculating the average speed of sound for the gas given the known types and percentages of each molecular compound comprising the gas mixture. It just so happens that gas composition analysis by chromatograph is something typically done for custody transfer flow measurement of natural gas anyway, for the primary purpose of calculating the gas's *heating value* as a fuel, and therefore no additional investment of instrumentation is necessary to calculate the gas's speed of sound in this application.

One potential problem with any ultrasonic flowmeter is being able to measure the true average fluid velocity when the flow profile changes with Reynolds number. If just one ultrasonic "beam" is used to probe the fluid velocity, the path this beam takes will likely see a different velocity profile as the flow rate changes (and the Reynolds number changes along with it). Recall the difference in fluid velocity profiles between low Reynolds number flows (left) and high Reynolds number flows (right):



A popular way to mitigate this problem is to use multiple sensor pairs, sending acoustic signals along multiple paths through the fluid (i.e. a *multipath* ultrasonic flowmeter), and to average the resulting velocity measurements. Dual-beam transit-time flowmeters have been in use for well over a decade at the time of this writing (2009), and one manufacturer even has a *five beam* ultrasonic flowmeter model which they claim maintains an accuracy of $\pm$ 0.15% through the laminar-to-turbulent flow regime transition[53]. A simplified illustration of a Daniel four-beam (or four "chord") ultrasonic flowmeter is shown here:



Multipath ultrasonic flowmeters, by virtue of measuring more than one sound wave path through the fluid, also have the ability to detect irregular flow profiles. Each sonic path between sensor pairs in a transit-time ultrasonic flowmeter, called a *chord*, measures flow velocity. The velocities reported for each chord may be compared to the flowmeter's calculated average flow velocity, and expressed as *velocity ratios*. A particular chord measuring a velocity greater than the flowmeter's average will report a velocity ratio greater than one (> 1), whereas a chord measuring a velocity less than the meter's average will report a velocity ratio less than one (< 1). In the Daniel four-chord ultrasonic flowmeter, two chords (B and C) measure velocity near the center of the pipe while the others (A and D) measure velocity closer to the pipe walls. In normal operation, the center chord velocity

[53] See page 10 of Friedrich Hofmann's *Fundamentals of Ultrasonic Flow Measurement for industrial applications* paper.

ratios should exceed the outer chord velocity ratios by a small amount, since the flow profiles of laminar and turbulent flow regimes alike exhibit a greater velocity at the center of a pipe than near the walls of a pipe.

Another parameter called *profile factor* expresses the chord velocity factors as a ratio, inner velocity factors over outer velocity factors: $\frac{B+C}{A+D}$, a number which should always exceed one (> 1). The exact value of this profile factor varies with the flowmeter's installation, and may shift over time if the flow profile shifts for some reason (e.g. partial blockage in a flow conditioner, piping change, accumulation of debris on pipe wall). For example, a pipe accumulating dirt or other solid material on its walls over time will tend to slow down the velocity of fluid near the walls compared to the velocity of fluid in the center. This has the effect of decreasing chord A and D velocity ratios while increasing chord B and C velocity ratios, which in turn increases the profile factor value. In this regard, the profile factor value for an operating flowmeter may be an important diagnostic tool, indicating some physical abnormality within the piping.

As previously mentioned, it is possible for a transit-time ultrasonic flowmeter to measure the speed of sound through the fluid from the sum of the upstream and downstream transit times. Multipath transit-time flowmeters may use this feature to cross-check calculated speed of sound values measured by each chord, as a self-diagnostic tool. Since each chord measures the same gas composition, each chord's calculated speed of sound should be exactly equal. Significant differences in calculated speed of sound between chords suggests a failure in the acoustic sensor(s) or electronics for the errant chord.

Some modern ultrasonic flowmeters have the ability to switch back and forth between Doppler and transit-time (counterpropagation) modes, automatically adapting to the fluid being sensed. This capability enhances the suitability of ultrasonic flowmeters to a wider range of process applications.

Ultrasonic flowmeters are adversely affected by swirl and other large-scale fluid disturbances, and as such may require substantial lengths of straight pipe upstream and downstream of the measurement flowtube to stabilize the flow profile.

Like magnetic flowmeters, ultrasonic flowmeters are completely non-obstructive, which means they exhibit extremely low permanent pressure loss and will not accumulate debris.

Advances in ultrasonic flow measurement technology have reached a point where it is now feasible to consider ultrasonic flowmeters for custody transfer measurement of natural gas. The American Gas Association has released a report specifying the use of multipath transit-time ultrasonic flowmeters in this capacity (Report #9). As with the AGA's #3 (orifice plate) and #7 (turbine) high-accuracy gas flow measurement standards, the AGA9 standard requires the addition of pressure and temperature instruments on the gas line to measure gas pressure and temperature in order to calculate flow either in units of mass or in units of standardized volume (e.g. SCFM). The measurement of temperature and pressure for a transit-time ultrasonic flowmeter has nothing to do with correcting errors within the meter itself, since we know transit-time flowmeters are inherently immune to changes in gas density or composition. Temperature and pressure measurements are necessary for custody transfer applications simply because ultrasonic flowmeters, like turbine flowmeters, measure only *volumetric* flow. The fair sale and purchase of a gas requires measurement of *molecular quantity*, not just volume, which is why a flow computer requires measurements of pressure and temperature in order to convert the ultrasonic flowmeter's volumetric output to either mass flow or standardized volumetric flow.

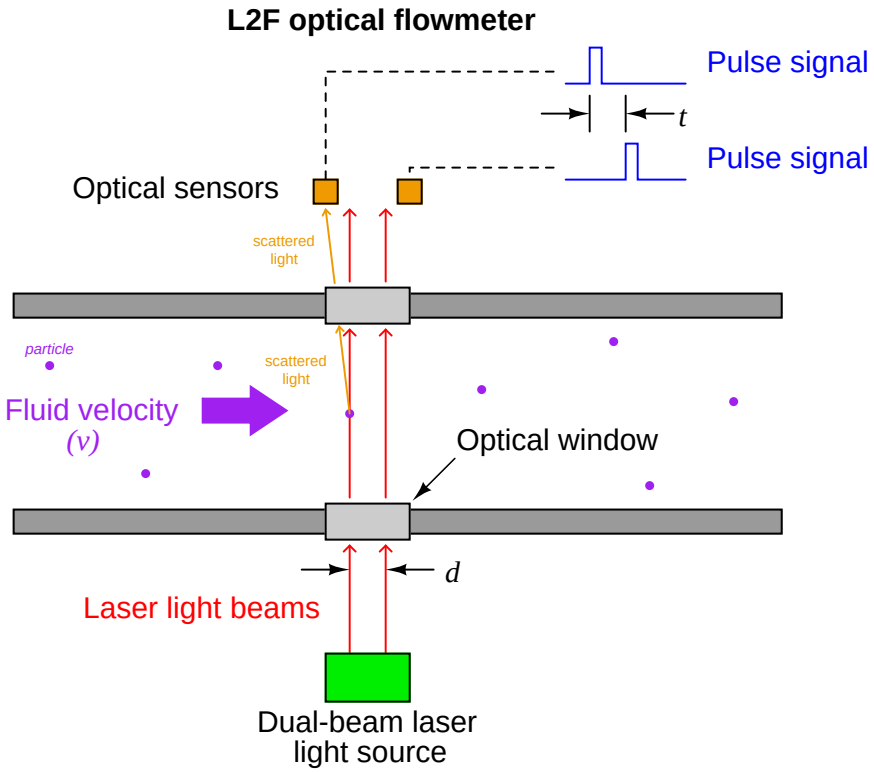
A unique advantage of ultrasonic flow measurement is the ability to measure flow through the use of temporary *clamp-on* sensors rather than a specialized flowtube with built-in ultrasonic transducers. While clamp-on sensors are not without their share of problems[54], they constitute an excellent solution for certain flow measurement applications.

An important criterion for successful application of a clamp-on flowmeter is that the pipe material be homogeneous in nature, to efficiently conduct sound waves between the process fluid and the clamp-on transducers. Porous pipe materials such as clay and concrete are therefore unsuitable for clamp-on ultrasonic flow measurement.

[54] Most notably, the problem of achieving good acoustic coupling with the pipe wall so signal transmission to the fluid and signal reception back to the sensor may be optimized. Also, there is the potential for sound waves to "ring around the pipe" instead of travel through the fluid with clamp-on ultrasonic flowmeters because the sound waves must travel through the full thickness of the pipe walls in order to enter and exit the fluid stream.

22.4.5 Optical flowmeters

A relatively recent development in industrial flow measurement is the use of light to measure the velocity of a fluid through a pipe. One such technology referred to as *Laser-Two-Focus* (L2F) uses two laser beams to detect the passage of any light-scattering particles carried along by the moving fluid:



$$v = \frac{d}{t}$$

Where,

v = Velocity of particle

d = Distance separating laser beams

t = Time difference between sensor pulses

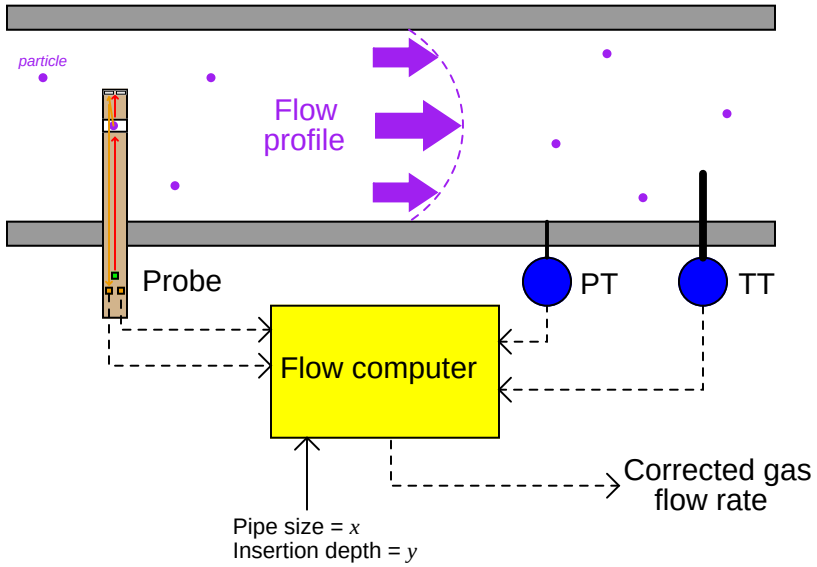
As a particle passes through each laser beam, it redirects the light away from its normal straight-line path in such a way that an optical sensor (one per beam) detects up the scattered light and generates a pulse signal. As that same particle passes through the second beam, the scattered light excites a second optical sensor to generate a corresponding pulse signal. The time delay between two successive pulses is inversely proportional to the velocity of that particle. This technique is analogous to the that used by law-enforcement officers to measure the speed of a vehicle on a highway when

viewed from an aircraft: measure how much time elapses as the vehicle passes between two marks on the road spaced a known distance from each other.

L2F flowmeters of course rely on the continual presence of light-scattering particles within the fluid. These particles could be either liquid droplets or solids within a gas stream, or they could be solid particles or bubbles in a liquid stream.

An alternative to passing laser beams across the entire width of the pipe is to shrink the assembly down to the size of a probe which may be inserted into a pipe. While minimizing installation and maintenance costs, this approach suffers the disadvantage of sensing velocity at only one point within the flow stream, much like a classic Pitot tube. In order to obtain a measurement of bulk (average) fluid velocity, the raw velocity measurement provided by the sensor must be corrected based on the expected Reynolds number for the process fluid, which is why insertion L2F flowmeters are equipped with pressure and temperature transmitters in addition to the optical probe. When the three measurements (pressure, temperature, and velocity at the probe) are combined, the Reynolds number may be calculated which then predicts how "flat" (consistent) the velocity profile is for the fluid stream:

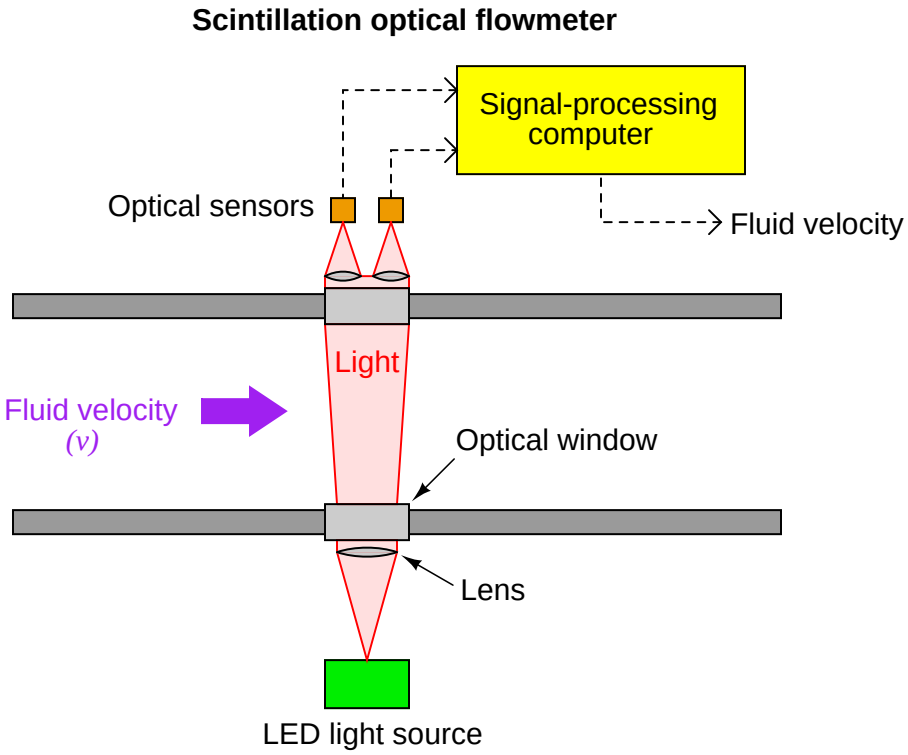
L2F insertion-style flowmeter



For example, if the probe velocity, pressure, temperature, and dimensional parameters indicate a high Reynolds number, it means the velocity profile will be relatively flat, and thus the single-point velocity measurement at the probe will be a fair representation of velocity across the entire width of the pipe. However, if the parameters indicate a low Reynolds number, it means the velocity profile will have a more pronounced "bullet" shape, and therefore the velocities near the pipe walls will be substantially less than the velocity at the center. The flow computer uses this predicted velocity profile to interpret the probe's single-point velocity measurement and calculate the average velocity of the entire flowstream.

A more sophisticated technique for optical flow measurement relies on the principle of *scintillation*, whereby the fluid itself warps the path of light passing through, rather than entrained particulate matter scattering the light. Scintillation is the same phenomenon responsible for the "twinkling" of stars and city lights viewed from a long distance: as air passes between the light and the observer, pockets of air having different density (due to differences in temperature) and/or sufficient turbulence cause some of the light to be refracted away from an otherwise straight-line path, making it appear as though the light source is randomly vibrating or oscillating. In fact, the phenomenon of scintillation has been used for the measurement of air velocity (anemometry) for many years before its application to industrial flow measurement.

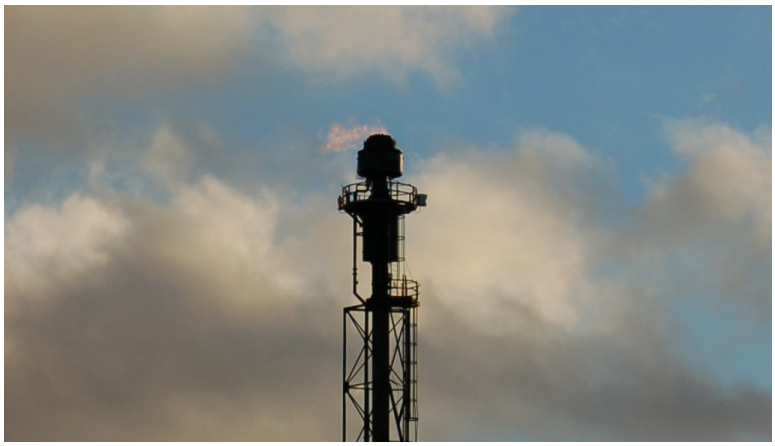
Velocity measurements are inferred by a scintillation flowmeter much the same as they are by an L2F optical flowmeter: measuring the time difference between two sensors' detection of the same scintillation pattern. Therefore, the scintillation flowmeter applies the same basic formula $v = \frac{d}{t}$ to calculate fluid velocity. A simplified diagram of a scintillation flowmeter is shown here:



Scintillation-style optical flowmeters require a long optical path in order to maximize the angle at which light will be refracted. Thus, they function best when used to measure across the full diameter of a pipe. An interesting feature of this flowmeter technology is that it functions *best* when the flow regime is highly turbulent, since increased fluid turbulence leads to greater scintillation.

Both L2F and scintillation flowmeters have phenomenal turndown ratios. At the low end of the velocity measurement range, an L2F meter is limited by the number of particles entrained in the flow stream and also by the random motions of particles which might be misinterpreted by the flowmeter as bulk motion of the fluid. Scintillation flowmeters are limited at the low end of their measurement range by loss of turbulence, which of course is one of the driving mechanisms of the scintillation effect. However, it should be noted that in both cases the low-velocity measuring limit is quite low in comparison to other types of flowmeters. With no moving parts and using light as the sensing medium, the upper limit for an optical flowmeter can extend into supersonic velocities. This combination of excellent low-velocity and high-velocity sensing yields practical turndown ratios of 1000:1 or more.

Optical flowmeters have been successfully used in one of the more challenging industrial flow measurement applications in existence: *flare gas flow metering*. A "flare" is a continually-ignited burner used as a safety relief point for flammable gases in industries such as petroleum production and refining where a need exists to occasionally vent these gases to the atmosphere:



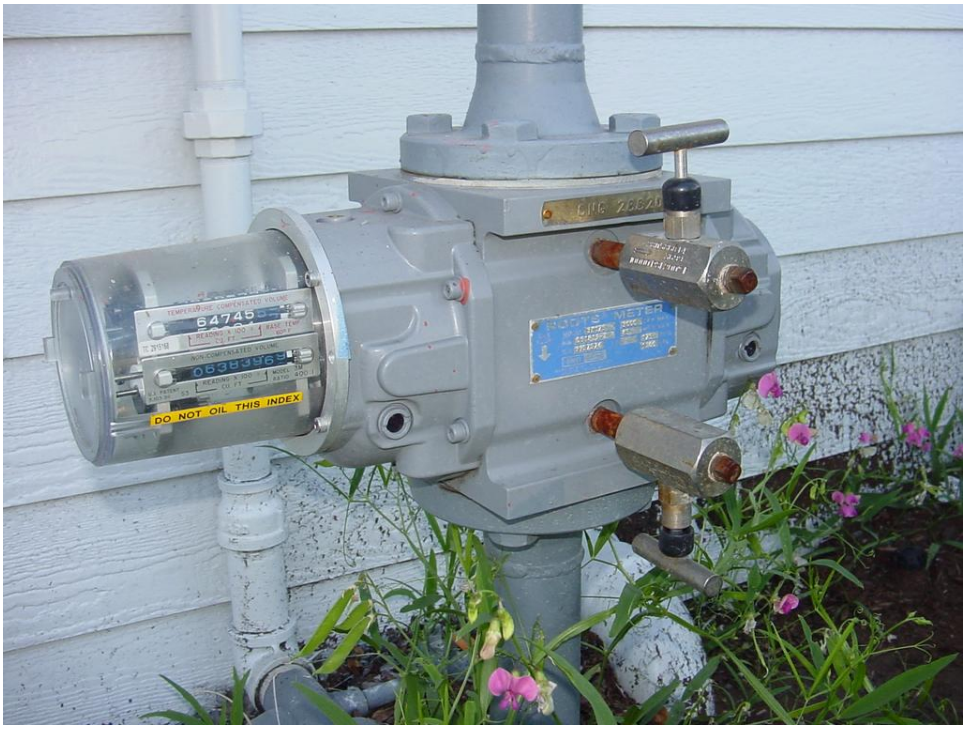
Ideally, flares operate with no flow of gas to them except during emergency conditions, and then when there is an emergency the flow rate can be very high. In order to accurately measure the low "quiescent" flow of gas to a flare during non-emergency operation (due to pressure relief valve leakage and other inefficiencies) while still being able to register full flow rates during emergencies, the flowmeter must have an extremely large turndown ratio (rangeability).

Gas sent to a flare typically varies widely in composition, temperature, and pressure, especially when multiple processing units share a common flare. The random variability of the gas eliminates many types of flowmeters from consideration. Pressure-based flowmeters such as orifice plates suffer from calibration errors due to density changes, as well as poor turndown ratio. Thermal mass flowmeters suffer from calibration errors due to changes in the specific heat value of the flare gas. Velocity-based gas flowmeters (e.g. turbine, vortex, ultrasonic) are not affected by compositional changes to the extent that pressure-based and thermal flowmeters are, but few exhibit the degree of turndown necessary to accurately measure the full range of flare gas flow. Multi-path transit-time ultrasonic flowmeters show promise in this challenging application, but their relatively high cost (especially at the large pipe diameters typical of flare headers) is a limiting factor.

22.5 Positive displacement flowmeters

A *positive displacement* flowmeter is a cyclic mechanism built to pass a fixed volume of fluid through with every cycle. Every cycle of the meter's mechanism *displaces* a precisely defined ("positive") quantity of fluid, so that a count of the number of mechanism cycles yields a precise quantity for the total fluid volume passed through the flowmeter. Many positive displacement flowmeters are rotary in nature, meaning each shaft revolution represents a certain volume of fluid has passed through the meter. Some positive displacement flowmeters use pistons, bellows, or expandable bags working on an alternating fill/dump cycle to measure discrete fluid quantities.

Positive displacement flowmeters have been the traditional choice for residential and commercial natural gas flow and water flow measurement in the United States (a simple application of *custody transfer* flow measurement, where the fluid being measured is a commodity bought and sold). The cyclic nature of a positive displacement meter lends itself well to total gas quantity measurement (and not just flow *rate*), as the mechanism may be coupled to a mechanical counter which is read by utility personnel on a monthly basis. A rotary gas flowmeter is shown in the following photograph. Note the odometer-style numerical display on the left-hand end of the meter, totalizing gas usage over time:



Positive displacement flowmeters rely on moving parts to shuttle quantities of fluid through them, and these moving parts must effectively seal against each other to prevent leakage past the mechanism (which will result in the instrument indicating less fluid passing through than there actually is). In fact, the defining characteristic of any positive displacement device is that fluid *cannot* move through without actuating the mechanism, and that the mechanism *cannot* move

without fluid passing through. This stands in contrast to machines such as centrifugal pumps and turbines, where it is possible for the moving part (the impeller or turbine wheel) to jam in place and still have fluid pass through the mechanism. If a positive displacement mechanism jams, fluid flow absolutely halts.

The finely-machined mechanical components of a positive displacement flowmeter will suffer damage from grit or other abrasive materials present in the fluid, which means these flowmeters are applicable only to clean fluid flowstreams. Even with clean fluid flowing through, the sealing surfaces of the mechanisms are subject to wear and accumulating inaccuracies over time. However, there is really nothing more definitive for measuring volumetric flow rate than an instrument built to measure individual volumes of fluid with each mechanical cycle. As one might guess, these instruments are completely immune to swirl and other large-scale fluid turbulence, and may be installed nearly anywhere in a piping system (i.e. no need for long sections of straight-length pipe upstream or downstream). So long as the fluid is clean, high viscosity poses no problem for a positive displacement flowmeter, which makes them well-suited to the measurement of such fluids as syrups, gels, and heavy oils. Positive displacement flowmeters are also very linear, since mechanism cycles are directly proportional to fluid volume.

A large positive displacement flowmeter used to measure the flow of liquid (registering total accumulated volume in units of gallons) is shown here, having been cut away for use as an instructional display:





The left-hand photograph shows the gear mechanism used to convert rotor motion into a visible total readout. The right-hand photograph shows a close-up of the interlocking rotors (one with three lobes, the other with four slots which those lobes mesh with). Both the lobes and slots are spiral-shaped, such that fluid passing along the spiral pathways must "push" the lobes out of the slots and cause the rotors to rotate. So long as there is no leakage between rotor lobes and slots, rotor turns will have a precise relationship to fluid volume passed through the flowmeter.

22.6 Standardized volumetric flow

The majority of flowmeter technologies operate on the principle of interpreting fluid flow based on the *velocity* of the fluid. Magnetic, ultrasonic, turbine, and vortex flowmeters are prime examples, where the sensing elements (of each meter type) respond directly to fluid velocity. Translating fluid velocity into volumetric flow is quite simple, following this equation:

$$Q = A\overline{v}$$

Where,

Q = Volumetric flow rate (e.g. cubic feet per minute)

A = Cross-sectional area of flowmeter throat (e.g. square feet)

$\overline{v}$ = Average fluid velocity at throat section (e.g. feet per minute)

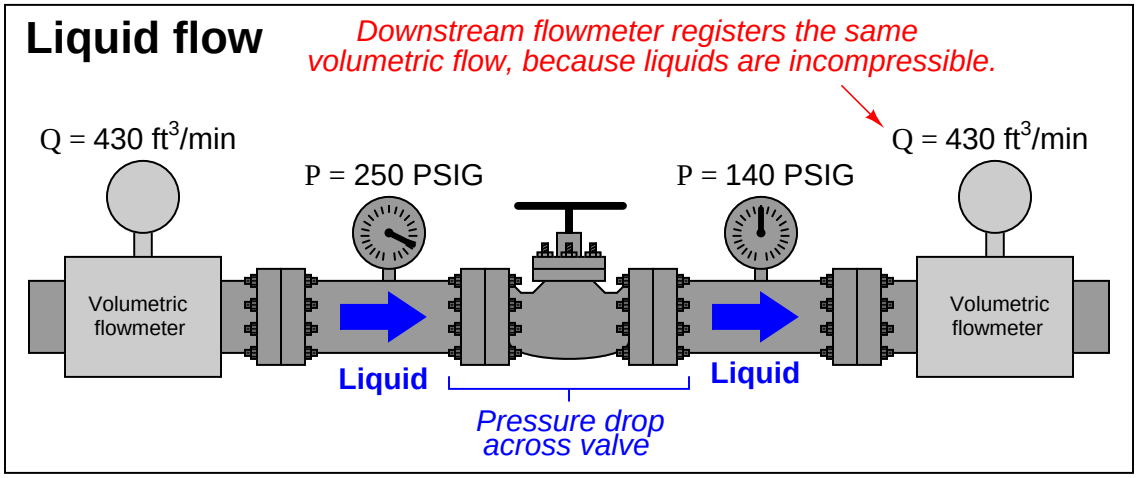
Positive displacement flowmeters are even more direct than velocity-sensing flowmeters. A positive displacement flowmeter directly measures volumetric flow, counting discrete volumes of fluid as it passes through the meter.

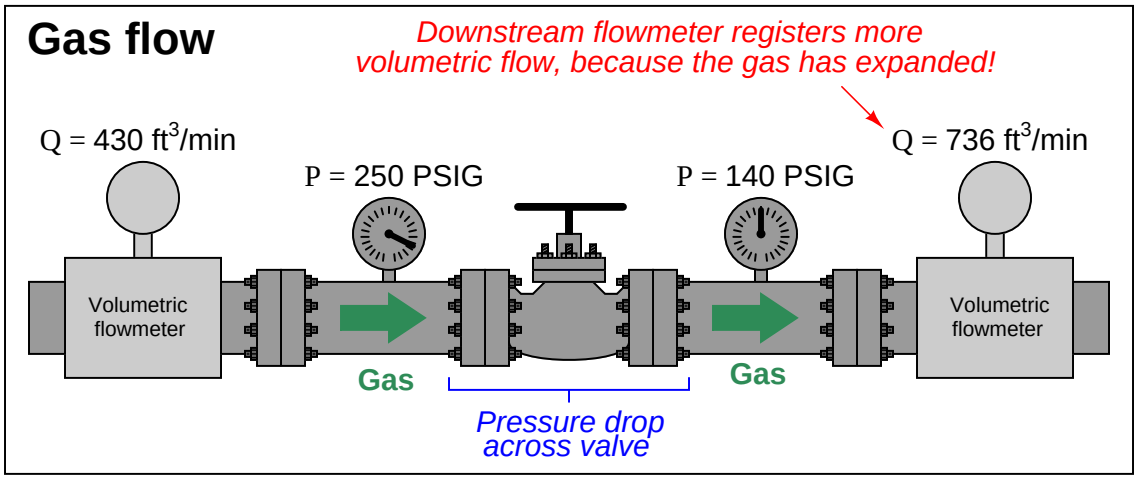
Even pressure-based flowmeters such as orifice plates and venturi tubes are usually calibrated to measure in units of volume over time (e.g. gallons per minute, barrels per hour, cubic feet per second, etc.). For a great many industrial fluid flow applications, measurement in volumetric units makes sense.

This is especially true if the fluid in question is a liquid. Liquids are essentially incompressible: that is, they do not easily yield in volume to applied pressure. This makes volumetric flow measurement relatively simple for liquids: one cubic foot of a liquid at high pressure inside a process vessel will occupy approximately the same volume (≈ 1 ft^3) when stored in a barrel at atmospheric pressure.

Gases and vapors, however, easily change volume under the influences of both pressure and temperature. In other words, a gas will yield to an increasing pressure by decreasing in volume as the gas molecules are forced closer together, and it will yield to a decreasing temperature by decreasing in volume as the kinetic energy of the individual molecules is reduced. This makes volumetric flow measurement more complex for gases than for liquids. One cubic foot of gas at high pressure and temperature inside a process vessel will *not* occupy one cubic foot under ambient pressure and temperature conditions.

The practical difference between volumetric flow measurement for liquids versus gases is easily seen through an example where we measure the volumetric flow rate before and after a pressure-reduction valve:



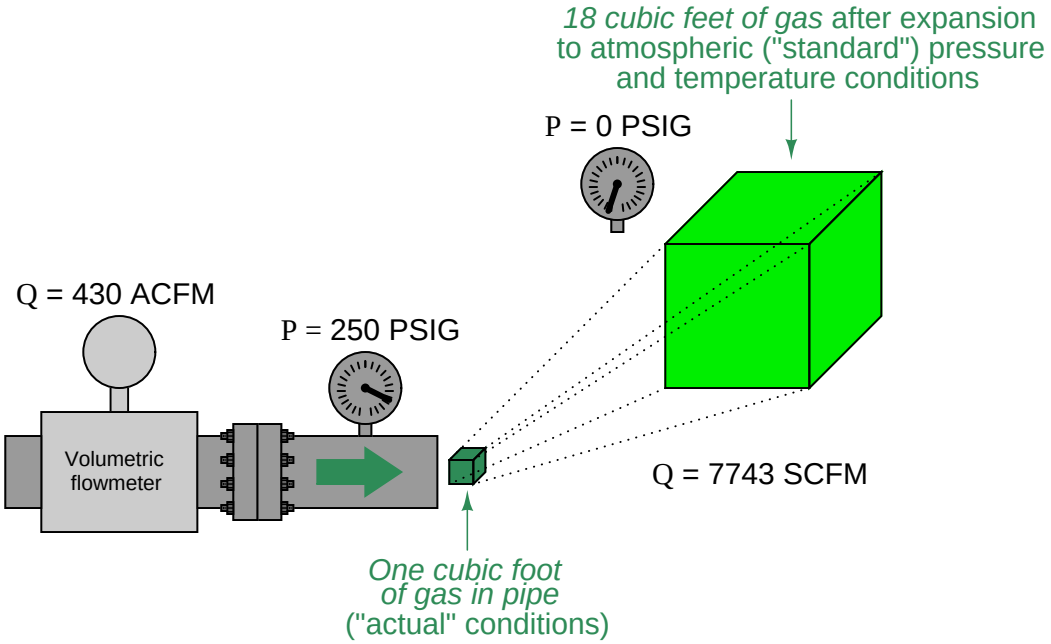


The volumetric flow rate of liquid before and after the pressure-reducing valve is the same, since the volume of a liquid does not depend on the pressure applied to it (i.e. liquids are *incompressible*). The volumetric flow rate of the gas, however, is significantly greater following the pressure-reducing valve than before, since the reduction in pressure allows the gas to expand (i.e. less pressure means the gas occupies a greater volume).

What this tells us is that volumetric flow measurement for gas is virtually meaningless without accompanying data on pressure and temperature. A flow rate of "430 ft^3/min" reported by a flowmeter measuring gas at 250 PSIG means something completely different than the same volumetric flow rate (430 ft^3/min) reported at a different line pressure.

One solution to this problem is to dispense with volumetric flow measurement altogether in favor of *mass flow measurement*, constructing the flowmeter in such a way that the actual *mass* of the gas molecules is measured as they pass through the instrument. This approach is explored in more detail in section 22.7 beginning on page 1712. A more traditional approach to this problem is to specify gas flow in volume units per time, at some agreed-upon (standardized) set of pressure and temperature conditions. This is known as *standardized* volumetric flow measurement.

Referring to our pictorial example previously shown, imagine if we took a sample of the gas flowing at a line pressure of 250 PSIG and let that sample expand to atmospheric pressure (0 PSIG), measuring its new volume under those new conditions. For the sake of keeping this example simple, we will consider only a change in pressure, but not a change in temperature. Obviously, one cubic foot of gas at 250 PSIG would expand to a far greater volume than 1 ft^3 at atmospheric pressure. This ratio of "standard volume" to "actual volume" (in the pressurized pipe) could then be used to *scale* the flowmeter's measurement, so that the flowmeter registers in *standard cubic feet per minute*, or *SCFM*:



Assuming the same temperature inside the pipe as outside, the expansion ratio for this gas will be approximately 18:1, meaning the "actual" flowing rate of 430 ft^3/min is equivalent to approximately 7743 ft^3/min of gas flow at "standard" atmospheric conditions. In order to unambiguously distinguish "actual" volumetric flow rate from "standardized" volumetric flow rate, we commonly preface each unit with a letter "A" or letter "S" (e.g. ACFM and SCFM).

If a volumetric-registering flowmeter is equipped with pressure and temperature sensors, it may automatically scale its own output signal to measure gas flow rate in standard volumetric units. All we need to determine is the mathematical procedure to scale actual conditions to standard conditions for a gas. To do this, we will refer to the Ideal Gas Law ($PV = nRT$), which is a fair approximation for most real gases at conditions far from their critical phase-change points. First, we shall write two versions of the Ideal Gas Law, one for the gas under "standard" atmospheric conditions, and one for the gas under "actual" flowing conditions (using the subscripts "S" and "A" to distinguish one set of variables from the other):

$$P_S V_S = nRT_S$$

$$P_A V_A = nRT_A$$

What we need to do is determine the *ratio* of standard volume to actual volume ($\frac{V_S}{V_A}$). To do this, we may divide one equation by the other[55]:

$$\frac{P_S V_S}{P_A V_A} = \frac{nRT_S}{nRT_A}$$

Seeing that both the n variables are identical and R is a constant, we may cancel them both from the right-hand side of the equation:

$$\frac{P_S V_S}{P_A V_A} = \frac{T_S}{T_A}$$

Solving for the ratio $\frac{V_S}{V_A}$:

$$\frac{V_S}{V_A} = \frac{P_A T_S}{P_S T_A}$$

Since we know the definition of volumetric flow (Q) is volume over time ($\frac{V}{t}$), we may divide each V variable by t to convert this into a volumetric flow *rate* correction ratio[56]:

$$\frac{\frac{V_S}{t}}{\frac{V_A}{t}} = \frac{P_A T_S}{P_S T_A}$$

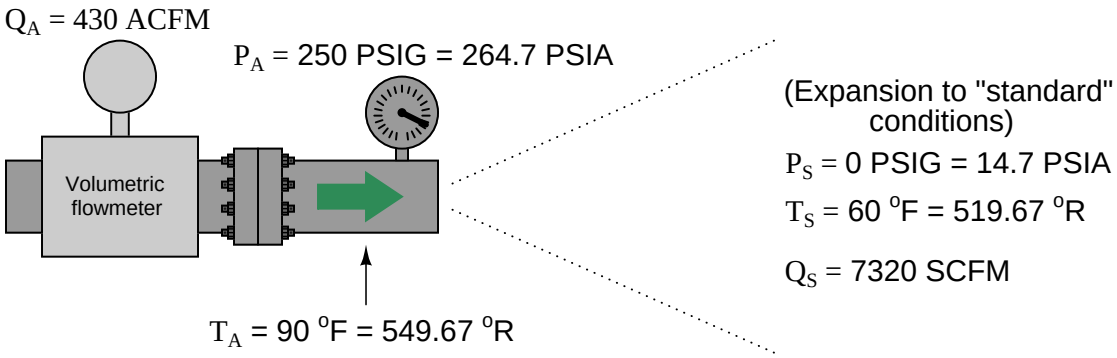
This leaves us with a ratio of "standardized" volumetric flow (Q_S) to "actual" volumetric flow (Q_A), for any known pressures and temperatures, standard to actual:

$$\frac{Q_S}{Q_A} = \frac{P_A T_S}{P_S T_A}$$

[55] Recall from algebra that we may perform any arithmetic operation we wish to any equation, so long as we apply that operation equally to both sides of the equation. Dividing one equation by another equation obeys this principle, because both sides of the second equation are equal. In other words, we could divide both sides of the first equation by $P_A V_A$ (although that would not give us the solution we are looking for), but dividing the left side by $P_A V_A$ and the right side by nRT_A is really doing the same thing, since nRT_A is identical in value to $P_A V_A$.

[56] Division by t does not alter the equation at all, since we are essentially multiplying the left-hand side by $\frac{t}{t}$ which is multiplication by 1. This is why we did not have to apply t to the right-hand side of the equation.

We may apply this to a practical example, assuming flowing conditions of 250 PSIG and 90 degrees Fahrenheit, and "standard" conditions of 0 PSIG and 60 degrees Fahrenheit. It is very important to ensure all values for pressure and temperature are expressed in *absolute* units (e.g. PSIA and degrees Rankine), which is what the Ideal Gas Law assumes:



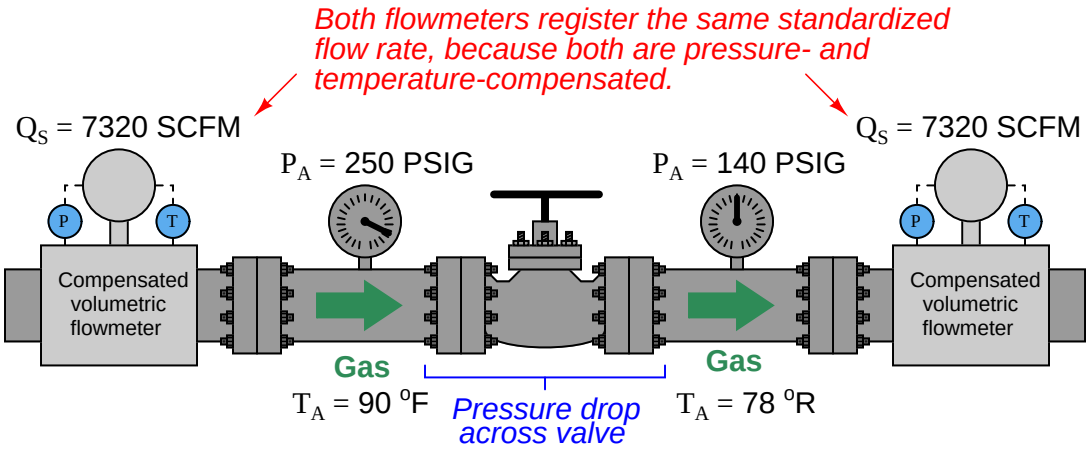
$$\frac{Q_S}{Q_A} = \frac{P_A T_S}{P_S T_A}$$

$$Q_S = Q_A \left(\frac{P_A T_S}{P_S T_A} \right)$$

$$7320 \text{ SCFM} = (430 \text{ ACFM}) \left(\frac{(264.7 \text{ PSIA})(519.67^o\text{R})}{(14.7 \text{ PSIA})(549.67^o\text{R})} \right)$$

This figure of 7320 SCFM indicates the volumetric flow rate of the gas through the pipe *had it been allowed to expand to atmospheric pressure and cool to ambient temperature*. Although we know these are definitely not the same conditions inside the gas pipe, the correction of actual volumetric flow measurement to these imagined conditions allows us to express gas flow rates in an equitable fashion regardless of the process line pressure or temperature. To phrase this in colloquial terms, standardized volumetric flow figures allow us to compare different process gas flow rates on an "apples to apples" basis, instead of the "apples to oranges" problem we faced earlier where flowmeters would register different volumetric flow values because their line pressures and/or flowing temperatures differed.

With pressure and temperature compensation integrated into volumetric flowmeters, we should be able to measure the exact same *standardized* flow rate at any point in a series gas piping system regardless of pressure or temperature changes:



An unfortunate state of affairs is the existence of multiple[57] "standard" conditions of pressure and temperature defined by different organizations. In the previous example, 14.7 PSIA was assumed to be the "standard" atmospheric pressure, and 60 degrees Fahrenheit (519.67 degrees Rankine) was assumed to be the "standard" ambient temperature. This conforms to the API (American Petroleum Institute) standards in the United States of America, but it does *not* conform to other standards in America or in Europe. The ASME (American Society of Mechanical Engineers) uses 14.7 PSIA and 68 degrees Fahrenheit (527.67 degrees Rankine) as their "standard" conditions for calculating SCFM. In Europe, the PNEUROP agency has standardized with the American CAGI (Compressed Air and Gas Institute) organization on 14.5 PSIA and 68 degrees Fahrenheit being the "standard" conditions.

For gas flows containing condensible vapors, the *partial pressures* of the vapors must be subtracted from the absolute pressures (P_A and P_S) in order that the correction factor accurately reflects gas behavior alone. A common application of standardized gas flow involving partial pressure correction is found in compressed air systems, where water vapor (relative humidity) is a factor. Here too, standards differ as to the humidity conditions of "standard" cubic feet. The API and PNEUROP/CAGI standards call for 0% humidity (perfectly dry air) as the "standard," while the ASME defines 35% relative humidity as "standard" for compressed air calculations.

[57] The wonderful thing about standards is that there are so many to choose from!

22.7 True mass flowmeters

Many traditional flowmeter technologies respond to the *volumetric flow rate* of the moving fluid. Velocity-based flowmeters such as magnetic, vortex, turbine, ultrasonic, and optical generate output signals proportional to the speed of fluid molecules and nothing else. This means that if the fluid flowing through one of these flowmeter types were to suddenly become denser (while still flowing by at the same number of volumetric units per minute), the flowmeter's response would not change at all.

The information provided by a volumetric flowmeter may not be what is actually best for the process being measured, however. If the flowmeter in question happens to be measuring the flow rate of feed into a chemical reactor vessel, for example, what we're really concerned with is *how many molecules per unit time* of feed is entering that reactor, not how many cubic meters or how many gallons. We know that changes in temperature will cause gases and liquids alike to change density, which means each volumetric unit will contain a different number of molecules after a temperature change than before. Pressure has a similar influence on gases: increased pressure means more gas molecules occupying each cubic foot (or other volumetric unit), all other factors being equal. If a process requires an accounting of molecular flow rate, a volumetric flowmeter will not provide relevant information.

In steam boiler control systems, the flow rate of water into the boiler and the flow rate of steam coming out of the boiler must be matched in order to maintain a constant quantity of water within the boiler tubes and drums. However, water is a liquid and steam is a vapor, so flow measurements based on volume are meaningless: a cubic foot of steam will *never* contain the same number of molecules as a cubic foot of water. The only reasonable way for the control system to balance both flow rates is to measure them as *mass* flows rather than volumetric flows. No matter what form (phase) the H_2O molecules take, every kilogram going into the boiler must be matched by a kilogram coming out of the boiler in accordance with the Law of Mass Conservation: every H_2O molecule entering the boiler must be matched by one H_2O molecule exiting the boiler in order to maintain an unchanging quantity of H_2O molecules within the boiler. This is why boiler feedwater and steam flowmeters alike are typically calibrated to measure in units of lbm (pounds mass) per unit time.

A similar problem arises in instances where the flowmeter is used for *custody transfer*. This term denotes scenarios where a particular material is being bought and sold, and where accuracy of flow measurement is a matter of monetary importance. Again, in such instances, it is usually the *number of molecules* being bought and sold that really matters, not how many cubic meters or gallons those molecules occupy[58]. Here, as with the chemical reactor feed flow application, a volumetric flowmeter does not provide the most relevant information.

We know from the study of chemistry that all elements have fixed mass values: one *mole*[59] of any element in monatomic form (single, unbound atoms) will have a mass equal to the atomic mass of that element. For example, one mole of carbon (C) atoms has a mass of 12 grams because the element carbon has an atomic mass of 12. Similarly, one mole of oxygen (O) atoms is guaranteed to have a mass of 16 grams[60] because 16 is the atomic mass for the element oxygen. Consequently, one

[58] In some applications, such as the custody transfer of natural gas, we are interested in something even more abstract: *heating value*. However, in order to calculate the gross heating value of a fuel gas stream, we must begin with an accurate mass flow measurement – volumetric flow is not really helpful.

[59] A "mole" is equal to a value of 6.022×10^{23} entities. Therefore, one mole of carbon atoms is 602,200,000,000,000,000,000,000 carbon atoms. For a more detailed examination of this subject, refer to section 3.7 beginning on page 262.

[60] I am purposely ignoring the fact that naturally occurring carbon has an average atomic mass of 12.011, and

mole of carbon monoxide (CO) molecules will have a mass of 28 grams (12 + 16), and one mole of carbon dioxide (CO_2) molecules will have a mass of 44 grams (12 + 16×2). These molecule/mass relationships are fixed regardless of how dense or sparse the substances are: one mole of CO_2 will have a mass of 44 grams regardless of pressure or temperature conditions affecting the density of that gas sample. The relationship between molecule count and mass for any given chemical compound is fixed, because mass is an intrinsic property of matter. If our desire is to account for the number of molecules passed through a pipe, and we happen to know the chemical composition of those molecules, measuring the *mass* of the fluid passing through is the most practical way to do it.

The mathematical relationship between volumetric flow (Q) and mass flow (W) is one of proportionality with mass density (ρ):

$$W = \rho Q$$

Dimensional analysis confirms this relationship. Volumetric flow is always measured in volume units (m^3, ft^3, cc, in^3, gallons, etc.) over time, whereas mass flow is always measured in mass units (g, kg, lbm[61], or slugs) over time. To use a specific example, a mass flow rate in pounds (mass) per minute will be obtained by multiplying a mass density in pounds per cubic foot by a volumetric flow rate in cubic feet per minute:

$$\left[\frac{\text{lbm}}{\text{min}}\right] = \left[\frac{\text{lbm}}{\text{ft}^3}\right]\left[\frac{\text{ft}^3}{\text{min}}\right]$$

For example, a volumetric flow rate of 1000 cubic feet per minute of water is equivalent to 62400 pounds (mass) per minute, or 1040 lbm/s, with water having a density of 62.4 lbm/ft^3.

With modern sensing and computational technology, it is possible to combine pressure, temperature, and volumetric flow measurements in such a way to *derive* a measurement of mass flow. This is precisely the goal with AGA3 flow measurement (orifice plates), AGA7 flow measurements (turbines), and AGA9 flow measurement (ultrasonic): "compensating" the fundamentally volumetric nature[62] of these flow-measuring elements with pressure and temperature data to calculate the flow rate in mass units over time.

However, compensated flowmeter systems require much more calibration effort to maintain their long-term accuracy, not to mention a significant capital investment in the multiple transmitters and

naturally occurring oxygen has an atomic mass of 15.9994.

[61] The British unit of the "pound" is technically a measure of *force* or *weight* and not *mass*. The proper unit of mass measurement in the British system is the "slug." However, for better or worse, the "slug" is rarely used, and so engineers have gotten into the habit of using "pound" as a mass measurement. In order to distinguish the use of "pound" to represent mass (an intrinsic property of matter) as opposed to the use of "pound" to represent weight (an incidental property of matter), the former is abbreviated *lbm* (literally, "pounds mass"). In Earth gravity, "lbm" and "lb" are synonymous. However, the standard Newtonian equation relating force, mass, and acceleration ($F = ma$) does not work when "lbm" is the unit used for mass and "lb" is used for force (it does when "slug" is used for mass and "lb" is used for force, though!). A weird unit of force invented to legitimize "pound" as an expression of mass is the *poundal* ("pdl"): one "poundal" of force is the reaction of one "pound" of mass (lbm) accelerated one foot per second squared. By this definition, a one-pound mass (1 lbm) in Earth gravity weighs 32 poundals!

[62] One could argue that orifice plates and other pressure-based flowmeters respond primarily to mass flow rather than volumetric flow, since their operation is based on the pressure created by *accelerating a mass*. However, fluid density does affect the relationship between mass flow rate and differential pressure (note how the density term ρ appears in the mass flow equation $W = k\sqrt{\rho(P_1 - P_2)}$, where it would not if differential pressure were a strict function of mass flow rate and nothing else), and so the raw output of these instruments must still be "compensated" by pressure and temperature measurements.

flow computer required to gather all the necessary data and perform the mass flow calculations. It would be much simpler if there existed flowmeter technologies naturally responsive to the mass flow rate of a fluid! Fortunately, such flowmeter technologies *do* indeed exist, which is the subject of this section.

For each of the following mass flowmeter technologies, it should be clearly understood that the instrument in question *naturally* responds to mass flow rate. To use our hypothetical example of a fluid stream whose density suddenly increases while the volumetric rate remains constant, a true mass flowmeter will immediately recognize the increase in mass flow (same volume rate, but more mass per unit volume) without the need for additional compensating measurements or computer calculations. True mass flowmeters operate on principles directly related to the mass of the fluid molecules passing through the meter, making them fundamentally different from other flowmeter types.

In the case of the Coriolis flowmeter, the instrument works on the principle of *inertia*: the force generated by an object when it is accelerated or decelerated. This basic property of mass (opposition to change in velocity) forms the basis of the Coriolis flowmeter's function. The inertial force generated inside a Coriolis flowmeter will thus double if the volumetric flowrate of a constant-mass fluid doubles; the inertial force will likewise double if the density of a constant volumetric flow of fluid doubles. Either way, the inertial force becomes a representation of *how fast mass is moving through the flowmeter*, and so the Coriolis flowmeter is a true mass flow instrument.

In the case of the thermal flowmeter, the instrument works on the principle of *convective heat transfer*: heat energy extracted from a hot object as cooler molecules pass by. The ability for fluid molecules to transport heat is a function of the *specific heat* of each molecule and the number of molecules moving past the warmer object. So long as the chemical composition of the fluid remains unchanged, the convective transfer of heat is a function of how many fluid molecules pass by in a given time. The heat transfer rate inside a thermal flowmeter will thus double if the volumetric flowrate of a given fluid doubles and all else remains constant; the heat transfer rate will likewise double if the density of a given fluid doubles and all else remains constant (i.e. twice the number of molecules passing by with each time interval). Either way, the convective heat transfer rate becomes a representation of *how many molecules of fluid are moving through the flowmeter*, which for any given fluid type is proportional to the fluid's mass flow rate. This makes the thermal flowmeter a true mass flow instrument for any (calibrated) fluid composition.

Some older, mechanical technologies[63] exist for measuring true mass flow, but these are being supplanted by Coriolis and thermal mass flowmeter technologies. Coriolis and thermal mass flowmeters are also fast becoming the technology of choice for applications formerly the domain of compensated orifice plate (e.g. AGA3) and turbine (e.g. AGA7) flowmeters.

[63] The impeller-turbine and twin-turbine mass flowmeter types are examples of mechanical true-mass flow technologies. Both work on the principle of fluid inertia. In the case of the impeller-turbine flowmeter, an impeller driven by a constant-speed electric motor imparts a "spin" to a moving fluid, which then impinges on a stationary turbine wheel to generate a measurable torque. The greater the mass flow rate, the greater the impulse force imparted to the turbine wheel. In the twin-turbine mass flowmeter, two rotating turbine wheels with different blade pitches are coupled together by a flexible coupling. As each turbine wheel attempts to spin at its own speed, the inertia of the fluid causes a differential torque to develop between the two wheels. The more mass flow rate, the greater the angular displacement (offset) between the two wheels.

22.7.1 Coriolis flowmeters

Simplified explanation

Coriolis flowmeters represent the state-of-the-art in mass flow measurement at the time of this writing (2010). While incredibly versatile and accurate, their internal operation can be difficult to understand. Put into very simple terms, a Coriolis flowmeter works by shaking one or more tubes carrying the flowing fluid, then precisely measuring the frequency and phase of that shaking. The back-and-forth shaking is driven by an electromagnetic coil, powered by an electronic amplifier circuit to shake the tube(s) at their mechanical resonant frequency. Since this frequency depends on the mass of each tube, and the mass of the tubes depends on the density of the fluid filling the fixed volume of the tubes, the resonant frequency becomes an inverse indication of fluid *density*[64] regardless of fluid flow through the tubes. As fluid begins to move through the tubes, the inertia of the moving fluid adds another dimension to the tubes' motion: the tubes begin to *undulate*[65], twisting slightly instead of just shaking back and forth. This twisting motion is directly proportional to the mass flow rate, and is internally measured by comparing the phase shift (θ) between motion at one point on the tube versus another point on the tube: the greater the undulation, the greater the phase shift between these two points' vibrations.

Expressed as proportionalities:

$$\text{Tube frequency} \propto \frac{1}{\text{Density}} \qquad f \propto \frac{1}{\rho}$$

$$\text{Tube twisting} \propto \text{Mass flowrate} \qquad \theta \propto W$$

[64] In fact, this density-measuring function of Coriolis flowmeters is so precise that they often find use *primarily* as density meters, and only secondarily as flowmeters!

[65] An interesting experiment to perform consists of holding a water hose in a U-shape and gently swinging the hose back and forth like a pendulum, then flowing water through that same hose while you continue to swing it. The hose will begin to undulate, its twisting motion becoming visually apparent.

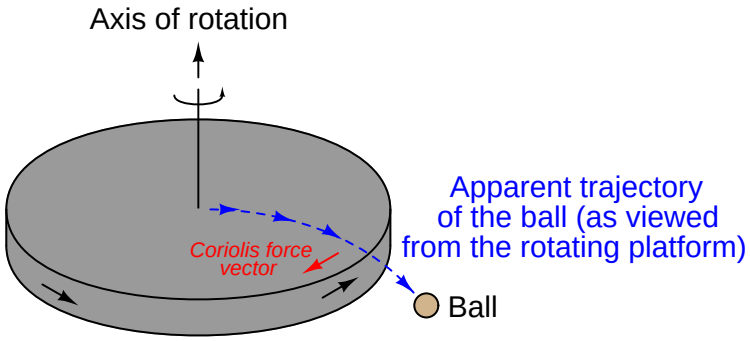
The Coriolis force

In physics, certain types of forces are classified as *fictitious* or *pseudoforces* because they only appear to exist when viewed from an accelerating perspective (called a *non-inertial reference frame*). The feeling you get in your stomach when you accelerate either up or down in an elevator, or when riding a roller-coaster at an amusement park, feels like a force acting against your body when it is really nothing more than the reaction of your body's inertia to being accelerated by the vehicle you are in. The real force is the force of the vehicle against your body, causing it to accelerate. What you perceive is merely a reaction to that force, and not the primary cause of your discomfort as it might appear to be.

Centrifugal force is another example of a "pseudoforce" because although it may appear to be a real force acting on any rotating object, it is in fact nothing more than an inertial reaction. Centrifugal force is a common experience to any child who has ever played on a "merry-go-round:" that perception of a force drawing you away from the center of rotation, toward the rim. The real force acting on any rotating object is toward the center of rotation (a *centripetal* force) which is necessary to make the object radially accelerate toward a center point rather than travel in a straight line as it normally would without any forces acting upon it. When viewed from the perspective of the spinning object, however, it would seem there is a force drawing the object away from the center (a *centrifugal* force).

Yet another example of a "pseudoforce" is the *Coriolis force*, more complicated than centrifugal force, arising from motion perpendicular to the axis of rotation in a non-inertial reference frame. The example of a merry-go-round works to illustrate Coriolis force as well: imagine sitting at the center of a spinning merry-go-round, holding a ball. If you gently toss the ball away from you and watch the trajectory of the ball, you will notice it curve rather than travel away in a straight line. In reality, the ball *is* traveling in a straight line (as viewed from an observer standing on the ground), but from your perspective on the merry-go-round, it appears to be deflected by an invisible force which we call the Coriolis force.

In order to generate a Coriolis force, we must have a mass moving at a velocity perpendicular to an axis of rotation:



The magnitude of this force is predicted by the following vector equation[66]:

$$\vec{F}_c = -2\vec{\omega} \times \vec{v}' m$$

Where,

$\vec{F}_c$ = Coriolis force vector

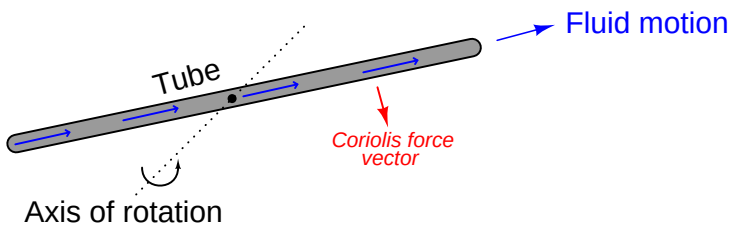
$\vec{\omega}$ = Angular velocity (rotation) vector

$\vec{v}'$ = Velocity vector as viewed from the rotating reference frame

m = Mass of the object

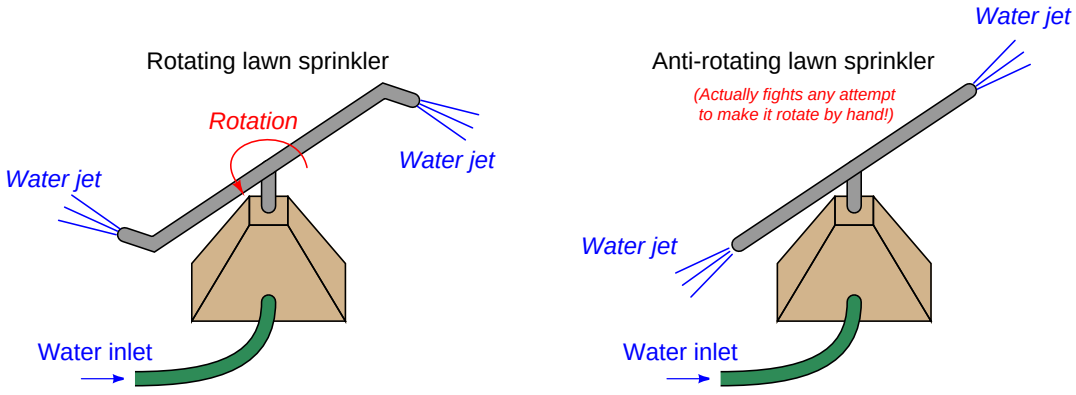
[66] This is an example of a vector *cross-product* where all three vectors are perpendicular to each other, and the directions follow the right-hand rule.

If we replace the ball with a fluid moving through a tube, and we introduce a rotation vector by tilting that tube around a stationary axis (a fulcrum), a Coriolis force develops on the tube in such a way as to oppose the direction of rotation just like the Coriolis force opposed the direction of rotation of the rotating platform in the previous illustration:



To phrase this in anthropomorphic terms, the fluid "fights" against this rotation because it "wants" to keep traveling in a straight line. For any given rotational velocity, the amount of "fight" will be directly proportional to the product of fluid velocity and fluid mass. In other words, the magnitude of the Coriolis force will be in direct proportion to the fluid's mass flow rate. This is the basis of a *Coriolis mass flowmeter*.

A demonstration of this Coriolis force may be made by modifying the nozzles on a rotary lawn sprinkler so they point straight out from the center rather than angle in one direction. As water squirts through the now-straight nozzles, they no longer generate a rotational reaction force to spin the nozzle assembly, and so the nozzles remain in place (this much should be obvious). However, if someone were to try rotating the nozzle assembly by hand, they would discover the Coriolis force *opposes* the rotation, acting to keep the nozzle assembly from rotating. The greater the mass flow rate of water through the nozzles, the stronger the inhibiting Coriolis force. Instead of a rotating lawn sprinkler, you are now the proud owner of an *anti*-rotating lawn sprinkler that actually *fights* any attempt to rotate it:

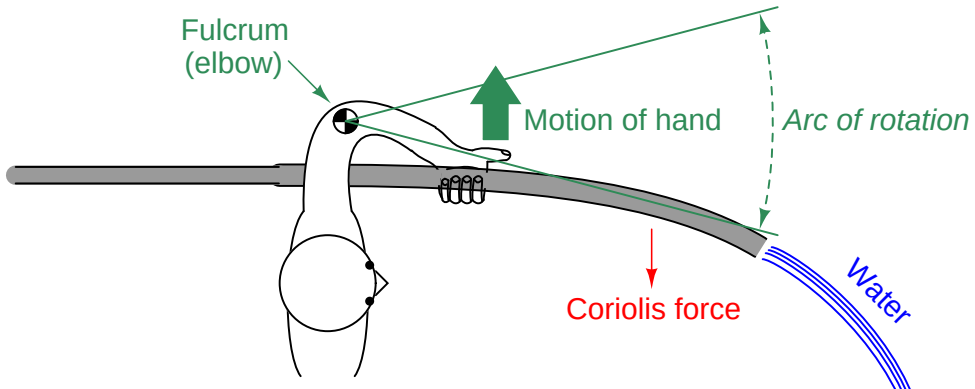


This is a very non-intuitive concept, so it deserves further explanation. The "anti-rotating" sprinkler doesn't just fail to rotate on its own – it actually *opposes* any attempt to rotate from an external force (e.g. a person trying to push the tubes by hand).

This opposition would not occur if the tubes were merely capped off at the ends and filled with stagnant water. If this were the case, the tubes would simply be heavy with the water's weight, and they would rotate freely about the axis just like any pair of heavy metal tubes would (whether hollow and filled with water, or solid metal). The tubes would have inertia, but they would not *actively* oppose any external effort to rotate.

Having liquid water *move* through the tubes is what makes the difference, and the reason becomes clear once we imagine what each water molecule experiences as it flows from the center (axis of rotation) to the nozzle at the tube tip. Each water molecule originating from the center begins with no lateral velocity, but must *accelerate* as it travels farther along the tube toward the circumference of the tips' rotation where the lateral velocity is at a maximum. The fact that new water molecules are continually making this journey from center to tip means there will always be a *new* set of water molecules requiring acceleration from center velocity (zero) to tip velocity (maximum). In capped tubes filled with stagnant water, the acceleration would only occur in getting the tubes' rotation up to speed – once there, the lateral velocity of each water molecule sitting stagnant inside the tubes would remain the same. However, with water *flowing* from center to tip, this process of acceleration from zero velocity to tip velocity must occur over and over again (continually) for each new water molecule flowing through. This continual acceleration of *new mass* is what generates the Coriolis force, and what actively opposes any external force trying to rotate the "anti-rotating" sprinkler.

As you might guess, it can be difficult to engineer a tubing system capable of spinning in circles while carrying a flowstream of pressurized fluid. To bypass the practical difficulties of building a spinning tube system, Coriolis flowmeters are instead built on the principle of a flexible tube that *oscillates* back and forth, producing the same effect in a cyclic rather than continuous fashion. The effect is not unlike shaking a hose[67] side to side as it carries a stream of water:



(This illustration is from a vertical view, looking down. The Coriolis force acts laterally, bending the hose to the side.)

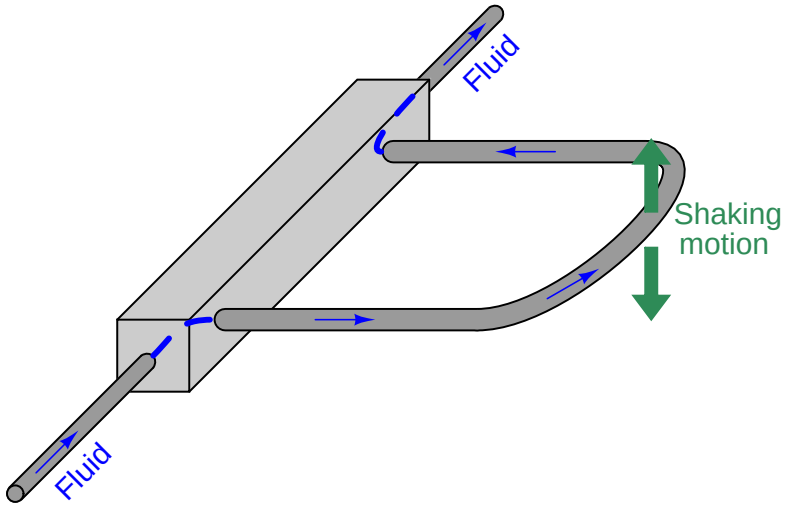
The Coriolis force opposes the direction of rotation. The greater the mass flow rate of water through the hose, the stronger the Coriolis force. If we had a way to precisely measure the Coriolis

[67] The Coriolis force generated by a flowing fire hose as firefighters work to point it in a different direction can be quite significant, owing to the high mass flow rate of the water as it flows through the hose and out the nozzle!

force imparted to the hose by the water stream, and to precisely wave the hose so its rotational velocity held constant for every wave, we could directly infer the water's mass flow rate.

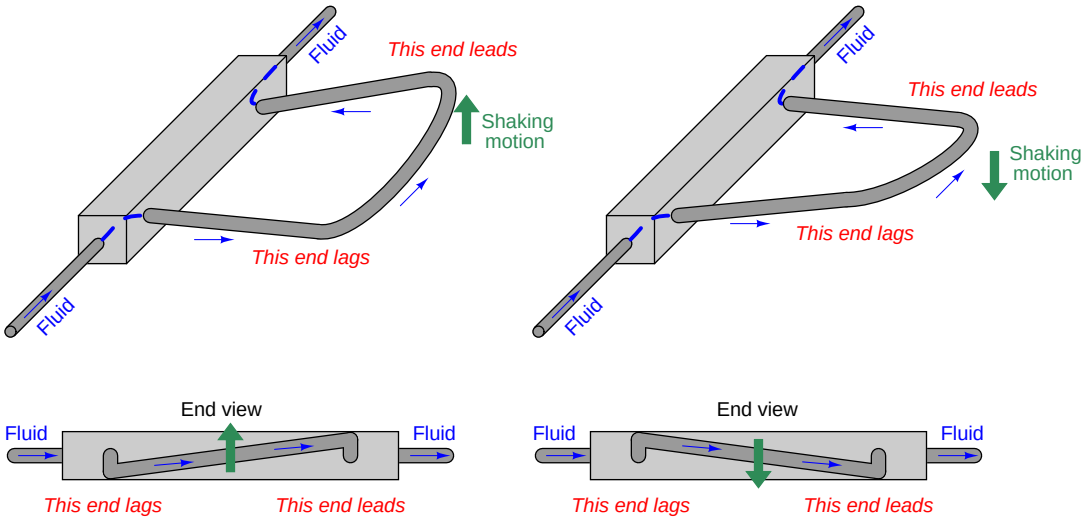
Practical Coriolis flowmeter construction

We cannot build a Coriolis flowmeter exactly like the water hose or lawn sprinkler unless we are willing to let the process fluid exit the tubing, so a common Coriolis flowmeter design uses a U-shaped tube that redirects the fluid flow back to the center of rotation. The curved end of the flexible U-tube is forced to shake back and forth by an electromagnetic force coil (like the force coil on an audio speaker) while the tube ends anchor to a stationary manifold:

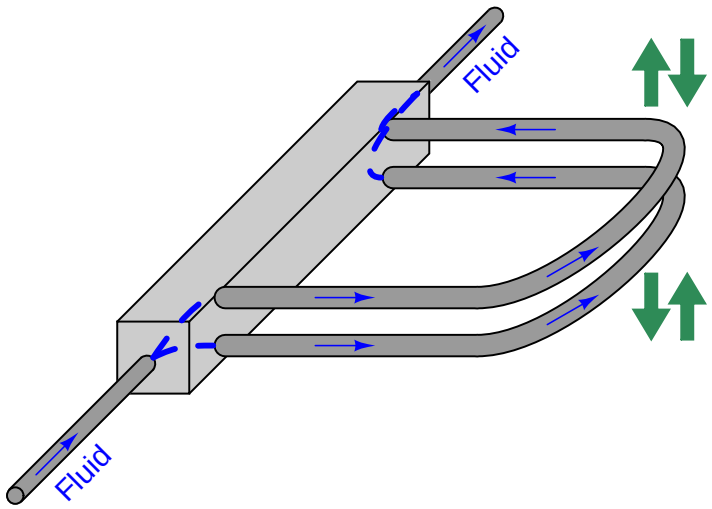


If fluid inside the tube is stagnant (no flow), the tube will simply vibrate back and forth with the applied force. However, if fluid *flows* through the tube, the moving fluid molecules will experience acceleration as they travel from the anchored base to the tube's rounded end, then experience *deceleration* as they travel back to the anchored base. This continual acceleration and subsequent deceleration of new mass generates a Coriolis force altering the tube's motion.

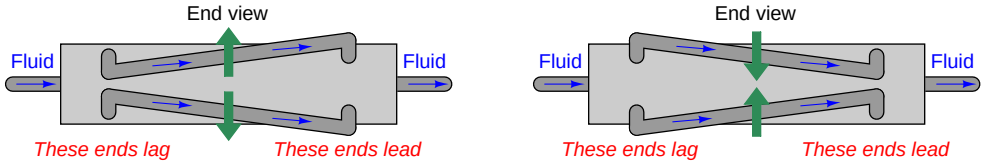
This Coriolis force causes the U-tube assembly to *twist*. The tube portion carrying fluid from the anchored base to the end tends to *lag* in motion because the fluid molecules in that section of the tube are being accelerated to a greater lateral velocity. The tube portion carrying fluid from the end back to the anchored base tends to *lead* in motion because those molecules are being decelerated back to zero lateral velocity. As mass flow rate through the tube increases, so does the degree of twisting. By monitoring the severity of this twisting motion, we may infer the mass flow rate of the fluid passing through the tube:



In order to reduce the amount of vibration generated by a Coriolis flowmeter, and more importantly to reduce the effect any external vibrations may have on the flowmeter, two identical U-tubes are built next to each other and shaken in complementary fashion (always moving in opposite directions)[68]. Tube twist is measured as *relative* motion from one tube to the next, not as motion between the tube and the stationary housing of the flowmeter. This (ideally) eliminates the effect of any common-mode vibrations on the inferred flow measurement:



Viewed from the end, the complimentary shaking and twisting of the tubes looks like this:

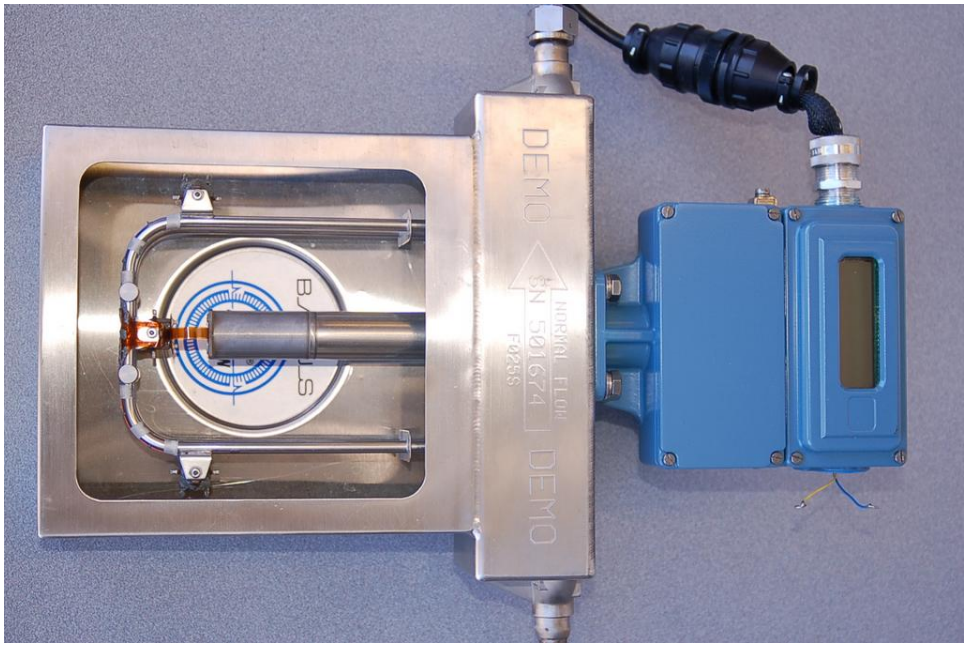


Great care is taken by the manufacturer to ensure the two tubes are as close to identical as possible: not only are their physical characteristics precisely matched, but the fluid flow is split very evenly between the tubes[69] so their respective Coriolis forces should be identical in magnitude.

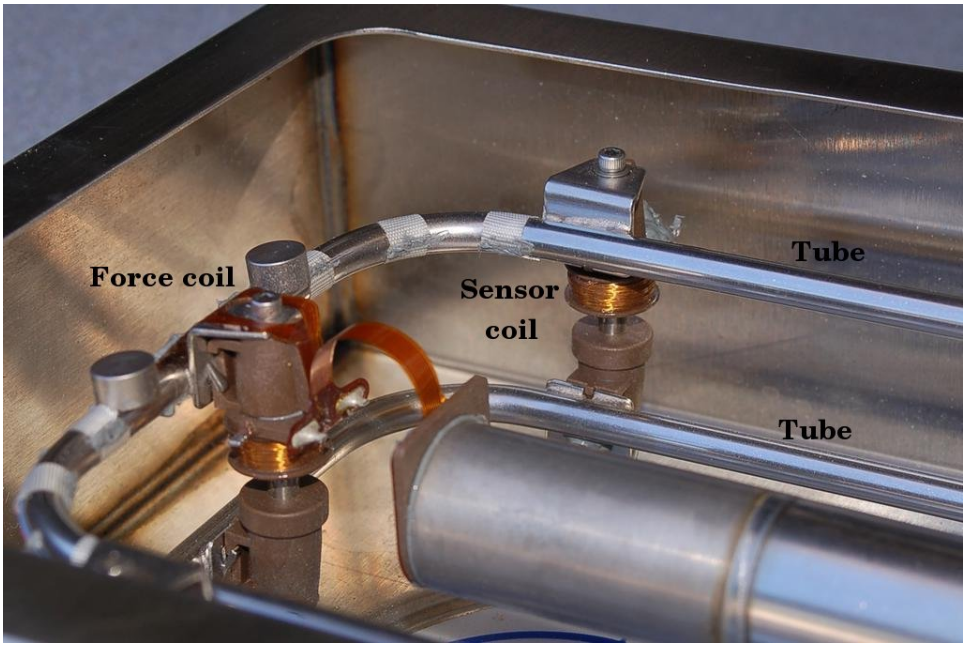
[68] For those readers with an automotive bent, this is the same principle applied in opposed-cylinder engines (e.g. Porsche "boxer" air-cooled 6-cylinder engine, Volkswagen air-cooled 4-cylinder engine, BMW air-cooled motorcycle twin engine, Citroen 2CV 2-cylinder engine, Subaru 4- and 6-cylinder opposed engines, etc.). Opposite piston pairs are *always* 180° out of phase for the purpose of maintaining mechanical balance: both moving away from the crankshaft or both moving toward the crankshaft, at any given time.

[69] An alternative to splitting the flow is to plumb the tubes in series so they *must* share the exact same flow rate, like series-connected resistors sharing the exact same amount of electrical current.

A photograph of a Rosemount (Micro-Motion) U-tube Coriolis flowmeter demonstration unit shows the U-shaped tubes (one tube is directly above the other in this picture, so you cannot tell there are actually two U-tubes):



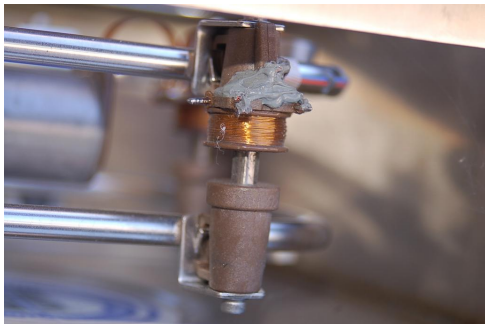
A closer inspection of this flowmeter shows that there are actually two U-tubes, one positioned directly above the other, shaken in complementary directions by a common electromagnetic force coil:



The force coil works on the same principle as an audio speaker: AC electric current passed through a wire coil generates an oscillating magnetic field, which acts against a permanent magnet's field to produce an oscillating force. In the case of an audio speaker, this force causes a lightweight cone to move, which then creates sound waves through the air. In the case of the Coriolis meter assembly, the force pushes and pulls between the two metal tubes, causing them to alternately separate and come together (shake in opposite directions).

Two magnetic displacement sensors monitor the relative motions of the tubes and transmit signals to an electronics module for digital processing. One of those sensor coils may be seen in the previous photograph. Both the force coil and the sensor coil are nothing more than permanent magnets surrounded by movable copper wire coils. The main difference between the force coil and the sensor coil is that the force coil is powered by an AC electric current to impart a vibratory force to the tubes, whereas the sensor coils are both unpowered so they can detect tube motion by generating AC voltage signals to be sensed by the electronics module. The force coil is shown in the left-hand photograph, while one of the two sensor coils appears in the right-hand photograph:





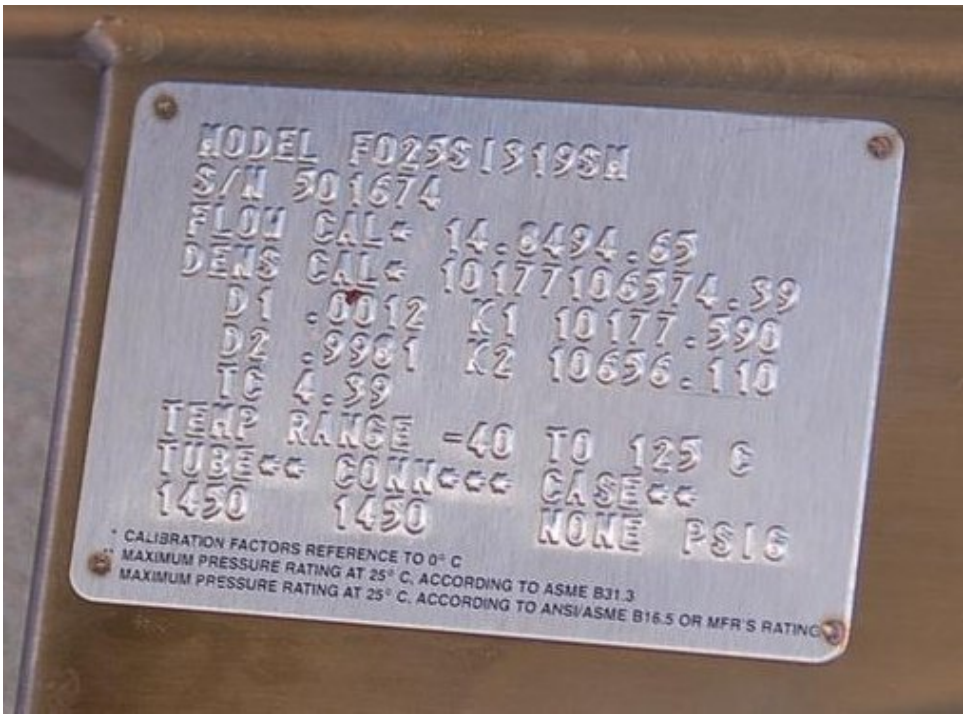
The two AC signals generated by the sensor coils provide data from which the electronics package may interpret fluid density and mass flow rate. The *frequency* of the two coils' signals is inversely related to fluid density, because a denser fluid will cause the tubes to have greater mass and therefore vibrate at a lower frequency[70]. The *phase shift* of the two coils' signals is directly related to mass flow rate, because a greater mass flow rate will cause the tubes to twist to a greater degree, causing the sensors' signals to shift further out of phase with each other.

Advances in sensor technology and signal processing have allowed the construction of Coriolis flowmeters employing straighter tubes than the U-tube unit previously illustrated and photographed. Straighter tubes are advantageous for reasons of reduced plugging potential and the ability to easily drain all liquids out of the flowmeter when needed. In straight-tube Coriolis flowmeters, we still find the same general design of a force coil flanked by matching sensor coils measuring vibration frequency (for density measurement) and phase shift (for mass flow measurement).

[70] The force coil is powered by an electronic amplifier circuit, which receives feedback from the sensor coils. Like any amplifier circuit given positive (regenerative) feedback, it will begin to oscillate at a frequency determined by the feedback network. In this case, the feedback "network" consists of the force coil, tubes, and sensor coils. The tubes, having both resilience and mass, naturally possess their own resonant frequency. This mechanical resonance dominates the feedback characteristic of the amplifier loop, causing the amplifier circuit to oscillate at that same frequency.

Matched tubes and electronics

The tubes inside a Coriolis flowmeter are not just conduits for fluid flow, they are also precision spring elements and volume chambers. As such, it is important to precisely know the spring characteristics and precise dimensions of these tubes so both mass flow and density may be inferred from tube motion. Every Coriolis flow element is factory-tested to determine the flow tubes' mechanical properties, then the electronic transmitter is programmed with these parameters describing the tubes' mechanical properties. The following photograph shows a close-up view of the nameplate on a Rosemount (Micro-Motion) Coriolis mass flowmeter, showing the physical constant values determined for that specific flowtube assembly at the time of manufacture:



This means every Coriolis flowmeter element (the tube and sensor assembly) is unique, with no two identical in behavior. Consequently, the transmitter (the electronics package outputting the process variable signals) must be programmed with values describing the element's behavior, and the complete flowmeter is shipped from the manufacturer as a *matched set*. You cannot interchange elements and transmitters without re-programming the transmitters with the new elements' physical constant values.

Density and temperature measurement

The tubes within a Coriolis flowmeter are shaken at their mechanical resonant frequency to maximize their shaking motion while minimizing electrical power applied to the force coil. The electronics module uses a feedback loop[71] between the sensor coils and the shaker coil to maintain the tubes in a continuous state of resonant oscillation. This resonant frequency changes with process fluid density, since the effective mass of the fluid-filled tubes changes with process fluid density[72], and mass is one of the variables influencing the mechanical resonant frequency of any elastic structure. Note the "mass" term in the following formula, describing the resonant frequency of a tensed string:

$$f = \frac{1}{2L}\sqrt{\frac{F_T}{\mu}}$$

Where,

f = Fundamental resonant frequency of string (Hertz)
L = String length (meters)
F_T = String tension (newtons)
μ = Unit mass of string (kilograms per meter)

A fluid-filled tube is a close analogue to a tensed string, with tube stiffness analogous to string tension and liquid density analogous to unit mass. So long as the spring constant (tube stiffness) is known, the resonant frequency of the tubes' vibration serves to indicate the unit mass of the tubes, which in turn represents fluid density given the known internal volume of the tubes.

Temperature changes have the potential to interfere with this density measurement, because temperature affects the elasticity of metal (Young's modulus) as well as the tubes' physical dimensions. This is why all Coriolis flowmeters are equipped with RTD temperature sensors to continuously monitor the temperature of the vibrating tubes. The flowmeter's microprocessor takes this tube temperature measurement and uses it to compensate for the resulting elasticity and dimensional changes based on a prior modeling of the tube metal characteristics. In other words, the flowmeter's microprocessor continuously updates the force variable (F_T) representing tube stiffness in the resonant frequency equation so that the frequency will always be a reliable indicator of unit mass (fluid density). This temperature measurement happens to be accessible as an auxiliary output signal, which means a Coriolis flowmeter may double as a (very expensive!) temperature[73] transmitter in addition to measuring mass flow rate and fluid density.

The ability to simultaneously measure three process variables (mass flow rate, temperature, and density) makes the Coriolis flowmeter a very versatile instrument indeed. This is especially true

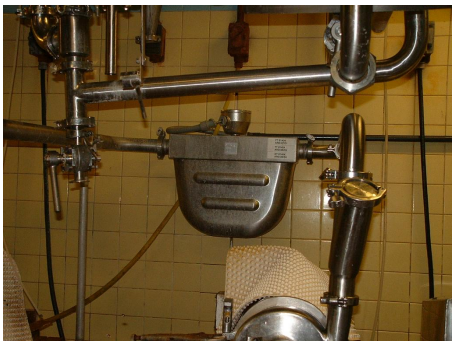
[71] This usually takes the form of a simple analog oscillator circuit, using the tubes and sensors as feedback elements. It is not unlike a *crystal* oscillator circuit where the mechanical resonance of a quartz crystal stabilizes the circuit's frequency at one value. The feedback system naturally finds and maintains resonance, just as a crystal oscillator circuit naturally finds and maintains the resonant frequency of the quartz crystal when provided with ample regenerative (positive) feedback. As fluid density inside the tubes changes, the tubes' mass changes accordingly, thus altering the resonant frequency of the system. The analog nature of this mechanical oscillator explains why some early versions of Coriolis flowmeters sometimes required a minor shake or tap to the flowtubes to start their oscillation!

[72] If you consider each tube as a container with a fixed volume capacity, a change in fluid density (e.g. pounds per cubic foot) must result in a change in mass for each tube.

[73] An important caveat is that the RTD sensing tube temperature in a Coriolis flowmeter really senses the tubes' *outside skin temperature*, which may not be exactly the same as the temperature of the fluid inside the tube. If the ambient temperature near the flowmeter differs substantially from the fluid's temperature, the tube skin temperature reading may not be accurate enough for the flowmeter to double as a fluid temperature transmitter.

when the flowmeter in question communicates digitally using a "fieldbus" standard rather than an analog 4-20 mA signal. Fieldbus communication allows multiple variables to be transmitted by the device to the host system (and/or to other devices on the same fieldbus network), allowing the Coriolis flowmeter to do the job of three instruments!

An example of a Coriolis mass flowmeter being used as a multi-variable transmitter appears in the following photographs. Note the instrument tag labels in the close-up photograph (FT, TT, and DT), documenting its use as a flow transmitter, temperature transmitter, and density transmitter, respectively:

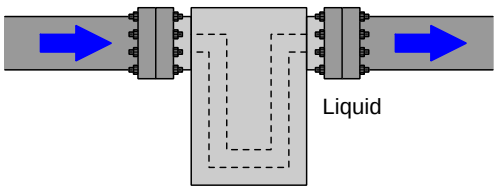


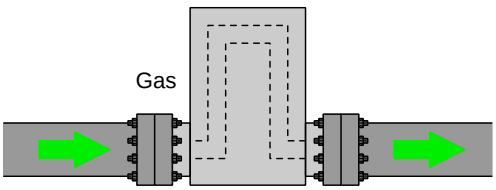


Proper installation

Although Coriolis flowmeters are immune to fluid turbulence and therefore have no upstream or downstream straight-pipe length requirements, they are still susceptible to other installation-related problems. One of these is vibration: attaching a Coriolis flowmeter a machine that vibrates, or to piping that vibrates from attachment to such a machine, can be problematic because sufficient external vibration may interfere with the resonant vibration of the flowmeter tubes, causing errors in density and/or flow measurement.

A problem unique to bent-tube Coriolis flowmeters is the entrapment of gas bubbles (in a liquid process) or liquid droplets (in a gas process). Either condition will create an uneven distribution of mass inside the flowmeter's tubes, potentially leading to measurement errors in density and/or flow. The bent tubes of a Coriolis flowmeter should be oriented such that bubbles or droplets cannot collect within them, similar to how a differential pressure sensor should be oriented in realtion to an orifice plate: for liquid processes, the bent tubes should be located below the pipe's centerline; for gas processes, the bent tubes should be located above the pipe's centerline:





Coriolis flowmeter capabilities and limitations

Even though a Coriolis flowmeter inherently measures *mass* flow rate, the continuous measurement of fluid density allows the meter to calculate *volumetric flow rate* if this is the preferred means of expressing fluid flow. The relationship between mass flow (W), volumetric flow (Q), and mass density (ρ) is quite simple:

$$W = \rho Q \qquad\qquad Q = \frac{W}{\rho}$$

All the flowmeter's computer must do to output a volumetric flow measurement is take the mass flow measurement value and divide that by the fluid's measured density. A simple exercise in dimensional analysis (performed with metric units of measurement) validates this concept for both forms of the equation shown above:

$$\left[\frac{\text{kg}}{\text{s}}\right] = \left[\frac{\text{kg}}{\text{m}^3}\right]\left[\frac{\text{m}^3}{\text{s}}\right] \qquad\qquad \left[\frac{\text{m}^3}{\text{s}}\right] = \frac{\left[\frac{\text{kg}}{\text{s}}\right]}{\left[\frac{\text{kg}}{\text{m}^3}\right]}$$

Coriolis mass flowmeters are very accurate and dependable. They are also completely immune to swirl and other fluid disturbances, which means they may be located nearly anywhere in a piping system with no need at all for straight-run pipe lengths upstream or downstream of the meter. Their natural ability to measure true mass flow, along with their characteristic linearity and accuracy, makes them ideally suited for custody transfer applications (where the flow of fluid represents product being bought and sold).

The American Gas Association (AGA) formalized the use of Coriolis mass flowmeters for the measurement of natural gas with their Report #11. This standard is equivalent to AGA #3 for orifice meters, AGA #7 for turbine meters, and AGA #9 for ultrasonic meters.

Perhaps the greatest disadvantage of Coriolis flowmeters is their high initial cost, especially for large pipe sizes. Coriolis flowmeters are also more limited in operating temperature than other types of flowmeters and may have difficulty measuring low-density fluids (gases) and mixed-phase[74] (liquid/vapor) flows. The bent tubes used to sense process flow may also trap process fluid inside to the point where it becomes unacceptable for hygienic (e.g. food processing, pharmaceuticals) applications. Straight-tube Coriolis flowmeter designs, and designs where the angle of the tubes is slight, fare better in this regard than the traditional U-tube Coriolis flowmeter design. However, a disadvantage of straight tubes is that they are stiffer than U-shaped tubes, and so straight-tube Coriolis flowmeters tend to be less sensitive to low flow rates than their U-tube counterparts.

[74] Significant technological progress has been made on mixed-phase Coriolis flow measurement, to the point where this may no longer be a serious consideration in the future.

22.7.2 Thermal flowmeters

Wind chill is a phenomenon common to anyone who has ever lived in a cold environment. When the ambient air temperature is substantially colder than the temperature of your body, heat will transfer from your body to the surrounding air. If there is no breeze to move air past your body, the air molecules immediately surrounding your body will begin to warm up as they absorb heat from your body, which will then decrease the rate of heat loss. However, if there is even a slight breeze of air moving past your body, your body will come into contact with more cool (unheated) air molecules than it would otherwise, causing a greater rate of heat loss. Thus, your perception of the surrounding temperature will be cooler than if there were no breeze.

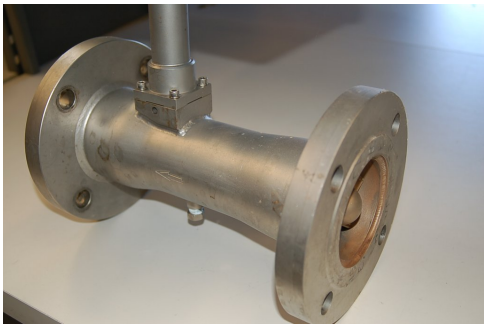
We may exploit this principle to measure mass flow rate, by placing a heated object in the midst of a fluid flowstream, and measuring how much heat the flowing fluid convects away from the heated object. The "wind chill" experienced by that heated object is a function of true mass flow rate (and not just volumetric flow rate) because the mechanism of heat loss is the rate at which fluid molecules contact the heated object, with each of those molecules having a definite mass.

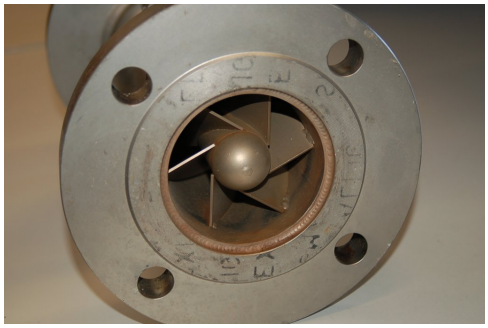
The simplest form of thermal mass flowmeter is the *hot-wire anemometer*, used to measure air speed. This flowmeter consists of a metal wire through which an electric current passes to heat it up. An electric circuit monitors the resistance of this wire (which is directly proportional to wire temperature because most metals have a definite temperature coefficient of resistance). If air speed past the wire increases, more heat will be drawn away from the wire and cause its temperature to drop. The circuit senses this temperature change and compensates by increasing current through the wire to bring its temperature back up to setpoint. The amount of electrical power required to maintain the hot wire at a constant elevated temperature is a direct function of mass air flow rate past the wire.

Most mass air flow sensors used in automotive engine control applications employ this principle. It is important for engine control computers to measure *mass* air flow and not just volumetric air flow because it is important to maintain proper air/fuel ratio even if the air density changes due to changes in altitude. In other words, the computer needs to know how many air molecules are entering the engine per second in order to properly meter the correct amount of fuel into the engine for complete and efficient combustion. The "hot wire" mass air flow sensor is simple and inexpensive to produce in quantity, which is why it finds common use in automotive applications.

Industrial thermal mass flowmeters usually consist of a specially designed "flowtube" with two temperature sensors inside: one that is heated and one that is unheated. The heated sensor acts as the mass flow sensor (cooling down as flow rate increases) while the unheated sensor serves to compensate for the "ambient" temperature of the process fluid.

A typical thermal mass flowtube appears in the following photographs (note the swirl vanes in the close-up photograph, designed to introduce large-scale turbulence into the flowstream to maximize the convective cooling effect of the fluid against the heated sensor element):





Thermal mass flowmeters lend themselves well to "insertion" style probes, sensing the passage of fluid molecules at one point within the flowstream. An example is shown in the next two photographs, where a thermal mass flowmeter (manufactured by Sage) senses the amount of gas sent to a flare. The insertion probe appears in the left-hand photo (mounted in the vertical flare pipe) while the transmitter head appears in the right-hand photo (located inside of a weather-sheltered building):





The simple construction of thermal mass flowmeters allows them to be manufactured in very small sizes. The following photograph shows a small device that is not only a mass flow meter, but also a mass flow *controller* with its own built-in throttling valve mechanism and control electronics. To give you a sense of scale, the tube fittings seen on the left- and right-hand sides of this device are 1/4 inch, making this photograph nearly full-size:



An important factor in the calibration of a thermal mass flowmeter is the *specific heat* of the process fluid. "Specific heat" is a measure of the amount of heat energy needed to change the temperature of a standard quantity of substance by some specified amount[75]. Some substances have much greater specific heat values than others, meaning those substances have the ability to absorb (or release) a lot of heat energy without experiencing a great temperature change. Fluids with high specific heat values make good *coolants*, because they are able to remove much heat energy from hot objects without experiencing great increases in temperature themselves. Since thermal mass

[75]For example, the specific heat of water is 1.00 kcal / kg $\cdot$ ^{o}C, meaning that the addition of 1000 calories of heat energy is required to raise the temperature of 1 kilogram of water by 1 degree Celsius, or that we must remove 1000 calories of heat energy to cool that same quantity of water by 1 degree Celsius. Ethyl alcohol, by contrast, has a specific heat value of only 0.58 kcal / kg $\cdot$ ^{o}C, meaning it is almost twice as easy to warm up or cool down as water (little more than half the energy required to heat or cool water needs to be transferred to heat or cool the same mass quantity of ethyl alcohol by the same amount of temperature).

flowmeters work on the principle of convective cooling, this means a fluid having a high specific heat value will elicit a greater response from a thermal mass flowmeter than the exact same mass flow rate of a fluid having a lesser specific heat value (i.e. a fluid that is not as good of a coolant).

This means we must know the specific heat value of whatever fluid we plan to measure with a thermal mass flowmeter, and we must be assured its specific heat value will remain constant. For this reason, thermal mass flowmeters are not suitable for measuring the flow rates of fluid streams whose chemical composition is likely to change over time. This limitation is analogous to that of a pressure sensor used to hydrostatically measure the level of liquid in a vessel: in order for this level-measurement technique to be accurate, we must know the density of the liquid and also be assured that density will be constant over time.

Thermal mass flowmeters are simple and reliable instruments. While not as accurate or tolerant of piping disturbances as Coriolis mass flowmeters, they are far less expensive.

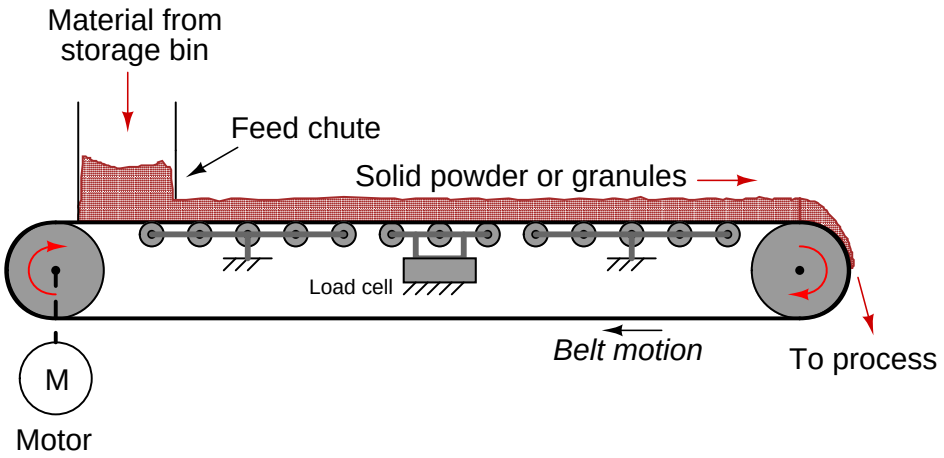
Perhaps the greatest disadvantage of thermal mass flowmeters is their sensitivity to changes in the specific heat of the process fluid. This makes the calibration of any thermal mass flowmeter specific for one composition of fluid only. In some applications such as automotive engine intake air flow, where the fluid composition is constant, this limitation is not a factor. In many industrial applications, however, this limitation is severe enough to prohibit the use of thermal mass flowmeters. Industrial applications for thermal mass flowmeters include natural gas flow measurement (non-custody transfer), and the measurement of purified gas flows (oxygen, hydrogen, nitrogen) where the composition is known to be very stable.

Another (potential) limitation of thermal mass flowmeters is the sensitivity of some designs to changes in flow regime. Since the measurement principle is based on heat transfer by fluid convection, any factor influencing the convective heat-transfer efficiency will translate into a perceived difference in mass flow rate. It is a well-known fact in fluid mechanics that turbulent flows are more efficient at convecting heat than laminar flows, because the "stratified" nature of a laminar flowstream impedes heat transfer across the fluid width[76]. In some thermal flowmeter designs, the walls of a heated metal *tube* serve as the "hot" element cooled by the fluid, and the difference between the rate of heat transferred by a laminar flowstream from the walls of a heated tube versus a turbulent flowstream can be great. Therefore, a change in flow regime (from turbulent to laminar, and vice-versa) will cause a calibration shift for this design of thermal mass flowmeter.

[76] In a laminar flowstream, individual molecules do not cross paths, but rather travel in parallel lines. This means only those molecules traveling near the wall of a tube will be exposed to the temperature of the wall. The lack of "mixing" in a laminar flowstream means molecules traveling in the inner portions of the stream never contact the tube wall, and therefore never serve to transfer heat directly to or from the wall. At best, those inner-path molecules transfer heat by *conduction* with adjacent molecules which is a less efficient transfer mechanism than convection.

22.8 Weighfeeders

A special type flowmeter suited for powdered or granular solids is the *weighfeeder*. One of the most common weighfeeder designs consists of a conveyor belt with a section supported by rollers coupled to one or more load cells, such that a fixed length of the belt is continuously weighed:



The load cell measures the weight of a fixed-length belt section, yielding a figure of material weight per linear distance on the belt. A tachometer (speed sensor) measures the speed of the belt. The product of these two variables is the mass flow rate of solid material "through" the weighfeeder:

$$W = \frac{Fv}{d}$$

Where,

W = Mass flow rate (e.g. pounds per second)

F = Force of gravity acting on the weighed belt section (e.g. pounds)

v = Belt speed (e.g. feet per second)

d = Length of weighed belt section (e.g. feet)

A small weighfeeder (about two feet in length) is shown in the following photograph, the weighfeeder being used to feed powdered soda ash into water at a municipal filtration plant to neutralize pH:

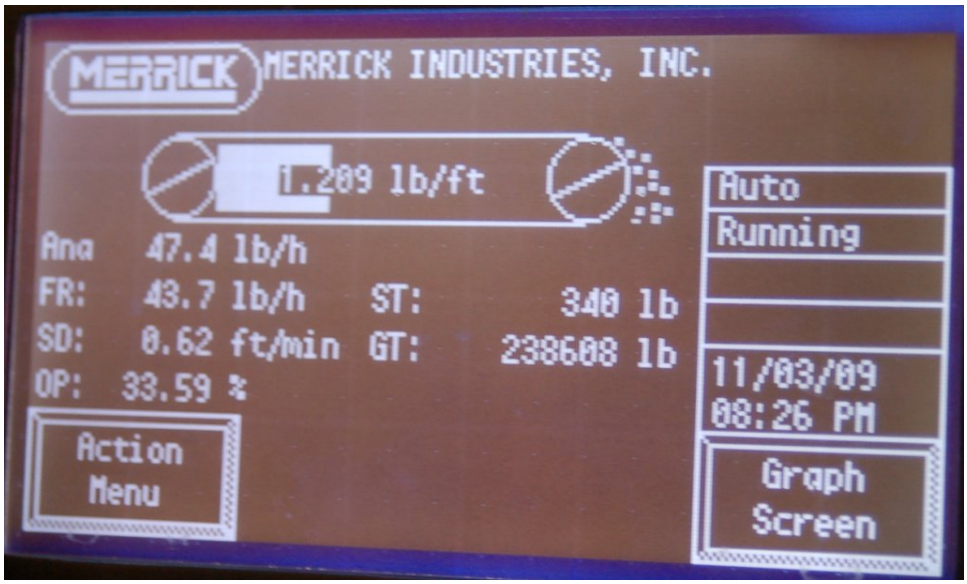


In the middle of the belt's span (hidden from view) is a set of rollers supporting the weight of the belt and of the soda ash piled on the belt. This load cell array provides a measurement of pounds material per foot of belt length (lb/ft).

As you can see in this next picture, the soda ash powder simply falls off the far end of the conveyor belt, into the water below:



The speed sensor measures belt speed in feet per minute (ft/min). This measurement, when multiplied by the pounds-per-foot measurement sensed by the load cells, translates into a mass flow rate (W) in units of pounds per minute (lb/min). A simple unit conversion ($\times 60$) expresses the mass flow rate in units of pounds per hour (lb/h). A photograph of this weighfeeder's display screen shows these variables:



Note that the belt loading of 1.209 lb/ft and the belt speed of 0.62 feet per minute do *not* exactly equate[77] to the displayed mass flow rate of 43.7 lb/h. The reason for this discrepancy is that the camera's snapshot of the weighfeeder display screen happened to capture an image where the values were not simultaneous. Weighfeeders often exhibit fluctuations in belt loading during normal operation, leading to fluctuations in calculated mass flow rate. Sometimes these fluctuations in measured and calculated variables do not coincide on the display screen, given the latency inherent to the mass flow calculation (delaying the flow rate value until after the belt loading has been measured and displayed).

[77] The proper mass flow rate value corresponding to these two measurements would be 45.0 lb/h.

22.9 Change-of-quantity flow measurement

Flow, by definition, is the passage of material from one location to another over time. So far this chapter has explored technologies for measuring flow rate en route from source to destination. However, a completely different method exists for measuring flow rates: measuring how much material has either departed or arrived at the terminal locations over time.

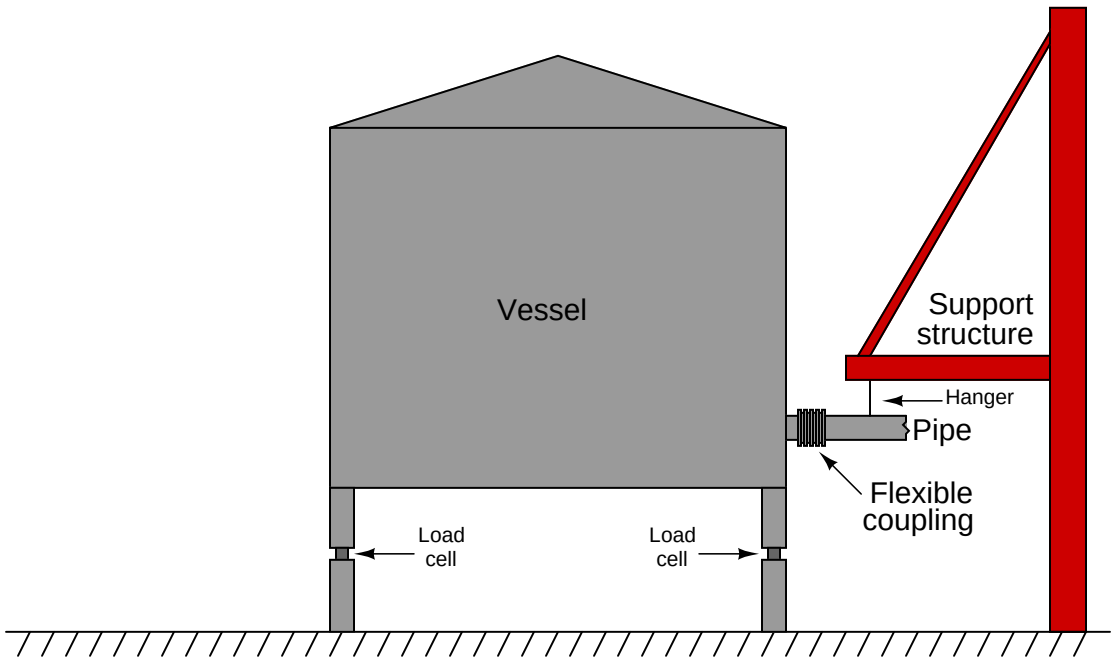
Mathematically, we may express flow as a ratio of quantity to time. Whether it is volumetric flow or mass flow we are referring to, the concept is the same: quantity of material moved per quantity of time. We may express average flow rates as ratios of changes:

$$\overline{W} = \frac{\Delta m}{\Delta t} \qquad\qquad \overline{Q} = \frac{\Delta V}{\Delta t}$$

Where,

$\overline{W}$ = Average mass flow rate
$\overline{Q}$ = Average volumetric flow rate
Δm = Change in mass
ΔV = Change in volume
Δt = Change in time

Suppose a water storage vessel is equipped with load cells to precisely measure weight (which is directly proportional to mass with constant gravity). Assuming only one pipe entering or exiting the vessel, any flow of water through that pipe will result in the vessel's total weight changing over time:



If the measured mass of this vessel decreased from 74688 kilograms to 70100 kilograms between 4:05 AM and 4:07 AM, we could say that the average mass flow rate of water leaving the vessel is 2294 kilograms per minute over that time span.

$$\overline{W} = \frac{\Delta m}{\Delta t} = \frac{70100 \text{ kg} - 74688 \text{ kg}}{4{:}07 - 4{:}05} = \frac{-4588 \text{ kg}}{2 \text{ min}} = -2294 \frac{\text{kg}}{\text{min}}$$

Note that this average flow measurement may be determined without any flowmeter of any kind installed in the pipe to intercept the water flow. All the concerns of flowmeters studied thus far (turbulence, Reynolds number, fluid properties, etc.) are completely irrelevant. We may measure practically any flow rate we desire simply by measuring stored weight (or volume) over time. A computer may do this calculation automatically for us if we wish, on practically any time scale desired.

Now suppose the practice of determining average flow rates every two minutes was considered too infrequent. Imagine that operations personnel require flow data calculated and displayed more often than just 30 times an hour. All we must do to achieve better time resolution is take weight (mass) measurements more often. Of course, each mass-change interval will be expected to be less with more frequent measurements, but the amount of time we divide by in each calculation will be proportionally smaller as well. If the flow rate happens to be absolutely steady, we may sample mass as frequently as we might like and we will still arrive at the same flow rate value as before (sampling mass just once every two minutes). If, however, the flow rate is not steady, sampling more often will allow us to better see the immediate "ups" and "downs" of flow behavior.

Imagine now that we had our hypothetical "flow computer" take weight (mass) measurements at an infinitely fast pace: an infinite number of samples per second. Now, we are no longer *averaging* flow rates over finite periods of time; instead we would be calculating *instantaneous* flow rate at any given *point* in time.

Calculus has a special form of symbology to represent such hypothetical scenarios: we replace the Greek letter "delta" (Δ, meaning "change") with the roman letter "d" (meaning *differential*). A simple way of picturing the meaning of "d" is to think of it as meaning an *infinitesimal* change in whatever variable follows the "d" in the equation[78]. When we set up two differentials in a quotient, we call the $\frac{d}{d}$ fraction a *derivative*. Re-writing our average flow rate equations in derivative (calculus) form:

$$W = \frac{dm}{dt} \qquad\qquad Q = \frac{dV}{dt}$$

Where,

W = Instantaneous mass flow rate

Q = Instantaneous volumetric flow rate

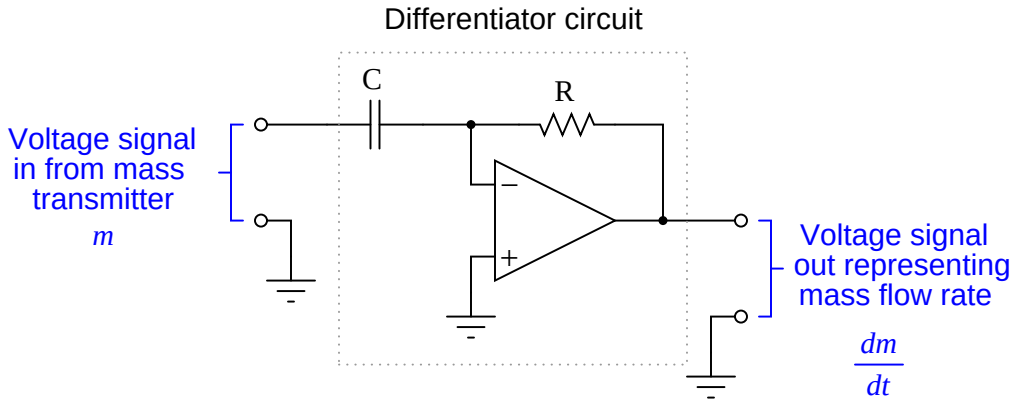
dm = Infinitesimal (infinitely small) change in mass

dV = Infinitesimal (infinitely small) change in volume

dt = Infinitesimal (infinitely small) change in time

[78] While this may seem like a very informal definition of differential, it is actually rooted in a field of mathematics called *nonstandard analysis*, and closely compares with the conceptual notions envisioned by calculus' founders.

We need not dream of hypothetical computers capable of infinite calculations per second in order to derive a flow measurement from a mass (or volume) measurement. Analog electronic circuitry exploits the natural properties of resistors and capacitors to essentially do this very thing in real time[79]:



In the vast majority of applications you will see digital computers used to calculate average flow rates rather than analog electronic circuits calculating instantaneous flow rates. The broad capabilities of digital computers virtually ensures they will be used somewhere in the measurement/control system, so the rationale is to use the existing digital computer to calculate flow rates (albeit imperfectly) rather than complicate the system design with additional (analog) circuitry. As fast as modern digital computers are able to process simple calculations such as these anyway, there is little practical reason to prefer analog signal differentiation except in specialized applications where high speed performance is paramount.

Perhaps the single greatest disadvantage to inferring flow rate by differentiating mass or volume measurements over time is the requirement that the storage vessel have only one flow path in and out. If the vessel has multiple paths for liquid to move in and out (simultaneously), any flow rate calculated on change-in-quantity will be a *net* flow rate only. It is impossible to use this flow measurement technique to measure one flow out of multiple flows common to one liquid storage vessel.

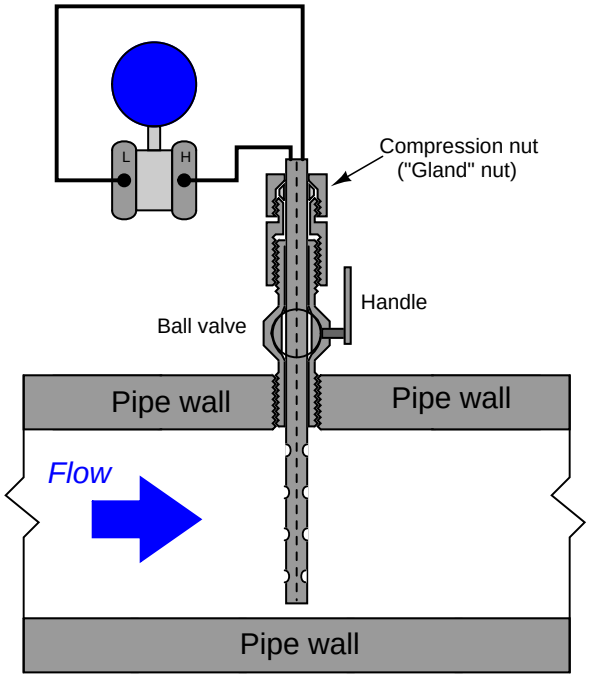
A simple "thought experiment" confirms this fact. Imagine a water storage vessel receiving a flow rate in at 200 gallons per minute. Next, imagine that same vessel emptying water out of a second pipe at the exact same flow rate: 200 gallons per minute. With the exact same flow rate both entering and exiting the vessel, the water level in the vessel will remain constant. Any change-of-quantity flow measurement system would register zero change in mass or volume over time, consequently calculating a flow rate of absolutely zero. Truly, the *net* flow rate for this vessel is zero, but this tells us nothing about the flow in each pipe, except that those flow rates are equal in magnitude and opposite in direction.

[79] To be precise, the equation describing the function of this analog differentiator circuit is: $V_{out} = -RC\frac{dV_{in}}{dt}$. The negative sign is an artifact of the circuit design – being essentially an inverting amplifier with negative gain – and not an essential element of the math.

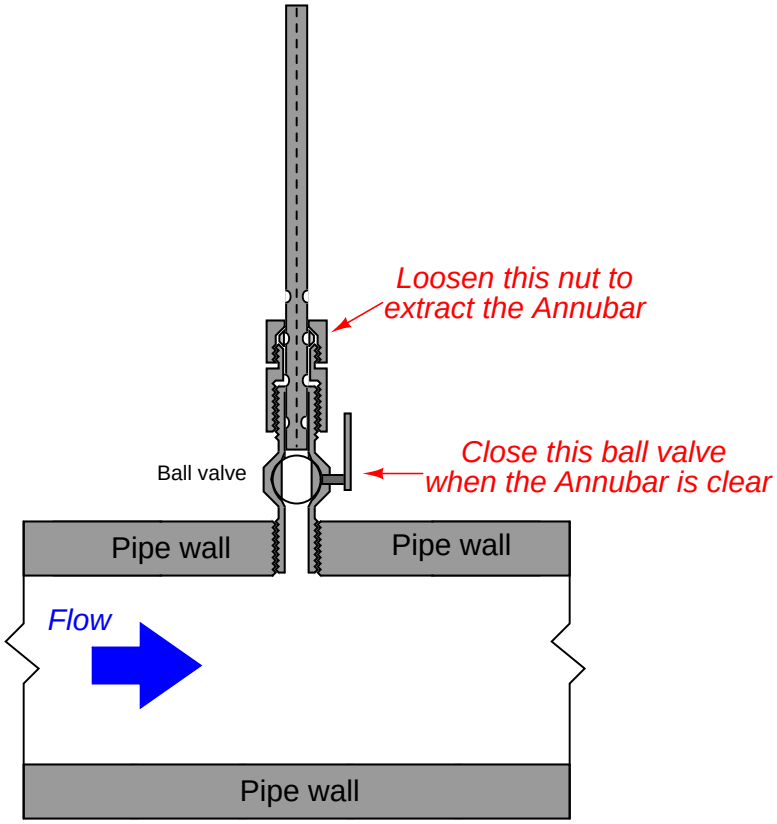
22.10 Insertion flowmeters

This section does not describe a particular type of flowmeter, but rather a design that may be implemented for several different kinds of flow measurement technologies. When the pipe carrying process fluid is large in size, it may be impractical or cost-prohibitive to install a full-diameter flowmeter to measure fluid flow rate. A practical alternative for many applications is the installation of an *insertion* flowmeter: a probe that may be inserted into or extracted from a pipe, to measure fluid velocity in one region of the pipe's cross-sectional area (usually the center).

A classic example of an insertion flowmeter element is the *Annubar*, a form of averaging pitot tube pioneered by the Dieterich Standard corporation. The Annubar flow element is inserted into a pipe carrying fluid where it generates a differential pressure for a pressure sensor to measure:

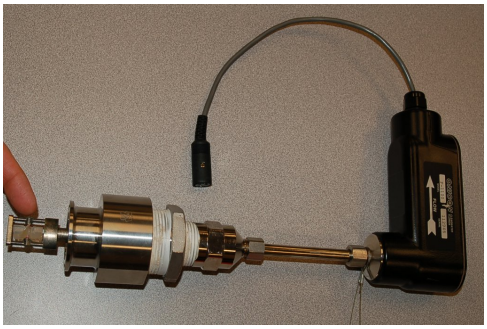


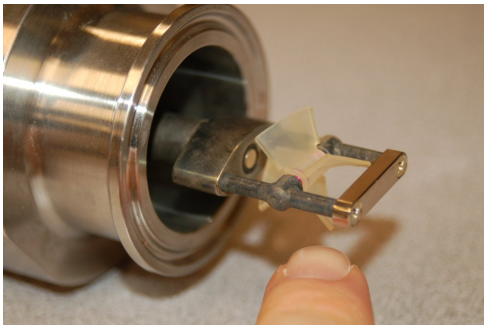
The Annubar element may be extracted from the pipe by loosening a "gland nut" and pulling the assembly out until the end passes through a hand ball valve. Once the element has been extracted this far, the ball valve may be shut and the Annubar completely removed from the pipe:



For safety reasons, a "stop" is usually built into the assembly to prevent someone from accidently pulling the element all the way out with the valve still open.

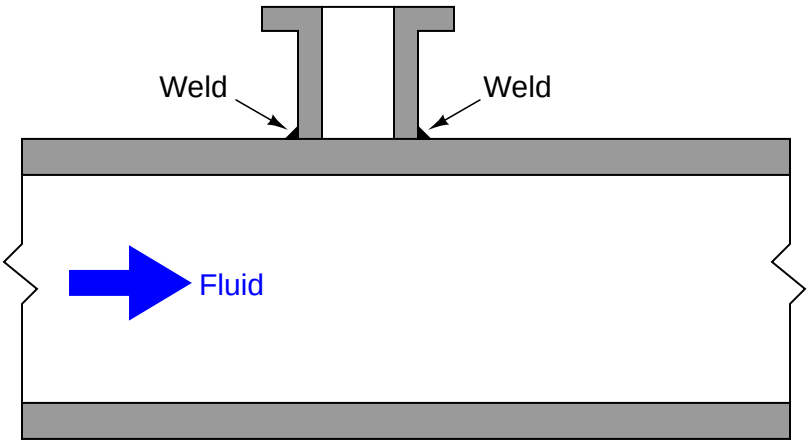
Other flowmeter technologies manufactured in insertion form include vortex, turbine, and thermal mass. An insertion-type turbine flowmeter appears in the following photographs:



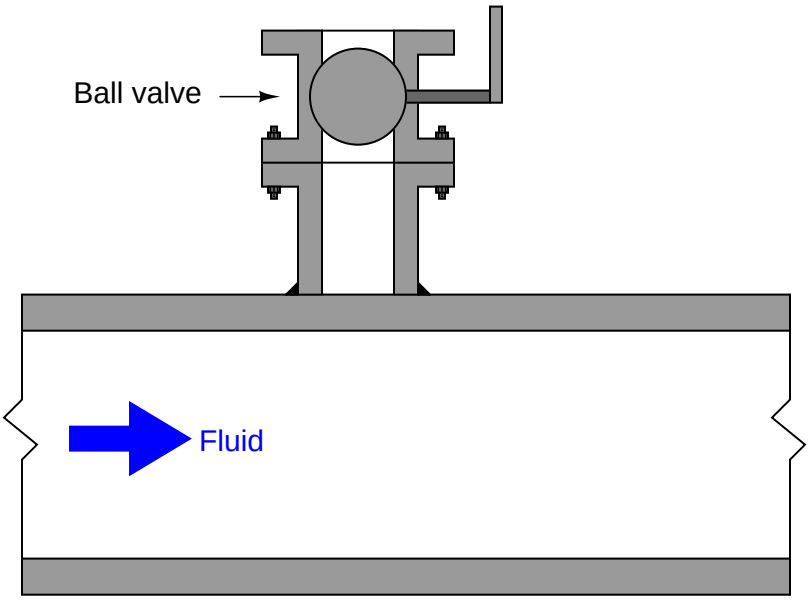


If the flow-detection element is compact rather than distributed (as is certainly the case with the turbine flowmeter shown above), care must be taken to ensure correct positioning within the pipe. Since flow profiles are never completely flat, any insertion meter element will register a greater flow rate at the center of the pipe than near the walls. Wherever the insertion element is placed in the pipe diameter, that placement must remain consistent through repeated extractions and re-insertions or else the effective calibration of the insertion flowmeter will change every time it is removed and re-inserted into the pipe. Care must also be taken to insert the flowmeter so the flow element points directly upstream, and not at an angle.

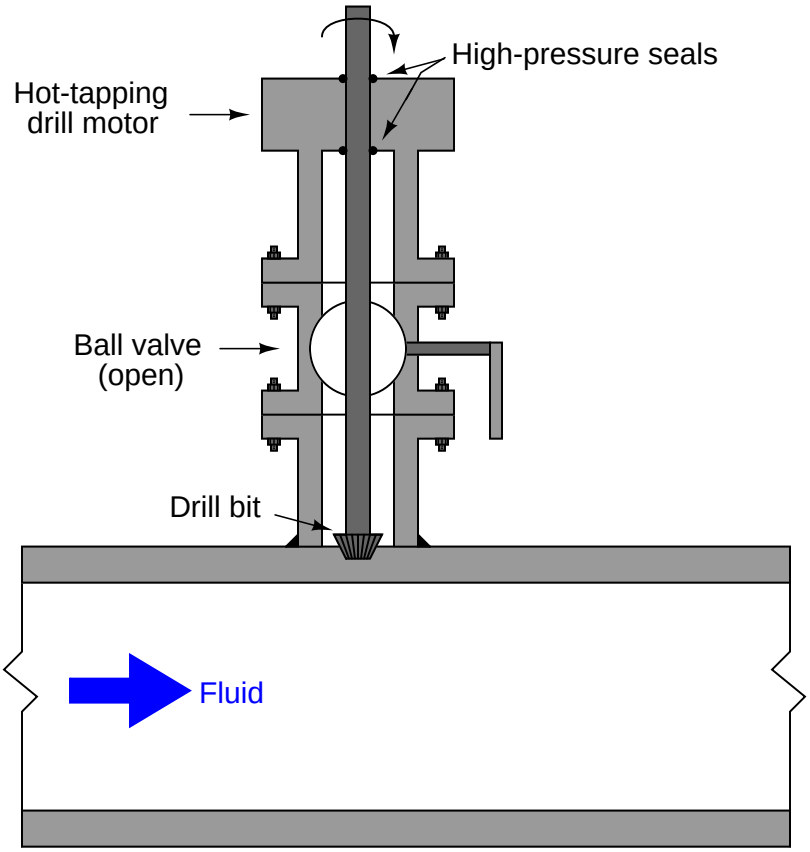
A unique advantage of insertion instruments is that they may be installed in an operating pipe by using specialized *hot-tapping* equipment. A "hot tap" is a procedure whereby a safe penetration is made into a pipe while the pipe is carrying fluid under pressure. The first step in a hot-tapping operation is to weld a "saddle tee" fitting on the side of the pipe:



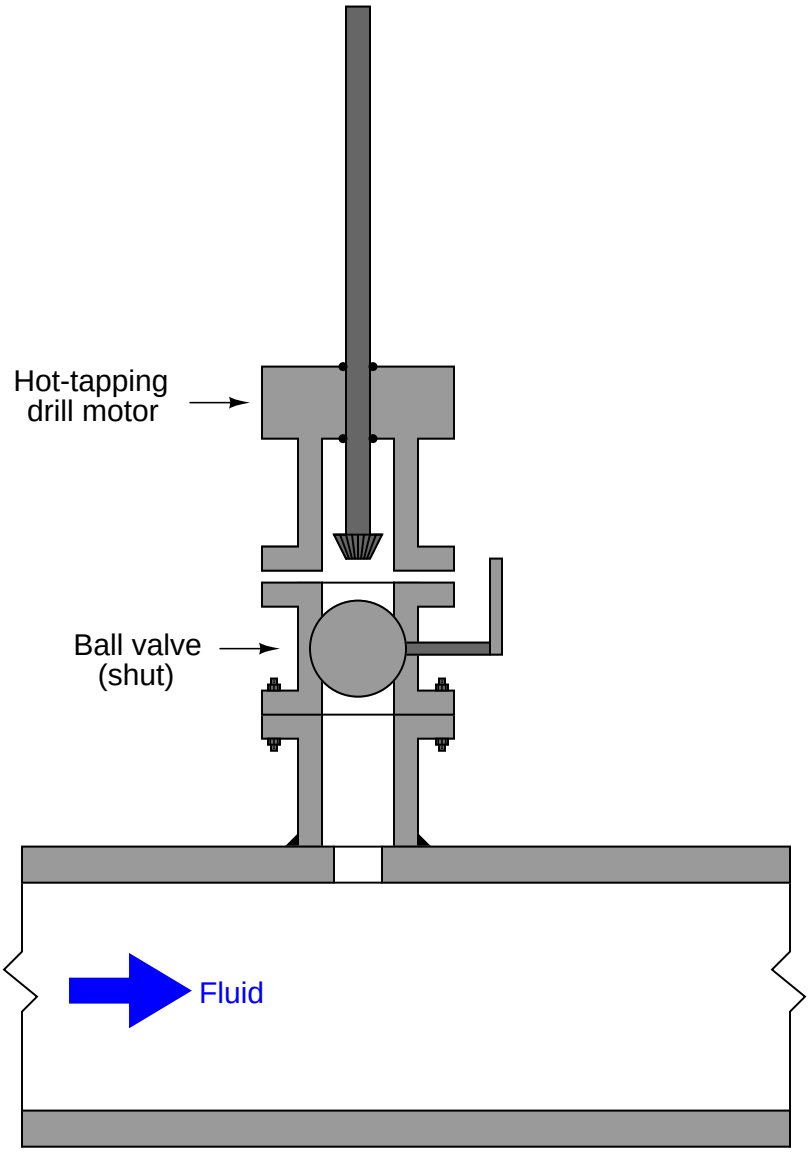
Next, a ball valve is bolted onto the saddle tee flange. This ball valve will be used to isolate the insertion instrument from the fluid pressure inside the pipe:



A special hot-tapping drill is then bolted to the open end of the ball valve. This drill uses a high-pressure seal to contain fluid pressure inside the drill chamber as a motor spins the drill bit. The ball valve is opened, then the drill bit is advanced toward the pipe wall where it cuts a hole into the pipe. Fluid pressure rushes into the empty chamber of the ball valve and hot-tapping drill as soon as the pipe wall is breached:



Once the hole has been completely drilled, the bit is extracted and the ball valve shut to allow removal of the hot-tapping drill:



Now there is a flanged and isolated connection into the "hot" pipe, through which an insertion flowmeter (or other instrument/device) may be installed.

Hot-tapping is a technical skill, with many safety concerns specific to different process fluids, pipe types, and process applications. This brief introduction to the technique is not intended to be instructional, but merely informational.

22.11 Process/instrument suitability

Every flow-measuring instrument exploits a physical principle to measure the flow rate of fluid stream. Understanding each of these principles as they apply to different flow-measurement technologies is the first and most important step in properly applying a suitable technology to the measurement of a particular process stream flow rate. The following table lists the specific operating principles exploited by different flow measurement technologies:

Flow measurement technology	**Operating principle**	**Linearity**	**2-way flow**
Differential pressure	Fluid mass self-acceleration, potential-kinetic energy exchange	$\sqrt{\Delta P}$	(some)
Laminar	Viscous fluid friction	linear	yes
Weirs & flumes	Fluid mass self-acceleration, potential-kinetic energy exchange	H^n	no
Turbine (velocity)	Fluid velocity spinning a vaned wheel	linear	yes
Vortex	von Kármán effect	linear	no
Magnetic	Electromagnetic induction	linear	yes
Ultrasonic	Sound wave time-of-flight	linear	yes
Coriolis	Fluid inertia, Coriolis effect	linear	yes
Turbine (mass)	Fluid inertia	linear	(some)
Thermal	Convective cooling, specific heat of fluid	linear	no
Positive displacement	Movement of fixed volumes	linear	(some)

A potentially important factor in choosing an appropriate flowmeter technology is energy loss caused by pressure drop. Some flowmeter designs, such as the common orifice plate, are inexpensive to install but carry a high price in terms of the energy lost in *permanent pressure drop* (the total, non-recoverable loss in pressure from the inlet of the device to the outlet, not the temporary pressure difference between inlet and vena contracta). Energy costs money, and so industrial facilities would be wise to consider the long-term cost of a flowmeter before settling on the one that is cheapest to install. It could very well be, for example, that an expensive venturi tube will cost less after years of operation than a cheap orifice plate[80].

In this regard, certain flowmeters stand above the rest: those with obstructionless flowtubes. Magnetic and ultrasonic flowmeters have no obstructions whatsoever in the path of the flow. This translates to (nearly) zero permanent pressure loss along the length of the tube, and therefore. Thermal mass and straight-tube Coriolis flowmeters are nearly obstructionless, while vortex and turbine meters are only slightly worse.

[80] This is not always the case, as primary elements are often found on throttled process lines. In such cases where a control valve normally throttles the flow rate, any energy dissipated by the orifice plate is simply less energy that the valve would otherwise be required to dissipate. Therefore, the presence or absence of an orifice plate has no net impact on energy dissipation when used on a process flow throttled by a control valve, and therefore does not affect cost over time due to energy loss.

22.12 Review of fundamental principles

Shown here is a partial listing of principles applied in the subject matter of this chapter, given for the purpose of expanding the reader's view of this chapter's concepts and of their general interrelationships with concepts elsewhere in the book. Your abilities as a problem-solver and as a life-long learner will be greatly enhanced by mastering the applications of these principles to a wide variety of topics, the more varied the better.

- **Basic geometrical quantities**: *distance* and *velocity* are both one-dimensional measurements. *Area* is a two-dimensional measurement. *Volume* is a three-dimensional measurement. Relevant to unit conversions for flow measurements. For example, fluid velocity in ft/s, fluid pressure in pounds per ft^2, volumetric flow in ft^3/s.

- **Density**: the ratio of mass to volume for a particular substance. Relevant to the kinetic energy of a moving fluid, as well as true-mass flow measurement.

- **Viscosity**: the resistance of a fluid to *shear*, which may be thought of as the "internal friction" of that fluid. Relevant to whether a fluid moves in *laminar* or *turbulent* fashion.

- **Laminar flow**: a condition where a fluid's molecules move in parallel paths, never crossing. Relevant to flowmeter selection, because many flowmeters require the flow regime to be turbulent rather than laminar (e.g. orifice plates, thermal mass). Also relevant to certain industrial processes such as mixing and heat transfer, because laminar flow impedes both these processes.

- **Turbulent flow**: a condition where a fluid's molecules move chaotically, randomly crossing pathways. Relevant to flowmeter selection, because many flowmeters require the flow regime to be turbulent rather than laminar (e.g. orifice plates, thermal mass). Also required for certain industrial processes such as mixing and heat transfer to efficiently occur.

- **Conservation of energy**: energy cannot be created or destroyed, only converted between different forms. Relevant to fluid velocities and pressures inside of flow elements such as venturi tubes, orifice plates, Pitot tubes, etc.

- **Bernoulli's equation**: $z_1 \rho g + \frac{v_1^2 \rho}{2} + P_1 = z_2 \rho g + \frac{v_2^2 \rho}{2} + P_2$, which is an application of the Law of Energy Conservation, stating that the sum of all forms of energy in a moving fluid stream (height, kinetic, and pressure) must remain the same. Relevant to calculations of pressure drop and pressure recovery across restrictions such as venturi tubes, orifice plates, etc.

- **Conservation of mass**: mass is an intrinsic property of matter, and as such cannot be created or destroyed. Relevant to the Continuity Principle for moving fluids, where the mass flow rate of a fluid entering a pipe must equal the mass flow rate exiting the pipe, assuming no accumulation or depletion (storage) of mass occurs within the pipe.

- **Reynolds number**: a unitless value representing the ratio of kinetic to viscous forces in a fluid. The greater the Reynolds number, the more turbulent the flow. The smaller the Reynolds number, the more likely the fluid will move in a *laminar* fashion. Relevant to many types of flowmeters, which operate accurately only within certain ranges of Reynolds number.

- **Flow profile or velocity profile**: the relative velocities of a fluid as it moves through a pipe, the velocity at the center being greater than the velocity at the pipe wall. Laminar flow is characterized by large differences in velocity along the profile, while turbulent flow exhibits a "flatter" profile with more consistent velocity across the pipe diameter. Relevant to insertion-type flowmeters such as Pitot tubes where the flowing velocity is sampled at only one point in the flowstream.

- **Ideal Gas Law**: $PV = nRT$, describing the relationship between gas pressure, chamber volume, gas quantity (in moles), and gas temperature. Relevant to measurements of gas flow rate at different pressures and temperatures (e.g. converting between "actual" or "flowing" units and "standard" units of gas measurement).

- **Newton's Second Law of motion**: $F = ma$, describing how the acceleration of an object is directly proportional to the amount of applied (resultant) force and inversely proportional to its mass. Relevant to the development of a pressure difference across a flow element where the fluid molecules must either accelerate (positive a) or decelerate (negative a).

- **Inverse mathematical functions**: an inverse function, when applied to the result of its counterpart function, "un-does" the operation and leaves you with the original quantity. Relevant to the application of "square-root" in DP-based flow measurements. The natural characteristic of an accelerating or decelerating flow element (e.g. orifice plate) is to generate a pressure drop proportional to the square of the flow rate. Therefore, we must "square-root" that pressure signal in order to infer flow rate.

- **Wavelength vs. frequency**: $v = \lambda f$, describing the relationship between wavelength (λ) and frequency (f) for a wave. Relevant to vortex-shedding flowmeters, where the frequency is directly proportional to the velocity of the fluid. Also relevant to ultrasonic flowmeters.

- **Time, velocity, and distance**: $x = vt$, describing the relationship between velocity (v), time of travel (t), and distance traveled (x). Relevant to ultrasonic flowmeters, where travel time of an ultrasonic sound wave is used to calculate fluid velocity.

- **Speed of sound through a substance**: varies directly with the bulk modulus of the substance and inversely with the mass density of the substance as described by the formula $c = \sqrt{\frac{B}{\rho}}$. Relevant to Doppler-style ultrasonic flowmeters, where the Doppler frequency shift depends on the speed of sound through the fluid.

- **Resonance**: when something oscillates at its natural frequency. Relevant to Coriolis flowmeters, whose tubes are made to vibrate at their resonant frequency in order to measure fluid density. For a vibrating string (which closely approximates a vibrating tube), resonant frequency is directly proportional to string tension and inversely proportional to both length and mass as described by the formula $f = \frac{1}{2L}\sqrt{\frac{F_T}{\mu}}$.

- **Specific heat**: the amount of heat necessary to change the temperature of a some substance per unit mass and per unit of temperature. Relevant to thermal flowmeters, which work on the principle of heat transfer from a heated object to the moving fluid. The greater the mass flow rate of the fluid, the greater the heat transfer rate for any given specific heat value of the fluid. Specific heat is a function of the fluid's chemical composition.

- **Scintillation**: the random and time-varying warping of light rays due to pockets of fluid having different refractive indices. Variations in temperature will cause this, as will turbulent motion of the fluid. Relevant to certain types of optical flowmeter, which pass light through a flowstream and look for patterns of scintillation after the light has traveled through the moving fluid.

- **Differentiation (calculus)**: where one variable is proportional to the rate-of-change of two others. Differentiation always results in a division (quotient) of units. Relevant to calculations of flow rate based on mass or volume. Volumetric flow rate (Q) is equal to the rate of change in fluid volume over time ($Q = \frac{dV}{dt}$). Mass flow rate (W) is equal to the rate of change in fluid mass over time ($W = \frac{dm}{dt}$)

- **Integration (calculus)**: where one variable is proportional to the accumulation of the product of two others. Integration always results in a multiplication of units. Relevant to calculations of mass or volume based on flow rate. Total volume of fluid passed by a point in a pipe (V) equal to the integral of volumetric flow rate times time: $V = \int Q\, dt$. Total mass of fluid passed by a point in a pipe (m) equal to the integral of mass flow rate times time: $m = \int W\, dt$.

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